United Nations Development Programme

<table>
<tr><td colspan="3">Project title: Facilitating Cleaner and Energy Efficient Phosphate Chemicals Industry in China (PhosChemEE) Project</td></tr>
<tr><td>Country(ies): People's Republic of China</td><td>Implementing Partner (GEF Executing Entity): In China – Ministry of Industry and Information Technology (MIIT)</td><td>Execution Modality: Full National Implementation Modality (NIM)</td></tr>
<tr><td colspan="3">Contributing Outcome (UNDAF/CPD, RPD, GPD): UN Development Assistance Framework China (2016-2020): More people enjoy a cleaner, healthier, and safer environment as a result of improved environmental protection and sustainable green growth (Outcome 2).
UNDP Strategic Plan Output: Output 1.5.1: Solutions adopted to achieve universal access to clean, affordable and sustainable energy. Output 2.5.1: Solutions developed, financed, and applied at scale for energy efficiency and transformation to clean energy and zero-carbon development, for poverty eradication and structural transformation.</td></tr>
<tr><td colspan="2">UNDP Social and Environmental Screening Category: High</td><td>UNDP Gender Marker: GEN2</td></tr>
<tr><td colspan="2">Atlas Award ID: 00113640</td><td>Atlas Project/Output ID: 00111713</td></tr>
<tr><td colspan="2">UNDP-GEF PIMS ID number: 6618</td><td>GEF Project ID number: 10722</td></tr>
<tr><td colspan="3">LPAC meeting date: 15 January 2022</td></tr>
<tr><td colspan="3">Last possible date to submit to GEF: 11 December 2021</td></tr>
<tr><td colspan="3">Latest possible CEO endorsement date: 11 June 2022</td></tr>
<tr><td colspan="3">Project duration in months: 60</td></tr>
<tr><td colspan="2">Planned start date: 1 July 2022</td><td>Planned end date: 30 June 2027</td></tr>
<tr><td colspan="2">Expected date of Mid-Term Review: 1 October 2024</td><td>Expected date of Terminal evaluation: 31 March 2027</td></tr>
<tr><td colspan="3">Brief project description: The objective of the Facilitating Cleaner and Energy Efficient Phosphate Chemicals Industry in China (PhosChemEE) Project is the facilitation of the extensive application of low carbon and energy efficient technologies in the phosphate chemicals industry (PCI) in China. The strategy to achieve this objective is to remove the barriers that hinder the promotion and deployment of these technologies, including applicable energy efficient techniques, measures and practices in cleaner production in the 3 sub-sectors of the PCI in China, namely: (1) Phosphate ore mining and refining; (2) Phosphate chemicals production; and (3) Processing of by-products and waste from phosphate mining/refining, and phosphate chemicals production. The project is comprised of three components, one for each sub-sector. Each project component is comprised of activities addressing</td></tr>
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the various barriers (i.e., policy/regulatory, institutional, technical, financial, information and awareness) to the application of low carbon and energy efficient technologies in each sub-sector of the Chinese PCI. The achievement of the project objective will contribute to the achievement of China's climate change mitigation targets and the achievement of the United Nations 2030 Sustainable Development Goals (SDGs), specifically SDG 3 "Good health and well-being"; SDG – 7 "Sustainable Energy"; SDG 9 "Industry, Innovation and Infrastructure"; and SDG 13 "Climate Action".	
FINANCING PLAN	
GEF Trust Fund grant	USD 9,343,379
(1) Total Budget administered by UNDP	**USD 9,343,379**
CO-FINANCIERS THAT WILL DELIVER PROJECT RESULTS INCLUDED IN THE PROJECT RESULTS FRAMEWORK (FUNDS NOT ADMINISTERED THROUGH UNDP ACCOUNTS)	
Ministry of Industry and Information Technology (MIIT) (grant)	USD 3,000,000
MIIT (in-kind)	USD 900,000
Ministry of Natural Resources (MNR) (grant)	USD 2,000,000
MNR (in-kind)	USD 650,000
Guizhou Provincial Department of Industry and Information Technology (grant)	USD 1,590,000
Guizhou Provincial Department of Industry and Information Technology (in-kind)	USD 397,500
Hubei Provincial Department of Economy and Information Technology (grant)	USD 1,590,000
Hubei Provincial Department of Economy and Information Technology (in-kind)	USD 397,500
Sichuan Provincial Department of Economy and Information Technology (grant)	USD 1,590,000
Sichuan Provincial Department of Economy and Information Technology (in-kind)	USD 397,500
Yunnan Provincial Department of Economy and Information Technology (grant)	USD 1,590,000
Yunnan Provincial Department of Economy and Information Technology (in-kind)	USD 397,500
Guizhou Provincial Department of Natural Resources (grant)	USD 1,272,000
Guizhou Provincial Department of Natural Resources (in-kind)	USD 318,000
Yunnan Provincial Department of Natural Resources (grant)	USD 1,272,000
Yunnan Provincial Department of Natural Resources (in-kind)	USD 318,000
Sichuan Provincial Department of Natural Resources (grant)	USD 1,696,000
Sichuan Provincial Department of Natural Resources (in-kind)	USD 424,000
Private sector (grant and in-kind)	USD 77,763,687
UNDP (in-kind)	USD 200,000

(2) Total confirmed co-financing		**USD 97,763,687**
(3) Grand-Total Project Financing (1)+(2)		**USD 107,107,066**
SIGNATURES:		
Signature: print name below	**Agreed by Government Development Coordination Authority**	**Date/Month/Year:** *within 6 months of GEF CEO endorsement*
Signature: print name below	**Agreed by Implementing Partner**	**Date/Month/Year:** *within 6 months of GEF CEO endorsement*
Signature: print name below	**Agreed by UNDP**	**Date/Month/Year:** *within 6 months of GEF CEO endorsement*

~~**Key GEF Project Cycle Milestones:**~~

~~**Project document signature**: within 25 days of GEF CEO endorsement~~
~~**First disbursement date**: within 40 days of GEF CEO endorsement~~
~~**Inception workshop date**: within 60 days of GEF CEO endorsement~~
~~**Operational closure:** expected end date as per the approved duration after Project Document signature~~
~~**Financial closure:** within 6 months of operational closure~~

List of Abbreviations & Acronyms

ASEAN	Association of South East Asian Nations
ACWF	All China Women's Federation
BAU	Business As Usual
BPPS	Bureau for Policy and Programme Support
CO	Country Office
CO_2	Carbon Dioxide
CO_{2eq}	Carbon Dioxide Equivalent
CTA	Chief Technical Advisor
EE	Energy Efficiency
ERC	Evaluation Resource Center
ESIA	Environmental and Social Impact Assessment
ESMF	Environmental Social Management Framework
ESMP	Environmental Social Management Plan
GDP	Gross Domestic Product
GEF	Global Environment Facility
GEFSEC	Global Environment Facility Secretariat
GHG	Greenhouse Gas
GOC	Government of China
IRR	Implementing Rules and Regulations
LC	Low Carbon
LCA	Life Cycle Analysis
LFA	Logical Framework Analysis
MEE	Ministry of Ecology and Environment
MIIT	Ministry of Industry and Information Technology
MNR	Ministry of Natural Resources
MOF	Ministry of Finance
MRV	Monitoring Reporting and Verification
Mton	Million tons
MTR	Mid-Term Review
MVR	Mechanical Vapor Recompression
M&E	Monitoring and Evaluation
NDC	Nationally Determined Contributions
NDRC	National Development and Reform Commission
NGO	Non-Governmental Organization
NIM	National Implementation Modality
PCI	Phosphate Chemicals Industry
PIR	Project Implementation Report
PMO	Project Management Office
PMU	Project Management Unit
POPP	Programme and Operations Policies and Procedures
PPG	Project Preparation Grant
RE	Renewable Energy
RMB	Ren Min Bi

RTA	Regional Technical Advisor
R&D	Research & Development
SDGs	Sustainable Development Goals
SEC	Specific Energy Consumption
SESP	Social and Environmental Screening Procedure
SME	Small and Medium Enterprise
tce	Tons of Coal Equivalent
tCO_2	Tons of Carbon Dioxide
TE	Terminal Evaluation
ToC	Theory of Change
TOR	Terms of Reference
UNFCCC	United Nations Framework Convention on Climate Change
UNDP	United Nations Development Programme
USD	United States Dollar
USGS	United States Geological Survey
WPA	Wet-process Phosphoric Acid

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II. DEVELOPMENT CHALLENGE

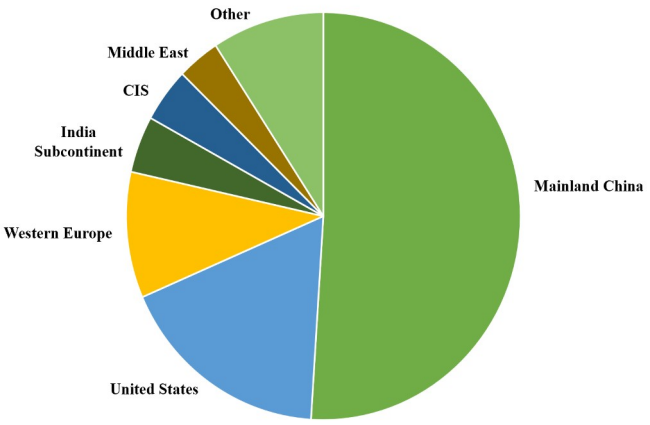
Development Challenge and Relevance to National Development Priorities, Global Environment and Sustainable Development Goals (SDGs)

Phosphorus is essential to life and the functioning of our society. Phosphorus is present in all living things and is required for cell growth. Together with nitrogen and potassium, phosphorus is one of the most important elements for plant life. In agriculture, the growth of crops is dependent on the phosphorus content of soil. There is no substitute for phosphorus and there will never be. Soil is often depleted of phosphorus by the plants that use it for their growth, or it is partially washed away by rain. Modern farming is therefore reliant on phosphate-derived fertilizers to enrich the soil in order to produce sufficient quantities of food. About 90% of global phosphate rock demand is used in the food production chain.

Phosphate rocks are used as raw material for manufacturing phosphoric acid, phosphate fertilizers and other phosphate chemicals. Phosphate rocks can be used directly in some applications, whereas beneficiation is required to upgrade the P_2O_5 ore to a concentration higher than 29% when used as a raw material for the production of fertilizers, phosphoric acid and many other phosphate compounds.

From 2015 to 2019, the industrial phosphates market experienced steady growth, but the recession resulting from the COVID-19 pandemic led to minimal growth in 2020, with some regions experiencing declines as large as 8%, also depending on the end-use applications. China is the largest consumer of phosphorus products, accounting for nearly half of the global market (Figure 1). Many of the markets for industrial phosphates were less-affected by the downturn, or have even grown year-over-year in 2020, including food and beverages, detergents, specialty agricultural products, and pharmaceuticals. Overall growth for industrial phosphates is expected to resume worldwide in line with average gross domestic product growth between 2020 and 2025.

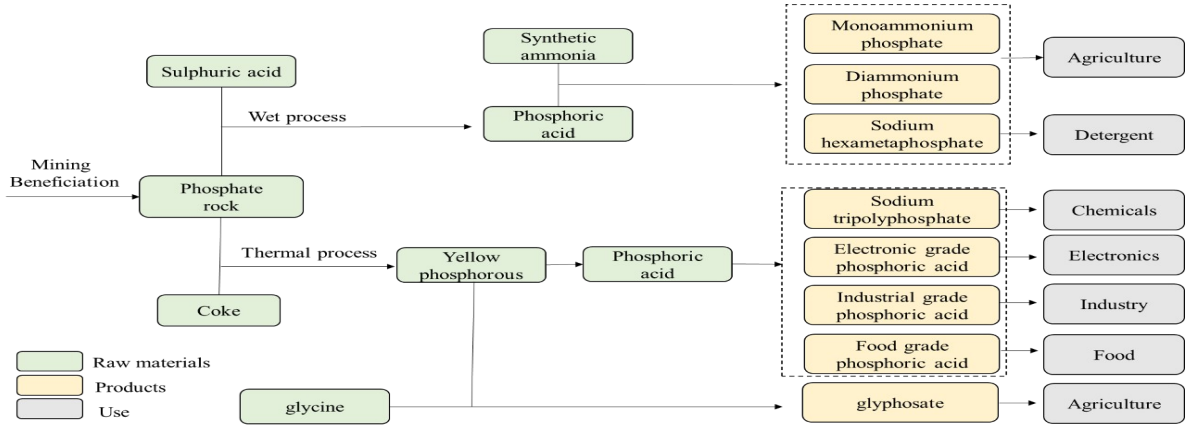
Figure 1. Phosphorus Products Consumption in the World



The phosphate chemical industry (PCI) is an important sub-sector of China's industry sector. Figure 2 shows the flowchart of the PCI chain in China. At present, the PCI is a relatively mature industry, which covers phosphate mining and refining and the production of phosphate chemical products. There are about 100 different phosphate chemical products that are produced by the industry, which can basically meet the domestic demand, and a large portion of the production is exported. The current trend of phosrock production and phosphate fertilizer consumption in China are expected to continue to grow at

the same rates in the past 5 years and are expected to peak during the period 2020-2025 where the demand for phosrock is estimated to reach 100 million tons/year.

Figure 2. Flowchart of PCI in China



The PCI in China faces several challenges in the process in its three main sub-sectors including phosphate mining, chemical production process, and utilization of the industry by-products. The carbon emission in the PCI in China is manifested in three aspects, which are also the key problems to be solved in this project.

Phosphate rock mining is facing severe challenges because of its ecological environment impact. China's phosphate rock resource purity is poor, and consequently its extraction and utilization requires high energy consumption, and the resource utilization rate is low. Collophanite is the main phosphate rock in China. This kind of resource is produced in layers, which has the characteristics of thin ore body, low grade of raw ore, high impurity content and difficult separation. Moreover, the phosrock mining areas are typically large, the land surface damage is serious, and the comprehensive utilization rate of waste rock and tailings is low. By end 2019, the phosrock mines in Yunnan, Guizhou, and Sichuan discharged 104.7 million tons of waste rock, and about 41.4 million tons of that will be used, with an average utilization rate of 39.5%. Large quantities of waste rocks and tailings are just stacked in the waste rock dump and tailings in tailings ponds for long periods of a long time.

Phosphate chemicals production is facing severe challenges because of high energy consumption and CO_2 emission. In terms of production capacity and output, China is the largest producer, exporter, and consumer of phosphate chemical products in the world, accounting for more than 80% of the world's total production capacity. In 2019, the national phosphate fertilizer production capacity was 22.5 million tons of P_2O_5. In 2019, the carbon emission for the production of phosphate chemicals was about 59 million tons CO_{2eq}, accounting for about 4.3% of the carbon emission of the chemical industry. The extensive production mode of the PCI causes a lot of energy waste.

The utilization of the industry by-products, particularly phosphogypsum, is insufficient. At present, the annual output of phosphogypsum in China is about 85 million tons. The comprehensive utilization rate of this byproduct is about 40%, and the annual additional storage of phosphogypsum is 51 million tons. The historical storage of phosphogypsum in China is more than 800 million tons, and the problems of environmental pressure and safety risks are increasingly prominent. Among them, nearly 650 million tons of phosphogypsum piles exist along the main stream and upstream tributaries of the Yangtze River in the provinces of Hubei, Yunnan, Guizhou, Sichuan, etc., which is the key and difficult problem in the ecological

environment restoration of the Yangtze River economic belt. In addition, the low quality of phosphate concentrate leads to high energy consumption in the downstream phosphoric acid production, and the disposal of by-product phosphogypsum is difficult. Phosphate rock needs to be enriched to obtain the proper phosphate concentrate before it can be used to produce phosphate fertilizer. At present, the grade of phosphate concentrate in China is about 28-30%, and the main impurity is silica. In other countries, phosphate concentrate is mainly extracted from apatite, and the highest grade of phosphate concentrate can reach 40%. When the grade of phosphate concentrate is increased by one percentage point, the consumption of sulfuric acid and phosphogypsum production can be reduced by 17.2 kg and 10.0 kg, respectively.

In this context, the green development of the PCI in China is based on higher penetration of low-carbon technologies in the three sub-sectors mentioned above. This shift towards green development in PCI the will contribute to the achievement of China's National Determined Contributions (NDCs) targets under the Paris Agreement.

The implementation of PhosChemEE project will contribute to the achievement of several Sustainable Development Goals (SDGs) as set by the United Nations General Assembly, specifically: SDG 7: “Ensure access to affordable, reliable, sustainable and modern energy for all”, SDG 9: " Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation"; SDG 12: “Ensure sustainable consumption and production patterns”; SDG 13: "Take urgent action to combat climate change and its impacts by regulating emissions and promoting developments in renewable energy “, and to a lesser extent SGD 8: “Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all”.

Barrier Analysis

The approach taken for the design of the PhosChemEE project is barrier removal. The barriers hindering the transition towards a cleaner and more energy efficient phosphate chemical industry in China have been identified in the PhosChemEE Project Information Form (PIF). The Logical Framework Analysis (LFA) Workshop was held virtually, because of the travel restrictions due to the outspread of the Covid-19 pandemic, in two sessions, on March 5, 2021 and on March 25-26, 2021, respectively. During the workshop the participant stakeholders have reassessed the validity of the identified barriers. The major identified barriers include:

Policy Barriers: ***The industry standards, systems and related policies are not sufficient.*** Like in any mineral resource development in China, the PCI involves many processes and operations such as phosrock extraction, enrichment/refining, phosphate product (chemicals) manufacturing, and waste management (e.g., recycling, reprocessing, reuse). The relatively low quality phosrock leads to higher energy consumption, and larger amounts of waste products, as compared to other countries where the phosrock quality is better. The waste produced in the downstream production processes is more difficult to use, resulting in large amounts of stacking, which poses a direct threat to the natural environment in the phosrock mining areas and in the phosphate, chemicals manufacturing enterprises. There are some standards for the utilization of the phosphogypsum by-product, but these are only limited to the building materials industry. Because of that, the market recognition for the large scale applications of this by-product as a low carbon building material is low.

Technology Barriers: ***Technology innovations in the PCI in China are insufficient.*** At present, there is a lack of technological innovation in improving the operations and processes in the PCI in China. There are

limited innovation efforts geared towards technologies on the phosrock resource development and utilization technology, including research and application of new, more energy efficient and less polluting processes, systems and equipment for phosrock mining and refining that reduce the discharge of solid waste such as waste rock and tailings, reduce energy consumption in the mining operations, improve the quality of refined phosrock and reduce the energy consumption and waste generation in the phosphate chemicals production. There are also limited applications of cleaner production technologies and by-product treatment technologies in the PCI. As the disposal of phosphogypsum has not been given enough attention, the current phosphogypsum disposal and utilization rates are relatively low. The current technologies that are applied do not help in further exploiting the high value-added utilization of the PCI's phosphogypsum by-product while at the same time reducing the environment risks that the ever-growing stacks of phosphogypsum poses to the surrounding areas of PCI enterprises. Technological innovations must be carried out by the industry in areas such as on-line quality control technology of phosphogypsum production and discharge, and pretreatment technology of low-cost secondary pollution.

Infrastructure Barriers: ***Inadequate transportation systems between the different phosphate products producing regions and the phosphate products consuming regions in China***. Due to the geographical distribution of regional phosrock resources and transportation requirements, the production, storage, and comprehensive utilization of phosphogypsum vary greatly in different regions. The discharge and storage of phosphogypsum are mainly concentrated in Yunnan, Guizhou, Sichuan, and parts of the southwest region of the country, while the consumption of gypsum is mainly concentrated in areas with large demand for building gypsum and cement, such as the eastern provinces and cities with developed economy. Due to the existing transportation infrastructure, the supply of gypsum in areas with large amount of gypsum is in short supply, while the area with concentrated emission can only be stored in large quantities. The obstacles due to the spatial mismatch between the areas where there is demand for gypsum and areas where there is gypsum production (from phosphogypsum) is worsened by insufficient coordination and promotion efforts to devise and implement a comprehensive phosphogypsum utilization policy among the different regional governments.

Financial Barriers: ***The existing extensive development mode of PCI chain.*** At present, China's phosphate chemical industry is geared towards extensive production in pursuit of economic benefits. Most enterprises within the industry do not take reducing carbon emissions and achieving cleaner production as their social responsibility, but to maximize their economic benefits while complying with industry regulations. This general attitude of enterprises pursuing short-term interests makes it difficult to implement and realize green and low-carbon development objectives. The financial barriers are coupled and exacerbated by a concurrent lack of awareness and inadequate capacity of the PCI companies' management teams concerning environmental issues and sustainable production.

The abovementioned barriers, if not properly and adequately addressed/removed will continue to prevent the transition towards a cleaner and more energy efficient PCI in China.

Business-As-Usual (BAU) Scenario

China possesses about 4.5% of the world's 71 billion tons total proven phosphate reserves, based on 2019 data[1], trailing by a long distance Morocco that has over 70% of the total phosphate resources. However, the situation completely reverses when it comes to phosphate production, with China firmly ranking at

[1] https://pubs.usgs.gov/periodicals/mcs2021/mcs2021-phosphate.pdf

number one with an annual production of 95 Mton of phosphate in 2019, or nearly 42% of the total, which is about 2.7 times larger than Morocco's production of 35 Mton ranked at number two[2].

Since China's average phosphate reserves have a low quality in terms of phosphate concentration compared to other countries, it requires to mining a larger quantity of phosrock to obtaining a ton of product, which also produces more by-products, such as phosphogypsum, as well as more waste, especially rock tailings. Currently only 40% of the 85 million tons of phosphogypsum produced annually is utilized, with the balance simply disposed of in piles, which have reached a total accumulated storage of over 800 million tons posing serious environmental hazards to the land and water basins and wasting valuable resources. Similarly, waste tailings are not recycled or reused, and they are largely discharged untreated in open landfills with the risk of polluting not only the air, but also the soil and water resources. In addition, the mined phosrock requires higher energy consumption per unit of product, and consequently cause larger GHG emissions, when compared to developed countries, because it needs to be enriched to achieve the proper concentration required for most phosphate chemicals production. Therefore, China's sizable phosphate production combined with the large energy consumption, GHG emissions, and by-products and waste generation per unit of product, assign an extremely high priority to the greening of the entire PCI value chain.

In 2016 the Government of China (GOC) has submitted its Nationally Determined Contributions (NDC) to the UNFCCC, which contains several climate change mitigation targets for 2030, and specifically:

1. Peaking CO_2 emissions by no later than 2030.
2. Lower CO_2 emissions per unit of gross domestic product (GDP) by 60-65% compared to the 2005 levels.
3. Increase the share of non-fossil fuels in primary energy consumption to approximately 20% (in 2019, the share of non-fossil fuel energy consumption was about 14.9%, including 2.2% of nuclear energy[3]).
4. Improve the CO_2 sink potential by increasing the forest stock volume by around 4.5 billion m^3 compared to the 2005 levels.

Although the GOC has not submitted its updated NDC yet, in his 2020 address to the United Nations General Assembly, President Xi has pledged that China will achieve carbon neutrality by 2060. The current NDC does not contain specific targets for the PCI, but it does set qualitative energy efficiency and low-carbon objectives for the entire industrial system, most notably:

a. Develop a circular economy while accelerating the elimination of inefficient production capacity.
b. Effectively control emissions from key industrial sectors, including the chemical industries.
c. Promote and increase recycling in the industrial system as well as utilization of renewable resources.
d. Promote low-carbon development of industrial sectors.

The pace and scale of initiatives devoted to the achievement of these goals have not reached yet a satisfactory level and more efforts are required to drift away from the current BAU scenario. The holistic approach proposed in this project, which starts from improvements of the policy and regulatory

[2] The same USGS (U.S. Geological Survey) document from the previous footnote reports preliminary data for 2020 and it show that although still firmly ranked number one in the world, China's phosphate production declined to 90 Mton largely due to the impact of the COVID-19 pandemic.

[3] https://www.eia.gov/international/analysis/country/CHN

frameworks, the strengthening of the technical capacity of key stakeholders and includes the demonstration of state-of-the-art energy efficient and low-carbon technologies will trigger a transformational shift towards a green and environmentally sustainable phosphate chemical industry.

Baseline Initiatives

The GOC as well as the private sector have embarked in several projects and initiatives aligned with China's objective of a cleaner and more energy efficient PCI. Table 1 summarizes the projects and programs that will be the baseline of the activities designed under this proposed PhosChemEE Project. These baseline projects and programs not only have been already approved and budgeted, but their execution is either ongoing or planned and they will be completed over the PhosChemEE Project implementation period.

Table 1. Ongoing and Planned Baseline Projects and Initiatives

Baseline Project/Program	Description	Available Budget, USD
MIIT: Green, low-carbon technology development and large-scale applications	The Ministry of Industry and Information Technology (MIIT) of the People's Republic of China attaches great importance to the promotion and implementation of green development in the phosphorus chemical industry. In recent years, funds have been invested in projects supporting green manufacturing. In the "14th five-year plan for Industrial Green Development", the GOC has devised a road map leading to the peaking of carbon emissions in the industrial system, promoting the reduction, intensification and greening of the chemical industry, including the PCI. The plan also aims at accelerating the comprehensive utilization of industrial solid waste on a large scale, such as tailings, and industrial by-products like phosphogypsum, particularly in the middle and upper reaches of the Yangtze River Economic Belt. The MIIT will support this green, low-carbon transformation of the PCI by providing an annual investment of USD 780,000 for a total of USD 3,900,000 over the five years of PhosChemEE implementation period. The activities that will be designed under this government program will include the introduction of relevant policies, and promotion and demonstration of key clean technologies. The proposed PhosChemEE project will also cooperate in the formulation of relevant policies, technical specifications, and demonstration of key low-carbon technologies to support the green transformation of the phosphorus chemical industry. **Timeline of Implementation: 2022-2027**	3,900,000
Hubei Province: Phosphate chemicals production optimization and comprehensive utilization of phosphogypsum	Hubei province attributes great importance to the green and low-carbon development of the phosphorus chemical industry. In particular great efforts have been made to optimize the manufacturing of phosphorus chemical products and the province is targeting, by 2025, the achievement of a world renowned phosphate industrial cluster with a market share of USD 50 billion (RMB 300 billion). Additionally, Hubei's efforts are focusing on accelerating the comprehensive utilization of phosphogypsum resources and the province is aiming to achieve a balance between phosphogypsum production and consumption by the end of 2025. Hubei province will disburse USD 1,987,500 over the next five years to support a number of low-carbon comprehensive phosphogypsum utilization projects as well as the construction of harmless phosphogypsum treatment	1,987,500

Baseline Project/Program	Description	Available Budget, USD
	facilities, other than that, Yichang, Xiangyang, Jingmen, Huanggang, Jingzhou, Xiaogan and other cities and towns will invest additional budget in this regard to jointly reach the target of 25 million tons of phosphogypsum utilization. **Timeline of Implementation: 2022-2027**	
Yunnan Province: Develop Green and low-carbon technology in phosphate chemical industry	In the "14th five-year development plan for the petrochemical and chemical industry", Yunnan Province is proposing to improve safety in the phosphorus chemical industry, strengthen environmental protection, and promote green and low-carbon development of chemical enterprises and industrial parks. Over the next five years Yunnan will support technological research and development demonstration projects by investing a total budget of USD 1,987,500. Manufacturers of phosphorus compound fertilizers and fine phosphorous chemicals will be encouraged by the local government of Yunnan to develop and implement new energy-saving technologies, as well as to adopt technologies that will allow to providing a more effective protection of the environment. These initiatives will include the demonstration of these green, low-carbon technologies for the advancement of the entire provincial phosphorus chemical industry. ***Timeline of Implementation: 2022-2027***	1,987,500
Guizhou Province: Formulation of green industry standards, and R&D of key low-carbon technologies	Guizhou Province is planning to: 1) strengthen the standardization of the specialty chemical industry, including the phosphorus chemical industry; 2) support the formulation of standards for new modern chemical and phosphogypsum products; and 3) promote the design and implementation of evaluation standards for green enterprises and green industrial parks. Concomitantly, significant efforts will be made to develop high-efficiency fertilizers suitable for modern agricultural development. The introduction of new standards for phosphogypsum products will be integrated with research and development activities concerning the application of key low-carbon technologies to produce high-purity phosphoric acid and improve the efficient utilization of phosphogypsum produced with the wet process phosphoric acid (WPA) method. Over the next five years, the provincial government of Guizhou will contribute a total of USD 1,987,500 co-financing for this green, low-carbon PCI development. ***Timeline of Implementation: 2022-2027***	1,987,500
Sichuan Province: Support green, low-carbon technological transformation of phosphorus chemical industry	Sichuan Province has integrated the greening of the chemical industry into one of the 16 key areas of the province's "5 + 1" modern industrial system, including the accelerated development of the phosphate chemicals manufacturing. The initiative aims at upgrading traditional enterprises and building green chemical industrial parks. To support these efforts, Sichuan Province will provide a budget of USD 1,987,500 over the next five years. The financed initiatives will include the implementation of projects and R&D activities in the following areas: a) energy savings and GHG emissions reduction; b) clean production; c) comprehensive utilization of resources; and d) other technical innovation projects. ***Timeline of Implementation: 2022-2027***	1,987,500

Baseline Project/Program	Description	Available Budget, USD
MNR and Provincial DNRs: Green, Low-carbon technology development and large-scale application	The Ministry of Natural Resources (MNR) of the People's Republic of China and provincial local governments are the promoters of green mine construction[4]. A total of USD 7,950,000 will be invested over the next five years for green phosphorus mine construction. This budget includes USD 2,650,000 provided by MNR and USD 5,300,000 funded by the local governments of the participating provinces. These activities will be implemented in the next five years will be the baseline for PhosChemEE project activities, including the introduction of relevant policies, demonstration projects, and key technology promotion. The proposed PhosChemEE project will also cooperate in the formulation of relevant policies, technical specifications, and demonstration of key low-carbon technologies to support the green transformation of the phosphorus chemical industry. **Timeline of Implementation: 2022-2027**	7,950,000
Private Sector: develop green, low-carbon phosphate mining & refining, phosphate chemicals production, and waste management demo projects	The private sector will participate in the design and demonstration of green, energy-saving and low-carbon technology projects in the entire PCI value chain, namely: a) phosrock mining and refining; b) phosphate chemicals production; and c) waste management. The demo projects, which are annexed to this project proposal, will include: construction, installation and operation of equipment and facilities to showcase green, low-carbon technology applications; and capacity building and training programs for key technical/design personnel and stakeholders, which will accelerate the promotion and development of a green PCI. **Timeline of Implementation: 2022-2027**	77,763,687

The successful completion of all the baseline projects and programs listed in Table 1 will bring China closer to enabling the extensive application of low carbon and energy efficient technologies in the phosphate chemicals industry in China and transitioning towards to achieving China dual-carbon target, but still they will not be enough to fully realize these goals. Improvements in the baseline efforts of the country are necessary. Such improvements call for the removal of the identified existing barriers. The removal of such barriers will not only achieve the desired transformation of the PCI development, but also contribute to the realization of the climate change mitigation NDC targets of the country.

III. STRATEGY

Applied Project Strategy and Incremental Activities

[4] Green mine refers to a mine that implements efficient mining in the whole process of mineral resources development, monitors and reduces the impact of the mining activities on the environment within a controllable range, and implements environmental ecology, scientific mining methods, efficient resource utilization, and digitization of management information, with minimal impact over the mining area community. The specific requirements include nine aspects: running mines according to law, standardized management, comprehensive utilization of resources, technological innovation, energy conservation and emission reduction, environmental protection, land reclamation, community harmony and corporate culture.

The PhosChemEE Project is designed based on a barrier removal approach. The proposed strategy calls for the design of *ad-hoc* incremental activities that by building on, and complementing, supplementing, or augmenting, the aforementioned baseline projects aim to remove all identified barriers. A barrier removal approach is the best available strategy because of its sustainability long after the implementation of the PhosChemEE project has been completed. The strategy is illustrated in the Theory of Change (ToC) diagram illustrated in Figure 3 below. The ToC visually show how the identified barriers lead to the main problem for China's PCI industry (yellow section of the figure), while the implementation of the PhosChemEE incremental activities will allow the removal of the barriers and the achievement of the main Project Objective: ***"Enabling the extensive application of low carbon and energy efficient technologies in the phosphate chemicals industry in China"*** (purple section of the figure). The following are the expected outcomes of the project and the proposed ways in which the barrier removal strategy will be carried out to realize them:

The value chain of the phosphate chemical industry in China is made of three main sub-sectors, namely: 1) the upstream phosphate mining and refining sub-sector; 2) the subsequent phosphate chemicals production sub-sector; and 3) the waste management sub-sector, which covers waste generated in both the previous two sub-sector. The project Log Frame is organized in three components, one for each sub-sector. Each of the three components is structured in a similar way and is made of two Outcomes, the first concerns provision of technical assistance while the second focuses on investment type activities. Therefore, the entire PhosChemEE project has a total of six Outcomes.

Component 1 – Green and Low Carbon Development and Operation of Phosphate Mines:

This project component is intended for the removal of barriers to the application of green, energy efficient low carbon technologies/techniques and practices in the mining and refining of phosphate ore. The delivery of the envisioned outputs from this project component will bring about two main outcomes: (1) Improved interest and commitment of the phosphate chemical industry in the green, low carbon and energy efficient operations of the phosphate mining sub-sector in China; and, (2) Enhanced confidence in the feasibility of the application of green, energy efficient low carbon technologies in phosphate ore mining and refining in China.

Outcome 1.1: Improved interest and commitment of the phosphate chemical industry in the green, low carbon and energy efficient operations of the phosphate mining sub-sector in China. This outcome is expected to be realized through the implementation of six sets of activities involving the: (1) Formulation and enforcement of policy and institutional frameworks supporting green, energy efficient low carbon development initiatives in phosphate ore mining and refining; (2) Formulation and enforcement of standards, policies and implementing rules and regulations on the promotion and practice of green, energy efficient, low carbon technologies/techniques, in the development and operations of phosphate mines; (3) Documentation of annual evaluation reports on the energy performance and environmental impacts of each demo activity and documented and disseminated demo results; (4) Design, conduct and post-evaluation of continuing education program in the PCI mining sub-sector in selected provinces, with regular capacity needs assessment in the area of green, energy efficient low carbon technologies applied in phosphate mining and refining; (5) Establishment of online monitoring, reporting and verification (MRV) system for energy-saving and GHG emission reduction from the application green, energy efficient low-carbon technologies in the PCI including information sharing scheme; and (6) Establishment and operationalization of an up-to-date information management and exchange system for all aspects of green, energy efficient low carbon phosphate mining and refining technologies and best practices.

Outcome 1.2: Enhanced confidence in the feasibility of the application of green, energy efficient low carbon technologies in phosrock mining and refining in China. This outcome is expected to be realized through the implementation of four sets of activities involving the: (1) Designing and planning of the demonstrations of green, energy efficient and low carbon technologies in phosphate mines development and operations in selected provinces; (2) Installation and operationalization of green, energy efficient low carbon technology application demos in phosphate ore mining and refining; (3) Establishment and operationalization of financial support schemes that are accessible to PCI mining companies to support their green, energy efficient and low carbon technology application initiatives; and (4) Approval of financed follow-up plan for the replication of the application of demonstrated green, energy efficient low carbon technologies in phosphate ore mining and refining in other localities.

The green, energy efficient and low carbon demos that will be implemented in the partner phosphate mining companies that mine and refine phosphate ore are the following:

- Demonstration 1.1: Application of Full Tailings Filling Technology in Underground Phosphate Mines
- Demonstration 1.2: Application of Underground Full-Bed Continuous Mining Technology in Phosphate Mines Using High Quality Mechanical Cutting Equipment
- Demonstration 1.3: Application of Phosphate Ore Pre-processing Sorting Technology for Discarding Tailings
- Demonstration 1.4: Improved Ore Flotation Efficiency through Automatic Control of Acid Mixtures used as Flotation Depressant

Component 2: Green and Low Carbon Design and Operation of Phosphate Chemicals Production Facilities

This component of the project will address the barriers to the application of green, energy efficient low carbon technologies/techniques and practices in phosphate chemicals production. This is comprised of activities that will deliver outputs that will collectively bring about two main outcomes: (1) Enhanced interest and commitment of the phosphate chemical industry in the green, low carbon and energy efficient production of phosphate chemicals in China; and, (2) Enhanced confidence in the feasibility of the application of green, energy efficient low carbon technologies in phosphate chemicals production in China.

Outcome 2.1: Enhanced adoption by the phosphate chemical industry of a green, low carbon and energy efficient development model in phosphate chemicals production. There are six sets of activities that will bring about this expected outcome and these involve the: (1) Formulation and enforcement of policy and institutional frameworks supporting green, energy efficient low carbon technology applications in phosphate chemicals production; (2) Formulation and enforcement of standards, policies and implementing rules and regulations on the promotion and practice of green, energy efficient, low carbon technologies/techniques, in phosphate chemicals production; (3) Documentation o annual evaluation reports on the energy performance and environmental impacts of each demo activity and documented and disseminated demo results based on the MRV system established under Component 1; (4) Design, conduct and post-evaluation of continuing education program in the PCI chemicals sub-sector in selected provinces, with regular capacity needs assessment in the area of green, energy efficient and low carbon technologies applied in phosphate chemicals manufacturing; (5) Conduct of a market analysis report on the international trade trends for phosrock and phosphate chemicals, and the indirect carbon emission and ecological impacts in the international trading of these products; and (6) Establishment and

operationalization of a system for managing and sharing available and accessible information on all aspects of green, energy efficient and cleaner phosphate chemicals production in China.

Outcome 2.2: Enhanced confidence in the feasibility of the application of green, energy efficient low carbon technologies in phosphate chemicals production in China. There are five sets of activities that will bring about this expected outcome and these involve the: (1) designing and planning of demonstrations of green, energy efficient low carbon technologies in phosphate chemicals production including: (a) Improved design for cleaner production demos; (b) Feasibility analyses of the application and operation of green, energy efficient and low carbon phosphate chemical production systems; and, (c) Implementation plans (including financing arrangements) for each green, energy efficient and low carbon technology application in the demo phosphate chemical companies; (2) Installation and operationalization of green, energy efficient low carbon technology application demos in phosphate chemicals production; (3) Design and approval of pilot financing scheme for small and medium enterprises (SMEs) that manufacture phosphate chemicals; (4) Operationalization of a pilot financing scheme for phosphate chemicals SMEs in 2 to 3 regions; and (5) Approval of financed follow-up plan for the replication of the application of demonstrated green, energy efficient low carbon technologies in phosphate chemicals production in other localities.
The green, energy efficient and low carbon demos that will be implemented in the partner phosphate chemical manufacturing companies are the following:

- Demonstration 2.1: Application of Advanced Purification Technology for the Production of Purified Wet-Process Phosphoric Acid (WPA)
- Demonstration 2.2: Application of Optimally Controlled Mechanical Vapor Recompression (MVR) System in the Evaporation Process of Ammonium Dihydrogen Phosphate Production
- Demonstration 2.3: Application of Improved Dihydrate-Hemihydrate Process for Wet Process Phosphoric Acid (WPA) Production

Component 3: Green and Low Carbon Design and Operation of Waste Management Systems in the Phosphate Chemicals Industry

This component of the project will address the barriers to the application of green, energy efficient low carbon technologies/techniques and practices in waste management systems in the phosphate chemicals industry. This focuses on the green and low carbon management of solid wastes generated in phosphate ore mining and refining and in phosphate chemicals production. It is comprised of activities that will deliver outputs that will collectively bring about two main outcomes: (1) Enhanced commitment of the phosphate chemical industry in green and low carbon waste management; (2) Increased confidence in the feasibility of the application of green and low carbon technologies in the management of waste in the phosphate chemicals industry in China.

Outcome 3.1: Enhanced commitment of, and institutional and technical arrangements for, the phosphate chemical industry in green and low carbon waste management. This expected outcome is to be realized through the implementation of six sets of activities involving the: (1) Formulation and enforcement of policies and institutional frameworks that supports green, low carbon initiatives in waste management in the PCI; (2) Enforcement of improved existing and new enforced schemes, standards, policies/regulations on the application of green and low carbon waste management technologies in the PCI operations and processes; (3) Documentation of annual evaluation reports on the energy performance and environmental impacts of each demo based on the MRV system developed under Component 1; (4)

Design, conducted and post-evaluation of continuing education program with regular capacity needs assessments in the area of green and low carbon technologies applied in waste management in the PCI; (5) Production and dissemination of research reports on: (a) annual volume of waste production in phosrock mining and refining, and in phosphate chemicals production in China; (b) green and low carbon waste management systems for waste recycling and reuse developed and implemented in other countries and their energy utilization performances; and, (c) potential commercial markets for processed waste, e.g., phosphogypsum; and (6) Establishment and operationalization of an information management system for sharing available and accessible information about market and research developments and R&D in the country and abroad on green and low carbon waste management systems in the PCI.

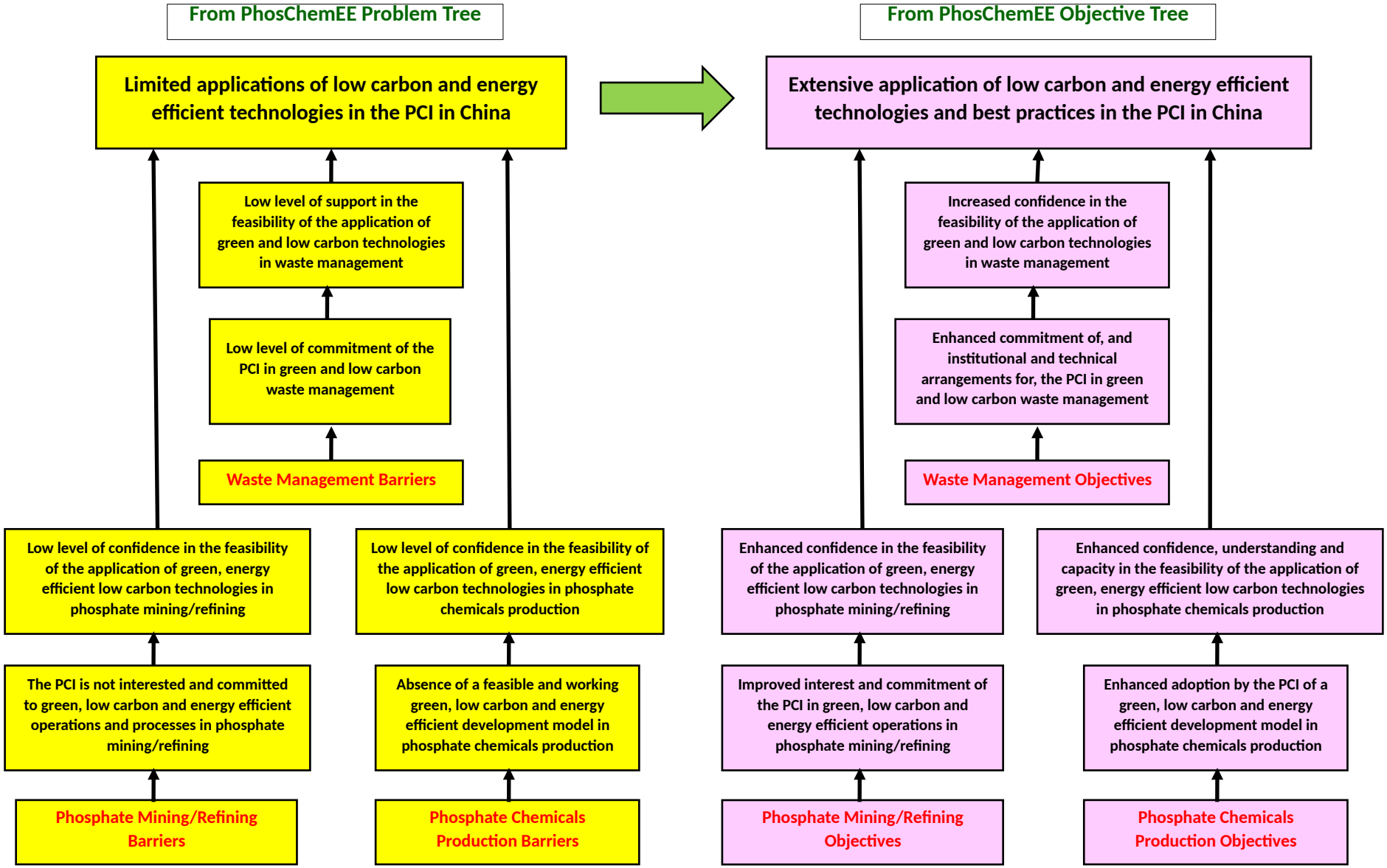
Outcome 3.2: Increased confidence in the feasibility of the application of green and low carbon technologies in the management of waste in the phosphate chemicals Industry in China. There are three sets of activities that will bring about this outcome and these involve the: (1) Designing and planning of demonstrations of the application of green and low carbon technologies/techniques in waste management in the phosphate chemicals industry, including: (a) improved phosphate ore mining tailings recycling and reuse; and (b) energy efficient processing of phosphogypsum and marketing of the gypsum products; (2) Installation and operationalization of green and low carbon waste management technology application demos in the phosphate chemicals industry; and, (3) Approval of financed follow-up plan for the replication of the application of demonstrated green, low carbon waste management technologies in the phosphate chemicals industry in other localities.

The green, energy efficient and low carbon demos on waste management that will be implemented in the partner phosphate ore mining and refining companies and phosphate chemical manufacturing companies are the following:

- Demonstration 3.1: Multi-Purpose Large-Scale Processing and Utilization of Phosphate Chemical Industry Waste
- Demonstration 3.2: Processing of Phosphogypsum for Use as Filler in the Production of Polymer Composite Plastic Products
- Demonstration 3.3: Recovery and Utilization of Yellow Phosphorus Tail Gas for Power Generation

Once successfully completed, PhosChemEE will enable China's phosphate chemicals industry to transform towards a sustainable, green, low carbon development, which will provide technical, financial and environmental benefits to the entire PCI and its associated value chain. Furthermore, the successful completion of the proposed project will also reduce air, soil and water pollution, create new jobs, and help China achieve its mitigation targets set in the NDC. The biggest innovation that the PhosChemEE project will introduce in China is the holistic approach taken, and the proposed activities have large potential for replication and scale-up in the whole country.

Figure 3. Theory of Change



Expected Results

Goal: Reduced annual growth rate of GHG emissions from the Phosphate Chemicals Industry in China.

Project Objective: Enabling the extensive application of low carbon and energy efficient technologies in the phosphate chemicals industry in China.

Component 1: Green and Low Carbon Development and Operation of Phosphate Mines

Outcome 1.1: Improved interest and commitment of the phosphate chemical industry in the green, low carbon and energy efficient operations of the phosphate mining sub-sector in China.

Output 1.1.1: Formulated, recommended, approved, and enforced policy and institutional frameworks supporting green, energy efficient low carbon development initiatives in phosphate rock (phosrock) mining and refining.

Activity	Description
Activity 1.1.1.1: Conduct of research to update existing policies and strengthen existing institutional mechanisms in China to enable the greening of the phosphate mining and refining sub-sector.	Research activities will be undertaken to analyze the existing policies and institutional mechanisms currently in place in China supporting environmentally friendly mining and refining activities. Relevant findings will be summarized to promote China's energy transition in phosphate mining and propose the development route to shift towards carbon neutral development in phosphate mining and refining.
Activity 1.1.1.2: Conduct of research studies on existing policies and institutional frameworks in other countries that can be used as reference for supporting and promoting low carbon development in phosrock mining and refining in China.	Analyze and evaluate the existing policies and institutional frameworks in other countries to define the pathway towards carbon neutral development in phosphate mining and refining. Based on the findings of the assessment, propose suggestions to support and promote the low-carbon development in phosphate mining and refining in China.
Activity 1.1.1.3: Formulation and recommendation of new policies on green, energy efficient low carbon development in phosrock mining and refining.	In accordance to the announcement of the ambitious goal to peak CO_2 emissions by 2030 and achieve carbon neutrality by 2060, and based on the results of Activities 1.1.1.1 and 1.1.1.2, new policies and regulations will be formulated, which will include the development and application of new low-carbon technologies, as well as new models and mechanisms that are conducive to the promotion of green and low-carbon development in phosphate mining and refining. Also, modifications on the existing policies will be proposed and recommended to meet the green and low-carbon development required in phosphate mining and refining.
Activity 1.1.1.4: Approval of the	Lobbying campaigns will be conducted to get the proposed new

newly formulated and recommended policies on green, energy efficient low carbon development in phosrock mining and refining.	policies and amended existing policies, as formulated under Activity 1.1.1.3, approved to support the green, energy efficient low carbon development in phosrock mining and refining.
Activity 1.1.1.5: Enforcement of the new approved policies on green, energy efficient low carbon development in phosrock mining and refining.	Assistance will be provided to government officials to support the enforcement of the policies formulated in Activity 1.1.1.3, and for the preparation of implementing rules and regulations (IRRs) to guide the enforcement. Concomitantly, capacity building of relevant personnel will be conducted to enhance the effect of polices enforcement.
Activity 1.1.1.6: Evaluation of the impact on the greening of the phosrock mining and refining sub-sector as a result of the enforcement of the newly developed policies and provision of follow-up recommendations to amend and improve the policies, if deemed necessary.	This activity will start with the formulation of an overall evaluation framework including evaluation contents, methods and processes to be implemented for this task. Afterwards, the developed evaluation method will be used to evaluate the impact that the new policies developed under PhosChemEE on the greening of the phosrock mining and refining sub-sector. In case the results will not be deemed satisfactory, follow-up recommendations will be provided to amend and strengthen the policies.
Activity 1.1.1.7: Formulation and recommendation of new institutional frameworks on green, energy efficient low carbon development in phosrock mining and refining.	Based on the results of Activities 1.1.1.1 and 1.1.1.2, new and updated institutional frameworks will be devised to advance the development and application of new green technologies, as well as new models and mechanisms that are conducive to the promotion of green and low-carbon development in phosphate mining and refining.
Activity 1.1.1.8: Approval of the newly formulated and recommended institutional frameworks on green, energy efficient low carbon development in phosrock mining and refining.	Lobbying campaigns will be conducted to get the proposed new and updated institutional frameworks, as devised under Activity 1.1.1.7, approved to facilitate the green, energy efficient low carbon development in phosrock mining and refining.
Activity 1.1.1.9: Enforcement of the new approved institutional frameworks on green, energy efficient low carbon development in phosrock mining and refining.	Assistance will be provided to government officials to support the enforcement of the institutional frameworks newly revised and updated in Activity 1.1.1.3, and for the preparation of supportive implementing rules and regulations (IRRs) to guide the enforcement. Concomitantly, capacity building of relevant personnel will be conducted to enhance the effect of institutional frameworks enforcement.
Activity 1.1.1.10: Evaluation of the effectiveness of the new institutional frameworks on green EE low carbon development in phosrock mining and refining, and provision of follow-up recommendations to amend and improve the institutional	This activity will start with the formulation of an overall evaluation framework including evaluation contents, methods and processes to be implemented for this task. Afterwards, the developed evaluation method will be used to evaluate the impact that the new and updated institutional frameworks developed under PhosChemEE on the greening of the phosrock mining and refining sub-sector. In case the results will not be deemed satisfactory, follow-up recommendations will be

frameworks, if deemed necessary.	provided to amend and strengthen the institutional frameworks.

GEF support is required for the incremental technical assistance in the conduct of the review of existing national and international policies and institutional frameworks, the formulation and enforcement of policies and institutional frameworks, and their evaluation followed by provision of recommendations to improve them.

Output 1.1.2: Formulated, recommended, and enforced standards, policies and implementing rules and regulations on the promotion and practice of green, energy efficient, low carbon technologies/techniques, in the development and operations of phosphate mines.

Activity	Description
Activity 1.1.2.1: Conduct of research to update existing standards, policies and implementing rules and regulations to promote green technologies in the development and operations of phosphate mines.	Research activities will be undertaken to analyze the existing standards, policies and implementing rules and regulations currently in place in China supporting the implementation of green technologies in mining and refining. Relevant findings will be summarized to promote China's adoption of low-carbon technologies in the development and operation of existing and new phosphate mines.
Activity 1.1.2.2: Formulation and recommendation of new standards/technical guidelines and implementing rules and regulations for promoting and adopting green technologies in phosphate mines.	Based on the results of Activity 1.1.2.1 new and updated standards, technical guidelines and implementing rules and regulations will be formulated, recommended and promoted for the advancement of the application of green technologies, as well as new models and mechanisms that are conducive to the increased implementation of low-carbon technologies in the development and operations of phosphate mines.
Activity 1.1.2.3: Approval of the newly formulated and recommended standards/technical guidelines and implementing rules and regulations for promoting and adopting green technologies in phosphate mines.	Lobbying campaigns will be conducted to get the proposed standards/technical guidelines and implementing rules and regulations, developed as part of Activity 1.1.2.2, approved by the government, to support the green, energy efficient low carbon development and operations of new and existing phosphate mines.
Activity 1.1.2.4: Enforcement of the new approved standards/technical guidelines and implementing rules and regulations for promoting and adopting green technologies in phosphate mines.	Assistance will be provided to government officials to support the enforcement of the standards/technical guidelines and implementing rules and regulations newly approved in Activity 1.1.2.3. Concomitantly, capacity building of relevant personnel will be conducted to enhance the effect schemes, standards, technical guidelines and implementing rules and regulations enforcement.
Activity 1.1.2.5: Evaluation of the impact and effectiveness of the new enforced standards/technical guidelines and implementing rules and regulations for promoting and	Comprehensively evaluate the efficacy and effects generated by the enforcement of the new and updated standards, technical guidelines and implementing rules and regulations for the promotion of widespread adoption of green technology innovations in phosphate mines. In case the assessment will not

adopting green technologies in phosphate mines, and provision of follow-up recommendations to amend and improve the standards/technical guidelines and implementing rules and regulations, if deemed necessary.	lead to satisfactory results, follow-up recommendations will be provided to amend and strengthen the standards/technical guidelines and implementing rules and regulations.

GEF support is required for the incremental technical assistance in the conduct of the review of existing standards, technical guidelines and implementing rules and regulations, the formulation and enforcement of new standards/technical guidelines and IRR, and their evaluation followed by provision of recommendations to improve them.

Output 1.1.3: Documented annual evaluation reports on the energy performance and environmental impacts of each demo activity and documented and disseminated demo results.

Activity	Description
Activity 1.1.3.1: Design and development of standardized methods for monitoring and data gathering and processing for the evaluation of the energy performance and environmental impacts of the demos and demo replications.	Conduct and in-depth study of the demo projects that have been developed for the mining and refining sub-sector during the project preparation stage. This study will allow to understanding what the important data and features are of each demo and use this information to design and develop standardized methods for monitoring the advancement of the demos, collecting the required data and then evaluate their energy performance and environmental impacts.
Activity 1.1.3.2: Establishment of standardized operational performance parameters to use as benchmark for the evaluation of the energy performance and environmental impacts of the demos and demo replications.	Based on available information for the same green technologies applied in mining and refining sectors from other countries and integrated with calculations and estimated performed specifically for the PhosChemEE demos, a set of standardized operational parameters and their expected target values will be determined and used as benchmark for the energy performance and environmental impact evaluation of the data. This set of parameters will include, but not limited to: specific energy consumption; energy footprint of materials/resources used and produced; energy savings; material loss reduction, and GHG emission reduction. This activity also includes the definition of the parameters for monitoring and assessing environmental risks of the demos and demo replications that are aligned with social and environmental screening procedures.
Activity 1.1.3.3: Conduct of annual evaluation of energy performance and environmental impacts of demos in the phosphate mining and refining sub-sector using the standardized methodology and benchmarks developed in Activities 1.1.3.1 and 1.1.3.2.	Based on the standardized methods developed under Activity 1.1.3.1 and using the set of parameters and target values established in Activity 1.1.3.2, an annual evaluation of the energy performance and environmental impacts of demos in the phosphate mining and refining will be conducted. This annual assessment will allow to determining if the demos implementation is on track and use the info for adaptive management for the demo's implementation activities.

Activity 1.1.3.4: Conduct of annual evaluation of financial performance of demos in the phosphate mining and refining sub-sector to promote the replication of the implemented technologies.	Financial experts will be hired to monitor and determine if the implemented demos in the phosphate mining and refining sub-sector achieve satisfactory financial performances. If required, the experts will provide recommendations to enhance the demos' financial performance up to required standards. This annual assessment will allow to promoting demos and attracting other phosphate mining companies to replicate similar low-carbon technologies and practices.
Activity 1.1.3.5: Preparation of annual and final reports on the operational performance (production, energy consumption, economic) and environmental impacts of demos in the phosphate mining and refining sub-sector.	Based on the results of activities 1.1.3.3 and 1.1.3.4 annual and final reports, at the end of the five-year implementation period, will be prepared reporting the technical and financial performances of the demos in phosphate mining and refining, also including their environmental impacts. The reports will be used as case studies and lessons learned to share knowledge to be used for similar projects in China and internationally.
Activity 1.1.3.6: Dissemination of Annual and final demo evaluation reports prepared in Activity 1.1.3.5.	This activity will focus on disseminating the technical, environmental and economic benefits of the demos, as drafted under Activity 1.1.3.5, to various target audiences, including other phosphate mining and refining companies, relevant government and industrial stakeholders, and potential investors. It will provide all stakeholders with relevant information in a timely manner.
Activity 1.1.3.7: Preparation and dissemination of case studies and lessons learned based on the application of demonstrated green, energy efficient and low carbon technologies.	This activity consists in drafting and promoting case studies and lessons learned based on energy savings projects and initiatives conducted in China, independently from PhosChemEE, and concerning green, energy efficient and low carbon technologies in the phosphate mining and refining sector. For the completion of this task the same experts involved in the design and implementation of the projects and initiatives on which the case studies are based will be hired. The case studies will be made available to relevant stakeholders, including the project management team, and will be used as source of additional information for the successful implementation of PhosChemEE.

GEF support is required for the incremental technical assistance in the design of energy performance evaluation methods, the conduct of annual energy and financial evaluation of the demos, preparation of reports and promotion of results to stakeholders, and the drafting of case studies from relevant project in the phosphate mining and refining not demonstrated under PhosChemEE.

Output 1.1.4: Designed, conducted and post-evaluated continuing education program in the PCI mining sub-sector in Weng'an county in Guizhou Province, Jinning county in Yunnan Province, and Mabian and Leibo counties in Sichuan Province, with regular capacity needs assessment in the area of green, energy efficient low carbon technologies applied in phosphate mining and refining.

Activity	Description
Activity 1.1.4.1: Assessment of the	Surveys and questionnaires will be prepared and administered

ability and awareness level of relevant stakeholders and evaluation of needs and gaps to fill about green, energy efficient low carbon development of phosphate mining and refining in Guizhou, Yunnan, and Sichuan Provinces.	to relevant stakeholders, followed by interviews and consultations, to assess their capabilities and awareness level concerning the technical, financial and environmental benefits of low carbon technologies applied in the phosphate mining and refining sub-sector. The information collected will be used to determine the needs to address and gaps to fill. This activity will focus on the cluster of regions where most phosphate mines are located, namely Guizhou, Yunnan and Sichuan provinces.
Activity 1.1.4.2: Design and implementation of a capacity building program for relevant government officials and industry management teams and key stakeholders on green and low-carbon development in the mining and refining.	Based on the assessment of needs to address and gaps to fill, and supplemented by information gathered from international best practices, in the area of green development in phosphate mining and refining, a capacity building program for relevant stakeholders from the public and private sector will be designed and implemented in the three selected provinces.
Activity 1.1.4.3: Evaluation of the capacity building program designed in Activity 1.1.4.2 a year after the completion to assess how many trainees are gainfully applying the acquired skills and determine the corrective actions to be taken to improve the program.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 1.1.4.2. This evaluation will use data collection tools similar to those implemented in Activity 1.1.4.1 to assess the new level of ability and awareness achieved. In addition, the evaluation will also include an assessment of how successfully the trained stakeholders are applying the newly acquired skills in their professional activities. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.
Activity 1.1.4.4: Design and implementation of a capacity building and information program on operation and maintenance of green, EE and low carbon technologies for workers in phosphate mining and refining factories.	Sector specialists will be engaged to design, develop and administer a training program for workers in the phosphate mining and refining sub-sector on how to operate and maintain the new energy efficient and low carbon technologies that will be implemented for the greening of the phosphate mines activities.
Activity 1.1.4.5: Evaluation of the capacity building program designed in Activity 1.1.4.4 a year after the completion to assess how many trainees are gainfully applying the acquired skills and determine the corrective actions to be taken to improve the program.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 1.1.4.4. This evaluation will determine the efficacy of the training program by assessing how successfully the trained phosphate mine workers are applying the newly acquired skills in their professional activities. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.

GEF support is required for the incremental technical assistance in the assessment of ability and awareness, the design and evaluation of two training programs.

Output 1.1.5: Established the online monitoring, reporting and verification (MRV) system for energy-saving and GHG emission reduction from the application green, energy efficient low-carbon technologies in the PCI including information sharing scheme.

Activity	Description
Activity 1.1.5.1: Design and development of an online monitoring, reporting and verification (MRV) system, including the development of an information sharing mechanism, for energy-saving and GHG emission reduction for the entire PCI.	This activity entails the design and development of a monitoring, reporting and verification (MRV) system that will cover the entire PCI value chain, and specifically: 1) mining and refining; 2) phosphate chemicals production; and 3) phosphate industry waste management. The MRV system will be used to evaluate the energy performance and the GHG emissions reductions for demos and projects for each of the three sub-sectors. The MRV system will be supplemented by the development of a supportive information sharing mechanism to interconnect key players from the entire PCI. Given the complexity of a MRV system that will be used throughout the PCI value chain a consulting firm or a team of experts will be retained, rather than an individual specialist.
Activity 1.1.5.2: Implementation and validation of the newly developed MRV system in the entire PCI.	The MRV system developed in Activity 1.1.5.1 will be implemented and its functioning will be verified by testing it on demos from each of the three PCI sub-sectors. Since the implementation of the MRV system will involve multiple ministries, provincial governments and phosphate industry sub-sectors, it will be housed in a consensually agreed upon entity that will be responsible for its operation and maintenance. At the same time, information sharing mechanism will also be tested.
Activity 1.1.5.3: Design and implementation of a capacity building program on operation and maintenance of the MRV system for companies operating in the entire PCI.	Sector specialists will be engaged to design, develop and implement a capacity building program for management teams of companies from the entire PCI value chain on how to operate and maintain the newly developed MRV system. The training program will also cover the functioning of the supportive information sharing mechanism.
Activity 1.1.5.4: Evaluation of the capacity building program designed in Activity 1.1.5.3 a year after the completion to assess how many trainees are gainfully applying the acquired skills and determine the corrective actions to be taken to improve the program.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 1.1.5.3. This evaluation will determine the efficacy of the training program by assessing the ability of the trained management teams to successfully utilize the MRV system as well as the information sharing mechanism. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.
Activity 1.1.5.5: Evaluation of the operation and performance of the MRV system and provision of recommendations on potential improvements and adjustments to	In addition to the evaluation of the training program in Activity 1.1.5.4, the operation and performance of the MRV system itself will also be assessed. This evaluation will be useful to determine all-encompassing applicability of the MRV system, the validity of the results obtained when using it for demos or

be made.	other projects, and finally to assure that the operation of the system is relatively manageable. If required, corrective interventions and adjustments will be made to enhance the operationalization of the MRV system.
Activity 1.1.5.6: Development and implementation of a greenhouse gases emission accounting and inventory system for the phosphate chemical industry.	Aiming to improve the reporting and the storage of greenhouse gases emissions data from the entire phosphate chemical industry, this activity will entail the development and implementation of: a) standardized GHG data collection methods, b) homogenization of the information collected; c) efficient data inventory systems; d) GHG data analysis procedures. Sector experts will be hired for the accomplishment of this activity and the data collected will be made available for the formulation of policies and regulations, for energy efficiency and low-carbon development in the PCI, for multi-lateral donor use, etc.

GEF support is required for the incremental technical assistance in the design, development, operationalization and evaluation of a MRV system for the entire PCI, the design, implementation and evaluation of training programs, and the design and establishment of a GHG accounting and inventory system.

Output 1.1.6: Established and operational up-to-date information management and exchange system for all aspects of green, energy efficient low carbon phosphate mining and refining technologies and best practices.

Activity	Description
Activity 1.1.6.1: Design and development of an up-to-date information management and exchange system on all aspects of low carbon phosphate mining and refining technologies and best practices.	Government officials, industry leaders and sector specialists will be engaged for the design and development of an up-to-date information management and exchange platform concerning all aspects of low carbon phosphate mining technologies and best practices in China. The platform will collect reliable data including phosphate mining and refining market information.
Activity 1.1.6.2: Establishment and operationalization of the devised up-to-date information communication platform for the exchange of data on all aspects of green phosphate mining and refining.	This activity concerns the establishment and operationalization of the information management and sharing system designed in activity 1.1.6.1. Interested parties will draw from the database of information and will be able to contribute their own data and interact and exchange information with other participants.
Activity 1.1.6.3: Evaluation of operation and performance of the information management and exchange system.	An independent third party will be engaged to evaluate the operation and performance of the information management and exchange system designed and established under activities 2.1.6.1 and 2.1.6.2. Adjustments and improvements will be made to the information platform based on the results of the evaluation.

GEF support is required for the incremental technical assistance in the design, development establishment and operationalization of an information management and sharing system, and the conduct of monitoring and evaluation activities of the information sharing system.

Outcome 1.2: Enhanced confidence in the feasibility of the application of green, energy efficient low carbon technologies in phosrock mining and refining in China

Output 1.2.1: Completed designs and plans of demonstrations of green, energy efficient and low carbon technologies in phosphate mines development and operations including: (a) Improved design for the mining and refining operations in the demo phosrock mines in Weng'an county in Guizhou province, and Mabian county in Sichuan province; (b) Feasibility analyses of the application and operation of green, energy efficient and low carbon phosrock mining and refining systems; and, (c) Implementation plans (including financing arrangements) for each green, energy efficient and low carbon technology application in the demo phosrock mines.

Activity	Description
Activity 1.2.1.1: Development and approval of engineering designs of green demos for the mining and refining operations in the demo phosrock mines in Weng'an county in Guizhou province, and Mabian county in Sichuan province.	This activity entails the development of detailed engineering designs of green, energy efficient low carbon technology applications in the phosphate mining and refining. The designs, based on the demo descriptions annexed to this project proposal, will include site selection (all participant mines will be located in one of the selected provinces: Guizhou and Sichuan), technology route to be adopted, investment estimate, and expected economic and social benefits. Since the SESP rating for the PhosChemEE project is high, the engineering designs must take carefully into consideration the identified risks associated with the demo activities. Afterwards the demo projects design will undergo a severe scrutiny and approval process by the competent authorities[5].
Activity 1.2.1.2: Review of the initial ESMF as well as an ESMP to manage and mitigate all identified risks associated with the design and implementation of the demo activities.	This involves the review of both the Environmental and Social Management Framework (ESMF) and Environmental and Social Management Plan (ESMP) to ascertain that the demos in the phosphate mining and refining sub-sector are designed and engineered in such a way that all identified risks associated with the implementation of the demos will be managed and mitigated. This includes the conduct of specific Environmental & Social Impact Assessments (ESIAs) in each host demo phosphate ore mining & refining company based on the ESMP/ESMF.
Activity 1.2.1.3: Development and approval of implementation plans, including financing arrangements,	In addition to the engineering designs, detailed implementation plans, financial arrangements, implementation schedule and work plans will also be developed and approved by national and

[5] The National Development and Reform Commission will approve the project application as well as the demonstration projects that will be submitted by the demonstration hosts; the administrative department of urban and rural development planning and the administrative department of land resources will approve the application for site selection also submitted by the demonstration enterprise; the meteorological bureau will approve the atmospheric environmental impact report; and the environmental protection bureau will approve the environmental assessment report.

for each designed demo.	provincial government bodies. The same engineering firm(s) procured for preparing the engineering designs will complete these additional tasks as well.

GEF support is required for the incremental technical assistance in the design, development and approval of engineering designs as well as implementation plans for the demo projects.

Output 1.2.2: Installed and operational green, energy efficient low carbon technology application demos in phosrock mining and refining in Weng'an county in Guizhou province, Jinning county in Yunnan province, and Mabian and Leibo counties in Sichuan province.

Activity	Description
Activity 1.2.2.1: Implementation and operation of demos on energy efficient technology of mining in Weng'an county in Guizhou province, Jinning county in Yunnan province, and Mabian and Leibo counties in Sichuan province.	In accordance with the engineering designs, implementation plans, and work plans formulated in activities 1.2.1.1 and 1.2.1.3, the demonstration applications of energy efficient technologies for mining activities of selected companies located in Weng'an county in Guizhou province, Jinning county in Yunnan province, and Mabian and Leibo counties in Sichuan province will be implemented and operated.
Activity 1.2.2.2: Implementation and operation of demos on green and low-carbon utilization technology and equipment in phosphate rock beneficiation and renewable energy use in Mabian, Sichuan province.	In accordance with the engineering designs, implementation plans, and work plans formulated in activities 1.2.1.1 and 1.2.1.3, the demonstration applications of green and low carbon technologies and renewable energy use for phosphate rock beneficiation activities of selected companies located in Mabian, Sichuan province will be implemented and operated.
Activity 1.2.2.3: Design and implementation of a capacity building program on green, energy efficient and low carbon technologies for phosphate mining and refining.	Based on the demo projects that will be implemented and operated under activities 1.2.2.1 and 1.2.2.2, a capacity building program for relevant personnel from the selected demo enterprises will be designed and implemented. The training program aims to familiarize the trainees with the operation and maintenance procedures pertinent to the new low carbon technologies and energy efficient measures that will be implemented with the demos.
Activity 1.2.2.4: Evaluation of the capacity building program a year after the completion and potential design and implementation of corrective actions to complete the training.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 1.2.2.3. This evaluation will determine the efficacy of the training program by assessing how successfully the trained phosphate mining and refining workers are applying the newly acquired skills in operating and maintaining the new low-carbon and energy efficient technologies. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.

GEF support is required for the incremental technical assistance in the implementation and operation of the demo projects, and the design and evaluation of a training program.

Output 1.2.3: Established, funded and operational financial support schemes that are accessible to PCI mining companies to support their green, energy efficient and low carbon technology application initiatives.

Activity	Description
Activity 1.2.3.1: Evaluation of existing financing mechanisms and assessment of needs and gaps to fill for the development of schemes for financially supporting PCI mining and refining companies' green initiatives.	Financial experts will be engaged to conduct a thorough evaluation of existing financial mechanisms currently in place in China as well as internationally and dedicated to supporting green initiatives of phosphate mining and refining companies. The findings will be summarized to assess needs and gaps to fill for the development of financial schemes solely accessible by PCI mining and refining companies.
Activity 1.2.3.2: Determination of the appropriate financing schemes based on the Activity 1.2.3.1, and design of the scheme(s) that will be promoted and implemented under the project.	Based on the findings of Activity 1.2.3.1, this activity entails the determination of optimal financing scheme(s) to support and accelerate the adoption of green and low carbon technologies by PCI companies operating in the phosphate mining and refining sub-sector. In addition to the assessment completed under activity 1.2.3.1, managers and key personnel from selected phosphate mining and refining companies will be consulted to better devise suitable financing scheme(s). Afterwards a financing scheme will be designed and promoted to the mining companies. This task will be accomplished by the same financial experts that have performed Activity 1.2.3.1.
Activity 1.2.3.3: Establishment, approval, and operationalization of financing schemes to support green and low carbon technology initiatives in the phosphate mining and refining sub-sector.	Lobbying campaigns will be conducted to get the proposed financing schemes, developed as part of Activity 1.2.3.2, established and approved by the central and provincial governments. Afterwards, assistance will be provided to government officials to select the appropriate financial institutions who will manage the approved financing schemes. Concomitantly, capacity building of relevant personnel from the financial institutions that will manage the financing schemes will be conducted to enhance their ability to effectively provide support to the phosphate mining and refining companies. The approved financing scheme(s) will facilitate and accelerate the shift towards green and low carbon development of the phosphate mining and refining sub-sector.
Activity 1.2.3.4: Promotion of the green financing scheme to PCI companies' management teams to enhance their involvement and participation to the scheme.	This activity consists in promoting the financing scheme operationalized under Activity 1.2.3.3 to PCI mining companies' management teams. The promotion initiatives (e.g., organization of workshops, and use of media such as radio, television, newspapers, pamphlets, etc.) are finalized to spur the participation of the mining companies in the pilot financing scheme. Technical assistance will also be provided to participant management teams to educate them about the procedures for accessing the financing scheme and provide information about the financial terms of the scheme and the benefits associated with low carbon technologies.

Activity 1.2.3.5: Evaluation of the operational performance of the financing scheme(s) and provision of recommendations to make improvements and adjustments on the implementation of the scheme(s), if deemed necessary.	The operational performance of the financing scheme for PCI companies operating in the phosphate mining and refining sub-sector will be assessed. This evaluation will be useful to determine the strength and well as the shortcomings of the proposed financing scheme and assess if the scheme allows the participant companies to achieve energy and cost savings while reducing GHG emissions. This task will be undertaken by financial experts and the findings will be summarized in a report. If required, corrective interventions and adjustments will be made to enhance the performance of the pilot financing scheme.

GEF support is required for the incremental technical assistance in the conduct of the review of existing financing mechanisms for phosphate mining and refining companies, and the design, approval, operationalization, promotion and evaluation of the financing scheme.

Output 1.2.4: Approved and financed follow-up plan for the replication of the application of demonstrated green, energy efficient low carbon technologies in phosrock mining and refining in other localities.

Activity	Description
Activity 1.2.4.1: Development, publication, and dissemination of guidelines for the replication and scaling-up of successfully implemented green and low carbon development and operation of phosphate mines demos.	Sector specialists will be hired to develop and publish guidelines concerning the green and low carbon development and operation in phosphate mining and refining. The guidelines will be then disseminated among public and private stakeholders (e.g., provincial government officials in provinces not included in PhosChemEE and phosphate mining and refining companies). These efforts are aimed at stimulating and facilitating the replication and scale-up of the demonstrated energy efficient and low carbon technologies.
Activity 1.2.4.2: Development of implementation plans and action plans for demos replication and scale-up.	This activity entails the engagement of sector experts to develop detailed implementation and action plans that will facilitate and accelerate the replication and scale-up of the demonstrated low carbon technologies for the phosphate mining and refining. These implementation and action plans will also cover regions not included in this proposed PhosChemEE project.
Activity 1.2.4.3: Conduct of lobbying and campaigning initiatives to promote and secure the approval and financing of the follow-up plans for the replication and scale-up of the demos.	Lobbying campaigns and initiatives will be devised and conducted to get the approval of the implementation and action plans formulated in Activity 1.2.4.2. The lobbying initiatives will also aim at securing sufficient funding from public and private stakeholders to finance the replication and scale-up of the demonstrated low carbon technologies in the phosphate mining and refining sub-sector.
Activity 1.2.4.4: Promotion of the application of successfully demonstrated green, EE, low carbon technologies in phosphate	This activity consists in promoting the widespread application of the successfully demonstrated green, energy efficient and low carbon technologies for phosphate mining and refining in other regions and provinces of China and will require the participation

mining and refining in different regions of the country.	of local governments. The promotion initiatives, besides the use of media and information material (e.g., organization of workshops, and use of media such as radio, television, newspapers, pamphlets, etc.), will also use the guidelines developed in Activity 1.2.4.1.
Activity 1.2.4.5: Design and implementation of a capacity building program on design, engineering, equipment manufacturing, operation and maintenance for service providing engineering firms in the phosphate mining and refining sub-sector.	Sector specialists will be engaged to design, develop and administer a capacity building program on the design, engineering, manufacturing of energy efficient equipment, and operation and maintenance of low carbon technologies for phosphate mining and refining. Participants to the training activities will include national consultants and sector specialists, engineering firms, manufacturers of PCI equipment, and service and repair providers.
Activity 1.2.4.6: Evaluation of the capacity building program designed in Activity 1.2.4.5 a year after the completion to assess how many trainees are gainfully applying the acquired skills and determine the corrective actions to be taken to improve the program.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 1.2.4.5. This evaluation will determine the efficacy of the training program by assessing how successfully the trained specialists are applying the newly acquired skills in their professional activities. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.

GEF support is required for the incremental technical assistance in the preparation of replication and scale-up guidelines, implementation and action plans, and promotional activities, and the design and evaluation of a training program.

Component 2: Green and Low Carbon Design and Operation of Phosphate Chemicals Production Facilities

Outcome 2.1: Enhanced adoption by the phosphate chemical industry of a green, low carbon and energy efficient development model in phosphate chemicals production.

Output 2.1.1: Formulated, recommended, approved, and enforced policy and institutional frameworks supporting green, energy efficient low carbon technology applications in phosphate chemicals production.

Activity	Description
Activity 2.1.1.1: Evaluation of existing policies and institutional frameworks on energy conservation, and green and low-carbon development in phosphate chemicals production from the perspective of the industrial chain to update and strengthen them.	Sector specialists will perform a detailed evaluation of existing policies and institutional frameworks currently in place in China on energy conservation and green development in the manufacturing of phosphate chemicals and its related industrial chain. Relevant findings will be summarized and later used to update and strengthen the assessed body of policies and institutional frameworks.
Activity 2.1.1.2: Completion of	This activity consists in completing a dual-carbon target analysis

dual-carbon target-oriented analysis of the energy-savings, and green and low-carbon development path in phosphate chemicals production.	of potential energy savings of the green and low-carbon development for the phosphate chemicals industry. The activity includes the description of the status quo of the green development of the phosphate chemicals industry, the prospects and difficulties for green and low-carbon development. The results of the analysis will provide new research ideas to strengthen the development and implementation path in the production of phosphate chemicals.
Activity 2.1.1.3: Conduct of research on inter-regional green, and low-carbon development policies, cooperation and information sharing arrangements in phosphate chemicals production in key regions of China	Research activities will be undertaken to analyze the inter-regional policies, cooperation mechanisms and information sharing arrangements currently in place and supporting green development in phosphate chemicals production among the key PCI provinces cluster of Yunnan, Guizhou, Sichuan and Hubei. Relevant findings will be summarized to promote and accelerate China's widespread adoption of low-carbon technologies in phosphate chemicals production fostering cooperation and sharing experiences.
Activity 2.1.1.4: Conduct of research on the inter-departmental and inter-industry policy on energy conservation, and green and low-carbon development in the phosphorus chemical industry.	Sector specialist will be engaged to analyze the present body of inter-departmental policies formulated to improve energy conservation and increase the implementation of low-carbon technologies in the manufacturing of phosphorous chemicals. Similarly to the previous activity, the focus will be on the phosphate industrial cluster of Yunnan, Guizhou, Sichuan and Hubei. The analysis will produce a report that will highlight all the shortcomings of the existing policies and the areas for their improvements.
Activity 2.1.1.5: Conduct of an international comparative study on energy conservation and green and low-carbon development policies in phosphate chemicals production.	This activity aims to provide a useful reference for the formulation and implementation of relevant domestic policies through international comparative research on the green and low-carbon development of the phosphate chemicals industry. The focus of this activity is to investigate the status quo of the development of green and low-carbon phosphate chemicals in foreign countries, combined with the development status of domestic phosphate chemicals, to compare and study international advanced experience, and pave the way for the follow-up activity 2.1.1.6.
Activity 2.1.1.6: Formulation and recommendation of new policies and institutional frameworks on green, energy efficient low carbon technology application in phosphate chemicals production.	Based on the results of the previous activities, new and updated policies and institutional frameworks on green, energy efficient and low carbon technology application in phosphate chemicals production will be formulated and recommended. The relevant policy reports drafted under this activity will be the basis for the follow-up activities 2.1.1.7 and 2.1.1.8.
Activity 2.1.1.7: Approval of the newly formulated and recommended policies and institutional frameworks on green, energy efficient low carbon	The focus of this activity will be to lobby for the approval of the policies formulated in Activity 2.1.1.6 and secure the commitments of various stakeholders. This lobbying campaign will be realized through the organization of seminars and meetings to promote the approval of policies related to the

technology application in phosphate chemicals production.	energy-saving and green and low-carbon development of the phosphate chemicals industry.
Activity 2.1.1.8: Enforcement of new approved policies and institutional frameworks on green, energy efficient low carbon technology application in phosphate chemicals production.	Assistance will be provided to government officials to support the enforcement of the policies and institutional frameworks newly formulated and/or updated in Activity 2.1.1.6, and for the preparation of supportive implementing rules and regulations (IRRs) to guide the enforcement. Concomitantly, capacity building of relevant personnel will be conducted to enhance the effect of polices and institutional frameworks enforcement.
Activity 2.1.1.9: Evaluation of the impact on the greening of the phosphate chemicals production sub-sector as a result of the enforcement of the newly developed policies and institutional frameworks, and provision of follow-up recommendations to amend and improve them, if deemed necessary.	This activity will start with the formulation of an overall evaluation framework including evaluation contents, methods and processes to be implemented for this task. Afterwards, the developed evaluation method will be used to evaluate the impact that the new and updated policies and institutional frameworks developed under PhosChemEE on the greening of the phosphate chemicals production sub-sector. In case the results will not be deemed satisfactory, follow-up recommendations will be provided to amend and strengthen policies and institutional frameworks.

GEF support is required for the incremental technical assistance in the conduct of the review of existing national and international policies and institutional frameworks, in the formulation and enforcement of policies and institutional frameworks, and in their evaluation followed by provision of recommendations to improve them.

Output 2.1.2: Formulated, recommended, and enforced standards, policies and implementing rules and regulations on the promotion and practice of green, energy efficient, low carbon technologies/techniques, in phosphate chemicals production.

Activity	Description
Activity 2.1.2.1: Conduct of research to update existing standards, policies and implementing rules and regulations to promote green technologies in phosphate chemicals production.	Research activities will be undertaken to analyze the existing standards, policies and implementing rules and regulations currently in place in China supporting the implementation of green technologies in phosphate chemicals production. Relevant findings will be summarized to promote China's adoption of low-carbon technologies in existing and new phosphate chemicals factories.
Activity 2.1.2.2: Formulation, recommendation, approval and enforcement of new standards and technical guidelines for promoting green and low-carbon development in phosphate chemicals production.	Based on the results of Activity 1.1.2.1 new and updated standards and technical guidelines will be formulated, recommended and promoted for the advancement of the application of green technologies in phosphate chemicals production. Lobbying campaigns will be then conducted to obtain the government approval of the standards and technical guidelines. Assistance will be provided to government officials to support their enforcement. Concomitantly, capacity building

	of relevant personnel will be conducted to enhance the efficacy of the enforcement.
Activity 2.1.2.3: Formulation, recommendation, approval and enforcement of new implementing rules and regulations for promoting green technologies in phosphate chemicals production.	Based on the results of Activity 1.1.2.1 new and updated implementing rules and regulations will be formulated, recommended and promoted for the advancement of the application of green technologies in phosphate chemicals production. Lobbying campaigns will be then conducted to obtain the government approval of the new and updated implementing rules and regulations. Assistance will be provided to government officials to support their enforcement. Concomitantly, capacity building of relevant personnel will be conducted to enhance the efficacy of the enforcement.
Activity 2.1.2.4: Evaluation of the impact and effectiveness of the new enforced standards/technical guidelines and implementing rules and regulations for promoting and adopting green technologies in phosphate chemicals production, and provision of follow-up recommendations to amend and improve the standards/technical guidelines and implementing rules and regulations, if deemed necessary.	Comprehensively evaluate the efficacy and effects generated by the enforcement of the new and updated standards, technical guidelines and implementing rules and regulations for the promotion of widespread adoption of green technology innovations in phosphate chemicals manufacturing. In case the assessment will not lead to satisfactory results, follow-up recommendations will be provided to amend and strengthen the standards/technical guidelines and implementing rules and regulations.

GEF support is required for the incremental technical assistance in the conduct of the review of existing standards, technical guidelines and implementing rules and regulations, the formulation and enforcement of new standards/technical guidelines and IRR, and their evaluation followed by provision of recommendations to improve them.

Output 2.1.3: Documented annual evaluation reports on the energy performance and environmental impacts of each demo activity and documented and disseminated demo results based on the MRV system (Output 1.1.5) established under Component 1.

Activity	Description
Activity 2.1.3.1: Design and implementation of methods for evaluation of the energy performance and environmental impacts of the demos for the phosphate chemicals production.	Sector specialists will be engaged to conduct and in-depth study of the demo projects that have been developed for the phosphate chemicals production sub-sector during the project preparation stage. This study will allow to understanding what the important data and features are of each demo and use this information to design and develop methods for evaluating the advancement of the demos, collecting the required data and then evaluate their energy performance and environmental impacts. This activity also includes the definition of the parameters for monitoring and assessing environmental risks of the demos and demo replications that are aligned with social

	and environmental screening procedures.
Activity 2.1.3.2: Conduct of annual evaluation of energy performance and environmental impacts of demos in phosphate chemicals production.	Based on the evaluation methods developed under Activity 2.1.3.1, and implementing the MRV system designed in Output 1.1.5, an annual evaluation of the energy performance and environmental impacts of demos in the phosphate chemicals manufacturing will be conducted. This annual assessment will allow to determining if the demos implementation is on track and use the info for adaptive management for the demo's implementation activities.
Activity 2.1.3.3: Conduct of annual evaluation of financial performance of demos in the phosphate chemicals production sub-sector to promote the replication of the implemented technologies.	Financial experts will be hired to monitor and determine if the implemented demos in the phosphate chemicals production achieve satisfactory financial performances. If required, the experts will provide recommendations to enhance the demos' financial performance up to required standards. This annual assessment will allow to promoting demos and attracting other phosphorous chemical companies to replicate similar low-carbon technologies and practices.
Activity 2.1.3.4: Preparation of annual and final reports on the operational performance (production, energy consumption, and economic), and environmental impacts of demos in phosphate chemicals production.	Based on the results of activities 2.1.3.2 and 2.1.3.3 annual and final reports, at the end of the five-year implementation period, will be prepared reporting the technical and financial performances of the demos in phosphate chemicals production, also including their environmental impacts. The reports will be used as case studies and lessons learned to share knowledge to be used for similar projects in China and internationally.
Activity 2.1.3.5: Dissemination of annual and final demos evaluation reports prepared in Activity 2.1.3.4.	This activity will focus on disseminating the technical, environmental and economic benefits of the demos, as drafted under Activity 2.1.3.4, to various target audiences, including other phosphorous chemical companies, relevant government and industrial stakeholders, and potential investors. It will provide all stakeholders with relevant information in a timely manner.

GEF support is required for the incremental technical assistance in the design of energy performance evaluation methods, the conduct of annual energy and financial evaluation of the demos, preparation of reports and promotion of results to stakeholders

Output 2.1.4: Designed, conducted and post-evaluated continuing education program in the PCI chemicals sub-sector in Yunnan, Guizhou, Sichuan, and Hubei, with regular capacity needs assessment in the area of green, energy efficient and low carbon technologies applied in phosphate chemicals manufacturing.

Activity	Description
Activity 2.1.4.1: Assessment of the ability and awareness level of relevant stakeholders and assessment of needs to address in	Surveys and questionnaires will be prepared and administered to relevant stakeholders, followed by interviews and consultations, to assess their capabilities and awareness level concerning the technical, financial and environmental benefits of low carbon

the area of green, energy efficient low carbon technologies in phosphate chemicals manufacturing in Yunnan, Guizhou, Sichuan, and Hubei Provinces.	technologies applied in the phosphate chemicals manufacturing. The information collected will be used to determine the needs to address and gaps to fill. This activity will focus on the cluster of regions where most phosphate chemicals companies are located, namely Yunnan, Guizhou, Sichuan, and Hubei provinces.
Activity 2.1.4.2: Design and implementation of a capacity building program for relevant government officials, industry management teams and key stakeholders on green and low-carbon development in the phosphate chemical industry, including international best practice.	Based on the assessment of needs to address and gaps to fill, and supplemented by information gathered from international best practices, in the area of green development in phosphate chemicals production, a capacity building program for relevant stakeholders from the public and private sector will be designed and implemented in the four selected provinces.
Activity 2.1.4.3: Evaluation of the capacity building program designed in Activity 2.1.4.2 a year after the completion to assess how many trainees are gainfully applying the acquired skills and determine the corrective action to improve the program.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 2.1.4.2. This evaluation will use data collection tools similar to those implemented in Activity 2.1.4.1 to assess the new level of ability and awareness achieved. In addition, the evaluation will also include an assessment of how successfully the trained stakeholders are applying the newly acquired skills in their professional activities. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.
Activity 2.1.4.4: Establishment of international exchanges and cooperation on energy conservation and green and low-carbon development in the phosphorus chemicals production.	Government officials, industry leaders and sector specialists will be engaged for the design and establishment of communication and information sharing channels as well as organization of potential cooperation initiatives. This will allow an osmotic exchange of knowledge, experience and information between Chinese stakeholders and their international counterparts concerning implementation of low carbon, energy efficient and energy conservation technologies and measures in the phosphorous chemicals production.

GEF support is required for the incremental technical assistance in the assessment of ability and awareness, the design and evaluation of a training program, and the establishment of international communication channels and cooperation initiatives.

Output 2.1.5: Completed market analysis report on the international trade trends for phosrock and phosphate chemicals, and the indirect carbon emission and ecological impacts in the international trading of these products.

Activity	Description
Activity 2.1.5.1: Conduct of market analysis on the international trade trends for phosrock and phosphate	Sector specialists will be hired to conduct an in-depth analysis of the international market for phosrock and phosphate chemicals. Based on historical trade trends, the study aims to estimate future

chemicals, and the indirect carbon emission and ecological impacts in the international trading of these products.	potential trends and anticipate market developments. The analysis will also cover environmental impacts of phosphate chemicals international trading activities, estimating its effect of the ecosystem as well as the amounts of indirect carbon emissions caused.
Activity 2.1.5.2: Preparation of a market analysis report based on the findings of Activity 2.1.5.1, including an analysis of the implications for the phosphate chemicals production sub-sector in China and share with the sub-sector stakeholders	Based on the findings of Activity 2.1.5.1, a market analysis report will be prepared for dissemination to all relevant stakeholders from the public and private sector. The report will also include an assessment of how, and to what extent, the international trading as well as its associated carbon emissions and environmental impacts influence the green development choices made by Chinese phosphate chemicals manufacturers (e.g., green production, carbon footprint, circular economy, etc.).
Activity 2.1.5.3: Organization and conduct of a seminar-workshop to present the market study to, and discuss appropriate future actions to take with, the phosphate chemicals production industry stakeholders and government.	The results of the analysis conducted in Activity 2.1.5.1 and the report prepared in Activity 2.1.5.2 will be presented to a wide audience of relevant government officials and key industry stakeholders by organizing and administering a seminar/workshop. The workshop will also provide a platform for the participants to discuss what are the appropriate future actions to undertake in order to enhance the greening of the phosphate chemicals production sub-sector.

GEF support is required for the incremental technical assistance in the conduct of the market analysis, the preparation of a report, and the organization of a workshop to present the findings of the market analysis and discuss future actions to take.

Output 2.1.6: Established and operational system for managing and sharing available and accessible information on all aspects of green, energy efficient and cleaner phosphate chemicals production in China.

Activity	Description
Activity 2.1.6.1: Design and development of an information management and sharing system on all aspects of low carbon phosphate chemicals production in China.	Government officials, industry leaders and sector specialists will be engaged for the design and development of an information management and sharing platform concerning all aspects of low carbon manufacturing of phosphate chemicals in China. The platform will collect reliable data including phosphate chemicals market information.
Activity 2.1.6.2: Establishment and operationalization of the devised information sharing system for the exchange of available data on all aspects of green phosphate chemicals production.	This activity concerns the establishment and operationalization of the information management and sharing system designed in activity 2.1.6.1. Interested parties will draw from the database of information and will be able to contribute their own data and interact and exchange information with other participants.
Activity 2.1.6.3: Monitoring and evaluation of the operation of the information management and exchange system.	An independent third party will be engaged to monitor and assess the operation and efficacy of the information management and sharing system designed and established under activities 2.1.6.1 and 2.1.6.2. Adjustments and improvements will be made to the

	information platform based on the results of the evaluation.
Activity 2.1.6.4: Conduct of international cooperation and communication on energy saving and green and low-carbon development in the phosphate chemicals production.	Appropriate international communication channels will be devised and established to foster the exchange of information and experience between relevant public and private stakeholders form the phosphate chemicals manufacturing sub-sectors in China and their international counterparts. The ultimate purpose of the communication channel is to propose and launch international cooperation to enhance energy savings and increase low-carbon technologies implementation in the phosphate chemicals production.
Activity 2.1.6.5: Drafting and promotion of representative cases of energy saving and green and low-carbon development in the phosphate chemicals production.	This activity consists in drafting and promoting case studies on energy savings projects and initiatives conducted in China, independently from PhosChemEE, and concerning cleaner production measures, techniques and practices aiming to reduce GHG emissions. If possible, for the completion of this task the same experts involved in the design and implementation of the projects and initiatives on which the case studies are based will be hired. The case studies will be made available to relevant stakeholders, including the project management team, and will be used as source of additional information for the successful implementation of PhosChemEE.

GEF support is required for the incremental technical assistance in the design, development establishment and operationalization of an information management and sharing system, the conduct of monitoring and evaluation activities of the information sharing system, and the drafting of case studies from relevant project in the phosphate chemicals production not demonstrated under PhosChemEE.

Outcome 2.2: Enhanced confidence in the feasibility of the application of green, energy efficient low carbon technologies in phosphate chemicals production in China

Output 2.2.1: Completed designs and plans of demonstrations of green, energy efficient low carbon technologies in phosphate chemicals production including: (a) Improved design for cleaner production demos; (2) Feasibility analyses of the application and operation of green, energy efficient and low carbon phosphate chemical production systems; and, (3) Implementation plans (including financing arrangements) for each green, energy efficient and low carbon technology application in the demo phosphate chemical companies.

Activity	Description
Activity 2.2.1.1: Development and approval of engineering designs of green, energy efficient low carbon technology demos in phosphate chemicals production.	This activity entails the development of detailed engineering designs of green, energy efficient low carbon technology applications in the phosphate chemicals production. The designs, based on the demo descriptions annexed to this project proposal, will include site selection, technology route to be adopted, investment estimate, and expected economic and social benefits.
Activity 2.2.1.2: Review of the initial ESMF as well as an ESMP to	This involves the review of both the Environmental and Social Management Framework (ESMF) and Environmental and Social

manage and mitigate all identified risks associated with the design and implementation of the demo activities.	Management Plan (ESMP) to ascertain that the demos in the phosphate chemicals manufacturing sub-sector are designed and engineered in such a way that all identified risks associated with the implementation of the demos will be managed and mitigated. This includes the conduct of specific Environmental & Social Impact Assessments (ESIAs) in each host demo phosphate chemical company based on the ESMP/ESMF.
Activity 2.2.1.3: Development and approval of implementation plans, including financing arrangements, for each designed demo.	In addition to the engineering designs, detailed implementation plans, financial arrangements, implementation schedule and work plans will also be developed and approved by national and provincial government bodies. The same engineering firm(s) procured for preparing the engineering designs will complete these additional tasks as well.

GEF support is required for the incremental technical assistance in the design, development and approval of engineering designs as well as implementation plans for the demo projects.

Output 2.2.2: Installed and operational green, energy efficient low carbon technology application demos in phosphate chemicals production.

Activity	Description
Activity 2.2.2.1: Implementation and operation of demonstrations on energy savings and green and low-carbon development in phosphate chemicals production.	In accordance with the engineering designs, implementation plans, and work plans formulated in activities 2.2.1.1 and 2.2.1.3, the technology application demonstrations on energy savings and low carbon development in the phosphate chemicals production will be implemented and then operated by the selected enterprises.
Activity 2.2.2.2: Design and implementation of a capacity building program on green, energy efficient and low carbon technologies for phosphate chemicals production.	Based on the demo projects that will be implemented and operated under activity 2.2.2.1, a capacity building program for relevant personnel from the selected demo enterprises will be designed and implemented. The training program aims to familiarize the trainees with the operation and maintenance procedures pertinent to the new low carbon technologies and energy efficient measures that will be implemented with the demos.
Activity 2.2.2.3: Evaluation of the capacity building program a year after the completion and potential design and implementation of corrective actions to complete the training.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 2.2.2.2. This evaluation will determine the efficacy of the training program by assessing how successfully the trained phosphate chemicals production workers are applying the newly acquired skills in operating and maintaining the new low-carbon technologies. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.

GEF support is required for the incremental technical assistance in the implementation and operation of the demo projects, and the design and evaluation of a training program.

Output 2.2.3: Designed and approved pilot financing scheme for small and medium enterprises (SMEs) that manufacture phosphate chemicals.

Activity	Description
Activity 2.2.3.1: Evaluation of existing financing mechanisms and assessment of needs and gaps to develop financial schemes for financially support SMEs that manufacture phosphate chemicals	Financial experts will be engaged to conduct a thorough evaluation of existing financial mechanisms currently in place in China as well as internationally and dedicated to Small and Medium Enterprises (SMEs) operating in the manufacturing of phosphate chemicals. The findings will be summarized to assess needs and gaps to fill for the development of financial schemes solely accessible by SMEs.
Activity 2.2.3.2: Determination of the appropriate financing schemes for SMEs that manufacture phosphate chemicals based on the findings of Activity 2.2.3.1, and design of a pilot scheme that will be promoted and implemented under the project.	Based on the findings of Activity 2.2.3.1, this activity entails the determination of optimal financing schemes to support and accelerate the adoption of green and low carbon technologies by SMEs operating in the phosphate chemicals production sub-sector. In addition to the assessment completed under activity 2.2.3.1, managers and key personnel from selected SMEs will be consulted to better devise suitable financing schemes. Afterwards a pilot financing scheme will be designed and promoted to the SMEs. This task will be accomplished by the same financial experts that have performed Activity 2.2.3.1.
Activity 2.2.3.3: Approval of the pilot financing scheme for SMEs that manufacture phosphate chemicals designed in Activity 2.2.3.2.	Lobbying campaigns will be conducted to get the proposed pilot financing scheme, developed as part of Activity 2.2.3.2, approved by the central and provincial governments. The approved pilot financing scheme will facilitate and accelerate the shift towards green and low carbon development of SMEs operating in the phosphate chemicals production sub-sector.

GEF support is required for the incremental technical assistance in the conduct of the review of existing financing mechanisms for SMEs, and the design and approval of pilot financing scheme.

Output 2.2.4: Operational pilot financing scheme for phosphate chemicals SMEs in 2 to 3 regions.

Activity	Description
Activity 2.2.4.1: Operationalization of the pilot financing scheme for SMEs that manufacture phosphate chemicals, approved under Activity 2.2.3.3, in 2 or 3 selected regions.	Assistance will be provided to government officials to select the appropriate financial institutions who will manage the pilot financing scheme for SMEs that manufacture phosphate chemicals approved in Activity 2.2.3.3. The financing scheme will be piloted in two or three regions offering the best potential for SMEs involvement. Concomitantly, capacity building of relevant personnel from the financial institutions that will manage the pilot financing scheme will be conducted to enhance their ability to effectively provide support to the SMEs.
Activity 2.2.4.2: Promotion of the pilot financing scheme and provision of technical assistance to phosphate chemicals SMEs'	This activity consists in promoting the pilot financing scheme operationalized under Activity 2.2.4.1 to SMEs manufacturing phosphate chemicals and operating in the 2 or 3 selected regions. The promotion initiatives (e.g., organization of workshops, and

management team in the 2 or 3 selected regions to enhance their involvement and participation in the pilot scheme.	use of media such as radio, television, newspapers, pamphlets, etc.) are finalized to spur the participation of the SMEs in the pilot financing scheme. Technical assistance will also be provided to SMEs' management teams to educate them about the procedures for accessing the financing scheme and provide information about the financial terms of the scheme and the benefits associated with low carbon technologies.
Activity 2.2.4.3: Evaluation of the operational performance of the pilot financing scheme for SMEs and provision of recommendations to make improvements and adjustments on the implementation of the scheme, if deemed necessary.	The operational performance of the pilot financing scheme for SMEs operating in the phosphate chemicals production will be assessed. This evaluation will be useful to determine the strength and well as the shortcomings of the proposed pilot financing scheme and assess if the scheme allows the participant SMEs to achieve energy and cost savings while reducing GHG emissions. This task will be undertaken by financial experts and the findings will be summarized in a report. If required, corrective interventions and adjustments will be made to enhance the performance of the pilot financing scheme.

GEF support is required for the incremental technical assistance in the operationalization, promotion and evaluation of the financing scheme for SMEs.

Output 2.2.5: Approved and financed follow-up plan for the replication of the application of demonstrated green, energy efficient low carbon technologies in phosphate chemicals production in other localities.

Activity	Description
Activity 2.2.5.1: Preparation and dissemination of case studies and lessons learned based on the application of demonstrated green, energy efficient and low carbon technologies and the annual and final reports in activity 2.1.3.4.	This activity consists in drafting case studies and lessons learned based on the application of low carbon technologies implemented in the demo projects in the phosphate chemicals production. The cases studies and lessons learned will be also based on the annual and final reports on the performance of the demos prepared in Activity 2.1.3.4.The formulated reports will be prepared by sector specialists and will be made available to relevant stakeholders. This activity will allow more phosphate chemicals manufacturers in China to familiarize themselves with green and low carbon technologies.
Activity 2.2.5.2: Development, publication, and dissemination of green and low carbon development and operation in phosphate chemicals production guidelines to replicate and scale-up the demos.	In addition to Activity 2.2.5.1., sector specialists will also develop and publish guidelines concerning the green and low carbon development and operation in manufacturing phosphate chemicals. The guidelines will be then disseminated among public and private stakeholders (e.g., provincial government officials in provinces not included in PhosChemEE and phosphate chemical production companies). These efforts are aimed at stimulating and facilitating the replication and scale-up of the demonstrated energy efficient and low carbon technologies.
Activity 2.2.5.3: Development of implementation plans and action	This activity entails the engagement of sector experts to develop detailed implementation and action plans that will facilitate and

plans for demos replication and scale-up to promote “global greening” in different regions of the country.	accelerate the replication and scale-up of the demonstrated low carbon technologies for the phosphate chemicals production. These implementation and action plans will cover regions not included in this proposed PhosChemEE project. This activity will also support the enactment of the “global greening” plan, which promotes the development of green factories, products and technologies starting from key provinces and reaching out to the whole country.
Activity 2.2.5.4: Conduct of lobbying and campaigning initiatives to promote and secure the approval and financing of the follow-up plans for the replication and scale-up of the demos.	Lobbying campaigns and initiatives will be devised and conducted to get the approval of the implementation and action plans formulated in Activity 2.2.5.3. The lobbying initiatives will also aim at securing sufficient funding from public and private stakeholders to finance the replication and scale-up of the demonstrated low carbon technologies in the phosphate chemicals production.
Activity 2.2.5.5: Promotion of the application of successfully demonstrated green, energy-efficient and low-carbon technologies in phosphate chemicals production in different regions of the country.	This activity consists in promoting the widespread application of the successfully demonstrated green, energy efficient and low carbon technologies for the manufacturing of phosphate chemicals in other regions and provinces of China and will require the participation of local governments. The promotion initiatives, besides the use of media and information material (e.g., organization of workshops, and use of media such as radio, television, newspapers, pamphlets, etc.), will also use the case studies and lessons learned prepared in Activity 2.2.5.1 and the guidelines developed in Activity 2.2.5.2.
Activity 2.2.5.6: Design and implementation of a capacity building program on design, engineering, equipment manufacturing, operation and maintenance for service providing engineering firms in the phosphate chemicals production sub-sector.	Sector specialists will be engaged to design, develop and administer a capacity building program on the design, engineering, manufacturing of energy efficient equipment, and operation and maintenance of low carbon technologies for the phosphate chemicals production. Participants to the training activities will include national consultants and sector specialists, engineering firms, manufacturers of PCI equipment, and service and repair providers.
Activity 2.2.5.7: Evaluation of the capacity building program designed in Activity 2.2.5.6 a year after the completion to assess how many trainees are gainfully applying the acquired skills and determine the corrective actions to be taken to improve the program.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 2.2.5.6. This evaluation will determine the efficacy of the training program by assessing how successfully the trained specialists are applying the newly acquired skills in their professional activities. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.

GEF support is required for the incremental technical assistance in the preparation of case studies, lessons learned, replication and scale-up guidelines, implementation and action plans, and promotional activities, and the design and evaluation of a training program.

Component 3: Green and Low Carbon Design and Operation of Waste Management Systems in the Phosphate Chemicals Industry

Outcome 3.1: Enhanced commitment of, and institutional and technical arrangements for, the phosphate chemical industry in green and low carbon waste management.

Output 3.1.1: Formulated, recommended, approved, and enforced policies and institutional frameworks that supports green, low carbon initiatives in waste management in the PCI.

Activity	Description
Activity 3.1.1.1: Conduct of research to update existing policies and strengthen existing institutional mechanisms in China to enable the greening of waste management in the phosphate mining and refining sub-sector.	Research activities will be undertaken to analyze the existing policies and institutional mechanisms currently in place in China supporting environmentally friendly waste management in the phosphate mining and refining sub-sector. Relevant findings will be summarized to promote China's energy transition in phosphate mining and refining waste management and propose the development route to shift towards carbon neutrality.
Activity 3.1.1.2: Conduct of research to update existing policies and strengthen existing institutional mechanisms in China to enable the greening of waste management in the phosphate chemicals production sub-sector.	Research activities will be undertaken to analyze the existing policies and institutional mechanisms currently in place in China supporting environmentally friendly waste management in the phosphate chemicals production sub-sector. Relevant findings will be summarized to promote China's energy transition in phosphate chemicals production waste management and propose the development route to shift towards carbon neutrality.
Activity 3.1.1.3: Conduct of research studies on existing policies and institutional frameworks in other countries that can be used as reference for supporting and promoting low carbon development in waste management in the phosphate mining and refining sub-sector.	Analyze and evaluate the existing policies and institutional frameworks in other countries to define the pathway towards carbon neutral development in waste management in the phosphate mining and refining sub-sector. Based on the findings of the assessment, propose suggestions to support and promote the low-carbon development in phosphate mining and refining waste management in China.
Activity 3.1.1.4: Conduct of research studies on existing policies and institutional frameworks in other countries that can be used as reference for supporting and promoting low carbon development in waste management in the phosphate chemicals production sub-sector.	Analyze and evaluate the existing policies and institutional frameworks in other countries to define the pathway towards carbon neutral development in waste management in the phosphate chemicals production sub-sector. Based on the findings of the assessment, propose suggestions to support and promote the low-carbon development in phosphate chemicals manufacturing waste management in China.
Activity 3.1.1.5: Formulation and recommendation of new policies and institutional framework on green, energy efficient low carbon	Based on the results of activities 3.1.1.1 and 3.1.1.3, new and updated policies and institutional frameworks on green, energy efficient and low carbon technology application in waste management in the phosphate mining and refining sub-sector will

development in waste management in the phosphate mining and refining sub-sector.	be formulated and recommended. The relevant policy reports drafted under this activity will be the basis for the follow-up activities 3.1.1.6 and 3.1.1.7.
Activity 3.1.1.6: Approval of the newly formulated and recommended policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate mining and refining sub-sector.	The focus of this activity will be to lobby for the approval of the policies formulated in Activity 3.1.1.5 and secure the commitments of various stakeholders. This lobbying campaign will be realized through the organization of seminars and meetings to promote the approval of policies related to the green and low-carbon development in waste management in the phosphate mining and refining sub-sector.
Activity 3.1.1.7: Enforcement of new approved policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate mining and refining sub-sector.	Assistance will be provided to government officials to support the enforcement of the policies and institutional frameworks newly formulated and/or updated in Activity 3.1.1.5, and for the preparation of supportive implementing rules and regulations (IRRs) to guide the enforcement. Concomitantly, capacity building of relevant personnel will be conducted to enhance the effect of polices and institutional frameworks enforcement.
Activity 3.1.1.8: Evaluation of the impact on the greening of the waste management in the phosphate mining and refining sub-sector as a result of the enforcement of the newly developed policies and institutional framework and provision of follow-up recommendations to amend and improve the policies and the institutional framework, if deemed necessary.	This activity will start with the formulation of an overall evaluation framework including evaluation contents, methods and processes to be implemented for this task. Afterwards, the developed evaluation method will be used to evaluate the impact that the new and updated policies and institutional frameworks developed under PhosChemEE on the greening of waste management in the phosphate mining and refining sub-sector. In case the results will not be deemed satisfactory, follow-up recommendations will be provided to amend and strengthen policies and institutional frameworks.
Activity 3.1.1.9: Formulation and recommendation of new policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate chemicals production sub-sector.	Based on the results of activities 3.1.1.2 and 3.1.1.4, new and updated policies and institutional frameworks on green, energy efficient and low carbon technology application in waste management in the phosphate chemicals production sub-sector will be formulated and recommended. The relevant policy reports drafted under this activity will be the basis for the follow-up activities 3.1.1.10 and 3.1.1.11.
Activity 3.1.1.10: Approval of the newly formulated and recommended policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate chemicals production sub-sector.	The focus of this activity will be to lobby for the approval of the policies formulated in Activity 3.1.1.9 and secure the commitments of various stakeholders. This lobbying campaign will be realized through the organization of seminars and meetings to promote the approval of policies related to the green and low-carbon development in waste management in the phosphate chemicals production sub-sector.

Activity 3.1.1.11: Enforcement of new approved policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate chemicals production sub-sector.	Assistance will be provided to government officials to support the enforcement of the policies and institutional frameworks newly formulated and/or updated in Activity 3.1.1.9, and for the preparation of supportive implementing rules and regulations (IRRs) to guide the enforcement. Concomitantly, capacity building of relevant personnel will be conducted to enhance the effect of polices and institutional frameworks enforcement.
Activity 3.1.1.12: Evaluation of the impact on the greening of the waste management in the phosphate chemicals production sub-sector as a result of the enforcement of the newly developed policies and institutional framework and provision of follow-up recommendations to amend and improve the policies and the institutional framework, if deemed necessary.	This activity will start with the formulation of an overall evaluation framework including evaluation contents, methods and processes to be implemented for this task. Afterwards, the developed evaluation method will be used to evaluate the impact that the new and updated policies and institutional frameworks developed under PhosChemEE on the greening of waste management in the phosphate chemicals production sub-sector. In case the results will not be deemed satisfactory, follow-up recommendations will be provided to amend and strengthen policies and institutional frameworks.

GEF support is required for the incremental technical assistance in the conduct of the review of existing national and international policies and institutional frameworks, in the formulation and enforcement of policies and institutional frameworks, and in their evaluation followed by provision of recommendations to improve them.

Output 3.1.2: Improved existing and new enforced schemes, standards, policies/regulations on the application of green and low carbon waste management technologies in the PCI operations and processes.

Activity	Description
Activity 3.1.2.1: Conduct of research to update existing schemes, standards and regulations to promote the application of green and low carbon waste management technologies in the phosphate mining and refining sub-sector.	Research activities will be undertaken to analyze the existing schemes, standards, and regulations currently in place in China promoting the implementation of green technologies in waste management in the phosphate mining and refining sub-sector. Relevant findings will be summarized to promote China's adoption of low-carbon technologies for waste management in existing and new phosphate mining and refining factories.
Activity 3.1.2.2: Conduct of research to update existing schemes, standards and regulations to promote the application of green and low carbon waste management technologies in the phosphate chemicals production sub-sector.	Research activities will be undertaken to analyze the existing schemes, standards, and regulations currently in place in China promoting the implementation of green technologies in waste management in the phosphate chemicals production sub-sector. Relevant findings will be summarized to promote China's adoption of low-carbon technologies for waste management in existing and new phosphate chemicals manufacturing factories.
Activity 3.1.2.3: Conduct of research on the cross-ministry inter-departmental and cross-inter-	Research activities will be undertaken to analyze the cross-ministry, inter-departmental and cross-inter-industry policy linkage mechanisms currently in place and supporting green

industry policy linkage mechanisms for waste management and comprehensive utilization of resources in the phosphate mining and refining sub-sector.	development for waste management and comprehensive resources utilization in the phosphate mining and refining sub-sector. Relevant findings will be summarized to promote and accelerate China's widespread adoption of low-carbon technologies for waste management and utilization in phosphate mining and refining fostering cooperation and sharing experiences.
Activity 3.1.2.4: Conduct of research on the cross-ministry inter-departmental and cross-inter-industry policy linkage mechanisms for waste management and comprehensive utilization of resources in the phosphate chemicals production sub-sector.	Research activities will be undertaken to analyze the cross-ministry, inter-departmental and cross-inter-industry policy linkage mechanisms currently in place and supporting green development for waste management and comprehensive resources utilization in the phosphate chemicals production sub-sector. Relevant findings will be summarized to promote and accelerate China's widespread adoption of low-carbon technologies for waste management and utilization in phosphate chemicals manufacturing fostering cooperation and sharing experiences.
Activity 3.1.2.5: Formulation and recommendation of new relevant schemes, standards, technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate mining and refining sub-sector.	Based on the results of activities 3.1.2.1 and 3.1.2.3 new and updated schemes, standards, technical guidelines and implementing rules and regulations will be formulated, recommended and promoted for the advancement of the application of green technologies, as well as new models and mechanisms that are conducive to the increased implementation of low-carbon technologies for waste management in the development and operations of phosphate mines.
Activity 3.1.2.6: Approval of the newly formulated and recommended relevant schemes, standards, technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate mining and refining sub-sector.	Lobbying campaigns will be conducted to get the proposed schemes, standards, technical guidelines and implementing rules and regulations, developed as part of Activity 3.1.2.5, approved by the government, to support the green, energy efficient low carbon development and operations in waste management of new and existing phosphate mines.
Activity 3.1.2.7: Enforcement of new approved relevant schemes, standards, technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate mining and refining sub-	Assistance will be provided to government officials to support the enforcement of the schemes, standards, technical guidelines and implementing rules and regulations newly approved in Activity 3.1.2.6. Concomitantly, capacity building of relevant personnel will be conducted to enhance the effect of schemes, standards, technical guidelines and implementing rules and regulations enforcement.

sector.	
Activity 3.1.2.8: Evaluation of the effectiveness of the schemes, standards, technical guidance, implementing rules and regulations on green EE low carbon development in waste management in the phosphate mining and refining sub-sector, and provision of follow-up recommendations to amend and improve the institutional frameworks, if deemed necessary.	Comprehensively evaluate the efficacy and effects generated by the enforcement of the new and updated schemes, standards, technical guidelines and implementing rules and regulations for the promotion of widespread adoption of green technology innovations for waste management in phosphate mines. In case the assessment will not lead to satisfactory results, follow-up recommendations will be provided to amend and strengthen the schemes, standards, technical guidelines and implementing rules and regulations.
Activity 3.1.2.9: Formulation and recommendation of new relevant schemes, standards, technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate chemicals production sub-sector.	Based on the results of activities 3.1.2.2 and 3.1.2.4 new and updated schemes, standards, technical guidelines and implementing rules and regulations will be formulated, recommended and promoted for the advancement of the application of green technologies, as well as new models and mechanisms that are conducive to the increased implementation of low-carbon technologies for waste management in the development and operations of phosphate chemicals manufacturers.
Activity 3.1.2.10: Approval of the newly formulated and recommended relevant schemes, standards, technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate chemicals production sub-sector.	Lobbying campaigns will be conducted to get the proposed schemes, standards, technical guidelines and implementing rules and regulations, developed as part of Activity 3.1.2.9, approved by the government, to support the green, energy efficient low carbon development and operations in waste management of new and existing phosphate chemicals manufacturers.
Activity 3.1.2.11: Enforcement of new approved relevant schemes, standards, technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate chemicals production sub-sector.	Assistance will be provided to government officials to support the enforcement of the schemes, standards, technical guidelines and implementing rules and regulations newly approved in Activity 3.1.2.10. Concomitantly, capacity building of relevant personnel will be conducted to enhance the effect of schemes, standards, technical guidelines and implementing rules and regulations enforcement.
Activity 3.1.2.12: Evaluation of the effectiveness of the schemes, standards, technical guidance,	Comprehensively evaluate the efficacy and effects generated by the enforcement of the new and updated schemes, standards, technical guidelines and implementing rules and regulations for

implementing rules and regulations on green EE low carbon development in waste management in the phosphate chemicals production sub-sector, and provision of follow-up recommendations to amend and improve the institutional frameworks, if deemed necessary.	the promotion of widespread adoption of green technology innovations for waste management in phosphate chemicals production. In case the assessment will not lead to satisfactory results, follow-up recommendations will be provided to amend and strengthen the schemes, standards, technical guidelines and implementing rules and regulations.

GEF support is required for the incremental technical assistance in the conduct of the review of existing national and international schemes, standards, technical guidelines and implementing rules and regulations, in the formulation and enforcement of schemes, standards, technical guidelines and implementing rules and regulations, and in their evaluation followed by provision of recommendations to improve them.

Output 3.1.3: Documented annual evaluation reports on the energy performance and environmental impacts of each demo based on the MRV system developed under Component 1.

Activity	Description
Activity 3.1.3.1: Design and implementation of methods for evaluation of the energy performance and environmental impacts of the waste management demos in the phosphate mining/refining sub-sector of the PCI.	Sector specialists will be engaged to conduct and in-depth study of the demo projects that have been developed for the waste management in the phosphate mining and refining sub-sector during the project preparation stage. This study will allow to understanding what the important data and features are of each demo and use this information to design and develop methods for evaluating the advancement of the demos, collecting the required data and then evaluate their energy performance and environmental impacts.
Activity 3.1.3.2: Design and implementation of methods for evaluation of the energy performance and environmental impacts of the waste management demos in the phosphate chemicals production sub-sector of the PCI.	Sector specialists will be engaged to conduct and in-depth study of the demo projects that have been developed for the waste management in the phosphate chemicals production sub-sector during the project preparation stage. This study will allow to understanding what the important data and features are of each demo and use this information to design and develop methods for evaluating the advancement of the demos, collecting the required data and then evaluate their energy performance and environmental impacts. This activity also includes the definition of the parameters for monitoring and assessing environmental risks of the demos and demo replications that are aligned with social and environmental screening procedures.
Activity 3.1.3.3: Conduct of annual evaluation of energy performance, environmental impacts and financial performance of demos in waste management of the phosphate mining and refining sub-	Based on the evaluation methods developed under Activity 3.1.3.1, and implementing the MRV system designed in Output 1.1.5, an annual evaluation of the energy performance and environmental impacts of demos on waste management in the phosphate mining and refining sub-sector will be conducted. In addition, financial experts will be hired to monitor and determine

sector to promote the replication of the implemented technologies.	if the implemented demos achieve satisfactory financial performances. If required, recommendations and corrective actions will be provided to enhance the demos' technical and financial performance. This annual assessment will allow to promoting demos and attracting other phosphate mines to replicate similar low-carbon technologies and practices.
Activity 3.1.3.4: Conduct of annual evaluation of energy performance, environmental impacts and financial performance of demos in waste management of the phosphate chemicals production sub-sector to promote the replication of the implemented technologies.	Based on the evaluation methods developed under Activity 3.1.3.2, and implementing the MRV system designed in Output 1.1.5, an annual evaluation of the energy performance and environmental impacts of demos on waste management in the phosphate chemicals production sub-sector will be conducted. In addition, financial experts will be hired to monitor and determine if the implemented demos achieve satisfactory financial performances. If required, recommendations and corrective actions will be provided to enhance the demos' technical and financial performance. This annual assessment will allow to promoting demos and attracting other phosphate chemicals manufacturers to replicate similar low-carbon technologies and practices.
Activity 3.1.3.5: Preparation of annual and final reports on energy performance, environmental impacts of demos and the evaluation of the results and effectiveness of the demonstration program in waste management of the phosphate mining and refining sub-sector.	Based on the results of Activity 3.1.3.3 annual and final reports, at the end of the five-year implementation period, will be prepared reporting the technical and financial performances of the demos waste management in the phosphate mining and refining sub-sector, also including their environmental impacts. The reports will be used as case studies and lessons learned to share knowledge to be used for similar projects in China and internationally.
Activity 3.1.3.6: Preparation of annual and final reports on energy performance, environmental impacts of demos and the evaluation of the results and effectiveness of the demonstration program in waste management of the phosphate chemicals production sub-sector.	Based on the results of Activity 3.1.3.4 annual and final reports, at the end of the five-year implementation period, will be prepared reporting the technical and financial performances of the demos waste management in the phosphate chemicals production sub-sector, also including their environmental impacts. The reports will be used as case studies and lessons learned to share knowledge to be used for similar projects in China and internationally.
Activity 3.1.3.7: Dissemination of annual and final demos evaluation reports prepared in Activity 3.1.3.5	This activity will focus on disseminating the technical, environmental and economic benefits of the demos, as drafted under Activity 3.1.3.5, to various target audiences, including other phosphate mining and refining companies, relevant government and industrial stakeholders, and potential investors. It will provide all stakeholders with relevant information in a timely manner.
Activity 3.1.3.8: Dissemination of annual and final demos evaluation reports prepared in Activity 3.1.3.6	This activity will focus on disseminating the technical, environmental and economic benefits of the demos, as drafted under Activity 3.1.3.6, to various target audiences, including other phosphate chemicals production companies, relevant

	government and industrial stakeholders, and potential investors. It will provide all stakeholders with relevant information in a timely manner.

GEF support is required for the incremental technical assistance in the design of energy performance evaluation methods, the conduct of annual energy and financial evaluation of the demos, preparation of reports and promotion of results to stakeholders

Output 3.1.4: Designed, conducted and post-evaluated continuing education program with regular capacity needs assessments in the area of green and low carbon technologies applied in waste management in the PCI.

Activity	Description
Activity 3.1.4.1: Assessment of the ability and awareness level of relevant stakeholders and assessment of needs and gaps to address in the area of waste management in the phosphate mining and refining sub-sector.	Surveys and questionnaires will be prepared and administered to relevant stakeholders, followed by interviews and consultations, to assess their capabilities and awareness level concerning the technical, financial and environmental benefits of low carbon technologies applied in waste management in the phosphate mining and refining sub-sector. The information collected will be used to determine the needs to address and gaps to fill.
Activity 3.1.4.2: Design and implementation of a capacity building program for relevant national and local government officials, management teams and key stakeholders on waste management in the phosphate mining and refining sub-sector based on the findings of Activity 3.1.4.1.	Based on the assessment of needs to address and gaps to fill determined in Activity 3.1.4.1, and supplemented by information gathered from international best practices, in the area of green development in waste management in phosphate mining and refining, a capacity building program for relevant stakeholders from the public and private sector will be designed and implemented.
Activity 3.1.4.3: Evaluation of the capacity building program of the waste management in the phosphate mining and refining sub-sector a year after the completion and potential design and implementation of corrective actions to complete the training.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 3.1.4.2. This evaluation will use data collection tools similar to those implemented in Activity 3.1.4.1 to assess the new level of ability and awareness achieved. In addition, the evaluation will also include an assessment of how successfully the trained stakeholders are applying the newly acquired skills in their professional activities. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.
Activity 3.1.4.4: Assessment of the ability and awareness level of relevant stakeholders and assessment of needs and gaps to address in the area of waste management in the phosphate chemicals production sub-sector.	Surveys and questionnaires will be prepared and administered to relevant stakeholders, followed by interviews and consultations, to assess their capabilities and awareness level concerning the technical, financial and environmental benefits of low carbon technologies applied in waste management in the phosphate chemicals production sub-sector. The information collected will be used to determine the needs to address and gaps to fill.

Activity 3.1.4.5: Design and implementation of a capacity building program for relevant national and local government officials, industry management teams and key stakeholders on waste management in the phosphate chemicals production sub-sector based on the findings of Activity 3.1.4.4.	Based on the assessment of needs to address and gaps to fill determined in Activity 3.1.1.4, and supplemented by information gathered from international best practices, in the area of green development in waste management in phosphate chemicals production, a capacity building program for relevant stakeholders from the public and private sector will be designed and implemented.
Activity 3.1.4.6: Evaluation of the capacity building program of the waste management in the phosphate chemicals production sub-sector a year after the completion and potential design and implementation of corrective actions to complete the training.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 3.1.4.5. This evaluation will use data collection tools similar to those implemented in Activity 3.1.4.4 to assess the new level of ability and awareness achieved. In addition, the evaluation will also include an assessment of how successfully the trained stakeholders are applying the newly acquired skills in their professional activities. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.
Activity 3.1.4.7: Design and implementation of a training program for professionals to build capacity concerning the new and updated policies, institutional frameworks, standards and green and low carbon technologies applied in waste management in the phosphate mining and refining sub-sector.	Sector specialists will be engaged to design, develop and administer a training program for workers in the phosphate mining and refining sub-sector on the new and updated policies and institutional frameworks (formulated in Activity 3.1.1.5) and standards and guidelines (formulated in Activity 3.1.2.5) concerning green and low carbon technologies applied in waste management in the phosphate mining and refining sub-sector.
Activity 3.1.4.8: Evaluation of the capacity building program of the waste management in the phosphate mining and refining sub-sector a year after the completion to gauge the effectiveness of the program and come up with improvements and/or expansion of the program.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 3.1.4.7. This evaluation will determine the efficacy of the training program by assessing how successfully the trained phosphate mine workers are applying the newly acquired skills in their professional activities. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.
Activity 3.1.4.9: Design and implementation of a training program for professionals to build capacity concerning the new and updated policies, institutional frameworks, standards and green and low carbon technologies applied in waste management in	Sector specialists will be engaged to design, develop and administer a training program for workers in the phosphate mining and refining sub-sector on the new and updated policies and institutional frameworks (formulated in Activity 3.1.1.9) and standards and guidelines (formulated in Activity 3.1.2.9) concerning green and low carbon technologies applied in waste management in the phosphate chemicals production sub-sector.

the phosphate chemicals production sub-sector.	
Activity 3.1.4.10: Evaluation of the capacity building program of the waste management in the phosphate chemicals production sub-sector a year after the completion to gauge the effectiveness of the program and come up with improvements and/or expansion of the program.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 3.1.4.9. This evaluation will determine the efficacy of the training program by assessing how successfully the trained phosphate chemicals production workers are applying the newly acquired skills in their professional activities. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.

GEF support is required for the incremental technical assistance in the assessment of ability and awareness, and the design and evaluation of four training programs.

Output 3.1.5: Completed research reports on: (1) annual volume of waste production in phosrock mining and refining, and in phosphate chemicals production in China; (2) green and low carbon waste management systems for waste recycling and reuse developed and implemented in other countries and their energy utilization performances; and, (3) potential commercial markets for processed waste, e.g., phosphogypsum.

Activity	Description
Activity 3.1.5.1: Drafting and finalization of research reports on annual volume of waste production in the phosphate mining and refining sub-sector.	Sector specialists will be engaged to perform an in-depth study of the waste production in the entire phosphate mining and refining sub-sector in China and prepare reports summarizing the findings. The study will cover all regions where phosphate mines are located, and for each region will be provided important information, such as: annual volumes of waste produced; chemical and physical properties of the waste produced; description of the transportation methods available; etc.
Activity 3.1.5.2: Drafting and finalization of research reports on annual volume of waste production in phosphate chemical production sub-sector.	Sector specialists will be engaged to perform an in-depth study of the waste production in the entire phosphate chemicals production sub-sector in China and prepare reports summarizing the findings. The study will cover all regions where phosphate manufacturers are located, and for each region will be provided important information, such as: annual volumes of waste produced; chemical and physical properties of the waste produced; description of the transportation methods available; etc.
Activity 3.1.5.3: Drafting and finalization of research reports on green and low carbon waste management systems for waste recycling and reuse in the phosphate mining and refining sub-sector developed and implemented in other countries and their energy	A thorough analysis will be undertaken to study and evaluate the green and low carbon technologies and methods implemented in other countries, particularly developed countries, to manage, recycle and reuse waste produced in the phosphate mining and refining activities. The study will include information concerning energy performance, cost analysis, environmental impacts, etc. The results obtained will be summarized in research reports that will be used by public and private stakeholders in China for

utilization performances.	designing future initiatives in the greening of the waste management in the phosphate mining and refining sub-sector.
Activity 3.1.5.4: Drafting and finalization of research reports on green and low carbon waste management systems for waste recycling and reuse in the phosphate chemicals production sub-sector developed and implemented in other countries and their energy utilization performances.	A thorough analysis will be undertaken to study and evaluate the green and low carbon technologies and methods implemented in other countries, particularly developed countries, to manage, recycle and reuse waste produced in the phosphate chemicals manufacturing activities. The study will include information concerning energy performance, cost analysis, environmental impacts, etc. The results obtained will be summarized in research reports that will be used by public and private stakeholders in China for designing future initiatives in the greening of the waste management in the phosphate chemicals production sub-sector.
Activity 3.1.5.5: Drafting and finalization of research reports on potential commercial markets for processed waste produced in the mining and refining of the PCI.	Based on the reports prepared in activity 3.1.5.1, which characterize the waste produced in phosphate mining and refining activities, and activity 3.1.5.3, which describe best practices in place in other countries, a complete and detailed market research will be performed to estimate the market potential in China for the products that can be obtained from processing the waste generated in the phosphate mining and refining sub-sector. At the end of this activities research reports will be prepared.
Activity 3.1.5.6: Drafting and finalization of research reports on potential commercial markets for processed waste produced in the production process of the PCI.	Based on the reports prepared in activity 3.1.5.2, which characterize the waste produced in phosphate chemicals manufacturing, and activity 3.1.5.4, which describe best practices in place in other countries, a complete and detailed market research will be performed to estimate the market potential in China for the products that can be obtained from processing the waste generated in the phosphate chemicals production sub-sector. At the end of this activities research reports will be prepared. research reports will be drafted that will illustrate the potential and the benefits that phosphate manufacturing companies can achieve by adopting green and low carbon technologies for their waste management.
Activity 3.1.5.7: Presentation in a seminar-workshop of the research findings to the PCI and discussions to come up with action plans for circular economy in phosphate mining/refining.	The results of the analyses conducted in activities 3.1.5.1 and 3.1.5.3, and the report prepared in Activity 3.1.5.5 will be presented to a wide audience of relevant government officials and key industry stakeholders by organizing and administering a seminar/workshop. During the workshop, the potential markets and the financial and environmental benefits that phosphate mines can achieve by adopting green and low carbon technologies for their waste management will be illustrated. The workshop will provide a platform for all participants to discuss what are the appropriate future actions to undertake in order to enhance the greening of waste management in the phosphate mining and refining sub-sector.
Activity 3.1.5.8: Presentation in a seminar-workshop of the research	The results of the analyses conducted in activities 3.1.5.2 and 3.1.5.4, and the report prepared in Activity 3.1.5.6 will be

findings to the PCI and discussions to come up with action plans for circular economy in phosphate chemicals production.	presented to a wide audience of relevant government officials and key industry stakeholders by organizing and administering a seminar/workshop. During the workshop, the potential markets and the financial and environmental benefits that phosphate chemicals manufacturers can achieve by adopting green and low carbon technologies for their waste management will be illustrated. The workshop will provide a platform for all participants to discuss what are the appropriate future actions to undertake in order to enhance the greening of waste management in the phosphate chemicals production sub-sector.

GEF support is required for the incremental technical assistance in the conduct of the market analyses, the preparation of reports, and the organization of workshops to present the findings of the market analysis and discuss future actions to take.

Output 3.1.6: Established and operational information management system for sharing available and accessible information about market and research developments and R&D in the country and abroad on green and low carbon waste management systems in the PCI.

Activity	Description
Activity 3.1.6.1: Design and development of an information management and sharing system on market developments and R&D on low carbon waste management systems in the phosphate mining and refining sub-sector in China and abroad.	Government officials, industry leaders and sector specialists will be engaged for the design and development of an information management and sharing platform concerning market development and R&D on all aspects of low carbon waste management systems in the phosphate mining and refining sub-sector in China and abroad.
Activity 3.1.6.2: Establishment and operationalization of the devised information sharing system for the exchange of available data on all aspects of market developments and R&D on green waste management systems in the phosphate mining and refining sub-sector in China and abroad.	This activity concerns the establishment and operationalization of the information management and sharing system designed in activity 3.1.6.1. Interested parties will draw from the database of information and will be able to contribute their own data and interact and exchange information with other participants.
Activity 3.1.6.3: Monitoring and evaluation of operation and performance of the information management and exchange system for the phosphate mining and refining sub-sector.	An independent third party will be engaged to monitor and assess the operation and efficacy of the information management and sharing system designed and established under activities 3.1.6.1 and 3.1.6.2. Adjustments and improvements will be made to the information platform based on the results of the evaluation.
Activity 3.1.6.4: Design and development of an information management and sharing system on market developments and R&D	Government officials, industry leaders and sector specialists will be engaged for the design and development of an information management and sharing platform concerning market development and R&D on all aspects of low carbon waste

on low carbon waste management systems in the phosphate chemicals production sub-sector in China and abroad.	management systems in the phosphate chemicals production sub-sector in China and abroad.
Activity 3.1.6.5: Establishment and operationalization of the devised information sharing system for the exchange of available data on all aspects of market developments and R&D on green waste management systems in the phosphate chemicals production sub-sector in China and abroad.	This activity concerns the establishment and operationalization of the information management and sharing system designed in activity 3.1.6.4. Interested parties will draw from the database of information and will be able to contribute their own data and interact and exchange information with other participants.
Activity 3.1.6.6: Monitoring and evaluation of operation and performance of the information management and exchange system for the phosphate chemicals production sub-sector.	An independent third party will be engaged to monitor and assess the operation and efficacy of the information management and sharing system designed and established under activities 3.1.6.4 and 3.1.6.5. Adjustments and improvements will be made to the information platform based on the results of the evaluation.
Activity 3.1.6.7: Drafting, promotion and dissemination of representative cases of energy saving and green and low-carbon development in waste management systems in the phosphate mining and refining sub-sector.	This activity consists in drafting, promoting and disseminating case studies on energy savings projects and initiatives conducted in China, independently from PhosChemEE, and concerning cleaner waste management measures, techniques and practices in the phosphate mining and refining sub-sector aiming to reduce GHG emissions. If possible, for the completion of this task the same experts involved in the design and implementation of the projects and initiatives on which the case studies are based will be hired. The case studies will be made available to relevant stakeholders, including the project management team, and will be used as source of additional information for the successful implementation of PhosChemEE.
Activity 3.1.6.8: Drafting, promotion and dissemination of representative cases of energy saving and green and low-carbon development in waste management systems in the phosphate chemicals production sub-sector.	This activity consists in drafting, promoting and disseminating case studies on energy savings projects and initiatives conducted in China, independently from PhosChemEE, and concerning cleaner waste management measures, techniques and practices in the phosphate chemicals production sub-sector aiming to reduce GHG emissions. If possible, for the completion of this task the same experts involved in the design and implementation of the projects and initiatives on which the case studies are based will be hired. The case studies will be made available to relevant stakeholders, including the project management team, and will be used as source of additional information for the successful implementation of PhosChemEE.

GEF support is required for the incremental technical assistance in the design, development establishment and operationalization of two information management and sharing systems, the conduct of monitoring

and evaluation activities of the information sharing systems, and the drafting of case studies from relevant project in the PCI not demonstrated under PhosChemEE.

Outcome 3.2: Increased confidence in the feasibility of the application of green and low carbon technologies in the management of waste in the phosphate chemicals Industry in China.

Output 3.2.1: Completed designs and plans of demonstrations of the application of green and low carbon technologies/techniques in waste management in the phosphate chemicals industry, including: (a) improved phosrock mining tailings recycling and reuse; and (b) energy efficient processing of phosphogypsum and marketing of the gypsum products.

Activity	Description
Activity 3.2.1.1: Development and approval of engineering designs of green demos for waste management in the phosphate mining and refining sub-sector to improve phosrock mining tailings recycling and reuse.	This activity entails the development of detailed engineering designs of green, energy efficient low carbon technology applications in waste management in the phosphate mining and refining sub-sector to improve phosrock tailings recycling and reuse. The designs, based on the demo descriptions annexed to this project proposal, will include site selection, technology route to be adopted, investment estimate, and expected economic and social benefits. Since the SESP rating for the PhosChemEE project is high, the engineering designs must take carefully into consideration the identified risks associated with the demo activities. Afterwards the demo projects design will undergo a severe scrutiny and approval process by the competent authorities[6].
Activity 3.2.1.2: Conduct of feasibility analysis to confirm the economic viability of the application and operation of new technologies for the improvement of green, energy efficient and low carbon in waste management in the phosphate chemicals production sub-sector.	In addition to the engineering designs, detailed implementation plans, financial arrangements, implementation schedule and work plans will also be developed and approved by national and provincial government bodies. The same engineering firm(s) procured for preparing the engineering designs will complete these additional tasks as well.
Activity 3.2.1.3: Development and approval of engineering designs of green demos for waste management in the phosphate chemicals production sub-sector to increase energy efficiency processing of phosphogypsum and marketing of the gypsum product.	This activity entails the development of detailed engineering designs of green, energy efficient low carbon technology applications in waste management in the phosphate chemicals production sub-sector to increase the marketing of the gypsum by-products. The designs, based on the demo descriptions annexed to this project proposal, will include site selection, technology route to be adopted, investment estimate, and expected economic and social benefits. Since the SESP rating for the PhosChemEE project is high, the engineering designs must take carefully into consideration the identified risks associated with the demo activities. Afterwards the demo projects design

[6] Ibid, p. 29.

	will undergo a severe scrutiny and approval process by the competent authorities[7].
Activity 3.2.1.4: Preparation of an ESMF as well as an ESMP to manage and mitigate all identified risks associated with the design and implementation of the demo activities for waste management in the phosphate chemicals production sub-sector.	This involves the review of both the Environmental and Social Management Framework (ESMF) and Environmental and Social Management Plan (ESMP) to ascertain that the waste management demos in both the phosphate mining and refining; and phosphate chemicals manufacturing sub-sectors are designed and engineered in such a way that all identified risks associated with the implementation of the demos will be managed and mitigated. This includes the conduct of specific Environmental & Social Impact Assessments (ESIAs) in each host demo PCI company based on the ESMP/ESMF.
Activity 3.2.1.5: Development and approval of implementation plans for the waste management in the phosphate mining and refining sub-sector, including financing arrangements, for each designed demo.	In addition to the engineering designs, detailed implementation plans, financial arrangements, implementation schedule and work plans will also be developed and approved by national and provincial government bodies for the waste management in the phosphate mining and refining sub-sector. The same engineering firm(s) procured for preparing the engineering designs will complete these additional tasks as well.
Activity 3.2.1.6: Development and approval of implementation plans for the waste management in the phosphate chemicals production sub-sector, including financing arrangements, for each designed demo.	In addition to the engineering designs, detailed implementation plans, financial arrangements, implementation schedule and work plans will also be developed and approved by national and provincial government bodies for the waste management in the phosphate chemicals production sub-sector. The same engineering firm(s) procured for preparing the engineering designs will complete these additional tasks as well.

GEF support is required for the incremental technical assistance in the design, development and approval of engineering designs as well as implementation plans for the demo projects.

Output 3.2.2: Installed and operational green and low carbon waste management technology application demos in the phosphate chemicals industry.

Activity	Description
Activity 3.2.2.1: Implementation and operation of demonstrations of green equipment/technologies for the comprehensive utilization of resources in waste management in the phosphate mining and refining sub-sector.	In accordance with the engineering designs, implementation plans, and work plans formulated in activities 3.2.1.1 and 3.2.1.5, the technology application demonstrations on energy savings and low carbon development in waste management in the phosphate mining and refining sub-sector will be implemented and then operated by the selected enterprises.
Activity 3.2.2.2: Implementation and operation of demonstrations of green equipment/technologies for the comprehensive utilization of	In accordance with the engineering designs, implementation plans, and work plans formulated in activities 3.2.1.3 and 3.2.1.6, the technology application demonstrations on energy savings and low carbon development in waste management in the phosphate

[7] Ibid, p. 29.

resources in waste management in the phosphate chemicals production sub-sector.	chemicals production sub-sector will be implemented and then operated by the selected enterprises.
Activity 3.2.2.3: Design and implementation of a capacity building program on green, energy efficient and low carbon technologies for waste management in the phosphate mining and refining sub-sector.	Based on the demo projects that will be implemented and operated under activity 3.2.2.1, a capacity building program for relevant personnel, from the selected demo phosphate mining and refining enterprises, will be designed and implemented. The training program aims to familiarize the trainees with the operation and maintenance procedures pertinent to the new low carbon technologies and energy efficient measures that will be implemented with the demos.
Activity 3.2.2.4: Evaluation of the capacity building program of the phosphate mining and refining sub-sector a year after the completion and potential design and implementation of corrective actions to complete the training.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 3.2.2.3. This evaluation will determine the efficacy of the training program by assessing how successfully the trained phosphate mining and refining workers are applying the newly acquired skills in operating and maintaining the new low-carbon technologies. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.
Activity 3.2.2.5: Design and implementation of a capacity building program on green, energy efficient and low carbon technologies for waste management in the phosphate chemicals production sub-sector.	Based on the demo projects that will be implemented and operated under activity 3.2.2.21, a capacity building program for relevant personnel, from the selected demo phosphate chemicals production enterprises, will be designed and implemented. The training program aims to familiarize the trainees with the operation and maintenance procedures pertinent to the new low carbon technologies and energy efficient measures that will be implemented with the demos.
Activity 3.2.2.6: Evaluation of the capacity building program of the phosphate chemicals production sub-sector a year after the completion and potential design and implementation of corrective actions to complete the training.	This follow up activity will be undertaken a year after the completion of the capacity building program conducted in Activity 3.2.2.5. This evaluation will determine the efficacy of the training program by assessing how successfully the trained phosphate chemicals production workers are applying the newly acquired skills in operating and maintaining the new low-carbon technologies. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.

GEF support is required for the incremental technical assistance in the implementation and operation of the demo projects, and the design and evaluation of two training programs.

Output 3.2.3: Approved and financed follow-up plan for the replication of the application of demonstrated green, low carbon waste management technologies in the phosphate chemicals industry in other localities.

Activity	Description
Activity 3.2.3.1: Preparation and dissemination of case studies and lessons learned based on the	This activity consists in drafting case studies and lessons learned based on the application of low carbon technologies implemented in the demo projects for waste management in the

application of demonstrated green, low carbon waste management technologies in the phosphate mining and refining sub-sector.	phosphate mining and refining sub-sector. The cases studies and lessons learned will be also based on the annual and final reports on the performance of the demos prepared in Activity 3.1.3.5.The formulated reports will be prepared by sector specialists and will be made available to relevant stakeholders. This activity will allow more phosphate mining and refining companies in China to familiarize themselves with green and low carbon technologies.
Activity 3.2.3.2: Preparation and dissemination of case studies and lessons learned based on the application of demonstrated green, low carbon waste management technologies in the phosphate chemicals production sub-sector.	This activity consists in drafting case studies and lessons learned based on the application of low carbon technologies implemented in the demo projects for waste management in the phosphate chemicals production sub-sector. The cases studies and lessons learned will be also based on the annual and final reports on the performance of the demos prepared in Activity 3.1.3.6.The formulated reports will be prepared by sector specialists and will be made available to relevant stakeholders. This activity will allow more phosphate chemicals production companies in China to familiarize themselves with green and low carbon technologies.
Activity 3.2.3.3: Development, publication, and dissemination of green and low carbon waste management technologies and operation guidelines in the phosphate mining and refining sub-sector to replicate and scale-up the demos.	In addition to Activity 3.2.3.1., sector specialists will also develop and publish guidelines concerning the green and low carbon development and operation in waste management in phosphate mining and refining. The guidelines will be then disseminated among public and private stakeholders (e.g., provincial government officials in provinces not included in PhosChemEE and phosphate mining and refining companies). These efforts are aimed at stimulating and facilitating the replication and scale-up of the demonstrated energy efficient and low carbon technologies.
Activity 3.2.3.4: Development, publication, and dissemination of green and low carbon waste management technologies and operation guidelines in the phosphate chemicals production sub-sector to replicate and scale-up the demos.	In addition to Activity 3.2.3.2., sector specialists will also develop and publish guidelines concerning the green and low carbon development and operation in waste management in phosphate chemicals manufacturing. The guidelines will be then disseminated among public and private stakeholders (e.g., provincial government officials in provinces not included in PhosChemEE and phosphate chemicals production companies). These efforts are aimed at stimulating and facilitating the replication and scale-up of the demonstrated energy efficient and low carbon technologies.
Activity 3.2.3.5: Development of implementation plans and action plans for demos replication and scale-up to facilitate the establishment of green factories, the production of green products and the promotion of demonstrated green, energy-efficient and low-carbon	This activity entails the engagement of sector experts to develop detailed implementation and action plans that will facilitate and accelerate the replication and scale-up of the demonstrated low carbon technologies for waste management in the phosphate mining and refining sub-sector. These implementation and action plans will cover regions not included in this proposed PhosChemEE project. This activity will also support the establishment of green factories and the development of green products.

technologies in the phosphate mining and refining sub-sector in different regions of the country.	
Activity 3.2.3.6: Approval and promotion of the application of the development and implementation plans formulated under Activity 3.2.3.5 to facilitate the establishment of green factories, the production of green products and the implementation of demonstrated green, energy-efficient and low-carbon technologies in the phosphate chemicals production sub-sector in different regions of the country.	Lobbying campaigns and initiatives will be devised and conducted to get the approval of the implementation and action plans formulated in Activity 3.2.3.5. The lobbying initiatives will also aim at securing sufficient funding from public and private stakeholders to finance the replication and scale-up of the demonstrated low carbon technologies. These successfully demonstrated green, energy efficient and low carbon technologies for waste management in the phosphate mining and refining sub-sector will be then promoted in other regions and provinces of China and will require the participation of local governments. The promotion initiatives, besides the use of media and information material (e.g., organization of workshops, and use of media such as radio, television, newspapers, pamphlets, etc.), will also use the case studies and lessons learned prepared in Activity 3.2.3.1 and the guidelines developed in Activity 3.2.3.3.
Activity 3.2.3.7: Development of implementation plans and action plans for demos replication and scale-up to facilitate the promotion of demonstrated green, energy-efficient and low-carbon technologies in the phosphate chemicals production sub-sector in different regions of the country.	This activity entails the engagement of sector experts to develop detailed implementation and action plans that will facilitate and accelerate the replication and scale-up of the demonstrated low carbon technologies for waste management in the phosphate chemicals production sub-sector. These implementation and action plans will cover regions not included in this proposed PhosChemEE project.
Activity 3.2.3.8: Approval and promotion of the application of the development and implementation plans formulated under Activity 3.2.3.7 to facilitate the establishment of green factories, the production of green products and the implementation of demonstrated green, energy-efficient and low-carbon technologies in the phosphate chemicals production sub-sector in different regions of the country.	Lobbying campaigns and initiatives will be devised and conducted to get the approval of the implementation and action plans formulated in Activity 3.2.3.7. The lobbying initiatives will also aim at securing sufficient funding from public and private stakeholders to finance the replication and scale-up of the demonstrated low carbon technologies. These successfully demonstrated green, energy efficient and low carbon technologies for waste management in the phosphate chemicals production sub-sector will be then promoted in other regions and provinces of China and will require the participation of local governments. The promotion initiatives, besides the use of media and information material (e.g., organization of workshops, and use of media such as radio, television, newspapers, pamphlets, etc.), will also use the case studies and lessons learned prepared in Activity 3.2.3.2 and the guidelines developed in Activity 3.2.3.4.
Activity 3.2.3.9: Enhancement of the capacity of the engineering service provision industry that are currently supporting the phosphate mining and refining sub-sector in	Sector specialists will be engaged to design, develop and administer a capacity building program on the design, engineering, manufacturing of energy efficient equipment, and operation and maintenance of low carbon technologies for the waste management in the phosphate mining and refining sub-

their waste management requirements.	sector. Participants to the training activities will include national consultants and sector specialists, engineering firms, manufacturers of PCI equipment, and service and repair providers. An evaluation of the training activities will be conducted a year after the completion of the program. This evaluation will determine the efficacy of the training program by assessing how successfully the trained specialists are applying the newly acquired skills in their professional activities. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.
Activity 3.2.3.10: Enhancement of the capacity of the engineering service provision industry that are currently supporting the phosphate chemicals production sub-sector in their waste management requirements.	Sector specialists will be engaged to design, develop and administer a capacity building program on the design, engineering, manufacturing of energy efficient equipment, and operation and maintenance of low carbon technologies for the waste management in the phosphate chemicals production sub-sector. Participants to the training activities will include national consultants and sector specialists, engineering firms, manufacturers of PCI equipment, and service and repair providers. An evaluation of the training activities will be conducted a year after the completion of the program. This evaluation will determine the efficacy of the training program by assessing how successfully the trained specialists are applying the newly acquired skills in their professional activities. If deemed necessary, corrective actions to improve the training program will be recommended and implemented.

GEF support is required for the incremental technical assistance in the preparation of case studies, lessons learned, replication and scale-up guidelines, implementation and action plans, and promotional activities, and the design and evaluation of training programs.

Partnerships:

The implementation of the PhosChemEE Project is in partnership with several stakeholders that also contribute to addressing the development challenges and support the application of low carbon and energy efficient technologies in the phosphate chemical industry in China.

Table 2: PhosChemEE Project Partners

Partners	Relevance to PhosChemEE and Description of Initiatives
Ministry of Industry and Information Technology (MIIT)	Its core responsibility is to undertake the management of the industry sector in China, among which is the Phosphate Chemicals Industry (PCI). It formulates and organizes the implementation in energy conservation, greenhouse gas emission reduction, comprehensive resources utilization and clean production promotion policy and organize and coordinate the promotion and application of major green

Partners	Relevance to PhosChemEE and Description of Initiatives
	and low-carbon demonstration projects and new products, new technologies, new equipment and new materials. It is responsible for providing guidance for the design of low-carbon phosphorus chemical projects, working with the team to solve the main problems of the PCI identified in the framework of this project, supporting the construction of national and provincial policy and institutional framework, participating in the whole project baseline activities and providing corresponding supporting funds, and responsible for the effective cooperation with UNDP, the Ministry of Finance and relevant departments in the energy conservation and emission reduction activities of PCI Coordination and communication.
Ministry of Natural Resources (MNR)	Its core responsibility is the management of natural resources such as mineral resources. It supports low-carbon and efficient utilization of phosphorus resources production areas, and is responsible for providing guidance for the design of low-carbon phosphorus mine projects and, working with the project team to solve the main problems identified in the framework of this project, supporting the establishment of national and provincial policy and institutional framework; clarifying its responsibilities, participating in the whole project baseline activities and providing corresponding supporting funds; and responsible for the effective coordination with UNDP, the Ministry of Finance and relevant departments in the phosphorus mine energy conservation and emission reduction activities and communication.
Ministry of Finance (MOF)	The Ministry of Finance is responsible for handling economic regulations and fiscal policies, and for administering government public expenditures, including the national annual budget. As the communication authority with GEF in China, the Ministry of finance provides high-level strategic guidance and participates in project proposal endorsement.
National Development and Reform Commission (NDRC)	In the preparation phase of this project, the NDRC participates in and guides the planning of this project to ensure that the output of the project can guide and encourage the participants of phosphorus resources development activities to promote the comprehensive utilization of phosphorus resources, and also put forward suggestions for the planning and policy measures of resource conservation and comprehensive utilization in China.
Ministry of Ecology and Environment (MEE)	Participate in and guide the planning of this project during the preparation period to ensure that the project's mechanism can provide intellectual support for China's participation in international climate change negotiations and the implementation of the United Nations Framework Convention on Climate Change.
Local Governments (Sichuan, Yunnan, Guizhou, and Hubei)	The provincial department of natural resources (development and reform, industry and information technology, environment) of the municipal and county-level governments directly participate in project design and baseline survey, provide technical support, guidance, and logistics support. During project implementation, provincial natural resources departments and industry and information technology departments are responsible for supervising municipal and county-level work, including technical support, monitoring, supporting, coordination with partners, and capacity-building.
Private Sector Enterprises	Low carbon and efficient comprehensive utilization of phosphorus resources, clean production of PCI and low-carbon reuse of phosphogypsum involve technology R&D, production, and service enterprises. Private sector investors and enterprises involved in PhosChemEE include: Guizhou Wengfu Group; Leibo County Huarui Mining Company; Mabian Fuma Phosphate Company; Sichuan Mabian Tianrui Mining Company; Guizhou Kailin group mineral fertilizer Co., Ltd.; Sichuan Lomom Phosphorus Chemical Co., Ltd.; Hubei Xiangyun (Group) Chemical Co., Ltd.; Yunnan Phosphate Group; Yunnan Xiangfeng Industrial Group Co., Ltd.; and Zhonglicheng

Partners	Relevance to PhosChemEE and Description of Initiatives
	Industry Co., Ltd. In order to assure the successful completion of all demonstrations and their associated activities, the selection process to identify the best private sector enterprises participating to the PhosChemEE Project has included an assessment of the companies' past, ongoing and planned initiatives to implement in their day-by-day operations good environmental and social management practices. All selected enterprises have shown strong commitment towards strengthening their adoption of sustainable environmental and social management practices and operations.

Risks:

As per standard UNDP requirements, the Project Manager will monitor risks quarterly and report on the status of risks to the UNDP Country Office. The UNDP Country Office will record progress in the UNDP ATLAS risk log. Risks will be reported as critical when the impact and probability are high (i.e. when impact is rated as 5, and when impact is rated as 4 and probability is rated at 3 or higher). Management responses to critical risks will also be reported to the GEF in the annual Project Implementation Review (PIR).

The risks that might prevent the project objectives from being achieved, during the project implementation, are listed as follows:

Table 3: PhosChemEE Project Risks

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
Implementation of the PhosChemEE Project (and most specifically Component 1 which focuses on the upstream Phosphate mining and refining sub-sector) could lead to adverse impacts on enjoyment of the human rights (civil, political, economic, social, or cultural) of the affected population, potentially including ethnic minorities and/~~or~~ vulnerable/disadvantaged groups~~,~~ .	Social Political Financial	I = 5 L = 4 High	In the design stage, the following safeguards documents have been prepared, covering this risk and all other risks, to meet the SES requirements of a **High Risk** project: · a stakeholder analysis and comprehensive Stakeholder Engagement Plan, · a gender analysis and Gender Action Plan, · an Environmental and Social Management Framework (ESMF) with initial FPIC procedures (if confirmed as required), · a Grievance Redress Mechanism (draft/outline). The ESMF will outline the procedures for the (site-specific) Environmental and Social Impact Assessments (ESIAs) and Environmental and Social Management Plans (ESMPs) that will be prepared during project implementation. To adequately assess impact from upstream/policy and strategy-level interventions (including potential cumulative impacts) the project shall undertake a **Strategic Environmental and Social Assessment (SESA).** The design and implementation of the project are consistent with the strategic requirements of the	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
This is most likely to occur under Outputs 1.2.2, 2.2.2 and 3.2.2 ~~in the rural areas where the on-the-ground project activities will be carried out.~~ Mining companies engaged in the project might not have the capacity and knowledge to uphold their duties as per UNDP SES requirements. This includes upholding duties and requirements on FPIC as per UNDP SES 6.			country's Law on ethnic unity and progress, which will ensure the enjoyment of the human rights of the potentially affected population in the demo places, especially ethnic minorities and/or vulnerable/disadvantaged groups. During the implementation of the project, the MIIT and MNR, which both have adequate experiences in GEF project implementation and have established long-term cooperative relationships with various ministries, local departments, industry-related supporting units and relevant organizations, can establish a scientific and efficient collaborative working mechanism to ensure the policy implementation of the project. Extensive negotiation will also be conducted with local people with the cooperation of the MIIT, MNR and provincial government.		
During the construction of demo activities (Output 2.2.2), and in support to the mining and refining sectors (under Component 1), ~~P~~potential adverse impacts to habitats and/or ecosystems and ecosystem services (e.g. non-hazardous waste from phosphogypsum reuse) may occur., ~~and some possible changes to the use of lands and resources in the phosphogypsum~~	Environmental	I = 4 L = 3 Substantial	The project has been designed such that demos involve the conduct of a detailed feasibility study and ESIA on the local area before the demo site selection, and a site-specific ESMP will be put in place before implementation of each demo. The detailed design of the demos and their engineering and construction shall be in accordance with national and international standards to ensure compliance with the SES and other relevant environmental ordinances and regulations, as to be outlined in the ESMF (and ESIA/ESMPs). Additionally, the Project will observe the established regulatory framework for monitoring and assessing such risks. China has regulations on the engineering of the project site for the restoration and protection of biodiversity, and manufactories are not allowed to be established in or around ecosystem function zone. In this case, the PhosChemEE Project will ensure that the demo sites will be located far away from ecological reserves and ecologically vulnerable areas. Second, demos will involve the conduct of a detailed feasibility study and ESIA on the local area before the demo site selection. For each demo, a site-specific ESMP will be put in	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
~~processing and reuse demo.~~			place before implementation. Moreover, the demos will be constructed and implemented in accordance with national standards and international general standards to prevent pollution and ensure compliance with the SES and other relevant environmental ordinances/regulations.		
Inadequate disposal of waste during the operation of the **phosphogypsum** processing reuse demo (i.e. Outputs 3.2.1 and 3.2.2) poses a threat to the environment as well as the health and safety of workers and the community. The associated construction, operation, or decommissioning of the demo installations may have potential health and safety risks to local communities and workers due to the transport, storage, and use and/or disposal of any hazardous or dangerous materials (e.g. explosives, fuel and other chemicals) that may be used during construction and operation.	Operational Environmental	I = 4 L = 3 Substantial	The ESMF and subsequent ESIA and ESMP will address safety risks to local communities. The required management measures will be included in site-specific ESMPs and be conducted before project construction. Per those forthcoming plans, appropriate training will be provided to the demonstration enterprises to ensure that risks are properly addressed and managed. Training will also be provided to local governments in order to enhance their regulation capacity to control potential risks in the demonstration. A Grievance Mechanism will be designed (see ESMF) and will be implemented by the project, allowing communities and workers to request for interventions when facing issues with health and safety risks. The associated construction, operation, or decommissioning of the demo installations will strictly comply with the national standards[8] and international general standards on construction and waste management. Site-specific environmental and social assessments will be conducted for each demo to identify the critical health and safety risks. Where public access is available, appropriate engineering and administrative controls (e.g., detours, traffic calming, signs) will be considered and implemented in advance during the construction and operation of demos. Per those forthcoming plans, appropriate training will be provided to workers to ensure that they install and operate the installed system correctly and safely, and properly control and manage the release or disposal of waste. Training will also be provided to local governments in order to enhance their regulation capacity to control potential risks in the demonstration. This will minimize or avoid any community health risks and safety issues for the communities regarding construction work involved in the installation of the demos, and the	PMU MIIT MNR	No Change

[8] Including Standard for pollution control on hazardous waste storage, Standard for pollution control on hazardous waste storage, etc.

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
			minimization and management of waste generated from these demos (e.g., explosives, fuel and other chemicals).		
The potential outcomes of the PhosChemEE Project will be sensitive or vulnerable to potential impacts of climate change. Yunnan, Sichuan, Guizhou, and Hubei, where the demos are located, have risks of extreme weather that are exacerbated by climate change. Thus, some of the demonstration activities of the project (located outdoors) can be impacted by weather events.	Environmental	I = 3 L = 3 Moderate	The ESIA (as required and outlined in the ESMF) assesses activities for impact and sustainability within China national context, including its climatic vulnerability and associated impacts~~e, and relevant ESMP has been outlined in the ESMF~~. The detailed design of the demos shall take climate-related events and risks into consideration by utilizing the findings of the risk assessment that shall be conducted as part of the ESIAs. Appropriate measures will be taken based on the specific climate-related events. For instance, the adverse impact of flood on demos can be eliminated by constructing diversion ditches, while sunshade roofs and cooling spray can reduce the impact of extreme high temperatures. The demos will also cover climate risk adaptation capacity building through training workshops (Activities 1.2.2.3, 2.2.2.2, 3.2.2.3). This training will be provided to the demo companies and local governments to enhance their capacities to deal with extreme climate events in general. Depending on the extent of the impacts of the adverse climate-related events, appropriate modifications in the installations (and budget) will be done. Potential reduction in the number of installations, or replacement with alternative demos will be done while considering the need to ensure the resulting interventions are still contributing to the realization of the project outcomes.	PMU MIIT MNR	No Change
~~The PhosChemEE Project will involve large-scale infrastructure development that might not be designed, built, operated or maintained safely.~~	~~Operational~~	~~I = 5 L = 2 Substantial~~	~~Aside from conducting the required site-specific ESIAs/ESMPs, the infrastructures (e.g., process buildings and access roads) shall be designed, constructed, operated and decommissioned following national legal requirements, good international practice, and any relevant international obligations and standards. The demo infrastructures shall be designed by qualified engineers and professionals who will be engaged by the PhosChemEE Project, and then be certified and approved by independent professionals not involved in the design process. During the construction phase of each demo, appropriate plans for construction supervision and quality assurance will be prepared and implemented. Construction personnel will also have appropriate qualifications and training.~~	~~PMU MIIT MNR~~	~~No Change~~

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
			Periodic safety inspections, monitoring and emergency preparedness will be included in demos in order to ensure the appropriate and safe operation and maintenance. The ESMF will outline detailed steps required to assess and manage this Substantial risk.		
The PhosChemEE Project might pose potential risks and vulnerabilities related to occupational health and safety due to physical, chemical, biological, and radiological hazards during new infrastructure or system construction, operation.	Operational Environmental	I = 4 L = 2 Moderate	This risk will be assessed and included in the scope of the ESMF and subsequent ESIAs/ESMPs. The occupational health impact assessment will be conducted prior to the construction phase of the demos, which will make detailed identification and assessment of potential hazards and risks, particularly those that could result in serious injury, ill health or death and those identified through worker health surveillance. The construction/implementation process of demos will adopt preventive and protective measures (e.g., dust and noise reduction measures) to eliminate or minimize occupational health hazards and risks. Safety reminder signs will be implemented at worksites where occupational health hazards may arise, and employees will be provided personal protective equipment at no cost. The demo will include the conduct of the safety and health training at no cost to workers, including on the appropriate operation of new and advanced facilities and the proper use of personal protective equipment. The emergency plan will be prepared for each demo, including emergency prevention, response arrangements and employment injury benefits and/or remedies for adverse occupational impacts.	PMU MIIT MNR	No Change
The Project includes activities both downstream/physical activities, (like phosphogypsum processing), as well as upstream/policy-level interventions that require/could lead to significant consumption of raw materials, energy, and/or	Environmental Operational	I = 4 L = 3 Substantial	Per the ESMF, the ESIA and ESMP will define the management measures for this risk. In the design of demos, life-cycle analysis (LCA) will be conducted before each industrial process to assess the raw materials, energy, and water used in new technologies and evaluate the potential of resource conservation. Based on LCA assessments, the project will be designed to seek cleaner and more energy-efficient phosphate chemicals production and processing of phosphogypsum. International practice for water and energy conservation will be applied in demos, especially the wastewater management and reuse technologies. Emergency accident pools will be designed in each demo in accordance with the requirement of contamination avoidance. Additionally, the demos will establish the resource	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
water; and involves significant extraction, diversion or containment of surface and/or ground water. That resource use, if not designed or implemented well, could lead to adverse impacts on ambient conditions in the project area and area of influence.			efficiency monitoring mechanism, including using benchmarking data to the relative level of efficiency and reporting water and energy consumption periodically. This will ensure that the new and advanced technologies of phosphate chemicals production and phosphogypsum processing will reduce or optimize resource consumption, so that the project's water consumption does not have significant adverse impacts on communities, other users or the environment and ecosystems. The potential for cumulative impacts relating to water and energy consumption will also be assessed in the project's SESA.		
The PhosChemEE Project result in significant greenhouse gas emissions or may exacerbate climate change due to massive energy consumption in phosrock mining and refining, and in phosphate chemicals production.	Environmental Operational	I = 4 L = 3 Substantial	As per the project's ESMF, ~~T~~this risk area will be included within the scope of the ESMF and ESIAs/ESMPs (for demo site related impacts) and the project's SESA for upstream and cumulative stanadard2 related impacts. LCA method will be conducted before project construction during the design of the demonstrations to identify potential project-related increases in GHG emissions that may exacerbate climate change. And then the potential sources of emissions related to project activities will be estimated to form a baseline for developing measures to reduce such emissions. The PhosChemEE Project aims to enable the extensive application of low carbon and energy-efficient technologies in the phosphate chemicals industry in China. To mitigate the potential increment of GHG emissions, the demos will apply these three solutions. First, the demos will promote the new, advanced and energy-efficient technologies, which will directly reduce emissions in phosphate ore processing and phosphorus chemical production. Second, Yunnan, Guizhou, Sichuan and Hubei, where the demos sites are located are rich in renewable energy resources. This will offset the potential extra GHG emissions from the overconsumption of energy induced by the new technologies. Last, the demos will follow the principle of circular economy, including avoiding waste discharge, promoting the recycling and reusing of waste. Such measures will reduce the indirect emissions from intermediate material	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
			production and waste management. Additionally, GHG emissions of demos will be tracked and reported in accordance with provisions of the UNFCCC and GHG minimization measures are implemented.		
Upstream project activities (primarily Outputs 1.1.1, 1.1.2, 2.1.1, 2.1.2, 2.1.6, 2.2.4, 2.2.5, 3.1.1, and 3.1.2) could result in adverse impacts to the local communities and the receptor environment if project-supported policy initiatives are not adequately screened and assessed. China accounts for more than 80% of the world's total production capacity of Phosphate. As such, the project could inadvertently lead to issues of global and/or cumulative environmental concern if not managed adequately	Environmental Operational	I-5 L-3 Substantial	The project will undertake the conduct of a SESA that shall cover all upstream activities, and those with potential cumulative impact related implications. Activities that will fall under the scope of the SESA include (but are not limited to): Outputs 1.1.1, 1.1.2, 2.1.1, 2.1.2, 2.1.6, 2.2.4, 2.2.5, 3.1.1, and 3.1.2)	PMU MIIT MNR	No Change
The phosphogypsum processing and reuse demos have been planned to be implemented in	Environmental	I-4 L-2 Moderate	The site-specific ESIAs will include an assessment of historical pollution/legacy issues at demo sites which are being planned for existing facilities. Any historical/legacy pollution issues that are deemed to have on-going environmental concerns will be remediated and managed within the framework of the site-specific ESMPs that are	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
the existing facility. Legacy pollution issues may exists that have on-going environmental impacts which will have to be managed in line with UNDP SES requirements.			to be developed as well.		
There is the possibility, that through some possible changes to the use of lands and resources in the phosphogypsum processing and reuse demo outputs of the project, access restrictions/economic displacement may occur as a result of indirect impacts emanating from the project's interventions.	Operational	I-4 L-2 Moderate.	Any access restrictions/economic displacement related issues are likely to occur only as an indirect result of project supported activities. However, SES5 related impacts cannot be ruled out at this stage of the project preparation. SES5 related risks will be assessed in detail at both a demonstration level (via the conduct of site specific ESIAs) and at a policy/upstream level (through the conduct of the project's SESA). Any impacts identified shall be managed through the relevant avenues. IF adverse SES5 related impacts are identified, then necessary SES tools may be required to be undertaken (this could include a Livelihood Action Plan, Land Acquisition Action Plan etc.).	PMU MIIT MNR	No Change
The project management office (PMO) cannot effectively organize and coordinate the participation of stakeholders (including relevant departments of local government, specific project implementation units, experts in various fields, etc.).	Organizational	I = 4 L = 3 Substantial	The MIIT and MNR have adequate experiences in GEF project implementation, and have established long-term cooperative relationships with various ministries, local departments, industry related supporting units and relevant organizations, and can establish a scientific and efficient collaborative working mechanism to ensure the policy implementation of the project. If coordination problems arise, the project team will have direct contact with the MNR and MIIT to ensure smooth coordination. Besides, investment from MIIT and MNR could facilitate building new program management team and increase management costs to ensure effective problem solving.	PMU MIIT MNR	No Change
Coordination of	Organizational	I = 4	The MNR, MIIT and implementation units	PMU	No

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
the relationship between project objectives and the implementation of comprehensive utilization technology.	Operational	L = 1 Low	maintain close contact with relevant departments and can apply necessary adjustment promptly according to the adjustment of national policies, so as to ensure the implementation of this project and achieve the optimal effect. Workshops shall be carried out on a regular basis to adjust project technology employment so that the overall goal will be achieved.	MIIT MNR	Change
Unemployment caused by the reduction of chemical phosphate fertilizer production (possibly in cement manufacturing too)	Social Financial	I = 5 L = 4 High	In the process of realizing green and low-carbon transformation of PCI, the United Nations '2030 agenda for sustainable development' was benchmarked to explore the formation of a multi-objective collaborative green development model to achieve ecological compensation, reduce poverty, improve ecological diversity, and ensure agricultural and food security. Focus on green employment action in the PCI, improve the employment quality of the PCI, create new employment growth points in the comprehensive utilization of phosphogypsum, and reduce the unemployment risk caused by industrial transformation. Training programs will be carried out for displaced workers to relocate them to emerging industrial enterprises that will be facilitated by the project, i.e. renewable energy, phosphogypsum processing, waste water treatment, natural phosphate fertilizer production, etc.	PMU MIIT MNR	No Change
Allocated funds cannot support project implementation in time and in full.	Operational Financial	I = 4 L = 3 Substantial	Energy saving, green and low-carbon development and green mine construction are the important contents of China's ecological civilization goals, which make an important basis for guaranteeing supporting funds. The MNR and MIIT attach great importance to the implementation of this project, and provide relevant technology, resources, funds, and system support. Relevant project provinces will also make commitment to supporting funds for the project in the early stage of implementation. In the process of project design and implementation, it will continue to promote and attract the establishment of diversified financing modes including private enterprises to ensure the supporting funds are fully in place. Constant follow-up with the pertinent co-financers will be conducted either to secure the committed co-financing or negotiate the amount of co-financing. Where required the reallocation of	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
			budget will be done to support the implementation of affected activities. This may entail the delivery of alternative outputs that are still contributing to the achievement of the relevant project outcome.		
Low level of commitment of stakeholders (including government agencies and PCI companies) in the implementation of project activities.	Political Strategic	I = 4 L = 2 Moderate	Policy regulations will be published to ensure stakeholders participate in the project. Market mechanism for phosphogypsum replacing cement will be established to encourage participation. Incremental costs will be applied to increase subsidy for cement substitution and low-carbon mining technologies.	PMU MIIT MNR	No Change
Difficulties in the replication of successful results of the project.	Organizational Operational Financial	I = 5 L = 2 Substantial	PMO will organize regional and national workshops to ensure timely spread of experience and knowledge. In these workshops other regions could share pilot projects results and experiences. Research and seminars will be carried out to identify the critical problems in replication and proper solutions. PMO, will carry out direct consultations with target private sector entities to determine their concerns and explore ways to fulfill their initial commitments and ensure end-users better understanding about the use and benefits of green, EE and LC technology applications	PMU MIIT MNR	No Change
Availability of co-financing is delayed and negatively affects the implementation of the project activities	Financial Political	I = 4 L = 1 Low	The MIIT and MNR will ensure that committed co-financing from the national and local governments is available and during the required time. Constant follow-up with the project partners in the private sector in the planning and implementation of the activities that they are co-financing will ensure that their financial support is available in a timely manner If financing problems arise, the project team will have direct contact with the MNR and MIIT to potentially secure additional funding from other sources of funds. If necessary, the project activities may have to be modified or replaced to adapt to the available funds.	PMU MIIT MNR	No Change
Varying vested interests and objectives of PCI companies as well as other stakeholders in	Operational Political	I = 3 L = 2 Moderate	Policy as a regulative measure will be carried out to ensure PCI companies stick to the project, and financial support will be provided to companies to ensure their participation. In regular workshops, companies could express	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
the local and central governments may prevent the effective organization and coordination of their participation and support of the project			their problems and together with the MNR and MIIT find proper solutions when faced with difficulties.		
The project would potentially result in the generation of waste (non-hazardous).	Environmental	I = 3 L = 5 Substantial	During the design and implementation of the project, strict environmental and social assessments will be conducted, and relevant recommendations will be made to reduce related environmental and safety risks. The project will promote the application of new and advanced technologies for cleaner production, which are expected to reduce emissions, process wastes, effluents compared to the current processes. The demonstration component of the proposed project will be designed taking into consideration the need to prevent a potential for the release, in the environment, of hazardous materials resulting from their handling, storage and use for the demonstrations and replications that will be carried out under the project. As the phosphate chemical industry is concentrated in Yunnan, Guizhou, Sichuan province, and other underdeveloped and ecologically fragile areas, the potential waste generation will be controlled and evaluated in the design and implementation stages. During operation, advanced waste disposal methods will be used to treat waste generated in the project to minimize the impact of waste on the environment. For example, solidification treatment of fluorine-containing waste and highly-charged heavy metal waste generated in the process of phosphogypsum reuse will minimize the impact of phosphate chemical waste on the environment.	PMU MIIT MNR	No Change
Low local and international prices of phosrock and phosphate chemicals will discourage PCI companies to implement green,	Financial Strategic	I = 4 L = 3 Substantial	MIIT and MNR will provide subsidies for green technologies and will carry out policies to favor companies in PCI with green technologies. Subsidizing policies should be flexible in order to provide proper incentives when market prices fluctuate.	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
EE, and LC technology application projects					
COVID-19 will negatively impact the social-economic development of the project areas.	Social Financial	I = 5 L = 4 High	The project development team will make use of China's effective institutional system to address the pandemic situation in a timely manner taking into consideration potential socio-economic impacts. Care will be taken in the design and implementation of the project especially when the pandemic persists to ensure that hardware installation works will be done safely in the demo sites with the health, safety, and welfare of the people in the demo sites and local communities in mind. If vaccination is not be ready, public health measures will be taken as is currently being done in China to ensure safety of the project participants and beneficiaries. If the pandemic persists during the project implementation phase, the project activities that will not involve on-the-ground work will be prioritized and implemented with the help of virtual means. The scheduling of, and action planning for, the hardware-related activities will be reviewed regularly by keeping abreast with the status of the pandemic and the public health situation in the country, in general, and particularly in the project demo sites.	PMU MIIT MNR	No Change
Extreme climate events in the project areas will delay and negatively affect the installation, operation and monitoring of the demonstration activities.	Environmental Operational	I = 3 L = 1 Low	Yunnan, Sichuan, Guizhou, and Hubei have higher risks of extreme weather compared to the other provinces. This make them through the years experienced in dealing with extreme climate risks. Local governments shall strengthen their capacities to deal with extreme climate events in general. The project will also cover climate risk adaptation capacity building sector in technology training workshops. Depending on the extent of the impacts of the adverse climate –related events, appropriate modifications in the installations (and budget) will be done. Potential reduction in the number of installations, or replacement with alternative demos will be done while considering the need to ensure the resulting interventions are still contributing to the realization of the project outcomes.	PMU MIIT MNR	No Change

The overall social and environmental risk rating of the project is ***High***. Because of this rating, an Environmental and Social Management Framework (ESMF) have been conducted during the PhosChemEE project preparation phase and are annexed together with the updated Social and Environmental Screening Procedure (SESP) to the proposal package. Environmental and social grievances (if any) will be reported to the GEF during the annual Project Implementation Review (PIR) reporting exercise.

Stakeholder Engagement

Based on the stakeholder analysis, the project's key players include MOF, UNDP, MNR, MIIT, NDRC, MEE, local governments, national research institutions, the private sector, and NGOs. They will take active part in the implementation of the PhosChemEE project activities while the others will assume either supporting or beneficiary roles.

Stakeholder	Roles and Responsibilities in Project Implementation
Ministry of Finance (MoF) of the People's Republic of China	As the GEF Operational Focal Point MOF will receive the GEF grant for this project on behalf of the Chinese Government. MOF will be a core member of the Project Steering Committee. MOF will have a core role in providing guidance, coordination and supervision on the disbursement and expense of the GEF grant, performance evaluation, documentation, and promotion of project results. The project team will provide project documentation and reporting to MOF through UNDP.
United Nations Development Programme (UNDP)	As the GEF Agency for the project it is overall responsible for the project oversight. UNDP will ensure project quality control to ensure that the Implementing Partner achieve the set project objective and outcomes. As part of its oversight work, UNDP will ensure that the project implementation and management follows UNDP/GEF rules and regulations. UNDP is a core member of the Project Steering Committee. The project Implementing Partner will provide project documentation and reporting to UNDP as outlined in the project document or otherwise agreed.
Ministry of Industry and Information Technology (MIIT) of the People's Republic of China	MIIT is the project implementing partner and serves as a member of the project steering committee. MIIT will lead the implementation of the project's Component 2 activities, which are in phosphate chemicals manufacturing; and part of the project's Component 3 activities on waste management in phosphate chemical production through its project office within the Energy Conservation and Comprehensive Utilization Division. It will ensure coordination between the relevant government agencies involved in the project at both national and local levels. MIIT will coordinate with the MNR, and supervise, the use of GEF funds allocated each year by the UNDP for all project components and carry out regular reporting of achievements in accordance with UNDP rules and procedures.
Ministry of Natural Resources (MNR) of the People's Republic of China	MNR is a responsible partner of the project and will assist the MIIT (project implementation partner). MNR will lead the implementation of the project's Component 1 activities, which are in phosphate mining and beneficiation; and part of the project's Component 3 activities in waste management in phosphate ore mining and refining through its project offices under the Division of International Cooperation and the Division of Mineral Resources Protection Supervision. It will assist the MIIT in ensuring close coordination between the relevant government agencies involved in the project at both national and local levels. MNR will coordinate with the MIIT the use of GEF funds allocated each year by the UNDP for Component 1 and part of Component 3 activities and carry out regular reporting of achievements through the MIIT in accordance with UNDP rules and procedures.
National Development and Reform Commission (NDRC)	As it has done during the preparation phase of this project, the NDRC will participate in and provide guidance in the planning of the proposed project activities to ensure that the project outputs can guide and encourage the

Stakeholder	Roles and Responsibilities in Project Implementation
	participating PCI entities to promote the comprehensive and conserving utilization of the country's phosphate ore resources, and also put forward suggestions for the planning and policy measures of resource conservation and comprehensive utilization in China.
Ministry of Ecology and Environment (MEE) of the People's Republic of China	As it has done during the PPG phase, MEE will participate in and guide the planning of this project activities during the implementation phase to ensure that the project outputs can provide intellectual support for China's participation in international climate change negotiations and the implementation of the United Nations Framework Convention on Climate Change (UNFCCC), and contributes to the achievement of the country's climate change mitigation and NDC targets.
Local Governments	The Provincial departments of natural resources (DNRs) and industry and information technology (DIITs) are to provide financial support and technical and policy advice in the detailed design and implementation of the demos that will be carried out under the project. Municipal and county-level governments will directly participate in the project activities providing support (e.g., logistical) in the implementation and monitoring and evaluation of the demos and demo replications that will be carried out in their respective localities. During project implementation, the provincial DNRs and DIITs are responsible for supervising municipal and county-level work, including providing technical support, monitoring support, coordination with partners, and capacity-building.
Relevant scientific research institutions[9]	These are expected to provide technical support in the implementation of research, study, evaluation activities of the project, making use of their expertise and scientific knowledge in the areas of energy saving and green low-carbon technologies as applied to the comprehensive and sustainable utilization of phosphate ore resources in the PCI. These entities are expected to assist in clearly defining and verifying the baseline situation and market prospects regarding the PCI and its value chain. It is also expected to assist in coordinating the research activities of the project with ongoing and planned similar research work both at the national and provincial levels.
Private sector	These are expected to provide technical (research, engineering, construction, operation and maintenance) support to the project in the areas of low carbon and efficient comprehensive utilization of phosphate ore resources, cleaner and energy efficient production in the PCI, as well as circular economy type measures such as low-carbon reuse of phosphogypsum and phosphate mining/refining tailings. The participation of enterprises and the private sector in the project will ensure participation of the target beneficiaries, i.e., PCI entities, in guiding the efforts of the project to remove the financial and market barriers to greener, cleaner and more energy efficient PCI.
Non-governmental organizations[10]	Non-governmental organizations at national, provincial and local levels shall participate in the stakeholder consultation processes during project implementation to ensure success of the promotional and capacity development activities of the project. They are also expected to facilitate effective coordination with the activities and projects of non-governmental organizations that contributes to the achievement of the project outcomes/outputs. These entities

[9] Includes Chinese Academy of Social Sciences, Institute of Mineral Resources Multi-Utilization of China Geological Survey, Sichuan University, Kunming University of science and technology, Guizhou Academy of Sciences and Wuhan University of Engineering, etc.

[10] Includes The Chemical Industry and Engineering Society of China, China Petroleum and Chemical Industry Federation, China chemical energy conservation technology association, China Phosphate and Compound Fertilizer Industry Association, local mining associations.

Stakeholder	Roles and Responsibilities in Project Implementation
	are also useful in the implementation of the capacity development and promotional activities of the project, where potential synergies can be used

South-South Cooperation:

To bring the voice of China to global and regional fora, the PhosChemEE project will explore opportunities for meaningful participation in specific events where UNDP could support engagement with the global development discourse on green and low carbon industrial development. The project will furthermore provide opportunities for regional and international cooperation with countries that are implementing similar initiatives on green and low carbon industrial development (especially the PCI industry) in geopolitical, social, and environmental contexts relevant to the proposed project in China. This international cooperation aspect is particularly important for the PhosChemEE project because of the leading role played by China towards countries along the "Belt and Road" initiative as well as the Association of Southeast Asian Nations (ASEAN). The advanced technology and successful practices of the PCI industry in China will help improve the green development capacity of the global PCI as well as in the countries included in the "Belt and Road" initiative and in the ASEAN economic zone.

Gender equality and Women's Empowerment

The project has been designed and it will be implemented guided by principles of gender equality and women's empowerment. Women are involved in both the management and technical departments of the national and local government agencies/institutions and play major roles in the top-level decision-making and strategic design process. On the ground level, the implementation of the proposed project will promote the welfare of, and offer equal opportunities to, both men and women, but particularly women in the areas where the project will be implemented. Firstly, implementation of this project will improve the technical level and reduce the intensity of labor in the PCI, which will in turn improve the work environment, job adaptability and enhanced chances of job advancement for women who are in some form or another involved in the PCI. Education and professional training opportunities that the proposed project will provide to PCI employees will contribute to improving women's professional competitiveness in the workplace and expand women's scope of employment in the underdeveloped and agriculture-based regions of the country, some of which include the planned demo sites of the proposed project. Secondly, the implementation of this project will directly improve the quality of local ecological environment and indirectly improve women's and children's health. Thirdly, the implementation of this project will help ensure agriculture and food security through the reduction of rural pollution, and thus reduce the risk in women's major source of income in these regions.

In general, the socio-economic condition of the project's demo sites is underdeveloped, and the ecological environment is very fragile. The implementation of this project will effectively improve the ecological environment quality and directly improve the health conditions of women and children. Furthermore, the implementation of this project helps to guarantee agriculture and food security, reduce rural pollution and the threat of crop failure, reduce women's risk of income loss. It is estimated that about 600,000 women and girls, and 600,000 to 1 million men would get benefits from the project through the implementation of the above measures.

During the project design and development (PPG) stage the following gender actions and considerations have been adopted and incorporated in the proposed activities:

- Ensure that the project is guided by principles of gender equality and women's empowerment. Women are involved in both the management and technical departments of the China's government agencies/institutions and play major roles in the top-level decision-making and strategic design process.
- Ensure the proposed policies, regulations, standards on the application of green, energy efficient, low carbon technologies in the PCI are gender responsive;
- Ensure gender equity is included as a criterion for providing/distribution of goods and services to communities in the project sites;
- Strengthen existing governance structures that promote gender equality and leadership in local governance structures in the PCI to support project implementation;
- Encourage information exchange and learning for men, women, and youths in the communities where the phosrock mines and phosphate refining and phosphate chemicals production facilities are situated to promote green, energy efficient and low carbon practices to other communities;
- Gender disaggregated data is collected during trainings, workshops, discussions, interviews, or focus group interaction;
- Consult both men and women during baseline data assessments, mid-term evaluations and terminal evaluations in the project sites;
- Ensure both men and women access information sharing sessions e.g. individuals with child-caring responsibilities or individuals with work obligations or disability;
- Support the development and promotion of service provision jobs in the relevant downstream activities of the PCI

A complete the Gender Action Plan has been already formulated during the PPG stage and it is annexed to the PhosChemEE project proposal package.

Innovativeness:

This project will promote and facilitate the greening and low carbon transformation of China's PCI through innovative (in the context of China) interventions that will address the current issues of the industry such as low quality phosphate chemicals produced, high resource (process material and energy) consumption and environment damaging operations and processes. Innovative production management approaches that are green, energy efficient, resource and environment-conserving will be introduced not only to respond to the needs to be productive, cost-effective, but also responsive to climate change issues. These innovative technology applications in the PCI are estimated to bring about improvements in the specific energy consumption (SEC) in the various sub-sectors of the PCI. The proper application of these technologies will reduce the overall SEC in phosrock mining and refining operations (e.g., improved mine recovery rate, mining dilution rate, and beneficiation recovery rate) as well as in phosphate chemicals production and concomitantly improve the total GHG emissions reduction. Please refer to the Core Indicators Annex for the potential energy savings and corresponding GHG emission reductions in the implementation of innovative circular economy technology applications in the management and utilization of the mining and refining wastes, and phosphate chemicals production by-product wastes from the PCI. Another innovative feature is the employment of comprehensive assessment methods of energy savings and emissions reduction, improved carbon accounting system of the industry, and formulation and enforcement of the standards for energy conservation and low-carbon development of the industry.

Sustainability:

The potential for a wider adoption of green, energy efficient low carbon technologies and techniques in the PCI is quite high, considering the support that the GOC is providing to the industry. The activities that will be carried out under this proposed project are barrier removal in nature. To ensure avoidance of the recurrence of the barriers and the continuance of the enabling environments that will be created and/or facilitated by the project, appropriate sustainable follow-up actions are planned as part of the project activities. Such action plan will be implemented after the project as per the institutional arrangements that will be developed for such purpose. Close coordination with the China Inorganic Salt Industry Association that is also fully committed to improving the PCI's energy utilization efficiency can also help sustain efforts of the phosphate chemicals SMEs to replicate the green, energy efficient low carbon energy technologies that are showcased and promoted by the project. Based on the number of companies in the PCI, it is estimated that the replication of the demos that will be implemented under the proposed project is in the order 20 times per demo.

Scale-up Potential:

The removal of barriers and the effective and seamless implementation of the procedures and regulatory and policy and institutional frameworks, as well as the financing schemes that will be established will influence the scaling-up of the project, e.g., to carry out the same interventions in other provinces with phosphate rock deposits. The proposed project will lay a solid basis for scaling up with the successful demonstration of the applicable green, energy efficient and low carbon technologies/techniques, as well as the financing models that will support such applications by the phosphate chemicals companies and SMEs. These, aside from the barrier removal activities, provide the possibility for scaling-up the application of these models/schemes in other provinces, as well as in other countries. The scale-up will also be made possible through the project's information sharing, technical assistance, and promotional activities. In the case of the EE and environmental footprint improvements in the phosrock mining and refining operations of the PCI and considering the number and production capacities of the big phosrock mining companies, the planned demo capacity in this proposed project can potentially be scaled up 20 times. In that case, the PCI in China can potentially be among the significant contributors to the achievement of the country's NDC targets.

V. PROJECT RESULTS FRAMEWORK

This project will contribute to the following Sustainable Development Goal (s): SDG 7: “Ensure access to affordable, reliable, sustainable and modern energy for all”, SDG 9: " Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation"; SDG 12: “Ensure sustainable consumption and production patterns”; SDG 13: "Take urgent action to combat climate change and its impacts by regulating emissions and promoting developments in renewable energy “, and to a lesser extent SGD 8: “Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all”.				
This project will contribute to the following country outcome (UNDAF/CPD, RPD, GPD): UN Development Assistance Framework China (2016-2020): More people enjoy a cleaner, healthier, and safer environment as a result of improved environmental protection and sustainable green growth (Outcome 2).				
	Objective and Outcome Indicators	**Baseline**	**Mid-term Target**	**End of Project Target**
Project Objective:	Enabling the extensive application of low carbon and energy efficient technologies in the phosphate chemicals industry in China.			
	Mandatory Indicator 1: Number of direct beneficiaries disaggregated by gender as co-benefit of GEF investment.	Female: 0 Male: 0	Female: 150,000 Male: 150,000-250,000	Female: 600,000 Male: 600,000-1,000,000
	Mandatory GEF Core Indicators: Cumulative amount of greenhouse gas emissions mitigated from the PCI (million tons CO_{2eq}).	0	0.429	35.987
	Cumulative amount of fossil fuels saved in all energy consuming operations and processes in the PCI (million tce).	0	0.178	14.779
Project Component 1	**Green and Low-carbon Development and Operation of Phosphate Mines**			
Project Outcome 1.1: Improved interest and commitment of the phosphate chemical industry in the green, low carbon and energy efficient operations of the phosphate mining sub-	Number of additional technologies and process/operation improvements implemented by phosphate ore mining companies that are facilitated by the effectively enforced policies and regulations that promote and support green, energy efficient and low carbon technologies/techniques.	0	3	6
	Number of additional certified technical and operations personnel of phosphate mining companies that are capable of designing and implementing projects on green, energy efficient	0	500	1600

sector in China.	and low carbon technology applications in phosphate ore mining and refining.			
Outputs to achieve Outcome 1.1	Output 1.1.1: Formulated, recommended, approved, and enforced policy and institutional frameworks supporting green, energy efficient low carbon development initiatives in phosphate rock (phosrock) mining and refining. Output 1.1.2: Formulated, recommended, and enforced standards, policies and implementing rules and regulations on the promotion and practice of green, energy efficient, low carbon technologies/techniques, in the development and operations of phosphate mines. Output 1.1.3: Documented annual evaluation reports on the energy performance and environmental impacts of each demo activity and documented and disseminated demo results. Output 1.1.4: Designed, conducted and post-evaluated continuing education program in the PCI mining sub-sector in Weng'an county in Guizhou Province, Jinning county in Yunnan Province, and Mabian and Leibo counties in Sichuan Province, with regular capacity needs assessment in the area of green, energy efficient low carbon technologies applied in phosphate mining and refining. Output 1.1.5: Established the online monitoring, reporting and verification (MRV) system for energy-saving and GHG emission reduction from the application green, energy efficient low-carbon technologies in the PCI including information sharing scheme. Output 1.1.6: Established and operational up-to-date information management and exchange system for all aspects of green, energy efficient low carbon phosphate mining and refining technologies and best practices.			
Outcome 1.2: Enhanced confidence in the feasibility of the application of green, energy efficient low carbon technologies in phosrock mining and refining in China.	Number of phosphate mining companies that have plans for projects on implementing green, low carbon and energy efficient technology applications in their operations & processes.	0	22	75
	Average overall specific energy consumption (SEC) in phosphate ore mining and refining, kgce per ton phosrock.	9.87 (2020)	9.67	9.38
Outputs to achieve Outcome 1.2	Output 1.2.1: Completed designs and plans of demonstrations of green, energy efficient and low carbon technologies in phosphate mines development and operations including: (a) Improved design for the mining and refining operations in the demo phosrock mines in Weng'an county in Guizhou Province, and Mabian county in Sichuan province; (b) Feasibility analyses of the application and operation of green, energy efficient and low carbon phosrock mining and refining systems; and, (c) Implementation plans (including financing arrangements) for each green, energy efficient and low carbon technology application in the demo phosrock mines. Output 1.2.2: Installed and operational green, energy efficient low carbon technology application demos in phosrock mining and refining in Weng'an county in Guizhou province, Jinning county in Yunnan province, and			

	Mabian and Leibo counties in Sichuan province. Output 1.2.3: Established, funded and operational financial support schemes that are accessible to PCI mining companies to support their green, energy efficient and low carbon technology application initiatives. Output 1.2.4: Approved and financed follow-up plan for the replication of the application of demonstrated green, energy efficient low carbon technologies in phosrock mining and refining in other localities.			
Project Component 2	**Green and Low-Carbon Design and Operation of Phosphate Chemicals Production Facilities**			
Outcome 2.1:	Number of additional technologies and process/operation improvements implemented in phosphate chemical companies that are facilitated by the effectively enforced policies and regulations that promote and support energy efficiency and cleaner production technologies/techniques.	0	10	20
	Number of additional certified technical and operations personnel of phosphate chemical companies that are capable of designing and implementing projects on energy efficiency and cleaner production technology applications in phosphate chemicals production.	0	500	1,800
Outputs to achieve Outcome 2.1	Output 2.1.1: Formulated, recommended, approved, and enforced policy and institutional frameworks supporting green, energy efficient low carbon technology applications in phosphate chemicals production. Output 2.1.2: Formulated, recommended, and enforced standards, policies and implementing rules and regulations on the promotion and practice of green, energy efficient, low carbon technologies/techniques, in phosphate chemicals production. Output 2.1.3: Documented annual evaluation reports on the energy performance and environmental impacts of each demo activity and documented and disseminated demo results based on the MRV system (Output 1.1.5) established under Component 1. Output 2.1.4: Designed, conducted and post-evaluated continuing education program in the PCI chemicals sub-sector in Yunnan, Guizhou, Sichuan, and Hubei, with regular capacity needs assessment in the area of green, energy efficient and low carbon technologies applied in phosphate chemicals manufacturing. Output 2.1.5: Completed market analysis report on the international trade trends for phosrock and phosphate chemicals, and the indirect carbon emission and ecological impacts in the international trading of these products. Output 2.1.6: Established and operational system for managing and sharing available and accessible information on all aspects of green, energy efficient and cleaner phosphate chemicals production in China.			
Outcome 2.2: Enhanced confidence, understanding and capacity in the	Number of phosphate chemical companies that have plans for projects for implementing energy efficiency and cleaner production technology applications in their operations &	0	15	50

feasibility of the application of green, energy efficient low carbon technologies in phosphate chemicals production in China.	processes.			
	Average overall specific energy consumption in phosphate chemicals production, tce/'000 CNY GDP (2020).	0.25 (base year 2020)	0.244	0.238
Outputs to achieve Outcome 2.2	Output 2.2.1: Completed designs and plans of demonstrations of green, energy efficient low carbon technologies in phosphate chemicals production including: (a) Improved design for cleaner production demos; (2) Feasibility analyses of the application and operation of green, energy efficient and low carbon phosphate chemical production systems; and, (3) Implementation plans (including financing arrangements) for each green, energy efficient and low carbon technology application in the demo phosphate chemical companies. Output 2.2.2: Installed and operational green, energy efficient low carbon technology application demos in phosphate chemicals production. Output 2.2.3: Designed and approved pilot financing scheme for small and medium enterprises (SMEs) that manufacture phosphate chemicals. Output 2.2.4: Operational pilot financing scheme for phosphate chemicals SMEs in 2 to 3 regions. Output 2.2.5: Approved and financed follow-up plan for the replication of the application of demonstrated green, energy efficient low carbon technologies in phosphate chemicals production in other localities.			
Project Component 3	**Green and Low Carbon Design and Operation of Waste Management Systems in the Phosphate Chemicals Industry**			
Outcome 3.1: Enhanced commitment of, and institutional and technical arrangements for, the phosphate chemical industry in green and low carbon waste management.	Number of additional technologies and process/operation improvements implemented in waste management in the PCI companies that are facilitated by the effectively enforced policies and regulations that promote and support green, energy efficient and low carbon technologies/techniques.	0	5	8
Outputs to achieve Outcome 3.1	Output 3.1.1: Formulated, recommended, approved, and enforced policies and institutional frameworks that support green, low carbon initiatives in waste management in the PCI. Output 3.1.2: Improved existing and new enforced schemes, standards, policies/regulations on the application of green and low carbon waste management technologies in the PCI operations and processes. Output 3.1.3: Documented annual evaluation reports on the energy performance and environmental impacts of each demo based on the MRV system developed under Component 1.			

	Output 3.1.4: Designed, conducted and post-evaluated continuing education program with regular capacity needs assessments in the area of green and low carbon technologies applied in waste management in the PCI. Output 3.1.5: Completed research reports on: (1) annual volume of waste production in phosrock mining and refining, and in phosphate chemicals production in China; (2) green and low carbon waste management systems for waste recycling and reuse developed and implemented in other countries and their energy utilization performances; and, (3) potential commercial markets for processed waste, e.g., phosphogypsum. Output 3.1.6: Established and operational information management system for sharing available and accessible information about market and research developments and R&D in the country and abroad on green and low carbon waste management systems in the PCI.			
Outcome 3.2: Increased confidence in the feasibility of the application of green and low carbon technologies in the management of waste in the phosphate chemicals Industry in China.	Number of additional projects on green, energy efficient low carbon technologies in waste management in the PCI that are supported by the national and/or local governments.	0	5	10
Outputs to achieve Outcome 3.2	Output 3.2.1: Completed designs and plans of demonstrations of the application of green and low carbon technologies/techniques in waste management in the phosphate chemicals industry, including: (a) improved phosrock mining tailings recycling and reuse; and (b) energy efficient processing of phosphogypsum and marketing of the gypsum products. Output 3.2.2: Installed and operational green and low carbon waste management technology application demos in the phosphate chemicals industry. Output 3.2.3: Approved and financed follow-up plan for the replication of the application of demonstrated green, low carbon waste management technologies in the phosphate chemicals industry in other localities.			

VI. MONITORING AND EVALUATION (M&E) PLAN

Project-level monitoring and evaluation will be undertaken in compliance with UNDP requirements as outlined in the UNDP POPP (including guidance on GEF project revisions) and UNDP Evaluation Policy **The UNDP Country Office is responsible for ensuring full compliance with all UNDP project M&E requirements including project monitoring, UNDP quality assurance requirements, quarterly risk management, and evaluation requirements.**

Additional mandatory GEF-specific M&E requirements will be undertaken in accordance with the GEF Monitoring Policy and the GEF Evaluation Policy and other relevant GEF policies[11]. The M&E plan and budget included below will guide the GEF-specific M&E activities to be undertaken by this project.

In addition to these mandatory UNDP and GEF M&E requirements, other M&E activities deemed necessary to support project-level adaptive management will be agreed – including during the Project Inception Workshop - and will be detailed in the Inception Report.

Minimum project monitoring and reporting requirements as required by the GEF:

Inception Workshop and Report: A project inception workshop will be held within 2 months from the First disbursement date, with the aim to:

a. Familiarize key stakeholders with the detailed project strategy and discuss any changes that may have taken place in the overall context since the project idea was initially conceptualized that may influence its strategy and implementation.
b. Discuss the roles and responsibilities of the project team, including reporting lines, stakeholder engagement strategies and conflict resolution mechanisms.
c. Review the results framework and monitoring plan.
d. Discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E budget; identify national/regional institutes to be involved in project-level M&E; discuss the role of the GEF OFP and other stakeholders in project-level M&E.
e. Update and review responsibilities for monitoring project strategies, including the risk log; SESP report, Social and Environmental Management Framework (where relevant) and other safeguard requirements; project grievance mechanisms; gender strategy; knowledge management strategy, and other relevant management strategies.
f. Review financial reporting procedures and budget monitoring and other mandatory requirements and agree on the arrangements for the annual audit.
g. Plan and schedule Project Board meetings and finalize the first-year annual work plan. Finalize the TOR of the Project Board.
h. Formally launch the Project.

GEF Project Implementation Report (PIR):
The annual GEF PIR covering the reporting period July (previous year) to June (current year) will be completed for each year of project implementation. UNDP will undertake quality assurance of the PIR before submission to the GEF. The PIR submitted to the GEF will be shared with the Project Board. UNDP will conduct a quality review of the PIR, and this quality review and feedback will be used to inform the preparation of the subsequent annual PIR.

[11] See https://www.thegef.org/gef/policies_guidelines

GEF Core Indicators:
The GEF and/or LDCF/SCCF Core indicators included as Annex 12 will be used to monitor global environmental benefits and will be updated for reporting to the GEF prior to MTR and TE. Note that the project team is responsible for updating the indicator status. The updated monitoring data should be shared with MTR/TE consultants prior to required evaluation missions, so these can be used for subsequent ground truthing. The methodologies to be used in data collection have been defined by the GEF and are available on the GEF website.

Independent Mid-term Review (MTR):
The evaluation will be 'independent, impartial and rigorous'. The evaluators that UNDP will hire to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project under review. The GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the evaluation process. Additional quality assurance support is available from the BPPS/NCE-VF Directorate. The final MTR report and MTR TOR will be publicly available in English and will be posted on the UNDP ERC by 1 October 2024. A management response to MTR recommendations will be posted in the ERC within six weeks of the MTR report's completion.

Terminal Evaluation (TE):
The evaluation will be 'independent, impartial and rigorous'. The evaluators that UNDP will hire to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project being evaluated. As in the MTR, the GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the terminal evaluation process. Additional quality assurance support is available from the BPPS/NCE-VF Directorate. The final TE report and TE TOR will be publicly available in English and posted on the UNDP ERC by 31 March 2027. A management response to the TE recommendations will be posted to the ERC within six weeks of the TE report's completion.

Final Report:
The project's terminal GEF PIR along with the terminal evaluation (TE) report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the Project Board during an end-of-project review meeting to discuss lesson learned and opportunities for scaling up.

Agreement on intellectual property rights and use of logo on the project's deliverables and disclosure of information**:** To accord proper acknowledgement to the GEF for providing grant funding, the GEF logo will appear together with the UNDP logo on all promotional materials, other written materials like publications developed by the project, and project hardware. Any citation on publications regarding projects funded by the GEF will also accord proper acknowledgement to the GEF. Information will be disclosed in accordance with relevant policies notably the UNDP Disclosure Policy[12] and the GEF policy on public involvement[13].

[12] See http://www.undp.org/content/undp/en/home/operations/transparency/information_disclosurepolicy/

[13] See https://www.thegef.org/gef/policies_guidelines

Monitoring Plan: The project results, corresponding indicators and mid-term and end-of-project targets in the project results framework will be monitored by the Project Management Unit annually, and will be reported in the GEF PIR every year, and will be evaluated periodically during project implementation. If baseline data for some of the results indicators is not yet available, it will be collected during the first year of project implementation. Project risks, as outlined in the risk register, will be monitored quarterly.

Results Monitoring	Indicators	Targets		Description of Indicators and Targets	Data source or Collection Methods[14]	Frequency	Responsible for Data Collection	Means of Verification	Critical Assumptions
		Mid-term	End of Project						
Project Objective: Enabling the extensive application of low carbon and energy efficient technologies in the phosphate chemicals industry in China.	**Indicator 1:** Number of direct beneficiaries disaggregated by gender as co-benefit of GEF investment.	Female: 150,000 Male: 150,000 - 250,000	Female: 600,000 Male: 600,000-1,000,000	Cumulative number of the workers in phosphate mining companies and residents living in the counties where demos and their replications and scale-ups will be implemented that will benefit from cleaner and healthier environment and socio-economic growth	Surveys Direct data collection	Annually Reported in DO tab of the GEF PIR	PMU Project Consultants National Office of Statistics UNDP CO	Project M&E and activity reports Annual reports submitted by relevant national and local GOC entities.	Continuous commitment, support & active participation of the national and local governments in the implementation of cleaner and energy-efficient technology. Realization of committed co-financing from the national and local governments as well as the private sector in the implementation of project activities and monitoring systems
	Indicator 2: Cumulative amount of greenhouse gas emissions mitigated from the PCI (million tons CO_{2eq})	0.429	35.987	Cumulative reduction of GHG emissions, over the implementation period of the PhosChemEE Project, attributable to the implementation of technologies proposed under the Alternative Scenario	Direct data collection using the MRV system, aligned with GEF GHG measurement methodology	Annually Reported in DO tab of the GEF PIR	PMU Project Consultants		
	Indicator 3: Cumulative amount of fossil fuels saved in all energy	0.178	14.779	Cumulative reduction of fossil fuel consumption, over the implementation period of the PhosChemEE Project, due to the application of	Direct data collection using the MRV system, aligned with GEF GHG	Annually Reported in DO tab of the GEF PIR	PMU Project Consultants		

[14] Data collection methods should outline specific tools used to collect data and additional information as necessary to support monitoring. The PIR cannot be used as a source of verification.

Results Monitoring	Indicators	Targets		Description of Indicators and Targets	Data source or Collection Methods	Frequency	Responsible for Data Collection	Means of Verification	Critical Assumptions
		Mid-term	End of Project						
	consuming operations and processes in the PCI (million tce).			technologies proposed under the Alternative Scenario	measurement methodology				
Outcome 1.1: Improved interest and commitment of the phosphate chemical industry in the green, low carbon and energy efficient operations of the phosphate mining sub-sector in China.	**Indicator 4:** Number of additional technologies and process/operation improvements implemented by phosphate ore mining companies that are facilitated by the effectively enforced policies and regulations that promote and support green, energy efficient and low carbon technologies or techniques.	3	6	This indicator will measure the number of additional low carbon technologies implemented by phosphate ore mining and refining companies that are spurred and facilitated by the enforcement of the policies that will be formulated under PhosChemEE and that will promote and support green, energy efficient and low carbon technologies.	Surveys Data reports from the demo companies	Annually Reported in DO tab of the GEF PIR	PMU Project Consultants	PhosChemEE Project M&E and activity reports Demo companies' annual reports Policies, regulations, and standards approved by the central and local governments Capacity building programs evaluation reports	Relevant entities will be fully cooperating in the sharing of data and information. The private sector fully supports and commits to the feasible replication of successful results of the PhosChemEE project demonstrations. The planned demonstrations are implemented and completed as per schedule
	Indicator 5: Number of additional certified technical and	500	1600	Number of relevant personnel of phosphate mining and refining companies that participate into capacity	Surveys Training programs assessment	Annually Reported in DO tab of the GEF PIR	PMU Project Consultants		

Results Monitoring	Indicators	Targets		Description of Indicators and Targets	Data source or Collection Methods	Frequency	Responsible for Data Collection	Means of Verification	Critical Assumptions
		Mid-term	End of Project						
	operations personnel of phosphate mining companies that are capable of designing and implementing projects on green, energy efficient and low carbon technology applications in phosphate ore mining and refining.			building programs and as a result of the training will become capable of designing and implementing projects on green, energy efficient and low carbon technology applications in the phosphate mining and refining sub-sector					
Outcome 1.2: Enhanced confidence in the feasibility of the application of green, energy efficient low carbon technologies in phosrock mining and refining in China.	**Indicator 6:** Number of phosphate mining companies that have plans for projects on implementing green, low carbon and energy efficient technology applications in their operations &	22	75	Number of phosphate mining and refining companies that during PhosChemEE implementation phase will have designed and developed plans for the implementation of projects on green, low carbon and energy efficient technology applications in their operations initiatives.	Surveys Companies' data reports	Annually Reported in DO tab of the GEF PIR	PMU Project Consultants National Office of Statistics		

Results Monitoring	Indicators	Targets		Description of Indicators and Targets	Data source or Collection Methods	Frequency	Responsible for Data Collection	Means of Verification	Critical Assumptions
		Mid-term	End of Project						
	processes.								
	Indicator 7: Average overall specific energy consumption in phosphate ore mining and refining, kgce per ton phosrock.	9.67	9.38	Taking 100% as basis for the current levels of average specific energy consumption, this indicator will measure the percentage of improvement of the SECs; in other words the percentage of energy saved referred to present levels.	Direct data collection using the MRV system	Annually Reported in DO tab of the GEF PIR	PMU Project Consultants Project consultant		
Outcome 2.1: Enhanced adoption by the phosphate chemical industry of a green, low carbon and energy efficient development model in phosphate chemicals production.	**Indicator 8:** Number of additional technologies and process/operation improvements implemented in phosphate chemical companies that are facilitated by the effectively enforced policies and regulations that promote and support energy efficiency and cleaner	10	20	This indicator will measure the number of additional low carbon technologies implemented by phosphate chemicals manufacturing companies that are spurred and facilitated by the enforcement of the policies that will be formulated under PhosChemEE and that will promote and support green, energy efficient and low carbon technologies.	Surveys Data reports from the demo companies	Annually Reported in DO tab of the GEF PIR	PMU Project Consultants	PhosChemEE Project M&E and activity reports Demo companies' annual reports Policies, regulations, and standards approved by the central and local governments Capacity building programs evaluation reports	Relevant entities will be fully cooperating in the sharing of data and information. The private sector fully support and commit to the feasible replication of successful results of the PhosChemEE project demonstrations. The planned demonstrations are implemented and completed as per schedule

Results Monitoring	Indicators	Targets		Description of Indicators and Targets	Data source or Collection Methods	Frequency	Responsible for Data Collection	Means of Verification	Critical Assumptions
		Mid-term	End of Project						
	production technologies or techniques.								
	Indicator 9: Number of additional certified technical and operations personnel of phosphate chemical companies that are capable of designing and implementing projects on energy efficiency and cleaner production technology applications in phosphate chemicals production.	500	1,800	Number of relevant personnel of phosphate chemicals production companies that participate into capacity building programs and as a result of the training will become capable of designing and implementing projects on green, energy efficient and low carbon technology applications in the phosphate chemicals production sub-sector	Surveys Training programs assessment	Annually Reported in DO tab of the GEF PIR	PMU Project Consultants		
Outcome 2.2: Enhanced confidence, understanding and capacity in the feasibility of the	**Indicator 10:** Cumulative number of phosphate chemicals companies that have plans for	15	50	Number of phosphate chemicals manufacturing companies that during PhosChemEE implementation phase will have designed and developed plans for the implementation of	Surveys Companies data reports	Annually Reported in DO tab of the GEF PIR	PMU Project Consultants National Office of Statistics		

Results Monitoring	Indicators	Targets		Description of Indicators and Targets	Data source or Collection Methods	Frequency	Responsible for Data Collection	Means of Verification	Critical Assumptions
		Mid-term	End of Project						
application of green, energy efficient low carbon technologies in phosphate chemicals production in China.	projects for implementing energy efficiency and cleaner production technology applications in their operations & processes.			projects on green, low carbon and energy efficient technology applications in their operations initiatives.					
	Indicator 11: Average overall specific energy consumption in phosphate chemicals production, tce/'000 CNY GDP (2020).	0.244	0.238	Average overall specific energy consumption in phosphate chemicals production, tce/'000 CNY GDP (2020). Taking 100% as basis for the current levels of average specific energy consumption, this indicator will measure the percentage of improvement of the SECs; in other words the percentage of energy saved referred to present levels.	Direct data collection using the MRV system	Annually Reported in DO tab of the GEF PIR	PMU Project Consultants Project consultant		
Outcome 3.1: Enhanced commitment of, and institutional and technical arrangements for, the	**Indicator 12:** Number of additional technologies and process and operation improvements implemented	5	8	This indicator will measure the number of additional low carbon technologies for waste management implemented by companies operating in the entire phosphate	Surveys Data reports from the demo companies	Annually Reported in DO tab of the GEF PIR	PMU Project Consultants	PhosChemEE Project M&E and activity reports Demo companies' annual reports	Continuous commitment and support by the private sector and general public on the establishment of a local market for

Results Monitoring	Indicators	Targets		Description of Indicators and Targets	Data source or Collection Methods	Frequency	Responsible for Data Collection	Means of Verification	Critical Assumptions
		Mid-term	End of Project						
phosphate chemical industry in green and low carbon waste management.	in waste management in the PCI companies that are facilitated by the effectively enforced policies and regulations that promote and support green, energy efficient and low carbon technologies or techniques.			chemicals industry that are spurred and facilitated by the enforcement of the policies that will be formulated under PhosChemEE and that will promote and support green, energy efficient and low carbon technologies.					cleaner and energy-efficient technologies and measures
Outcome 3.2: Increased confidence in the feasibility of the application of green and low carbon technologies in the management of waste in the phosphate chemicals Industry in China.	**Indicator 13:** Number of additional projects on green, energy efficient low carbon technologies in waste management in the PCI that are supported by the national and/or local governments.	5	10	Number of PCI companies that during PhosChemEE implementation phase will have approved projects for the implementation of projects on green, low carbon and energy efficient technology applications in their operations initiatives and supported by the national and local governments.	Surveys Companies data reports	Annually Reported in DO tab of the GEF PIR	PMU Project Consultants National Office of Statistics		

Monitoring and Evaluation Budget for Project Execution:		
GEF M&E Requirements to be undertaken by Project Management Unit (PMU)	**Indicative costs (US$)**	**Time frame**
Inception Workshop and Report	30,000	Inception Workshop within 2 months of the First Disbursement
M&E required to report on progress made in reaching Project Log Frame Indicators and GEF Core indicators	5,000 per year	Annually and at mid-point and closure.
Preparation of the annual GEF Project Implementation Report (PIR)	0	Annually typically between June-August
Monitoring all risks	3,000 per year	On-going each year
Monitoring of social and environmental safeguard screening	2,000 per year	On-going each year
Monitoring of stakeholder engagement plan	1,500 per year	Ongoing each year
Monitoring of gender action plan	1,500 per year	On-going each year
Project Board Meetings	5,000 per year	Annually
Independent Mid-term Review (MTR)*	40,000	*1 October 2024*
Independent Terminal Evaluation (TE):	40,000	*31 March 2027*
TOTAL Indicative COST	**200,000**	

*costs associated with conducting the independent evaluation to be commissioned by UNDP not the Implementing Partner or the PMU.

VII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

1. General Roles and Responsibilities of the Project's Governance Mechanism:

The PhosChemEE Project will be implemented in accordance with the UNDP's national implementation modality (NIM). In this implementation modality, the project is implemented by the government. UNDP identifies a suitable national entity as Implementing Partner(IP) to implement the project. Where necessary other national/local government entity/entities can be designated as responsible partner(s) to assist the IP. The IP directly assumes the responsibility for the delivery of the project outputs and carries out all activities towards the achievement of these outputs. UNDP has no role in the execution of the project activities. Its role is limited to overseeing the project implementation on behalf of the GEF. In NIM, the government regulations, rules and procedures shall apply to project implementation to the extent that they do not contravene the principles of the financial regulations and rules of UNDP and GEF, and unless requested by the IP, in which case UNDP rules shall apply.

UNDP (as the GEF Agency) oversees the implementation of the project. It will not be executing any project activity apart from the required GEF Agency project cycle management services. It will ensure compliance with the specific instructions that must be followed regarding the financial management requirements to be complied with the IP, as well as ensure that the GEF Minimum Fiduciary Standards Requirements are met by the IP and other designated responsible party/parties at all stages of project implementation.

Implementing Partner: The Implementing Partner for this project is the Ministry of Industry and Information Technology (MIIT) ;
The Implementing Partner is the entity to which the UNDP Administrator has entrusted the implementation of UNDP assistance specified in this signed project document along with the assumption of full responsibility and accountability for the effective use of UNDP resources and the delivery of outputs, as set forth in this document.

The Implementing Partner is responsible for executing this project. Specific tasks include:

- Project planning, coordination, management, monitoring, evaluation, and reporting. This includes providing all required information and data necessary for timely, comprehensive, and evidence-based project reporting, including results and financial data, as necessary. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes and is aligned with national systems so that the data used and generated by the project supports national systems.
- Responsibility in the implementation of the project activities in phosphate chemicals production (Outcomes 2.1 & 2.2 and parts of Outcomes 3.1 & 3.2). It will be responsible for coordinating with the PMO and the participating phosphate chemicals manufacturing companies on the implementation of planned technical assistance activities, particularly those on the demonstration activities.
- Risk management as outlined in this Project Document;
- Procurement of goods and services, including human resources;
- Financial management, including overseeing financial expenditures against project budgets;
- Approving and signing the multiyear work plan;
- Approving and signing the combined delivery report at the end of the year; and,
- Signing the financial report or the funding authorization and certificate of expenditures.

Responsible Parties: During the 5-year implementation period of the PhosChemEE Project, the MIIT, as the implementing partner will require the cooperation and services of several responsible parties to implement or support the implementation of the project activities.

MOF will receive the GEF grant for the implementation of the PhosChemEE Project on behalf of the Chinese Government. MOF will be a member of the Project Steering Committee, and it will have a key role in providing guidance, coordination and supervision on the disbursement and expense of GEF grant, performance evaluation, summary, and promotion of project results.

MNR will be responsible in the implementation of the project activities in phosphate ore mining and refining (Outcomes 1.1 & 1.2 and parts of Outcomes 3.1 & 3.2). It will be responsible for coordinating with the PMO and the participating phosphate mining and refining companies on the implementation of planned technical assistance activities, particularly those on the demonstration activities.

MEE will be involved in promotion activities to stimulate rural communities and other key players in the provinces where the project activities will be carried out, particularly the demonstrations in the host PCI companies. MEE experience and expertise will guarantee the environmental sustainability of the project activities and will assure that they comply with the UN Framework Convention on Climate Change requirements.

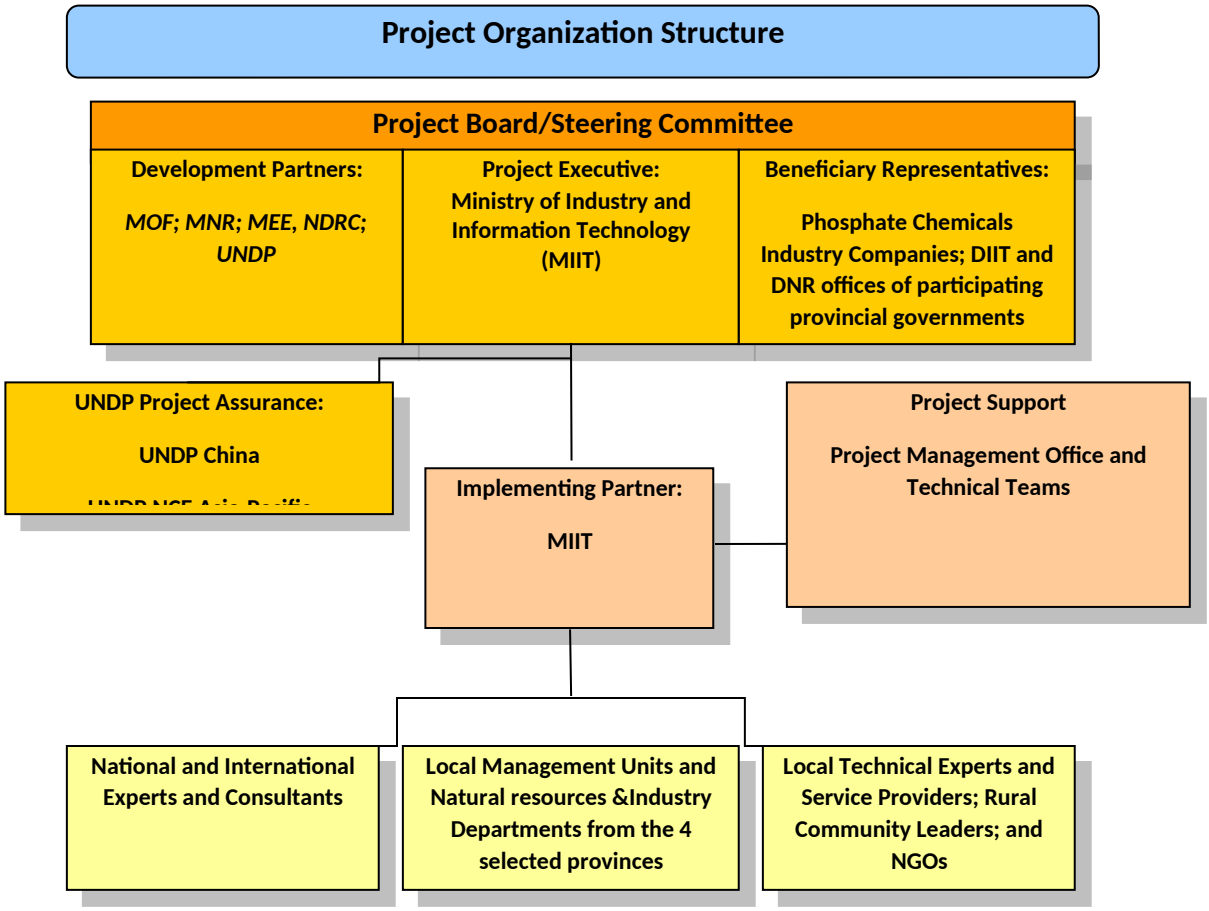
NDRC's main contributions to the PhosChemEE Project will be the promotion of low-carbon and energy efficient technologies and cleaner production measures to the PCI companies to stimulate their active participation, and provision of advice to the government on energy resources conservation, comprehensive energy utilization planning, and policy measures.

Local governments at provincial and county levels, and particularly their natural resources and industry and information technology departments, will be actively engaged for the implementation of baseline and demonstration projects and subsequent monitoring and evaluation activities. Furthermore, local government officials will participate into capacity building programs and awareness raising initiatives.

Project stakeholders and target groups: For project stakeholders and target groups, please refer to the Stakeholder Engagement Plan in Section IV and to Annex 7.

UNDP: UNDP is accountable to the GEF for the implementation of this project. This includes overseeing project execution undertaken by the Implementing Partner to ensure that the project is being carried out in accordance with UNDP and GEF policies and procedures and the standards and provisions outlined in the Delegation of Authority (DOA) letter for this project. **The UNDP GEF Executive Coordinator, in consultation with UNDP Bureaus and the Implementing Partner, retains the right to revoke the project DOA, suspend or cancel this GEF project.** UNDP is responsible for the Project Assurance function in the project governance structure and presents to the Project Board and attends Project Board meetings as a non-voting member.

2. Project Organization Structure:



Second line of defense

- Regional Bureau oversees RR and Country Office compliance at portfolio level.
- BPPS NCE RTA oversees technical quality assurance and GEF compliance. BPPS NCE PTA oversees RTA function.
- UNDP GEF Executive Coordinator and Regional Bureau Deputy Director can revoke DOA/cancel suspend project or provide enhanced oversight.

The UNDP Resident Representative assumes full responsibility and accountability for oversight and quality assurance of this Project and ensures its timely implementation in compliance with the GEF-specific requirements and UNDP's Programme and Operations Policies and Procedures (POPP), its Financial Regulations and Rules and Internal Control Framework. A representative of the UNDP Country Office will assume the assurance role and will present assurance findings to the Project Board, and therefore attends Project Board meetings as a non-voting member.

3. Segregation of Duties and Firewalls vis-à-vis UNDP Representation on the Project Board:

As noted in the Minimum Fiduciary Standards for GEF Partner Agencies, in cases where a GEF Partner Agency (i.e. UNDP) carries out both implementation oversight and execution of a project, the GEF Partner

Agency (i.e. UNDP) must separate its project implementation oversight and execution duties, and describe in the relevant project document a: 1) Satisfactory institutional arrangement for the separation of implementation oversight and executing functions in different departments of the GEF Partner Agency; and 2) Clear lines of responsibility, reporting and accountability within the GEF Partner Agency between the project implementation oversight and execution functions.

In this case, UNDP is only performing an implementation oversight role in the project vis-à-vis our role in the project board and in the project assurance function and therefore a full separation of project implementation oversight and execution duties has been assured.

4. Roles and Responsibilities of the Project Organization Structure:

a) Project Board: All UNDP projects must be governed by a multi-stakeholder board or committee established to review performance based on monitoring and evaluation, and implementation issues to ensure quality delivery of results. The Project Board (also called the Project Steering Committee) is the most senior, dedicated oversight body for a project.

The two main (mandatory) roles of the project board are as follows:

1) **High-level oversight of the execution of the project by the Implementing Partner** (as explained in the "Provide Oversight" section of the POPP). This is the primary function of the project board and includes annual (and as-needed) assessments of any major risks to the project, and decisions/agreements on any management actions or remedial measures to address them effectively. The Project Board reviews evidence of project performance based on monitoring, evaluation and reporting, including progress reports, evaluations, risk logs and the combined delivery report. The Project Board is responsible for taking corrective action as needed to ensure the project achieves the desired results.
2) **Approval of strategic project execution decisions of the Implementing Partner** with a view to assess and manage risks, monitor and ensure the overall achievement of projected results and impacts and ensure long term sustainability of project execution decisions of the Implementing Partner (as explained in the "Manage Change" section of the POPP).

Requirements to serve on the Project Board:

- ✓ Agree to the Terms of Reference of the Board and the rules on protocols, quorum and minuting.
- ✓ Meet annually; at least once.
- ✓ Disclose any conflict of interest in performing the functions of a Project Board member and take all measures to avoid any real or perceived conflicts of interest. This disclosure must be documented and kept on record by UNDP.
- ✓ Discharge the functions of the Project Board in accordance with UNDP policies and procedures.
- ✓ Ensure highest levels of transparency and ensure Project Board meeting minutes are recorded and shared with project stakeholders.

Responsibilities of the Project Board:

- ✓ Consensus decision making:
  - The project board provides overall guidance and direction to the project, ensuring it remains within any specified constraints, and providing overall oversight of the project implementation.

- Review project performance based on monitoring, evaluation and reporting, including progress reports, risk logs and the combined delivery report;
- The project board is responsible for making management decisions by consensus.
- In order to ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency and effective international competition.
- In case consensus cannot be reached within the Board, the UNDP representative on the board will mediate to find consensus and, if this cannot be found, will take the final decision to ensure project implementation is not unduly delayed.

✓ Oversee project execution:
- Agree on project manager's tolerances as required, within the parameters outlined in the project document, and provide direction and advice for exceptional situations when the project manager's tolerances are exceeded.
- Appraise annual work plans prepared by the Implementing Partner for the Project; review combined delivery reports prior to certification by the implementing partner.
- Address any high-level project issues as raised by the project manager and project assurance;
- Advise on major and minor amendments to the project within the parameters set by UNDP and the donor and refer such proposed major and minor amendments to the UNDP BPPS Nature, Climate and Energy Executive Coordinator (and the GEF, as required by GEF policies);
- Provide high-level direction and recommendations to the project management unit to ensure that the agreed deliverables are produced satisfactorily and according to plans.
- Track and monitor co-financed activities and realization of co-financing amounts of this project.
- Approve the Inception Report, GEF annual project implementation reports, mid-term review and terminal evaluation reports.
- Ensure commitment of human resources to support project implementation, arbitrating any issues within the project.

✓ Risk Management:
- Provide guidance on evolving or materialized project risks and agree on possible mitigation and management actions to address specific risks.
- Review and update the project risk register and associated management plans based on the information prepared by the Implementing Partner. This includes risks related that can be directly managed by this project, as well as contextual risks that may affect project delivery or continued UNDP compliance and reputation but are outside of the control of the project. For example, social and environmental risks associated with co-financed activities or activities taking place in the project's area of influence that have implications for the project.
- Address project-level grievances.

✓ Coordination:
- Ensure coordination between various donor and government-funded projects and programmes.
- Ensure coordination with various government agencies and their participation in project activities.

Composition of the Project Board: The composition of the Project Board must include individuals assigned to the following three roles:

1. **Project Executive:** This is an individual who represents ownership of the project and chairs (or co-chairs) the Project Board. The Executive usually is the senior national counterpart for nationally implemented projects (typically from the same entity as the Implementing Partner), and it must be UNDP for projects that are direct implementation (DIM). In exceptional cases, two individuals from different entities can co-share this role and/or co-chair the Project Board. If the project executive co-chairs the project board with representatives of another category, it typically does so with a development partner representative. The Project Executive is the Ministry of Industrial and Information Technology,
2. **Beneficiary Representative(s):** Individuals or groups representing the interests of those groups of stakeholders who will ultimately benefit from the project. Their primary function within the board is to ensure the realization of project results from the perspective of project beneficiaries. Often representatives from civil society, industry associations, or other government entities benefiting from the project can fulfil this role. There can be multiple beneficiary representatives in a Project Board. The Beneficiary representative (s) is/are: Phosphate Chemicals Industry Companies; DIIT and DNR offices of participating provincial governments, civil society related to this industry.
3. **Development Partner(s):** Individuals or groups representing the interests of the parties concerned that provide funding, strategic guidance and/or technical expertise to the project. The Development Partner(s) is/are: MOF, UNDP, MIIT, MNR

b) **Project Assurance:** Project assurance is the responsibility of each project board member; however, UNDP has a distinct assurance role for all UNDP projects in carrying out objective and independent project oversight and monitoring functions. UNDP performs quality assurance and supports the Project Board (and Project Management Unit) by carrying out objective and independent project oversight and monitoring functions, including compliance with the risk management and social and environmental standards of UNDP. The Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. Project assurance is totally independent of project execution.

A designated representative of UNDP playing the project assurance role is expected to attend all board meetings and support board processes as a non-voting representative. It should be noted that while in certain cases UNDP's project assurance role across the project may encompass activities happening at several levels (e.g. global, regional), at least one UNDP representative playing that function must, as part of their duties, specifically attend board meeting and provide board members with the required documentation required to perform their duties. The UNDP representative playing the main project assurance function is Ms. Shijun Liu, Programme Analyst.

c) **Project Management – Execution of the Project:** The Project Manager (PM) (also called project coordinator) is the senior most representative of the Project Management Unit (PMU) and is responsible for the overall day-to-day management of the project on behalf of the Implementing Partner, including the mobilization of all project inputs, supervision over project staff, responsible parties, consultants and sub-contractors. The project manager typically presents key deliverables and documents to the board for their review and approval, including progress reports, annual work plans, adjustments to tolerance levels and risk registers.

A designated representative of the PMU is expected to attend all board meetings and support board processes as a non-voting representative.

The primary PMU representative attending board meetings is the appointed Project Manager.

VIII. FINANCIAL PLANNING AND MANAGEMENT

The total cost of the project is *USD 107,107,066.* This is financed through a GEF grant of *USD* 9,343,379 administered by UNDP, and *USD* 97,763,687 co-financing (USD 85,954,543 cash, and USD 11,809,144 in kind). UNDP, as the GEF Implementing Agency, is responsible for the oversight of the GEF resources and the cash co-financing transferred to UNDP bank account only.

Co-financing: The actual realization of project co-financing amounts will be monitored by the UNDP Country Office and the PMU on an annual basis in the GEF PIR and will be reported to the GEF during the *mid-term review* and terminal evaluation process as follows:

Co-financing source	Co-financing type	Co-financing amount (USD)
MIIT	Grant	3,000,000
	In-kind	900,000
MNR	Grant	2,000,000
	In-kind	650,000
Guizhou Province Local Government	Grant	2,862,000
	In-kind	715,500
Hubei Province Local Government	Grant	1,590,000
	In-kind	397,500
Sichuan Province Local Government	Grant	3,286,000
	In-kind	821,500
Yunnan Province Local Government	Grant	2,862,000
	In-kind	715,500
Private Investors	Grant	70,354,543
	In-kind	7,409,144
UNDP	In-Kind	200,000
Sub-total Grant		85,954,543
Sub-total In-kind		11,809,144
Total		97,763,687

Budget Revision and Tolerance: As per UNDP POPP, the project board may agree with the project manager on a tolerance level for each detailed plan under the overall multi-year workplan. The agreed tolerance should be written in the project document or approved project board meeting minutes. It should normally not exceed 10 percent of the agreed annual budget at the activity level, but within the overall approved multi-year workplan at the activity level. Within the agreed tolerances, the project manager can operate without intervention from the project board. Restrictions apply as follows:

Should the following deviations occur, the Project Manager/IP through UNDP Country Office will seek the approval of the BPPS/NCE-VF team to ensure accurate reporting to the GEF. It is **strongly encouraged** to maintain the expenditures within the approved budget at the budgetary account and at the component level:

a) Budget reallocations must prove that the suggested changes in the budget will not lead to material changes in the results to be achieved by the project. A strong justification is required and will be approved on an exceptional basis. Budget re-allocations among the components (including PMC) of the approved Total Budget and Work Plans (TBWP) that represent a value greater than 10% of the total GEF grant.
b) Introduction of new outputs/activities (i.e. budget items) that were not part of the agreed project document and TBWP that represent a value greater than 5% of the total GEF grant. The new budget items must be eligible as per the GEF and UNDP policies.
c) Project management cost (PMC): budget under PMC component is capped and cannot be increased.

Any over expenditure incurred beyond the available GEF grant amount must be absorbed by non-GEF resources (e.g. UNDP TRAC or cash co-financing).

Project extensions: The UNDP Resident Representative and the UNDP-GEF Executive Coordinator must approve all project extension requests. Note that all extensions incur costs and the GEF project budget cannot be increased. A single extension may be granted on an exceptional basis and subject to the conditions and maximum durations set out in the UNDP POPP; the project management costs during the extension period must remain within the originally approved amount, and any increase in PMC costs will be covered by non-GEF resources; the additional UNDP oversight costs during the extension period must be covered by non-GEF resources, in accordance with UNDP's guidance set out in UNDP POPP.

Audit: The project will be audited as per UNDP Financial Regulations and Rules and applicable audit policies. Audit cycle and process must be discussed during the Inception workshop. If the Implementing Partner is an UN Agency, the project will be audited according to that Agencies applicable audit policies.

Project Closure: Project closure will be conducted as per UNDP requirements outlined in the UNDP POPP. All costs incurred to close the project must be included in the project closure budget and reported as final project commitments presented to the Project Board during the final project review. The only costs a project may incur following the final project review are those included in the project closure budget.

Operational completion: The project will be operationally completed when the last UNDP-financed inputs have been provided and the related activities have been completed. This includes the final clearance of the Terminal Evaluation Report (that will be available in English) and the corresponding management response, and the end-of-project review Project Board meeting. **Operational closure must happen at the end date calculated by the approved duration after the Project Document signature or at the revised operational closure date as approved in the project extension. Any expected activity after the operational date requires project extension approval.** The Implementing Partner through a Project Board decision will notify the UNDP Country Office when operational closure has been completed. At this time, the project should have completed the transfer or disposal of any equipment that is still the property of UNDP.

Transfer or disposal of assets: In consultation with the Implementing Partner and other parties of the project, UNDP is responsible for deciding on the transfer or other disposal of assets. Transfer or disposal of assets is recommended to be reviewed and endorsed by the project board following UNDP rules and regulations. Assets may be transferred to the government for project activities managed by a national institution at any time during the life of a project (it is strongly encouraged to be done before the

operational closure date). In all cases of transfer, a transfer document must be prepared and kept on file[15]. The transfer should be done before Project Management Unit complete their assignments.

Financial completion (closure): The project will be financially closed when the following conditions have been met: a) the project is operationally completed or has been cancelled; b) the Implementing Partner has reported all financial transactions to UNDP; c) UNDP has closed the accounts for the project; d) UNDP and the Implementing Partner have certified a final Combined Delivery Report (which serves as final budget revision).

The project will be financially completed **within 6 months of operational closure or after the date of cancellation**. If Operational Closure is delayed for any justified and approved reason, the Country Office should do all efforts to Financially Close the project within 9 months after TE is completed. Between operational and financial closure, the implementing partner will identify and settle all financial obligations and prepare a final expenditure report. The UNDP Country Office will send the final signed closure documents including confirmation of final cumulative expenditure and unspent balance to the BPPS/NCE-VF Unit for confirmation before the project will be financially closed in Atlas by the UNDP Country Office.

Refund to GEF: Should a refund of unspent funds to the GEF be necessary, this will be managed directly by the BPPS/NCE-VF Directorate in New York. No action is required by the UNDP Country Office on the actual refund from UNDP project to the GEF Trustee.

[15] See https://popp.undp.org/_layouts/15/WopiFrame.aspx?sourcedoc=/UNDP_POPP_DOCUMENT_LIBRARY/Public/PPM_Project%20Management_Closing.docx&action=default.

IX. TOTAL BUDGET AND WORK PLAN

Atlas Award ID:	00113640	**Atlas Output Project ID:**	00111713
Atlas Proposal or Award Title:	*PhoschemEE-FS*		
Atlas Business Unit	CHN10		
Atlas Primary Output Project Title	*PhoschemEE-FS*		
UNDP-GEF PIMS No.	6618		
Implementing Partner	Ministry of Industry and Information Technology (MIIT)		

Atlas Activity (GEF Component)	ATLAS Implementing Agent	Atlas Fund ID	Donor Name	Atlas Budgetary Account Code	ATLAS Budget Account Description	Amount Year 1 (USD)	Amount Year 2 (USD)	Amount Year 3 (USD)	Amount Year 4 (USD)	Amount Year 5 (USD)	Total (USD)	Budget Note
Component 1: Green and Low Carbon Development and Operation of Phosphate Mines	MNR	62000	GEF	71200	International Consultants	35,000	0	0	0	0	35,000	1
				71300	Local consultants	17,000	5,000	500	8,500	9,000	40,000	2
				71300	Local consultant	40,000	40,000	40,000	40,000	40,000	200,000	3
				71600	Travel	15,000	10,000	10,000	10,000	10,000	55,000	4
				72100	Contractual Services-Companies	65,000	58,500	21,500	19,500	23,500	188,000	5
				72200	Equipment and Furniture	0	1,520,080	380,020	0	0	1,900,100	6
				72800	Information Tech. Equipment	10,500	0	0	0	0	10,500	7
				72500	Supplies	200	200	200	100	100	800	8
				75700	Training, Workshops and Confer	0	36,000	5,000	9,000	12,000	62,000	9
	Total Component 1					**182,700**	**1,669,780**	**457,220**	**87,100**	**94,600**	**2,491,400**	
Component 2: Green and Low Carbon	MIIT	62000	GEF	71200	International consultants	0	2,500	4,500	2,000	8,500	17,500	10
				71300	Local consultants	5,500	9,500	65,000	80,000	70,000	230,000	11

Atlas Activity (GEF Component)	ATLAS Implementing Agent	Atlas Fund ID	Donor Name	Atlas Budgetary Account Code	ATLAS Budget Account Description	Amount Year 1 (USD)	Amount Year 2 (USD)	Amount Year 3 (USD)	Amount Year 4 (USD)	Amount Year 5 (USD)	Total (USD)	Budget Note
Design and Operation of Phosphate Chemicals Production Facilities				71600	Travel	50,250	158,625	152,625	51,125	57,875	470,500	12
				72100	Contractual Services - Companies	99,900	91,500	60,500	95,500	133,000	480,400	13
				72200	Equipment and Furniture	0	1,660,925	415,231	0	0	2,076,156	14
				72800	Information Tech. Equipment	5,200	2,500	2,500	2,500	2,500	15,200	15
				74200	Audio Visual & Print Prod Costs	2,500	2,500	2,500	2,500	6,000	16,000	16
				72500	Supplies	150	150	150	150	44	644	17
				75700	Training, Workshops and Confer	81,000	105,750	114,000	60,000	70,250	431,000	18
	Total Component 2					**244,500**	**2,033,950**	**817,006**	**293,775**	**348,169**	**3,737,400**	
Component 3: Green and Low Carbon Design and Operation of Waste Management Systems in the Phosphate Chemicals Industry	MNR	62000	GEF	71300	Local Consultants	13,000	2,000	1,000	4,500	4,500	25,000	19
				71600	Travel	2,000	2,000	2,000	2,000	2,000	10,000	20
				72100	Contractual Services - Companies	40,000	56,500	32,500	20,982	27,000	176,982	21
				75700	Training, workshops and Confer	8,000	14,000	5,000	7,000	6,000	40,000	22
				74200	Audio Visual & Print Prod Costs	3,000	3,000	3,000	3,000	3,000	15,000	23
				72500	Supplies	200	200	200	200	200	1,000	24
				72200	Equipment and Furniture	0	640,000	160,000	0	0	800,000	25
				Sub-total Component 3 (MNR)		**66,200**	**717,700**	**203,700**	**37,682**	**42,700**	**1,067,982**	
	MIIT	62000	GEF	71200	International	0	6,250	1,250	1,250	13,750	22,500	26

Atlas Activity (GEF Component)	ATLAS Implementing Agent	Atlas Fund ID	Donor Name	Atlas Budgetary Account Code	ATLAS Budget Account Description	Amount Year 1 (USD)	Amount Year 2 (USD)	Amount Year 3 (USD)	Amount Year 4 (USD)	Amount Year 5 (USD)	Total (USD)	Budget Note
					consultants							
				71300	Local consultants	12,500	20,400	700	3,200	5,700	42,500	27
				71600	Travel	27,000	29,900	18,950	22,700	31,450	130,000	28
				72100	Contractual Services - Companies	57,500	111,500	37,250	67,500	61,250	335,000	29
				72200	Equipment and Furniture	0	770830	190,708	0	0	961,538	30
				72500	Supplies	136	0	0	0	0	136	31
				75700	Training, Workshops and Confer	14,500	17,500	24,000	22,000	32,000	110,000	32
				Sub-total Component 3 (MITT)		**111,636**	**956,380**	**272,858**	**116,650**	**144,150**	**1,601,674**	
	Total Component 3					**177,836**	**1,674,080**	**476,558**	**154,332**	**186,850**	**2,669,656**	
M&E	UNDP	62000	GEF	71200	International consultants	0	0	20,000	0	0	20,000	33
				71300	Local consultants	0	0	20,000	0	0	20,000	34
	MNR	62000		71300	Local consultants	2,000	2,000	2,000	2,000	2,000	10,000	34
				71800	Contractual Services - IP	5,000	5,000	5,000	5,000	5,000	25,000	37
				75700	Training, Workshops and Confer	5,000	5,000	5,000	5,000	5,000	25,000	38
	UNDP	62000		71200	International consultants	0	0	0	0	20,000	20,000	39
				71300	Local Consultants	0	0	0	0	20,000	20,000	40
	MIIT	62000		71800	Contractual services – IP	6,000	6,000	6,000	6,000	6,000	30,000	42
				75700	Training, Workshops and Confer	30,000	0	0	0	0	30,000	43

Atlas Activity (GEF Component)	ATLAS Implementing Agent	Atlas Fund ID	Donor Name	Atlas Budgetary Account Code	ATLAS Budget Account Description	Amount Year 1 (USD)	Amount Year 2 (USD)	Amount Year 3 (USD)	Amount Year 4 (USD)	Amount Year 5 (USD)	Total (USD)	Budget Note
	Total M&E					**48,000**	**18,000**	**58,000**	**18,000**	**58,000**	**200,000**	
Project Management Unit	MNR	62000	62000	71300	Local consultants	4000	4000	4,000	4000	4000	20,000	34
	UNDP	62000		74100	Professional services	2,500	2,500	2,000	2,500	2,500	12,000	35
	MNR	62000		71600	Travel	10,000	10,000	13,000	10,000	10,000	53,000	36
				75700	Training, Workshops and Confer	1,800	1,700	1,700	1,700	1,569	8,469	38
	MIIT	62000		71300	Local Consultants	6,000	6,000	6,000	6,000	11,000	35,000	40
				71600	Travel	13,291	13,291	13,291	13,291	13,290	66,454	41
				71800	Contractual services – IP	1,000	11,000	11,000	11,000	11,000	45,000	42
				72800	Information Tech. Equipment	5,000	0	0	0	0	5,000	44
	Total Project Management Unit Cost					**43,591**	**48,491**	**50,991**	**48,491**	**53,359**	**244,923**	
Grand Total						**696,627**	**5,444,301**	**1,859,775**	**601,698**	**740,978**	**9,343,379**	

Summary of Funds

Source of Fund	Amount (Year 1), US$	Amount (Year 2), US$	Amount (Year 3), US$	Amount (Year 4), US$	Amount (Year 5), US$	Total, US$
GEF	696,627	5,444,301	1,859,775	601,698	740,978	9,343,379
UNDP	40,000	40,000	40,000	40,000	40,000	200,000
Government of China	3,960,000	3,960,000	3,960,000	3,960,000	3,960,000	19,800,000
Private Sector	15,552,737	15,552,737	15,552,737	15,552,738	15,552,738	77,763,687
TOTAL	20,249,364	24,997,038	21,412,512	20,154,436	20,293,716	107,107,066

Budget Notes

No.	Budget Note
1	**International consultants: USD 35,000** (including professional fees, travel expenses and DSA) for 50 working days at USD 700 per working day (multiple activities could be conducted by one consultant):
	50 working days in total for activity 1.1.1.2 under Output 1.1.1.
2	**Local consultants: USD 40,000** (including professional fees, travel expenses and DSA) for 80 working days at USD 500 per working day (multiple activities could be conducted by one consultant):
	28 working days for Output 1.1.1, and specifically: 10 days for activity 1.1.1.1; 8 days for activity 1.1.1.3; 4 days for activity 1.1.1.6; 2 days for activity 1.1.1.7; 4 days for activity 1.1.1.10.
	24 working days for Output 1.1.2, and specifically: 10 days for activity 1.1.2.1; 10 days for activity 1.1.2.2; 4 days for activity 1.1.2.5;
	4 working days for activity 1.1.3.6 under Output 1.1.3.
	12 working days for Output 1.1.4, and specifically: 4 days for activity 1.1.4.1; 4 days for activity 1.1.4.3; 4 days for activity 1.1.4.5;
	8 working days for Output 1.1.5, and specifically: 4 days for activity 1.1.5.4; 4 days for activity 1.1.5.5;
	4 working days for activity 1.1.6.3 under Output 1.1.6.
3	**Local consultants: USD 200,000 (including professional fees, travel expenses and DSA) for 400 working days at USD 500 per working day (multiple activities could be conducted by one consultant):**
4	**Travel: USD 55,000:** the travel and DSA expenses for 3 PMO officers participating to each of the 6 training sessions (3 sessions for Activity 1.1.4.4 and 3 session for Activity 1.1.5.4) and to the 7 workshops (Activities 1.1.1.6, 1.1.1.10, 1.1.2.5, 1.1.3.3, 1.1.4.3, 1.1.5.6 and 1.1.6.3), and for 8 stakeholders to the same 2 training sessions, and 2 workshops (55 travels at USD 1,000 each), (The other participants to the 3 training sessions and 2 workshops will be connected via videoconference)
5	**Contractual Services: USD 188,000: All sub-contract budgets include: national and international experts, travel, meetings, workshops and DSA for**

No.	Budget Note
	the subcontractor to undertake the activities and contribute to the outputs (multiple activities could be conducted by one company):
	Sub-contract 1 – USD 40,000: Study domestic and foreign related policies and make policy recommendations.(Activity 1.1.1.1, 1.1.1.2, 1.1.1.3, 1.1.1.7, 1.1.2.1 and 1.1.2.2)
	Sub-contract 2 – USD 30,000: Establish and apply standardized methods to assess demonstrated energy consumption and GHG emissions. (Activity 1.1.3.1, 1.1.3.2)
	Sub-contract 3 – USD 13,000: Preparation and dissemination of reports. (Activity 1.1.3.6 and 1.1.3.7)
	Sub-contract 4 – USD 20,000: Preparation of two capacity-building and two assessment meetings. (Activity 1.1.4.1, 1.1.4.2, 1.1.4.4 and 1.1.5.4)
	Sub-contract 5 – USD 50,000: Design and develop PCI energy saving and emission reduction online monitoring and technology sharing system and verify the system. (Activity 1.1.5.1, 1.1.5.2,and 1.1.6.1)
	Sub-contract 6 – USD 13,000: Feasibility analysis, design, risk identification and implementation plan for the demonstrated project.
	Sub-contract 7 – USD 10,000: Demonstration project financing plan establishment and approval, as well as propaganda and promotion.
	Sub-contract 8 – USD 8,000: Promotion and publicity of demonstration projects.
	Sub-contract 9 – USD 4,000: Green and Low Carbon Capacity Building in Phosphate Mining and Mineral Processing Design
6	**Equipment and Furniture: USD 1,900,100:**
	USD 1,166,700 for Activity 1.2.2.1 (Implementation and operation of demos on energy efficient technology of mining in Weng'an, Guizhou province and Leibo, Sichuan province.)
	USD 733,400 for Activity 1.2.2.2 (Implementation and operation of demos on green and low-carbon utilization technology and equipment in phosphate rock beneficiation in Mabian, Sichuan Province.)
7	**Information Tech. Equipment: USD 10,500 (for related activities under component 1)**
8	**Supplies and stationery: USD 800** Supplies and stationery related to Component 1.
9	**Training, workshops: USD 62,000:**
	Workshops on related policy approvals (Activities 1.1.1.4, 1.1.1.8 and 1.1.2.3). **6 sessions: USD 2,000*6 = USD 12,000**.
	Capacity-building programs in low-carbon and green phosphorus chemical industry. **3 session: USD 10,000*2 + 2000 = USD 22,000.**
	Policy and project evaluation meetings. **12 session: USD 2,000*12 = USD 24,000**.
	Promotion and implementation of the PCI greenhouse gas emissions accounting and inventory system. **1 session: USD 4,000**
10	**International consultants: USD 17,500** (including professional fees, travel expenses and DSA) for 35 working days at USD 500 per working day (multiple activities could be conducted by one consultant):
	5 working days for Output 2.1.6 activity 2.1.6.4.
	5 working days for Output 2.2.2 activity 2.2.2.3;
	5 working days for Output 2.2.3 activity 2.2.3.2;
	20 working days for Output 2.2.4, and specifically: 10 days for activity 2.2.4.2; 10 days for activity 2.2.4.3;
11	**Local consultants: USD 230,000 (including professional fees, travel expenses and DSA) for 920 working days at USD 250 per working day (multiple activities could be conducted by one consultant):**
	20 working days for Output 2.1.1 activity 2.1.1.2.
	10 working days for Output 2.1.6 activity 2.1.6.4;

No.	Budget Note
	20 working days for Output 2.2.1 activity 2.2.1.1;
	10 working days for output 2.2.2 activity 2.2.2.3;
	30 working days for Output 2.2.3; in which 10 working days for activity 2.2.3.2 and 20 working days for activity 2.2.3.3;
	For Output 2.2.4, USD 100,000 will be paid to full time project manager from activity 2.2.4.1's budget; USD 100,000 will be paid to full time project manager from activity 2.2.4.2's budget; 10 working days for activity 2.2.4.3;
	20 working days for Output 2.2.5 activity 2.2.5.5.
12	**Travel: USD 300,000:** The travel and DSA expenses for 3 PMO officers participating to each of the 7 national and/or international training sessions for Activity 2.1.4.4 and to the 8 workshops (Activities 2.1.1.1, 2.1.1.7, 2.1.1.8, 2.1.2.3, 2.1.4.1, 2.1.4.2, 2.1.5.1 and 2.1.6.4), and for 15 stakeholders to the same 7 training sessions, and 8 workshops (50 domestic travels at USD 1,000 each and 25 international travels at USD 10,000 each), (The other participants to the 7 training sessions and 8 workshops will be connected via videoconference)
	Travel: USD 168,000: The travel and DSA expenses for 3 PMO officers participating to the 14 workshops (Activity 2.2.1.1, 2.2.2.3, 2.2.3.1, 2.2.3.2, 2.2.3.3, 2.2.4.1, 2.2.4.2, 2.2.4.3, 2.2.5.1, 2.2.5.2, 2.2.5.3, 2.2.5.4, 2.2.5.5, and 2.2.5.7), (168 domestic travels at USD 1,000 each).
	Travel: others related travel, in total USD 2,500.
13	**Contractual Services: USD 480,400: All sub-contract budgets include: national and international experts, travel, meetings, workshops and DSA for the subcontractor to undertake the activities and contribute to the outputs (multiple activities could be conducted by one company):**
	Sub-contract 1 – USD 85,200: Formulated, recommended, approved, and enforced policy and institutional frameworks supporting green, energy efficient low carbon technology applications in phosphate chemicals production (Activity 2.1.1.1, 2.1.1.2, 2.1.1.4, 2.1.1.5, 2.1.1.6, 2.1.1.9)
	Sub-contract 2 – USD 60,000: Formulated, recommended, and enforced standards, policies and implementing rules and regulations on the promotion and practice of green, energy efficient, low carbon technologies/techniques, in phosphate chemicals production. (Activity 2.1.2.1,2.1.2.2 and 2.1.2.4)
	Sub-contract 3 – USD 40,000: Documented annual evaluation reports on the energy performance and environmental impacts of each demo activity and documented and disseminated demo results based on the MRV system (Output 1.1.5) established under Component 1 (Activity 2.1.3.1, 2.1.3.2, 2.1.3.3 and 2.1.3.4)
	Sub-contract 4 – USD 25,000: Designed, conducted and post-evaluated continuing education program in the PCI chemicals sub-sector in Yunnan, Guizhou, Sichuan, and Hubei, with regular capacity needs assessment in the area of green, energy efficient and low carbon technologies applied in phosphate chemicals manufacturing. (Activity 2.1.4.1 and 2.1.4.3)
	Sub-contract 5 - USD 10,000: Conduct of market analysis on the international trade trends for phosrock and phosphate chemicals, and the indirect carbon emission and ecological impacts in the international trading of these products. (activity 2.1.5.1)
	Sub-contract 6 - USD 150,000: Established and operational system for managing and sharing available and accessible information on all aspects of green, energy efficient and cleaner phosphate chemicals production in China. (activity 2.1.6.1,2.1.6.2,2.1.6.3,2.1.6.5)
	Sub-contract 7 – USD 50,200: Completed designs and plans of demonstrations of green, energy efficient low carbon technologies in phosphate chemicals production including: (a) Improved design for cleaner production demos; (2) Feasibility analyses of the application and operation of green, energy efficient and low carbon phosphate chemical production systems; and, (3) Implementation plans (including financing arrangements) for each green, energy efficient and low carbon technology application in the demo phosphate chemical companies (Activity 2.2.1.2, 2.2.1.3).
	Sub-contract 8 – USD 10,000: Installed and operational green, energy efficient low carbon technology application demos in phosphate chemicals production. (Activity 2.2.2.3).
	Sub-contract 9 – USD 15,000: Designed and approved pilot financing scheme for small and medium enterprises (SMEs) that manufacture phosphate

No.	Budget Note
	chemicals. (Activity 2.2.3.1 and 2.2.3.2).
	Sub-contract 10 – USD 10,000: Operational pilot financing scheme for phosphate chemicals SMEs in 2 to 3 regions. (Activity 2.2.4.3).
	Sub-contract 11 – USD 25,000: Approved and financed follow-up plan for the replication of the application of demonstrated green, energy efficient low carbon technologies in phosphate chemicals production in other localities. (Activity 2.2.5.1, 2.2.5.2, 2.2.5.7).
14	**Equipment and Furniture: USD 2,076,156:**
	For Activity 2.2.2.1 (Implementation and operation of demonstrations on energy savings and green and low-carbon development in phosphate chemicals production.)
15	**Information Tech. Equipment: USD 15,200 (for activities related to Component 2)**
16	**Publication and printing: USD 16,000**
17	**Supplies and stationery: USD 644** Supplies and stationery related to Component2.
18	**Training, workshops:** Workshops for enhanced adoption by the phosphate chemical industry of a green, low carbon and energy efficient development model in phosphate chemicals production. (Activities 2.1.1.2, 2.1.1.3, 2.1.1.4, 2.1.1.6, 2.1.1.7, 2.1.1.8, 2.1.1.9, 2.1.2.1, 2.1.2.3, 2.1.3.5, 2.1.4.2, 2.1.5.2, 2.1.5.3, 2.1.6.5). 14 sessions: USD 15,000*4 + USD 40,000*2 + USD 5,000*21 + USD 6,000*1 = USD 251,000.
	Training, workshops: Workshops for enhanced confidence in the feasibility of the application of green, energy efficient low carbon technologies in phosphate chemicals production in China.(Activities 2.2.1.2, 2.2.2.2, 2.2.3.1, 2.2.3.2, 2.2.3.3, 2.2.4.1, 2.2.4.2, 2.2.4.3, 2.2.5.1, 2.2.5.3, 2.2.5.4, 2.2.5.5, 2.2.5.6, 2.2.5.7). 14 sessions: USD 5,000*33 + USD 7,000*2 = USD 179,000.
	Training, workshops: other related training, workshop, and confer, in total USD 1,000.
19	**Local Consultants: USD 25,000 (including professional fees, travel expenses and DSA) for 50 working days at USD 500 per working day (multiple activities could be conducted by one consultant):**
	19 working days for Output 3.1.1, and specifically: 6 days for activity 3.1.1.1; 5 days for activity 3.1.1.3; 4 days for activity 3.1.1.5; 4 days for activity 3.1.1.8;
	15 working days for Output 3.1.2, and specifically: 6 days for activity 3.1.2.1; 5 days for activity 3.1.2.5; 4 days for activity 3.1.2.8;
	4 working days for activity 3.3.3.7 under Output 3.1.3.
	12 working days for Output 3.1.4, and specifically: 5 days for activity 3.1.4.1; 4 days for activity 3.1.4.3; 4 days for activity 3.1.4.8;
20	**Travel: USD 10,000:** the travel and DSA expenses for 2 PMO officers participating to each of the 1 training session and the 4 workshops (Activity 3.1.1.8, 3.1.2.8, 3.1.4.8 and 3.1.5.7), (10 travels at USD 1,000 each).
21	**Contractual Services: USD 176,982:** All sub-contract budgets include: national and international experts, travel, meetings, workshops and DSA for the subcontractor to undertake the activities and contribute to the outputs (multiple activities could be conducted by one company):
	Sub-contract 1 – USD 20,000: Formulated, recommended, approved, and enforced policies and institutional frameworks that support green, low carbon initiatives in waste management in the PCI. (Activity 3.1.1.1, 3.1.1.3, 3.1.1.5 and 3.1.1.7)
	Sub-contract 2 – USD 16,000: Improved existing and new enforced schemes, standards, policies/regulations on the application of green and low carbon waste management technologies in the PCI operations and processes. (Activity 3.1.2.1, 3.1.2.3, 3.1.2.5 and 3.1.2.7)
	Sub-contract 3 – USD 24,000: Documented annual evaluation reports on the energy performance and environmental impacts of each demo based on the MRV system developed under Component 1. (Activity 3.1.3.1, 3.1.3.5 and 3.1.3.7)

No.	Budget Note
	Sub-contract 4 – USD 48,000: Assessment of the ability and awareness level of relevant stakeholders and assessment of needs and gaps to address in the area of waste management in the phosphate mining and refining sub-sector.(Activity 3.1.4.1).
	Sub-contract 5 – USD 12,000: Completed research reports on: (1) annual volume of waste production in phosrock mining and refining, and in phosphate chemicals production in China; (2) green and low carbon waste management systems for waste recycling and reuse developed and implemented in other countries and their energy utilization performances; and, (3) potential commercial markets for processed waste, e.g., phosphogypsum.(Activity 3.1.5.1, 3.1.5.3 and 3.1.5.5)
	Sub-contract 6 – USD 32,982: Established and operational information management system for sharing available and accessible information about market and research developments and R&D in the country and abroad on green and low carbon waste management systems in the PCI. (Activity 3.1.6.1, 3.1.6.2, 3.1.6.3, 3.1.6.7 and 3.1.6.8)
	Sub-contract 7 – USD 8,000: Conduct of feasibility analysis to confirm the economic viability of the application and operation of new technologies for the improvement of green, energy efficient and low carbon in waste management in the phosphate mining and refining sub-sector. (Activity 3.2.1.1)
	Sub-contract 8 – USD 8,000: Design and implementation of a capacity building program on green, energy efficient and low carbon technologies for waste management in the phosphate mining and refining sub-sector. (Activity 3.2.2.3)
	Sub-contract 9 – USD 2,000: Evaluate the effectiveness of capacity-building activities.(Activity 3.2.2.4)
	Sub-contract 10– USD 6,000: Prepare, publish and promote green and low carbon waste management cases and empirical studies phosphorus chemical industry.(Activity 3.2.3.1, 3.2.3.3, 3.2.3.5 and 3.2.3.7)
22	**Training, workshops: USD 40,000:**
	Workshops on related policy approvals (Activities 3.1.1.6 and 3.1.1.8). **2 sessions: USD 3,000*2 = USD 6,000**.
	Capacity-building programs in low-carbon and green phosphorus chemical industry. **1 session: USD 2,000*2 = USD 4,000**
	Policy and project evaluation meetings. **7 session: USD 2,000*4 + 1000*5 = USD 13,000**.
	Promotion and implementation of the PCI greenhouse gas emissions accounting and inventory system. **1 session: USD 2,000**
	Workshops on development and approve project design, work plan and financing plan.(Activities 3.2.1.2). **1 sessions: USD 5,000**
	Prepare for ESMF and ESMP meetings. **2 session: USD 2,500*2 = USD 5,000.**
	Conduct capacity building training and evaluate it after one year. **2 session: USD 2,500*2 = USD 5,000**.
23	**Publication and printing: USD 15,000**
24	**Supplies and stationery: USD 1,000:** Supplies and stationery related to Component 3.
25	**Equipment and Furniture: USD 800,000:**
	USD 800,000 for Activity 3.2.2.1 (Implementation and operation of demonstrations of green equipment/technologies for the comprehensive utilization of resources in waste management in the phosphate mining and refining sub-sector.)
26	**International consultants: USD 22,500** (including professional fees, travel expenses and DSA) for 45 working days at USD 500 per working day (multiple activities could be conducted by one consultant):
	10 working days for Output 3.1.1 activity 3.1.1.11.
	5 working days for Output 3.1.2 activity 3.1.2.12.
	10 working days for Output 3.1.3 activity 3.1.3.2.
	10 working days for Output 3.2.2 activity 3.2.2.6.

No.	Budget Note
	10 working days for Output 3.2.3 activity 3.2.3.8.
27	**Local consultants: USD 42,500 (including professional fees, travel expenses and DSA) for 170 working days at USD 250 per working day (multiple activities could be conducted by one consultant):**
	50 working days for Output 3.1.1, in which 10 working days for activity 3.1.1.10; 10 working days for activity 3.1.1.11; 10 working days for activity 3.1.1.4; 20 working days for activity 3.1.1.9.
	60 working days for Output 3.1.2, in which 20 working days for activity 3.1.2.10; 10 working days for activity 3.1.2.12; 20 working days for activity 3.1.2.2; 20 working days for activity 3.1.2.4.
	20 working days for Output 3.1.3 activity 3.1.3.2.
	10 working days for Output 3.1.4 activity 3.1.4.6.
	20 working days for Output 3.1.6 activity 3.1.6.4.
	10 working days for Output 3.2.3 activity 3.2.3.8.
28	**Travel: USD 98,000:** The travel and DSA expenses for 2 PMO officers participating to each of the 5 international training sessions for Activity 3.1.4.9 and to the 11 workshops (Activities 3.1.1.10, 3.1.1.11, 3.1.1.2, 3.1.1.4, 3.1.2.10, 3.1.2.12, 3.1.2.2, 3.1.2.4, 3.1.3.4, 3.1.4.6 and 3.1.6.4), and for 16 stakeholders to the same 5 training session, and 11 workshops (78 domestic travels at USD 1,000 each and 2 international travels at USD 10,000 each), (The other participants to the 7 training sessions and 8 workshops will be connected via videoconference)
	Travel: USD 23,000: The travel and DSA expenses for 1 PMO officers participating to 1 training sessions for Activity 3.2.2.5 and to the 2 workshops (Activities 3.2.2.6 and 3.2.3.8), and for 3 stakeholders to the same 1 training session, and 2 workshops (23 domestic travels at USD 1,000 each).
	Travel: others travel cost related to component 3, in total USD 9,000
29	**Contractual Services: USD 335,000: All sub-contract budgets include: national and international experts, travel, meetings, workshops and DSA for the subcontractor to undertake the activities and contribute to the outputs (multiple activities could be conducted by one company):**
	Sub-contract 1 – USD 45,000: Formulated, recommended, approved, and enforced policies and institutional frameworks that support green, low carbon initiatives in waste management in the PCI. (Activity 3.1.1.10, 3.1.1.11, 3.1.1.12, 3.1.1.2 and 3.1.1.9).
	Sub-contract 2 – USD 40,000: Improved existing and new enforced schemes, standards, policies/regulations on the application of green and low carbon waste management technologies in the PCI operations and processes. (Activity 3.1.2.10, 3.1.2.11, 3.1.2.2, 3.1.2.4 and 3.1.2.9).
	Sub-contract 3 – USD 20,000: Documented annual evaluation reports on the energy performance and environmental impacts of each demo based on the MRV system developed under Component 1. (Activity 3.1.3.2, 3.1.3.4, 3.1.3.6 and 3.1.3.8).
	Sub-contract 4 – USD 40,000: Designed, conducted and post-evaluated continuing education program with regular capacity needs assessments in the area of green and low carbon technologies applied in waste management in the PCI. (Activity 3.1.4.10, 3.1.4.4, 3.1.4.5 and 3.1.4.6).
	Sub-contract 5 – USD 40,000: Completed research reports on: (1) annual volume of waste production in phosrock mining and refining, and in phosphate chemicals production in China; (2) green and low carbon waste management systems for waste recycling and reuse developed and implemented in other countries and their energy utilization performances; and, (3) potential commercial markets for processed waste, e.g., phosphogypsum. (Activity 3.1.5.2, 3.1.5.4, 3.1.5.6 and 3.1.5.8).
	Sub-contract 6 – USD 30,000: Established and operational information management system for sharing available and accessible information about market and research developments and R&D in the country and abroad on green and low carbon waste management systems in the PCI. (Activity 3.1.6.4, 3.1.6.5 and 3.1.6.6).
	Sub-contract 7 – USD 50,000: Completed designs and plans of demonstrations of the application of green and low carbon technologies/techniques in

No.	Budget Note
	waste management in the phosphate chemicals industry, including: (a) improved phosrock mining tailings recycling and reuse; and (b) energy efficient processing of phosphogypsum and marketing of the gypsum products. (Activity 3.2.1.4, 3.2.1.5, 3.2.1.6 and 3.2.1.8).
	Sub-contract 8 – USD 10,000: Installed and operational green and low carbon waste management technology application demos in the phosphate chemicals industry. (Activity 3.2.2.6).
	Sub-contract 9 – USD 60,000: Approved and financed follow-up plan for the replication of the application of demonstrated green, low carbon waste management technologies in the phosphate chemicals industry in other localities. (Activity 3.2.3.2, 3.2.3.4, 3.2.3.8, 3.2.2.7 and 3.2.2.8 and 3.2.3.10).
30	**Equipment and Furniture: USD 961,538:** For Activity 3.2.2.2 (Implementation and operation of demonstrations of green equipment/technologies for the comprehensive utilization of resources in waste management in the phosphate chemicals production sub-sector.)
31	**Supplies and stationery: USD 136:** Supplies and stationery related to Component 3.
32	
	Training, workshops: Workshops for enhanced commitment of, and institutional and technical arrangements for, the phosphate chemical industry in green and low carbon waste management (Activities 3.1.1.2, 3.1.1.4, 3.1.1.9, 3.1.2.12, 3.1.2.2, 3.1.3.2, 3.1.5.8). 7 sessions: USD 5,000*9 = USD 45,000.
	Training, workshops: Workshop for increased confidence in the feasibility of the application of green and low carbon technologies in the management of waste in the phosphate chemicals Industry in China (Activities 3.2.2.5, 3.2.3.4, 3.2.3.7, 3.2.3.8). 4 sessions: USD 15,000*1 +USD 5,000*8 = USD 55,000.
	Training, workshops: other related training, workshop, and confer, in total USD 10,000.
33	**International consultants: USD 20,000:** For mid-term review of the project (M&E)
34	**Local consultants: USD 50,000: (**USD 20000 For project management; **USD 2,000/yr.*5= USD 10,000 (M&E)** for Monitoring of social and environmental safeguard screening; **USD 500/day * 40 = USD 20,000 for mid-term review (M&E)**
35	**Professional services: USD 12,000:** For project audit
36	**Travel: USD 53,000:** Travel and DSA of 2 officials from the PMO to the demonstration site
37	**Contractual services: USD 25,000** for M&E required to report on progress made in reaching Project Log Frame Indicators and GEF Core indicators
38	**Training, Workshops and Confer: USD 5,000/yr.*5=25,000 (M&E)** for Project Board Meetings) and **USD 8,469:** for other meetings and conferences for project management office
39	**International Consultants: USD 20,000 will be paid in Year 5 for Termination Evaluation (M&E)**
40	**Local consultants: USD 55,000** **USD 20,000 will be paid in Year 5 for Termination Evaluation. (M&E)** **USD 6,000/yr. *4+11,000*1 =35,000 for project management office.**
41	**Travel:** others including PMC related travel, in total USD 66,454
42	**Contractual Services: USD75,000** **USD 6,000/yr. *5 (Monitoring all risks USD3,000/yr., Monitoring of stakeholder engagement plan USD 1,500/yr., Monitoring of gender action plan USD 1,500/yr.), 1,000/yr.*1+11,000*4 for project management office.**
43	**USD 30,000 in Year 1 is for inception workshop (M&E)**
44	**Information Tech. Equipment:** USD 5,000 (for office equipment.)

X. LEGAL CONTEXT

This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of China and UNDP, signed on 29 June 1979. All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

This project will be implemented by [name of entity] ("Implementing Partner") in accordance with its financial regulations, rules, practices, and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

The designations employed and the presentation of material on this map do not imply the expression of any opinion whatsoever on the part of the Secretariat of the United Nations or UNDP concerning the legal status of any country, territory, city or area or its authorities, or concerning the delimitation of its frontiers or boundaries.

XI. RISK MANAGEMENT

Implementing Partner is a Government Entity (NIM)

1. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:

    a. put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
    b. assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

2. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document.

3. The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via http://www.un.org/sc/committees/1267/aq_sanctions_list.shtml.

4. The Implementing Partner acknowledges and agrees that UNDP will not tolerate sexual harassment and sexual exploitation and abuse of anyone by the Implementing Partner, and each of its responsible parties, their respective sub-recipients and other entities involved in Project implementation, either as contractors or subcontractors and their personnel, and any individuals performing services for them under the Project Document.
   (a) In the implementation of the activities under this Project Document, the Implementing Partner, and each of its sub-parties referred to above, shall comply with the standards of conduct set forth in the Secretary General's Bulletin ST/SGB/2003/13 of 9 October 2003, concerning "Special measures for protection from sexual exploitation and sexual abuse" ("SEA").
   (b) Moreover, and without limitation to the application of other regulations, rules, policies and procedures bearing upon the performance of the activities under this Project Document, in the implementation of activities, the Implementing Partner, and each of its sub-parties referred to above, shall not engage in any form of sexual harassment ("SH"). SH is defined as any unwelcome conduct of a sexual nature that might reasonably be expected or be perceived to cause offense or humiliation, when such conduct interferes with work, is made a condition of employment or creates an intimidating, hostile or offensive work environment.

5. a) In the performance of the activities under this Project Document, the Implementing Partner shall (with respect to its own activities), and shall require from its sub-parties referred to in paragraph 4 (with respect to their activities) that they, have minimum standards and procedures in place, or a plan

to develop and/or improve such standards and procedures in order to be able to take effective preventive and investigative action. These should include: policies on sexual harassment and sexual exploitation and abuse; policies on whistleblowing/protection against retaliation; and complaints, disciplinary and investigative mechanisms. In line with this, the Implementing Partner will and will require that such sub-parties will take all appropriate measures to:

i. Prevent its employees, agents or any other persons engaged to perform any services under this Project Document, from engaging in SH or SEA;
ii. Offer employees and associated personnel training on prevention and response to SH and SEA, where the Implementing Partner and its sub-parties referred to in paragraph 4 have not put in place its own training regarding the prevention of SH and SEA, the Implementing Partner and its sub-parties may use the training material available at UNDP;
iii. Report and monitor allegations of SH and SEA of which the Implementing Partner and its sub-parties referred to in paragraph 4 have been informed or have otherwise become aware, and status thereof;
iv. Refer victims/survivors of SH and SEA to safe and confidential victim assistance; and
v. Promptly and confidentially record and investigate any allegations credible enough to warrant an investigation of SH or SEA. The Implementing Partner shall advise UNDP of any such allegations received and investigations being conducted by itself or any of its sub-parties referred to in paragraph 4 with respect to their activities under the Project Document, and shall keep UNDP informed during the investigation by it or any of such sub-parties, to the extent that such notification (i) does not jeopardize the conduct of the investigation, including but not limited to the safety or security of persons, and/or (ii) is not in contravention of any laws applicable to it. Following the investigation, the Implementing Partner shall advise UNDP of any actions taken by it or any of the other entities further to the investigation.

b) The Implementing Partner shall establish that it has complied with the foregoing, to the satisfaction of UNDP, when requested by UNDP or any party acting on its behalf to provide such confirmation. Failure of the Implementing Partner, and each of its sub-parties referred to in paragraph 4, to comply of the foregoing, as determined by UNDP, shall be considered grounds for suspension or termination of the Project.

6. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

7. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

8. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

9. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, responsible parties, subcontractors and sub-recipients in implementing the project or using UNDP funds. The Implementing Partner will ensure that its financial management, anti-corruption and anti-fraud policies are in place and enforced for all funding received from or through UNDP.

10. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

11. In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP projects and programmes in accordance with UNDP's regulations, rules, policies and procedures. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.

12. The signatories to this Project Document will promptly inform one another in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

    Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). The Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

13. UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail the Implementing Partner's obligations under this Project Document.

    Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

    Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

14. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

15. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

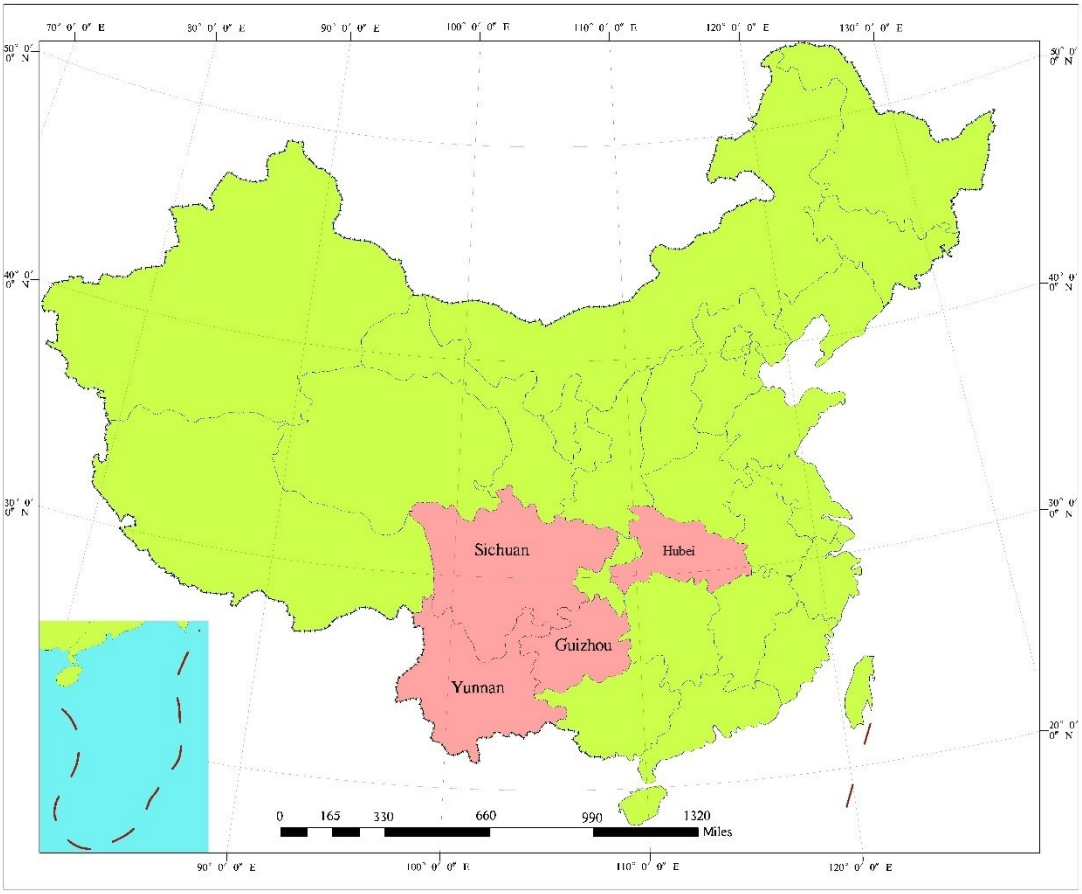
16. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled "Risk Management Standard Clauses" are included, *mutatis mutandis*, in all sub-contracts or sub-agreements entered into further to this Project Document.

XII. MANDATORY ANNEXES

1. GEF Budget Template
2. Project Map and geospatial coordinates of the project area
3. Multiyear Workplan
4. Social and Environmental Screening Procedure (SESP)
5. UNDP Atlas Risk Register
6. Overview of technical consultancies/subcontracts
7. Stakeholder Engagement Plan
8. Gender Analysis and Gender Action Plan
9. Procurement Plan – for first year of implementation especially
10. GEF focal area specific annexes (e.g. METT, GHG calculations, target landscape profile, feasibility study, other technical reports)
11. Additional agreements: such as cost sharing agreements, project cooperation agreements signed with NGOs (where the NGO is designated as the "executing entity"), letters of financial commitments etc.
12. GEF and/or LDCF/SCCF Core indicators (see template below)
13. GEF Taxonomy (see template below)

Annex 1: GEF Budget Template
To be submitted as a separated annex

Annex 2: Project Map and Geospatial Coordinates of Project Sites



Project Site	Geographical Coordinates
Fuquan City, Guizhou	107°31′E,26°38′N
Leibo County, Sichuan	103°34′E,28°16′N
Mabian County, Sichuan	103°32′E,28°50′N
Mabian County, Sichuan	103°29'E ,28°42′N
Jinzhong Town, Kaiyang County, Guiyang City	106°86 'E 27°19' N
Mianzhu City, Sichuan	104°19'E 31°22'N
Pantang, Wuxue, Huanggang City, Hubei	115°47'E 29°88'N
Jinning District, Kunming City, Yunnan	102°36′E,24°40′N
Anning City, Yunnan	102°36'E 24°96'N
Panzhihua City, Sichuan	101°73' E 26°56' N

Annex 3: Multi Year Work Plan

Activities	Year 1				Year 2				Year 3				Year 4				Year 5			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Component 1: Green and Low Carbon Development and Operation of Phosphate Mines																				
Activity 1.1.1.1: Conduct of research to update existing policies and strengthen existing institutional mechanisms in China to enable the greening of the phosphate mining and refining sub-sector.																				
Activity 1.1.1.2: Conduct of research studies on existing policies and institutional frameworks in other countries that can be used as reference for supporting and promoting low carbon development in phosrock mining and refining in China.																				
Activity 1.1.1.3: Formulation and recommendation of new policies on green, energy efficient low carbon development in phosrock mining and refining.																				
Activity 1.1.1.4: Approval of the newly formulated and recommended policies on green, energy efficient low carbon development in phosrock mining and refining.																				
Activity 1.1.1.5: Enforcement of the new approved policies on green, energy efficient low carbon development in phosrock mining and refining.																				
Activity 1.1.1.6: Evaluation of the impact on the greening of the phosrock mining and refining sub-sector as a result of the enforcement of the newly developed policies and provision of follow-up recommendations to amend and improve the policies, if deemed necessary.																				
Activity 1.1.1.7: Formulation and recommendation of new institutional frameworks on green, energy efficient low carbon development in phosrock mining and refining.																				
Activity 1.1.1.8: Approval of the newly formulated and recommended institutional frameworks on green, energy efficient low carbon development in phosrock mining and refining.																				
Activity 1.1.1.9: Enforcement of the new approved institutional frameworks on green, energy efficient low																				

carbon development in phosrock mining and refining.																				
Activity 1.1.1.10: Evaluation of the effectiveness of the new institutional frameworks on green EE low carbon development in phosrock mining and refining, and provision of follow-up recommendations to amend and improve the institutional frameworks, if deemed necessary.																				
Activity 1.1.2.1: Conduct of research to update existing standards, policies and implementing rules and regulations to promote green technologies in the development and operations of phosphate mines.																				
Activity 1.1.2.2: Formulation and recommendation of new standards/technical guidelines and implementing rules and regulations for promoting and adopting green technologies in phosphate mines.																				
Activity 1.1.2.3: Approval of the newly formulated and recommended standards/technical guidelines and implementing rules and regulations for promoting and adopting green technologies in phosphate mines.																				
Activity 1.1.2.4: Enforcement of the new approved standards/technical guidelines and implementing rules and regulations for promoting and adopting green technologies in phosphate mines.																				
Activity 1.1.2.5: Evaluation of the impact and effectiveness of the new enforced standards/technical guidelines and implementing rules and regulations for promoting and adopting green technologies in phosphate mines, and provision of follow-up recommendations to amend and improve the standards/technical guidelines and implementing rules and regulations, if deemed necessary.																				
Activity 1.1.3.1: Design and development of standardized methods for monitoring and data gathering and processing for the evaluation of the energy performance and environmental impacts of the demos and demo replications.																				
Activity 1.1.3.2: Establishment of standardized operational performance parameters to use as benchmark for the evaluation of the energy performance and environmental impacts of the demos																				

and demo replications.																				
Activity 1.1.3.3: Conduct of annual evaluation of energy performance and environmental impacts of demos in the phosphate mining and refining sub-sector using the standardized methodology and benchmarks developed in Activities 1.1.3.1 and 1.1.3.2.																				
Activity 1.1.3.4: Conduct of annual evaluation of financial performance of demos in the phosphate mining and refining sub-sector to promote the replication of the implemented technologies.																				
Activity 1.1.3.5: Preparation of annual and final reports on the operational performance (production, energy consumption, economic) and environmental impacts of demos in the phosphate mining and refining sub-sector.																				
Activity 1.1.3.6: Dissemination of Annual and final demo evaluation reports prepared in Activity 1.1.3.5.																				
Activity 1.1.3.7: Preparation and dissemination of case studies and lessons learned based on the application of demonstrated green, energy efficient and low carbon technologies.																				
Activity 1.1.4.1: Assessment of the ability and awareness level of relevant stakeholders and evaluation of needs and gaps to fill about green, energy efficient low carbon development of phosphate mining and refining in Guizhou, Yunnan, and Sichuan Provinces.																				
Activity 1.1.4.2: Design and implementation of a capacity building program for relevant government officials and industry management teams and key stakeholders on green and low-carbon development in the mining and refining.																				
Activity 1.1.4.3: Evaluation of the capacity building program designed in Activity 1.1.4.2 a year after the completion to assess how many trainees are gainfully applying the acquired skills and determine the corrective actions to be taken to improve the program.																				
Activity 1.1.4.4: Design and implementation of a capacity building and information program on operation and maintenance of green, EE and low carbon technologies for workers in phosphate mining and refining factories.																				

Activity 1.1.4.5: Evaluation of the capacity building program designed in Activity 1.1.4.4 a year after the completion to assess how many trainees are gainfully applying the acquired skills and determine the corrective actions to be taken to improve the program.																				
Activity 1.1.5.1: Design and development of an online monitoring, reporting and verification (MRV) system, including the development of an information sharing mechanism, for energy-saving and GHG emission reduction for the entire PCI.																				
Activity 1.1.5.2: Implementation and validation of the newly developed MRV system in the entire PCI.																				
Activity 1.1.5.3: Design and implementation of a capacity building program on operation and maintenance of the MRV system for companies operating in the entire PCI.																				
Activity 1.1.5.4: Evaluation of the capacity building program designed in Activity 1.1.5.3 a year after the completion to assess how many trainees are gainfully applying the acquired skills and determine the corrective actions to be taken to improve the program.																				
Activity 1.1.5.5: Evaluation of the operation and performance of the MRV system and provision of recommendations on potential improvements and adjustments to be made.																				
Activity 1.1.5.6: Development and implementation of a greenhouse gases emission accounting and inventory system for the phosphate chemical industry.																				
Activity 1.1.6.1: Design and development of an up-to-date information management and exchange system on all aspects of low carbon phosphate mining and refining technologies and best practices.																				
Activity 1.1.6.2: Establishment and operationalization of the devised up-to-date information communication platform for the exchange of data on all aspects of green phosphate mining and refining.																				
Activity 1.1.6.3: Evaluation of operation and performance of the information management and exchange system.																				

Activity 1.2.1.1: Development and approval of engineering designs of green demos for the mining and refining operations in the demo phosrock mines in Weng'an county in Guizhou province, and Mabian county in Sichuan province.																				
Activity 1.2.1.2: Preparation of an ESMF as well as an ESMP to manage and mitigate all identified risks associated with the design and implementation of the demo activities.																				
Activity 1.2.1.3: Development and approval of implementation plans, including financing arrangements, for each designed demo.																				
Activity 1.2.2.1: Implementation and operation of demos on energy efficient technology of mining in Weng'an county in Guizhou province, and Leibo county in Sichuan province.																				
Activity 1.2.2.2: Implementation and operation of demos on green and low-carbon utilization technology and equipment in phosphate rock beneficiation and renewable energy use in Mabian, Sichuan province.																				
Activity 1.2.2.3: Design and implementation of a capacity building program on green, energy efficient and low carbon technologies for phosphate mining and refining.																				
Activity 1.2.2.4: Evaluation of the capacity building program a year after the completion and potential design and implementation of corrective actions to complete the training.																				
Activity 1.2.3.1: Evaluation of existing financing mechanisms and assessment of needs and gaps to fill for the development of schemes for financially supporting PCI mining and refining companies' green initiatives.																				
Activity 1.2.3.2: Determination of the appropriate financing schemes based on the Activity 1.2.3.1, and design of the scheme(s) that will be promoted and implemented under the project.																				
Activity 1.2.3.3: Establishment, approval, and operationalization of financing schemes to support green and low carbon technology initiatives in the																				

phosphate mining and refining sub-sector.																				
Activity 1.2.3.4: Promotion of the green financing scheme to PCI companies' management teams to enhance their involvement and participation to the scheme.																				
Activity 1.2.3.5: Evaluation of the operational performance of the financing scheme(s) and provision of recommendations to make improvements and adjustments on the implementation of the scheme(s), if deemed necessary.																				
Activity 1.2.4.1: Development, publication, and dissemination of guidelines for the replication and scaling-up of successfully implemented green and low carbon development and operation of phosphate mines demos.																				
Activity 1.2.4.2: Development of implementation plans and action plans for demos replication and scale-up.																				
Activity 1.2.4.3: Conduct of lobbying and campaigning initiatives to promote and secure the approval and financing of the follow-up plans for the replication and scale-up of the demos.																				
Activity 1.2.4.4: Promotion of the application of successfully demonstrated green, EE, low carbon technologies in phosphate mining and refining in different regions of the country.																				
Activity 1.2.4.5: Design and implementation of a capacity building program on design, engineering, equipment manufacturing, operation and maintenance for service providing engineering firms in the phosphate mining and refining sub-sector.																				
Activity 1.2.4.6: Evaluation of the capacity building program designed in Activity 1.2.4.5 a year after the completion to assess how many trainees are gainfully applying the acquired skills and determine the corrective actions to be taken to improve the program.																				
Component 2: Green and Low Carbon Design and Operation of Phosphate Chemicals Production Facilities																				
Activity 2.1.1.1: Evaluation of existing policies and institutional frameworks on energy conservation, and green and low-carbon development in phosphate																				

chemicals production from the perspective of the industrial chain to update and strengthen them.																				
Activity 2.1.1.2: Completion of dual-carbon target-oriented analysis of the energy-savings, and green and low-carbon development path in phosphate chemicals production.																				
Activity 2.1.1.3: Conduct of research on inter-regional green, and low-carbon development policies, cooperation and information sharing arrangements in phosphate chemicals production in key regions of China																				
Activity 2.1.1.4: Conduct of research on the inter-departmental and inter-industry policy on energy conservation, and green and low-carbon development in the phosphorus chemical industry.																				
Activity 2.1.1.5: Conduct of an international comparative study on energy conservation and green and low-carbon development policies in phosphate chemicals production.																				
Activity 2.1.1.6: Formulation and recommendation of new policies and institutional frameworks on green, energy efficient low carbon technology application in phosphate chemicals production.																				
Activity 2.1.1.7: Approval of the newly formulated and recommended policies and institutional frameworks on green, energy efficient low carbon technology application in phosphate chemicals production.																				
Activity 2.1.1.8: Enforcement of new approved policies and institutional frameworks on green, energy efficient low carbon technology application in phosphate chemicals production.																				
Activity 2.1.1.9: Evaluation of the impact on the greening of the phosphate chemicals production sub-sector as a result of the enforcement of the newly developed policies and institutional frameworks, and provision of follow-up recommendations to amend and improve them, if deemed necessary.																				
Activity 2.1.2.1: Conduct of research to update existing standards, policies and implementing rules and regulations to promote green technologies in phosphate chemicals production.																				

Activity 2.1.2.2: Formulation, recommendation, approval and enforcement of new standards and technical guidelines for promoting green and low-carbon development in phosphate chemicals production.																				
Activity 2.1.2.3: Formulation, recommendation, approval and enforcement of new implementing rules and regulations for promoting green technologies in phosphate chemicals production.																				
Activity 2.1.2.4: Evaluation of the impact and effectiveness of the new enforced standards/technical guidelines and implementing rules and regulations for promoting and adopting green technologies in phosphate chemicals production, and provision of follow-up recommendations to amend and improve the standards/technical guidelines and implementing rules and regulations, if deemed necessary.																				
Activity 2.1.3.1: Design and implementation of methods for evaluation of the energy performance and environmental impacts of the demos for the phosphate chemicals production.																				
Activity 2.1.3.2: Conduct of annual evaluation of energy performance and environmental impacts of demos in phosphate chemicals production.																				
Activity 2.1.3.3: Conduct of annual evaluation of financial performance of demos in the phosphate chemicals production sub-sector to promote the replication of the implemented technologies.																				
Activity 2.1.3.4: Preparation of annual and final reports on the operational performance (production, energy consumption, and economic), and environmental impacts of demos in phosphate chemicals production.																				
Activity 2.1.3.5: Dissemination of annual and final demos evaluation reports prepared in Activity 2.1.3.4.																				
Activity 2.1.4.1: Assessment of the ability and awareness level of relevant stakeholders and assessment of needs to address in the area of green, energy efficient low carbon technologies in phosphate chemicals manufacturing in Yunnan, Guizhou, Sichuan, and Hubei Provinces.																				

Activity 2.1.4.2: Design and implementation of a capacity building program for relevant government officials, industry management teams and key stakeholders on green and low-carbon development in the phosphate chemical industry, including international best practice.																				
Activity 2.1.4.3: Evaluation of the capacity building program designed in Activity 2.1.4.2 a year after the completion to assess how many trainees are gainfully applying the acquired skills and determine the corrective action to improve the program.																				
Activity 2.1.4.4: Establishment of international exchanges and cooperation on energy conservation and green and low-carbon development in the phosphorus chemicals production.																				
Activity 2.1.5.1: Conduct of market analysis on the international trade trends for phosrock and phosphate chemicals, and the indirect carbon emission and ecological impacts in the international trading of these products.																				
Activity 2.1.5.2: Preparation of a market analysis report based on the findings of Activity 2.1.5.1, including an analysis of the implications for the phosphate chemicals production sub-sector in China and share with the sub-sector stakeholders																				
Activity 2.1.5.3: Organization and conduct of a seminar-workshop to present the market study to, and discuss appropriate future actions to take with, the phosphate chemicals production industry stakeholders and government.																				
Activity 2.1.6.1: Design and development of an information management and sharing system on all aspects of low carbon phosphate chemicals production in China.																				
Activity 2.1.6.2: Establishment and operationalization of the devised information sharing system for the exchange of available data on all aspects of green phosphate chemicals production.																				
Activity 2.1.6.3: Monitoring and evaluation of the operation of the information management and																				

exchange system.																				
Activity 2.1.6.4: Conduct of international cooperation and communication on energy saving and green and low-carbon development in the phosphate chemicals production.																				
Activity 2.1.6.5: Drafting and promotion of representative cases of energy saving and green and low-carbon development in the phosphate chemicals production.																				
Activity 2.2.1.1: Development and approval of engineering designs of green, energy efficient low carbon technology demos in phosphate chemicals production.																				
Activity 2.2.1.2: Preparation of an ESMF as well as an ESMP to manage and mitigate all identified risks associated with the design and implementation of the demo activities.																				
Activity 2.2.1.3: Development and approval of implementation plans, including financing arrangements, for each designed demo.																				
Activity 2.2.2.1: Implementation and operation of demonstrations on energy savings and green and low-carbon development in phosphate chemicals production.																				
Activity 2.2.2.2: Design and implementation of a capacity building program on green, energy efficient and low carbon technologies for phosphate chemicals production.																				
Activity 2.2.2.3: Evaluation of the capacity building program a year after the completion and potential design and implementation of corrective actions to complete the training.																				
Activity 2.2.3.1: Evaluation of existing financing mechanisms and assessment of needs and gaps to develop financial schemes for financially support SMEs that manufacture phosphate chemicals																				
Activity 2.2.3.2: Determination of the appropriate financing schemes for SMEs that manufacture phosphate chemicals based on the findings of Activity 2.2.3.1, and design of a pilot scheme that will be																				

promoted and implemented under the project.																				
Activity 2.2.3.3: Approval of the pilot financing scheme for SMEs that manufacture phosphate chemicals designed in Activity 2.2.3.2.																				
Activity 2.2.4.1: Operationalization of the pilot financing scheme for SMEs that manufacture phosphate chemicals, approved under Activity 2.2.3.3, in 2 or 3 selected regions.																				
Activity 2.2.4.2: Promotion of the pilot financing scheme and provision of technical assistance to phosphate chemicals SMEs' management team in the 2 or 3 selected regions to enhance their involvement and participation in the pilot scheme.																				
Activity 2.2.4.3: Evaluation of the operational performance of the pilot financing scheme for SMEs and provision of recommendations to make improvements and adjustments on the implementation of the scheme, if deemed necessary.																				
Activity 2.2.5.1: Preparation and dissemination of case studies and lessons learned based on the application of demonstrated green, energy efficient and low carbon technologies and the annual and final reports in activity 2.1.3.4.																				
Activity 2.2.5.2: Development, publication, and dissemination of green and low carbon development and operation in phosphate chemicals production guidelines to replicate and scale-up the demos.																				
Activity 2.2.5.3: Development of implementation plans and action plans for demos replication and scale-up to promote "global greening" in different regions of the country.																				
Activity 2.2.5.4: Conduct of lobbying and campaigning initiatives to promote and secure the approval and financing of the follow-up plans for the replication and scale-up of the demos.																				
Activity 2.2.5.5: Promotion of the application of successfully demonstrated green, energy-efficient and low-carbon technologies in phosphate chemicals production in different regions of the country.																				
Activity 2.2.5.6: Design and implementation of a																				

capacity building program on design, engineering, equipment manufacturing, operation and maintenance for service providing engineering firms in the phosphate chemicals production sub-sector.																				
Activity 2.2.5.7: Evaluation of the capacity building program designed in Activity 2.2.5.6 a year after the completion to assess how many trainees are gainfully applying the acquired skills and determine the corrective actions to be taken to improve the program.																				
Component 3: Green and Low Carbon Design and Operation of Waste Management Systems in the Phosphate Chemicals Industry																				
Activity 3.1.1.1: Conduct of research to update existing policies and strengthen existing institutional mechanisms in China to enable the greening of waste management in the phosphate mining and refining sub-sector.																				
Activity 3.1.1.2: Conduct of research to update existing policies and strengthen existing institutional mechanisms in China to enable the greening of waste management in the phosphate chemicals production sub-sector.																				
Activity 3.1.1.3: Conduct of research studies on existing policies and institutional frameworks in other countries that can be used as reference for supporting and promoting low carbon development in waste management in the phosphate mining and refining sub-sector.																				
Activity 3.1.1.4: Conduct of research studies on existing policies and institutional frameworks in other countries that can be used as reference for supporting and promoting low carbon development in waste management in the phosphate chemicals production sub-sector.																				
Activity 3.1.1.5: Formulation and recommendation of new policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate mining and refining sub-sector.																				
Activity 3.1.1.6: Approval of the newly formulated and recommended policies and institutional framework on green, energy efficient low carbon development in																				

waste management in the phosphate mining and refining sub-sector.																				
Activity 3.1.1.7: Enforcement of new approved policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate mining and refining sub-sector.																				
Activity 3.1.1.8: Evaluation of the impact on the greening of the waste management in the phosphate mining and refining sub-sector as a result of the enforcement of the newly developed policies and institutional framework and provision of follow-up recommendations to amend and improve the policies and the institutional framework, if deemed necessary.																				
Activity 3.1.1.9: Formulation and recommendation of new policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate chemicals production sub-sector.																				
Activity 3.1.1.10: Approval of the newly formulated and recommended policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate chemicals production sub-sector.																				
Activity 3.1.1.11: Enforcement of new approved policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate chemicals production sub-sector.																				
Activity 3.1.1.12: Evaluation of the impact on the greening of the waste management in the phosphate chemicals production sub-sector as a result of the enforcement of the newly developed policies and institutional framework and provision of follow-up recommendations to amend and improve the policies and the institutional framework, if deemed necessary.																				
Activity 3.1.2.1: Conduct of research to update existing schemes, standards and regulations to promote the application of green and low carbon waste management technologies in the phosphate mining and refining sub-sector.																				
Activity 3.1.2.2: Conduct of research to update existing																				

schemes, standards and regulations to promote the application of green and low carbon waste management technologies in the phosphate chemicals production sub-sector.																			
Activity 3.1.2.3: Conduct of research on the cross-ministry inter-departmental and cross-inter-industry policy linkage mechanisms for waste management and comprehensive utilization of resources in the phosphate mining and refining sub-sector.																			
Activity 3.1.2.4: Conduct of research on the cross-ministry inter-departmental and cross-inter-industry policy linkage mechanisms for waste management and comprehensive utilization of resources in the phosphate chemicals production sub-sector.																			
Activity 3.1.2.5: Formulation and recommendation of new relevant schemes, standards, technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate mining and refining sub-sector.																			
Activity 3.1.2.6: Approval of the newly formulated and recommended relevant schemes, standards, technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate mining and refining sub-sector.																			
Activity 3.1.2.7: Enforcement of new approved relevant schemes, standards, technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate mining and refining sub-sector.																			
Activity 3.1.2.8: Evaluation of the effectiveness of the schemes, standards, technical guidance, implementing rules and regulations on green EE low carbon development in waste management in the phosphate mining and refining sub-sector, and provision of follow-up recommendations to amend and improve the institutional frameworks, if deemed necessary.																			
Activity 3.1.2.9: Formulation and recommendation of																			

new relevant schemes, standards, technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate chemicals production sub-sector.																				
Activity 3.1.2.10: Approval of the newly formulated and recommended relevant schemes, standards, technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate chemicals production sub-sector.																				
Activity 3.1.2.11: Enforcement of new approved relevant schemes, standards, technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate chemicals production sub-sector.																				
Activity 3.1.2.12: Evaluation of the effectiveness of the schemes, standards, technical guidance, implementing rules and regulations on green EE low carbon development in waste management in the phosphate chemicals production sub-sector, and provision of follow-up recommendations to amend and improve the institutional frameworks, if deemed necessary.																				
Activity 3.1.3.1: Design and implementation of methods for evaluation of the energy performance and environmental impacts of the waste management demos in the phosphate mining/refining sub-sector of the PCI.																				
Activity 3.1.3.2: Design and implementation of methods for evaluation of the energy performance and environmental impacts of the waste management demos in the phosphate chemicals production sub-sector of the PCI.																				
Activity 3.1.3.3: Conduct of annual evaluation of energy performance, environmental impacts and financial performance of demos in waste management of the phosphate mining and refining sub-sector to promote the replication of the implemented technologies.																				
Activity 3.1.3.4: Conduct of annual evaluation of energy																				

performance, environmental impacts and financial performance of demos in waste management of the phosphate chemicals production sub-sector to promote the replication of the implemented technologies.																				
Activity 3.1.3.5: Preparation of annual and final reports on energy performance, environmental impacts of demos and the evaluation of the results and effectiveness of the demonstration program in waste management of the phosphate mining and refining sub-sector.																				
Activity 3.1.3.6: Preparation of annual and final reports on energy performance, environmental impacts of demos and the evaluation of the results and effectiveness of the demonstration program in waste management of the phosphate chemicals production sub-sector.																				
Activity 3.1.3.7: Dissemination of annual and final demos evaluation reports prepared in Activity 3.1.3.5																				
Activity 3.1.3.8: Dissemination of annual and final demos evaluation reports prepared in Activity 3.1.3.6																				
Activity 3.1.4.1: Assessment of the ability and awareness level of relevant stakeholders and assessment of needs and gaps to address in the area of waste management in the phosphate mining and refining sub-sector.																				
Activity 3.1.4.2: Design and implementation of a capacity building program for relevant national and local government officials, management teams and key stakeholders on waste management in the phosphate mining and refining sub-sector based on the findings of Activity 3.1.4.1.																				
Activity 3.1.4.3: Evaluation of the capacity building program of the waste management in the phosphate mining and refining sub-sector a year after the completion and potential design and implementation of corrective actions to complete the training.																				
Activity 3.1.4.4: Assessment of the ability and awareness level of relevant stakeholders and assessment of needs and gaps to address in the area of waste management in the phosphate chemicals																				

production sub-sector.																				
Activity 3.1.4.5: Design and implementation of a capacity building program for relevant national and local government officials, industry management teams and key stakeholders on waste management in the phosphate chemicals production sub-sector based on the findings of Activity 3.1.4.4.																				
Activity 3.1.4.6: Evaluation of the capacity building program of the waste management in the phosphate chemicals production sub-sector a year after the completion and potential design and implementation of corrective actions to complete the training.																				
Activity 3.1.4.7: Design and implementation of a training program for professionals to build capacity concerning the new and updated policies, institutional frameworks, standards and green and low carbon technologies applied in waste management in the phosphate mining and refining sub-sector.																				
Activity 3.1.4.8: Evaluation of the capacity building program of the waste management in the phosphate mining and refining sub-sector a year after the completion to gauge the effectiveness of the program and come up with improvements and/or expansion of the program.																				
Activity 3.1.4.9: Design and implementation of a training program for professionals to build capacity concerning the new and updated policies, institutional frameworks, standards and green and low carbon technologies applied in waste management in the phosphate chemicals production sub-sector.																				
Activity 3.1.4.10: Evaluation of the capacity building program of the waste management in the phosphate chemicals production sub-sector a year after the completion to gauge the effectiveness of the program and come up with improvements and/or expansion of the program.																				
Activity 3.1.5.1: Drafting and finalization of research reports on annual volume of waste production in the phosphate mining and refining sub-sector.																				
Activity 3.1.5.2: Drafting and finalization of research																				

reports on annual volume of waste production in phosphate chemical production sub-sector.																				
Activity 3.1.5.3: Drafting and finalization of research reports on green and low carbon waste management systems for waste recycling and reuse in the phosphate mining and refining sub-sector developed and implemented in other countries and their energy utilization performances.																				
Activity 3.1.5.4: Drafting and finalization of research reports on green and low carbon waste management systems for waste recycling and reuse in the phosphate chemicals production sub-sector developed and implemented in other countries and their energy utilization performances.																				
Activity 3.1.5.5: Drafting and finalization of research reports on potential commercial markets for processed waste produced in the mining and refining of the PCI.																				
Activity 3.1.5.6: Drafting and finalization of research reports on potential commercial markets for processed waste produced in the production process of the PCI.																				
Activity 3.1.5.7: Presentation in a seminar-workshop of the research findings to the PCI and discussions to come up with action plans for circular economy in phosphate mining/refining.																				
Activity 3.1.5.8: Presentation in a seminar-workshop of the research findings to the PCI and discussions to come up with action plans for circular economy in phosphate chemicals production.																				
Activity 3.1.6.1: Design and development of an information management and sharing system on market developments and R&D on low carbon waste management systems in the phosphate mining and refining sub-sector in China and abroad.																				
Activity 3.1.6.2: Establishment and operationalization of the devised information sharing system for the exchange of available data on all aspects of market developments and R&D on green waste management systems in the phosphate mining and refining sub-sector in China and abroad.																				

Activity 3.1.6.3: Monitoring and evaluation of operation and performance of the information management and exchange system for the phosphate mining and refining sub-sector.																				
Activity 3.1.6.4: Design and development of an information management and sharing system on market developments and R&D on low carbon waste management systems in the phosphate chemicals production sub-sector in China and abroad.																				
Activity 3.1.6.5: Establishment and operationalization of the devised information sharing system for the exchange of available data on all aspects of market developments and R&D on green waste management systems in the phosphate chemicals production sub-sector in China and abroad.																				
Activity 3.1.6.6: Monitoring and evaluation of operation and performance of the information management and exchange system for the phosphate chemicals production sub-sector.																				
Activity 3.1.6.7: Drafting, promotion and dissemination of representative cases of energy saving and green and low-carbon development in waste management systems in the phosphate mining and refining sub-sector.																				
Activity 3.1.6.8: Drafting, promotion and dissemination of representative cases of energy saving and green and low-carbon development in waste management systems in the phosphate chemicals production sub-sector.																				
Activity 3.2.1.1: Development and approval of engineering designs of green demos for waste management in the phosphate mining and refining sub-sector to improve phosrock mining tailings recycling and reuse.																				
Activity 3.2.1.2: Preparation of an ESMF as well as an ESMP to manage and mitigate all identified risks associated with the design and implementation of the demo activities for waste management in the phosphate mining and refining sub-sector.																				
Activity 3.2.1.3: Development and approval of																				

engineering designs of green demos for waste management in the phosphate chemicals production sub-sector to increase energy efficiency processing of phosphogypsum and marketing of the gypsum product.																				
Activity 3.2.1.4: Preparation of an ESMF as well as an ESMP to manage and mitigate all identified risks associated with the design and implementation of the demo activities for waste management in the phosphate chemicals production sub-sector.																				
Activity 3.2.1.5: Development and approval of implementation plans for the waste management in the phosphate mining and refining sub-sector, including financing arrangements, for each designed demo.																				
Activity 3.2.1.6: Development and approval of implementation plans for the waste management in the phosphate chemicals production sub-sector, including financing arrangements, for each designed demo.																				
Activity 3.2.2.1: Implementation and operation of demonstrations of green equipment/technologies for the comprehensive utilization of resources in waste management in the phosphate mining and refining sub-sector.																				
Activity 3.2.2.2: Implementation and operation of demonstrations of green equipment/technologies for the comprehensive utilization of resources in waste management in the phosphate chemicals production sub-sector.																				
Activity 3.2.2.3: Design and implementation of a capacity building program on green, energy efficient and low carbon technologies for waste management in the phosphate mining and refining sub-sector.																				
Activity 3.2.2.4: Evaluation of the capacity building program of the phosphate mining and refining sub-sector a year after the completion and potential design and implementation of corrective actions to complete the training.																				
Activity 3.2.2.5: Design and implementation of a capacity building program on green, energy efficient and low carbon technologies for waste management in the phosphate chemicals production sub-sector.																				

Activity 3.2.2.6: Evaluation of the capacity building program of the phosphate chemicals production sub-sector a year after the completion and potential design and implementation of corrective actions to complete the training.																				
Activity 3.2.3.1: Preparation and dissemination of case studies and lessons learned based on the application of demonstrated green, low carbon waste management technologies in the phosphate mining and refining sub-sector.																				
Activity 3.2.3.2: Preparation and dissemination of case studies and lessons learned based on the application of demonstrated green, low carbon waste management technologies in the phosphate chemicals production sub-sector.																				
Activity 3.2.3.3: Development, publication, and dissemination of green and low carbon waste management technologies and operation guidelines in the phosphate mining and refining sub-sector to replicate and scale-up the demos.																				
Activity 3.2.3.4: Development, publication, and dissemination of green and low carbon waste management technologies and operation guidelines in the phosphate chemicals production sub-sector to replicate and scale-up the demos.																				
Activity 3.2.3.5: Development of implementation plans and action plans for demos replication and scale-up to facilitate the establishment of green factories, the production of green products and the promotion of demonstrated green, energy-efficient and low-carbon technologies in the phosphate mining and refining sub-sector in different regions of the country.																				
Activity 3.2.3.6: Approval and promotion of the application of the development and implementation plans formulated under Activity 3.2.3.5 to facilitate the establishment of green factories, the production of green products and the implementation of demonstrated green, energy-efficient and low-carbon technologies in the phosphate chemicals production sub-sector in different regions of the country.																				

Activity 3.2.3.7: Development of implementation plans and action plans for demos replication and scale-up to facilitate the promotion of demonstrated green, energy-efficient and low-carbon technologies in the phosphate chemicals production sub-sector in different regions of the country.																				
Activity 3.2.3.8: Approval and promotion of the application of the development and implementation plans formulated under Activity 3.2.3.7 to facilitate the establishment of green factories, the production of green products and the implementation of demonstrated green, energy-efficient and low-carbon technologies in the phosphate chemicals production sub-sector in different regions of the country.																				
Activity 3.2.3.9: Enhancement of the capacity of the engineering service provision industry that are currently supporting the phosphate mining and refining sub-sector in their waste management requirements.																				
Activity 3.2.3.10: Enhancement of the capacity of the engineering service provision industry that are currently supporting the phosphate chemicals production sub-sector in their waste management requirements.																				

Annex 4: UNDP Social and Environmental Screening Procedure (SESP)

Project Information	
[1.] **Project Title**	Facilitating Cleaner and Energy Efficient Phosphate Chemicals Industry in China (PhosChemEE) Project
[2.] **Project Number (i.e. Atlas project ID, PIMS+)**	PIMS 6618
[3.] **Location (Global/Region/Country)**	China, People's Republic
[4.] **Project stage (Design or Implementation)**	Design (ProDoc)
[5.] **Date**	14 November 2021

Part A. Integrating Programming Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Programming Principles in Order to Strengthen Social and Environmental Sustainability?
Briefly describe in the space below how the project mainstreams the human rights-based approach
As in other UNDP-GEF projects, the PhosChemEE Project will be designed and implemented with a human-rights-based approach. This project will be based in the underdeveloped western regions of Yunnan, Guizhou, Sichuan, and Hubei, and will focus on promoting the economic and social welfare of local enterprises and communities. It is envisaged that as in the discussions with partner provincial governments, during the project design, consultations will be carried out with the local communities in selected areas where the demonstrations showcase the applications of green utilization technologies. The project will create employment opportunities and reduce poverty among the local communities through the potential phosphate mining with beneficiation technology applications, upgrade of existing phosphate chemical industry and new industry on the integrated processing of phosphate chemical byproducts such as phosphogypsum. The depletion of phosphogypsum stacks and reduction in phosphogypsum output in these regions will also reduce phosphogypsum-related pollutants' adverse impact on local water and soil quality, directly and indirectly contributing to the local agricultural production, food security and public health. In the project development phase, local communities and enterprises would be involved in the consultation process on demonstration project design and new industrial layout. Such consultations will be a continuous process even during project implementation to ensure that the project activities, particularly the demonstrations seriously consider the welfare of indigenous peoples and all other residents of the communities. Full respect of the rights of local communities and according to UNDP Standard 6, an FPIC ("Free Prior Informed Consent") will be carried out and documented according to UNDP requirements at each demonstration site. This work will be a part of site-specific environmental and social ("E/S") assessments to be carried out for each demonstration on technology upgrading and solid waste utilization under the project. In the project implementation phase, professional training and education would be provided to local governments, enterprises and employees of the phosphate chemical industry, and new employment opportunities would be offered in advance to indigenous communities. Additionally, a grievance redress mechanism would be established to ensure that access to work opportunities is equitable and that any adverse environmental and social effects of industrial activities be addressed promptly.
Briefly describe in the space below how the project is likely to improve gender equality and women's empowerment
The design of the proposed project will be guided by principles of gender equality and women's empowerment. Women are involved in both the management and technical departments of China's government agencies/institutions and play major roles in the top-level decision-making and strategic design process. On the ground level, the implementation of the proposed project will promote the welfare of and offer equal opportunities to both men and women, but particularly women in the underdeveloped regions. The design of the project activities will also consider further enhancing the role of women in the deployment of low carbon technologies and climate change mitigation options and come up with gender-sensitive policies in the resource sector and the downstream sectors of the country. Education and professional training would be provided in particular to local employees to improve women's professional competitiveness in the workplace and expand women's scope of employment in the underdeveloped and

agriculture-based region. The detailed and confidential staff information database can be established based on big data analysis, which records both men's and women's information on occupation, payment, etc. This will provide information for job mobility and welfare distribution in PCI enterprises following national gender-related law and good international practice. Additionally, qualified and capable women consultants/experts will be targeted to work in the implementation of the designed project. The implementation of this project will, first, improve the technical level and reduce the intensity of labor in the mining and beneficiation technology as well as phosphate chemical industry, which will in turn improve women's work environment, job adaptability and change of employment. Secondly, the implementation of this project will directly improve the quality of the local ecological environment and indirectly improve women's and children's health. Thirdly, implementation of this project will help ensure agriculture and food security through the reduction of rural pollution, and thus reduce the risk in women's major source of income in these regions.
Briefly describe in the space below how the project mainstreams sustainability and resilience
The central mission of this project is to promote systematic sustainable transformation of China's phosphate mining and chemical industry. This will be pursued through major top-level strategies: strengthening of low-carbon policy guidance on phosphate chemical industry, construction of low-carbon demonstration system, establishment of low-carbon development standard system, development of technology and equipment in mining and beneficiation sector, along with solid waste comprehensive utilization, implementation of low-carbon phosphate chemical technologies promotion action, and popularization of low-carbon education. The strategic mechanisms would improve the utilization of phosphate resource and mining area environment, reduce the energy consumption in production, promote waste utilization technology in the related areas, energy consumption, and solid waste output reduction will be facilitated in the phosphate chemical industry during and after implementation of the proposed project. The promotion of clean and low-carbon processes and technologies within the whole life cycle of phosphate chemical production, the integrated utilization of phosphogypsum, and the phosphate mine tailing production of phosphate fertilizers are three focus areas of the project's carbon-emission reduction initiative. According to estimates, carbon emission reduction in phosphate mining can be at an average of 0.05 million tons of CO2eq per year; the annual emission reduction in phosphate chemical production is about 1.19 million tons of CO2eq, and the management of phosphate waste can reduce CO2eq emissions of approximately 5.93 million tons per year (including 4.97 million tons in the integrated use of phosphogypsum to produce cement, and 0.96 million tons in phosphate mine tailing to produce phosphate-containing biological fertilizers). In total, the phosphate chemical industry can form a carbon dioxide emission reduction capacity of 7.39 million tons/year, or 36.97 million tons CO2eq during the five-year project implementation period. As most of China's phosphate mines and chemical production facilities are concentrated in Sichuan, Yunnan, Guizhou, Hubei and other Yangtze River main streams and upstream tributary areas, resulting in severe pollution of water resources in the upper and middle reaches of the Yangtze River, the project's targeted implementation in the Yangtze Economic Belt would not only address environmental pollution and facilitate ecological repair, but also enhance environmental resiliency of the underdeveloped communities through job creation and poverty reduction in these ecologically fragile areas.
Briefly describe in the space below how the project strengthens accountability to stakeholders
The project will be designed and implemented guided by principles of accountability. This would naturally ensure constant engagement with stakeholders from design, implementation, and completion of the project. It is envisaged that the potential project-affected people will be informed of potential risks and UNDP's Accountability Mechanism during the project design and further strengthened in the implementation process. The demos will support meaningful participation and inclusion of all stakeholders, in particular marginalized individuals and groups, in processes that may impact them including design, implementation and monitoring of the project. Provision of timely, accessible and functional information regarding supported activities, including potential environmental and social risks and impacts and management measures (SESP, ESMF) will be disclosed on UNDP China websites to ensure transparency of programming interventions. The Project Management Office (PMO) shall establish a project-level Grievance Redress Mechanism (GRM) during project inception. The full details of the GRM will be agreed upon during the project's inception phase. The Mechanism (SRM) that ensures individuals, people, and communities affected by projects have access to appropriate grievance resolution procedures for hearing and jointly addressing complaints and disputes related to the social and/or environmental impacts of the proposed projects. Interested stakeholders may raise a grievance at any time to the PMO, the Executing Agency, the Implementing Agency (UNDP), or the GEF. A monitoring and evaluation procedure is also designed to ensure effective monitoring and where appropriate, participatory monitoring with stakeholders—and reporting on implementation of social and environmental risk management measures.

Project Information	
1. Project Title	Facilitating Cleaner and Energy Efficient Phosphate Chemicals Industry in China (PhosChemEE) Project
2. Project Number (i.e. Atlas project ID, PIMS+)	PIMS 6618

3. Location (Global/Region/Country)	China, People's Republic
4. Project stage (Design or Implementation)	Design (ProDoc)
5. Date	May 2022

Part A. Integrating Programming Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Programming Principles in Order to Strengthen Social and Environmental Sustainability?

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As in other UNDP-GEF projects, the PhosChemEE Project will be designed and implemented with a human-rights-based approach. This project will be based in the underdeveloped western regions of Yunnan, Guizhou, Sichuan, and Hubei, and will focus on promoting the economic and social welfare of local enterprises and communities. It is envisaged that as in the discussions with partner provincial governments, during the project design, consultations will be carried out with the local communities in selected areas where the demonstrations showcase the applications of green utilization technologies. The project will create employment opportunities and reduce poverty among the local communities through the potential phosphate mining with beneficiation technology applications, upgrade of existing phosphate chemical industry and new industry on the integrated processing of phosphate chemical byproducts such as phosphogypsum. The depletion of phosphogypsum stacks and reduction in phosphogypsum output in these regions will also reduce phosphogypsum-related pollutants' adverse impact on local water and soil quality, directly and indirectly contributing to the local agricultural production, food security and public health. In the project development phase, local communities and enterprises would be involved in the consultation process on demonstration project design and new industrial layout. Such consultations will be a continuous process even during project implementation to ensure that the project activities, particularly the demonstrations seriously consider the welfare of indigenous peoples and all other residents of the communities. Full respect of the rights of local communities and according to UNDP Standard 6, an FPIC ("Free Prior Informed Consent") will be carried out and documented according to UNDP requirements at each demonstration site. This work will be a part of site-specific environmental and social ("E/S") assessments to be carried out for each demonstration on technology upgrading and solid waste utilization under the project. In the project implementation phase, professional training and education would be provided to local governments, enterprises and employees of the phosphate chemical industry, and new employment opportunities would be offered in advance to indigenous communities. Additionally, a grievance redress mechanism would be established to ensure that access to work opportunities is equitable and that any adverse environmental and social effects of industrial activities be addressed promptly.

Briefly describe in the space below how the project is likely to improve gender equality and women's empowerment

The design of the proposed project will be guided by principles of gender equality and women's empowerment. Women are involved in both the management and technical departments of China's government agencies/institutions and play major roles in the top-level decision-making and strategic design process. On the ground level, the implementation of the proposed project will promote the welfare of and offer equal opportunities to both men and women, but particularly women in the underdeveloped regions. The design of the project activities will also consider further enhancing the role of women in the deployment of low carbon technologies and climate change mitigation options and come up with gender-sensitive policies in the resource sector and the downstream sectors of the country. Education and professional training would be provided in particular to local employees to improve women's professional competitiveness in the workplace and expand women's scope of employment in the underdeveloped and agriculture-based region. The detailed and confidential staff information database can be established based on big data analysis, which records both men's and women's information on occupation, payment, etc. This will provide information for job mobility and welfare distribution in PCI enterprises following national gender-related law and good international practice. Additionally, qualified and capable women consultants/experts will be targeted to work in the implementation of the designed project.
The implementation of this project will, first, improve the technical level and reduce the intensity of labor in the mining and beneficiation technology as well as phosphate chemical industry, which will in turn improve women's work environment, job adaptability and change of employment. Secondly, the implementation of this project will directly improve the quality of the local ecological environment and indirectly improve women's and children's health. Thirdly, implementation of this project will help ensure agriculture and food security through the reduction of rural pollution, and thus reduce the risk in women's major source of income in these regions.

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The central mission of this project is to promote systematic sustainable transformation of China's phosphate mining and chemical industry. This will be pursued through major top-level strategies: strengthening of low-carbon policy guidance on phosphate chemical industry, construction of low-carbon demonstration system, establishment of low-carbon development standard system, development of technology and equipment in mining and beneficiation sector, along with solid waste comprehensive utilization, implementation of low-carbon phosphate chemical technologies promotion action, and popularization of low-carbon education. The strategic mechanisms would improve the utilization of phosphate resource and mining area environment, reduce the energy consumption in production, promote waste utilization technology in the related areas, energy consumption, and solid waste output reduction will be facilitated in the phosphate chemical industry during and after implementation of the proposed project. The promotion of clean and low-carbon processes and technologies within the whole life cycle of phosphate chemical production, the integrated utilization of phosphogypsum, and the phosphate mine tailing production of phosphate fertilizers are three focus areas of the project's carbon-emission reduction initiative. According to estimates, carbon emission reduction in phosphate mining can be at an average of 0.05 million tons of CO2eq per year; the annual emission reduction in phosphate chemical production is about 1.19 million tons of CO2eq, and the management of phosphate waste can reduce CO2eq emissions of approximately 5.93 million tons per year (including 4.97 million tons in the integrated use of phosphogypsum to produce cement, and 0.96 million tons in phosphate mine tailing to produce phosphate-containing biological fertilizers). In total, the phosphate chemical industry can form a carbon dioxide emission reduction capacity of 7.39 million tons/year, or 36.97 million tons CO2eq during the five-year project implementation period. As most of China's phosphate mines and chemical production facilities are concentrated in Sichuan, Yunnan, Guizhou, Hubei and other Yangtze River main streams and upstream tributary areas, resulting in severe pollution of water resources in the upper and middle reaches of the Yangtze River, the project's targeted implementation in the Yangtze Economic Belt would not only address environmental pollution and facilitate ecological repair, but also enhance environmental resiliency of the underdeveloped communities through job creation and poverty reduction in these ecologically fragile areas.
Briefly describe in the space below how the project strengthens accountability to stakeholders
The project will be designed and implemented guided by principles of accountability. This would naturally ensure constant engagement with stakeholders from design, implementation, and completion of the project. It is envisaged that the potential project-affected people will be informed of potential risks and UNDP's Accountability Mechanism during the project design and further strengthened in the implementation process. The demos will support meaningful participation and inclusion of all stakeholders, in particular marginalized individuals and groups, in processes that may impact them including design, implementation and monitoring of the project. Provision of timely, accessible and functional information regarding supported activities, including potential environmental and social risks and impacts and management measures (SESP, ESMF) will be disclosed on UNDP China websites to ensure transparency of programming interventions. The Project Management Office (PMO) shall establish a project-level Grievance Redress Mechanism (GRM) during project inception. The full details of the GRM will be agreed upon during the project's inception phase. The Mechanism (SRM) that ensures individuals, people, and communities affected by projects have access to appropriate grievance resolution procedures for hearing and jointly addressing complaints and disputes related to the social and/or environmental impacts of the proposed projects. Interested stakeholders may raise a grievance at any time to the PMO, the Executing Agency, the Implementing Agency (UNDP), or the GEF. A monitoring and evaluation procedure is also designed to ensure effective monitoring-and where appropriate, participatory monitoring with stakeholders—and reporting on implementation of social and environmental risk management measures.

Part B. Identifying and Managing Social and Environmental Risks

QUESTION 2: What are the Potential Social and Environmental Risks? *Note: Complete SESP Attachment 1 before responding to Question 2.*	**QUESTION 3: What is the level of significance of the potential social and environmental risks?** *Note: Respond to Questions 4 and 5below before proceeding to Question 5*			**QUESTION 6: Describe the assessment and management measures for each risk rated Moderate, Substantial or High**
Risk Description (broken down by event, cause, impact)	***Impact and Likelihood (1-5)***	***Significance (Low, Moderate Substantial, High)***	***Comments (optional)***	***Description of assessment and management measures for risks rated as Moderate, Substantial or High***
1. The PhosChemEE Project (and most specifically Component 1 which focuses on the upstream Phosphate mining and refining sub-sector) could lead to adverse impacts on enjoyment of the human rights (civil, political, economic, social, or cultural) of the affected population, including ethnic minorities and vulnerable/disadvantaged groups. This is most likely to occur under Outputs 1.2.2, 2.2.2 and 3.2.2. Mining companies engaged in the project might not have the capacity and knowledge to uphold their duties as per UNDP SES requirements. This includes upholding duties and requirements on FPIC as per UNDP SES 6. *Human Rights: P.3, P.4, P.5* *Gender Equality: P.10:* *Accountability: P.13, P.14* *Standard 6: 6.1, 6.3, 6.4*	I = 5 L = 4	High	This project will be based in the underdeveloped western regions of Yunnan, Guizhou, Sichuan, and Hubei. Yunnan, Guizhou, Sichuan and Hubei have widespread minority cultures (that might trigger SES Standard 6), but factories are mainly located in industrialized regions where cultural heritage do not concentrate. PCI industries do not geographically overlap with indigenous settlements, though impacts (positive or negative) on such communities cannot be ruled out at this early stage of project design.	In the design stage, the following safeguards documents have been prepared, covering this risk and all other risks, to meet the SES requirements of a **High risk** project: • a stakeholder analysis and comprehensive Stakeholder Engagement Plan, • a gender analysis and Gender Action Plan, • an Environmental and Social Management Framework (ESMF) with initial FPIC procedures (to be applied if confirmed as required), • a Grievance Redress Mechanism (draft/outline). The ESMF outlines the procedures for the (site-specific) Environmental and Social Impact Assessments (ESIAs) and Environmental and Social Management Plans (ESMPs) that will assess impacts directly related to the sphere of influence of the demo sites. These will be prepared during project implementation. To adequately assess impact from upstream/policy and strategy-level interventions (including potential cumulative impacts) the project shall undertake a Strategic Environmental and Social Assessment (SESA). The design and implementation of the project are consistent with the strategic requirements of the country's Law on ethnic unity and progress, which will

				ensure the enjoyment of the human rights of the potentially affected population in the demo places, especially ethnic minorities and/or vulnerable/disadvantaged groups. During the implementation of the project, extensive consultations will also be conducted with local people with the cooperation of the MIIT, MNR and provincial government.
2. During the construction of demo activities (Output 2.2.2), and in support to the mining and refining sectors (under Component 1), potential adverse impacts to habitats and/or ecosystems and ecosystem services (e.g. non-hazardous waste from phosphogypsum reuse) may occur. *Standard 1: 1.1, 1.2., 1.3, 1.7, 1.11, 1.14* *Standard 8: 8.1, 8.2, 8.3, 8.6*	I = 4 L = 3	Substantial	The demos (i.e. outputs 1.2.2, 2.2.2, 3.2.2) may generate construction waste during the construction of supporting facilities or new facilities and generate non-hazardous waste from phosphogypsum reuse. These may have an impact on the surrounding environment that includes ecologically fragile areas. The proposed interventions may not be effective at the start and will continue or may even further negatively impact the surrounding ambient environment , bodies of what or land areas in the project areas.	The project has been designed such that demos will involve the conduct of a detailed feasibility study and ESIA on the local area before the demo site selection, and a site-specific ESMP will be put in place before implementation of each demo. The demos will be designed and constructed in accordance with national and international standards to ensure compliance with the SES and other relevant environmental ordinances/regulations, as outlined in the ESMF (and to be in the forthcoming ESIA/ESMPs). Additionally, the Project will observe the established regulatory framework for monitoring and assessing such risks that are aligned with the SES. China has regulations on the engineer project site for the restoration and protection of biodiversity, and manufactories are not allowed to be established in or around ecosystem function zone. In this case, the PhosChemEE Project will ensure that the demo sites will be located far away from ecological reserves and ecologically vulnerable areas.
3. Inadequate disposal of waste during the operation of the phosphogypsum processing reuse demo (I.e. Outputs 3.2.1 and 3.2.2) poses a threat to the environment as well as the health and safety of workers and the community. The associated construction, operation, or decommissioning of the demo installations may have potential health and safety risks to local	I = 4 L = 3	Substantial	Specifically, if fluorine and arsenic contained in phosphogypsum are released into the atmosphere and soil environment, they will have adverse effects on surface/underground water plants and animals and produce health risks to local communities.	The ESMF and subsequent ESIA and ESMP will address all health and safety risks. The required management measures will be included in site-specific ESMPs and be conducted before project construction. Per those forthcoming plans, appropriate training will be provided to the demonstration enterprises to ensure that risks are properly addressed and managed. Training will also be provided to local governments in order to enhance their regulation capacity to control potential risks in the demonstration. A Grievance Mechanism will be designed (see ESMF) and will be implemented by the project, allowing communities and

communities and workers due to the transport, storage, and use and/or disposal of any hazardous or dangerous materials (e.g. explosives, fuel and other chemicals) that may be used during construction and operation. *Standard 1: 1.1, 1.2., 1.3, 1.7, 1.11, 1.14* *Standard 3: 3.1, 3.2, 3.5, 3.6, 3.7* *Standard 4: 4.2, 4.3* *Standard 6: 6.1, 6.3* *Standard 7: 7.1, 7.5, 7.6* *Standard 8: 8.1, 8.2, 8.3, 8.6*				workers to request for interventions when facing issues with health and safety risks. The associated construction, operation, or decommissioning of the demo installations will strictly comply with the national standards[16] and international general standards on construction and waste management, as noted in the ESMF. Site-specific environmental and social assessments will be conducted for each demo to identify the critical health and safety risks. Where public access is available, appropriate engineering and administrative controls (e.g., detours, traffic calming, signs) will be considered and implemented in advance during the construction and operation of demos. All contractors (i.e., undertaken engineering and infrastructure works as part of the project) will be required to develop, submit and adhere to a Labour Management Plan that meets the requirements of both UNDP SES 7 and relevant national/host country law and regulation. Appropriate training will be provided to workers to ensure that they install and operate the installed system correctly and safely, and properly control and manage the release or disposal of waste. Training will also be provided to local governments in order to enhance their regulation capacity to control potential risks in the demonstration. This will minimize or avoid any community health risks and safety issues for the communities regarding construction work involved in the installation of the demos, and the minimization and management of waste generated from these demos (e.g., explosives, fuel and other chemicals).
4. The potential outcomes of the PhosChemEE Project will be sensitive or vulnerable to potential impacts of climate change. Yunnan, Sichuan, Guizhou, and Hubei, where the demos are located, have risks of	I = 3 L = 3	Moderate	Some that are sheltered may also be impacted by such climate change-aggravated weather events that can cause extreme heat during summers or maybe flooding during rainy	The ESIAs (as required and outlined in the ESMF) will assess activities for impact and sustainability within China national context, including its climatic vulnerability and associated impacts. The design of the demo will take climate-related events

[16] Including Standard for pollution control on hazardous waste storage, Standard for pollution control on hazardous waste storage, etc.

extreme weather that are exacerbated by climate change. Thus, some of the demonstration activities of the project (located outdoors) can be impacted by weather events *Standard 2: 2.2, 2.3, 2.4* *Standard 3: 3.1*			seasons.	and risks into consideration by utilizing the findings of the risk assessment that shall be conducted as part of the ESIAs. environment risk assessment. Appropriate measures will be taken based on the specific climate-related events. For instance, the adverse impact of flood on demos can be eliminated by constructing diversion ditches, while sunshade roofs and cooling spray can reduce the impact of extreme high temperatures. The demos will also cover climate risk adaptation capacity building through training workshops (Activities 1.2.2.3, 2.2.2.2, 3.2.2.3). This training will be provided to the demo companies and local governments to enhance their capacities to deal with extreme climate events in general. Depending on the extent of the impacts of the adverse climate-related events, appropriate modifications in the installations (and budget) will be done. Potential reduction in the number of installations, or replacement with alternative demos will be done while considering the need to ensure the resulting interventions are still contributing to the realization of the project outcomes.
5. The Project includes activities both downstream/physical activities (like phosphogypsum processing), as well as upstream/policy-level interventions that require/could lead to significant consumption of raw materials, energy, and/or water; and involves significant extraction, diversion or containment of surface and/or ground water. That resource use, if not designed or implemented well, could lead to adverse impacts on ambient conditions in the project area and area of influence. *Standard 1: 1.1, 1.2., 1.3, 1.7, 1.11, 1.14* *Standard 8: 8.1, 8.2, 8.3, 8.6*	I = 4 L = 3	Substantial	Phosphate chemicals production, as well as phosphogypsum processing consume significant amounts of water. Existing processes are energy inefficient, have considerable material losses, and wastewater generation. While the project will promote cleaner and more energy efficient phosphate chemicals production and processing of phosphogypsum that will either reduce or optimize the use of water, more processing of phosphogypsum may increase the water demand, and depending on where water is sourced, can result in significant extraction, diversion or	Per the ESMF, the ESIA and ESMP will define the management measures for this risk. In the design of demos, life-cycle analysis (LCA) will be conducted before each industrial process to assess the raw materials, energy, and water used in new technologies and evaluate the potential of resource conservation. Based on LCA assessments, the project will be designed to seek cleaner and more energy-efficient phosphate chemicals production and processing of phosphogypsum. International practice for water and energy conservation will be applied in demos, especially the wastewater management and reuse technologies. Emergency accident pools will be designed in each demo in accordance with the requirement of contamination avoidance. Additionally, the demos will establish the resource efficiency monitoring mechanism, including using benchmarking data to the relative level of efficiency and reporting water and

			containment of surface and/or groundwater.	energy consumption periodically. This will ensure that the new and advanced technologies of phosphate chemicals production and phosphogypsum processing will reduce or optimize resource consumption, so that the project's water consumption does not have significant adverse impacts on communities, other users or the environment and ecosystems. The potential for cumulative impacts relating to water and energy consumption will also be assessed in the project's SESA.
6. The PhosChemEE Project result in significant greenhouse gas emissions or may exacerbate climate change due to massive energy consumption in phosrock mining and refining, and in phosphate chemicals production. *Standard 2: 2.3, 2.4* *Standard 8: 8.6*	I = 4 L = 3	Substantial	Improving the efficiency of energy usage in phosrock mining and refining, and in phosphate chemicals production will reduce energy consumption and reduce GHG emissions. Increased processing of phosphogypsum can be made energy efficient but will make use of extra energy that will bring about extra GHG emissions (unless the energy source is renewable).	As per the project's ESMF, this risk area will be included within the scope of the ESIAs/ESMPs (for demo site related impacts) and the project's SESA for upstream and cumulative STnadard2 related impacts. LCA method will be conducted before project construction during the design of the demonstrations to identify potential project-related increases in GHG emissions that may exacerbate climate change. And then the potential sources of emissions related to project activities will be estimated to form a baseline for developing measures to reduce such emissions. The PhosChemEE Project aims to enable the extensive application of low carbon and energy-efficient technologies in the phosphate chemicals industry in China. To mitigate the potential increment of GHG emissions, the demos will apply these three solutions. First, the demos will promote the new, advanced and energy-efficient technologies, which will directly reduce emissions in phosphate ore processing and phosphorus chemical production. Second, Yunnan, Guizhou, Sichuan and Hubei, where the demos sites are located are rich in renewable energy resources. This will offset the potential extra GHG emissions from the overconsumption of energy induced by the new technologies. Lastly, most of the demos will follow the principle of circular economy, including avoiding waste discharge, promoting the recycling and reusing of waste. Such measures will reduce the indirect emissions from intermediate material production and waste management. Additionally, GHG emissions of demos will be tracked and reported in

				accordance with provisions of the UNFCCC and GHG minimization measures are implemented.
7. Upstream project activities (primarily Outputs 1.1.1, 1.1.2, 2.1.1, 2.1.2, , 2.1.6, 2.2.4, 2.2.5, 3.1.1, and 3.1.2,) could result in adverse impacts to the local communities and the receptor environment if project-supported policy initiatives are not adequately screened and assessed. China accounts for more than 80% of the world's total production capacity of Phosphate. As such, the project could inadvertently lead to issues of global and/or cumulative environmental concern if not managed adequately *All SES Principles and Standards*	I-5 L-3	Substantial	Given the scope of the project, upstream activities are likely to have SES-related impacts associated with them. In addition, China's position as the main source of Phosphate production, potential SES-impacts could result in issues of global/cumulative concern.	The project will undertake the conduct of a SESA that shall cover all upstream activities, and those with potential cumulative impact related implications. Activities that will fall under the scope of the SESA include (but are not limited to): Outputs 1.1.1, 1.1.2, 2.1.1, 2.1.2, , 2.1.6, 2.2.4, 2.2.5, 3.1.1, and 3.1.2,)
8. The phosphogypsum processing and reuse demos have been planned to be implemented in the existing facility. Legacy pollution issues may exists that have on-going environmental impacts which will have to be managed in line with UNDP SES requirements. *Standard 8: 8.1, 8.2, 8.3, 8.6*	I-4 L-2	Moderate	The phosphogypsum processing and reuse demo is mainly implemented in the existing facility. However	The site-specific ESIAs will include an assessment of historical pollution/legacy issues at demo sites which are being planned for existing facilities. Any historical/legacy pollution issues that are deemed to have on-going environmental concerns will be remediated and managed within the framework of the site-specific ESMPs that are to be developed as well.
9. There is the possibility, that through some possible changes to the use of lands and resources in the phosphogypsum processing and reuse demo outputs of the project, access restrictions/economic displacement may occur as a result of indirect impacts emanating from the project's interventions. *Standard 5: 5.4*	I-4 L-2	Moderate	The project may in directly lead to changes in land use/tenure status for affected populations.	Any access restrictions/economic displacement related issues are likely to occur only as an indirect result of project supported activities. However, SES5 related impacts cannot be ruled out at this stage of the project preparation. SES5 related risks will be assessed in detail at both a demonstration level (via the conduct of site specific ESIAs) and at a policy/upstream level (through the conduct of the project's SESA). Any impacts identified shall be managed through the relevant avenues. IF adverse SES5 related impacts are identified, then necessary SES tools may be required to be undertaken (this could include a Livelihood Action Plan, Land

					Acquisition Action Plan etc.).	
	QUESTION 4: What is the overall project risk categorization?					
	Low Risk	☐				
	Moderate Risk	☐				
	Substantial Risk	☐				
	High Risk	√			The overall risk is High. An ESMF has been prepared during the PPG stage to ensure that the necessary steps are taken to assess and manage both (i) the site-specific risks during implementation (during which site-specific assessments and management plans will be required), and (ii) upstream/policy-level interventions that may result in adverse impacts (including cumulative impacts) through the conduct of a SESA.	
-	**QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are triggered? (check all that apply)**					
	Question only required for Moderate, Substantial and High Risk projects					
	Is assessment required? (check if "yes")	√				***Status? (completed, planned)***
	if yes, indicate overall type and status		√		Targeted assessment(s)	Completed: stakeholder analysis, gender analysis Planned: LCA
			√		ESIA (Environmental and Social Impact Assessment)	Site specific ESIAs (Planned)
			√		SESA (Strategic Environmental and Social Assessment)	Planned

	Are management plans required? (check if "yes)	√			
	If yes, indicate overall type		√	Targeted management plans (e.g. Gender Action Plan, Emergency Response Plan, Waste Management Plan, others)	Completed: Gender Action Plan, Stakeholder Engagement Plan has been completed Planned: Labor Management Plans
			√	ESMP (Environmental and Social Management Plan which may include range of targeted plans)	Site specific ESM (Planned)
			√	ESMF (Environmental and Social Management Framework)	Completed
	Based on identified risks, which Principles/Project-level Standards triggered?		**Comments (not required)**		
	Overarching Principle: Leave No One Behind				
	Human Rights	√	Risk 1		
	Gender Equality and Women's Empowerment	√	Risk 1		
	Accountability	√	Risk 1		
	1. Biodiversity Conservation and Sustainable Natural Resource Management	√	Risk 2 Risk 3 Risk 5 Risk 7		
	2. Climate Change and Disaster Risks	√	Risk 4 Risk 6 Risk 7		
	3. Community Health, Safety and Security	√	Risk 4 Risk 4 Risk 7		
	4. Cultural Heritage	√	Risk 1		

			Risk 7
	5. Displacement and Resettlement	√	Risk 7 Risk 9
	6. Indigenous Peoples	√	Risk 1 Risk 3 Risk 7
	7. Labour and Working Conditions	√	Risk 3 Risk 7
	8. Pollution Prevention and Resource Efficiency	√	Risk 2 Risk 3 Risk 5 Risk 6 Risk 7 Risk 8

QUESTION 2: What are the Potential Social and Environmental Risks? *Note: Complete SESP Attachment 1 before responding to Question 2.*	**QUESTION 3: What is the level of significance of the potential social and environmental risks?** *Note: Respond to Questions 4 and 5below before proceeding to Question 5*			**QUESTION 6: Describe the assessment and management measures for each risk rated Moderate, Substantial or High**
Risk Description (broken down by event, cause, impact)	***Impact and Likelihood (1-5)***	***Significance (Low, Moderate Substantial , High)***	***Comments (optional)***	***Description of assessment and management measures for risks rated as Moderate, Substantial or High***
[1.] The PhosChemEE Project could lead to adverse impacts on enjoyment of the human rights (civil, political, economic, social, or cultural) of the affected population, potentially including ethnic minorities and/or vulnerable/disadvantaged groups, in the rural areas where the on-the-ground project activities (e.g., demos and demo replications, which includes detailed site inspections, engineering, planning, installation and operation) will be carried out. These can potentially happen in the demonstration	I = 5 L = 4	High	This project will be based in the underdeveloped western regions of Yunnan, Guizhou, Sichuan, and Hubei. The development of detailed engineering designs of green, energy efficient low carbon technology applications in the phosphate mining and refining in these region is planned in these regions. This major activity is included in the 3 main components of the project (Outputs 1.2.1, 2.2.1 and 3.2.1). In general, the socio-economic condition of the project's demo sites is underdeveloped, and the	In the design stage, the following safeguards documents have been prepared, covering this risk and all other risks, to meet the SES requirements of a **High risk** project: - a stakeholder analysis and comprehensive Stakeholder Engagement Plan, - a gender analysis and Gender Action Plan, - an Environmental and Social Management Framework (ESMF) with initial FPIC procedures (to be applied if confirmed as required), - a Grievance Redress Mechanism (draft/outline). The ESMF outlines the procedures for the comprehensive and site-specific Environmental and Social Impact Assessments (ESIAs) and Environmental and Social Management Plans (ESMPs)

activities of the project in all 3 main components of the project, particularly those that will deliver Outputs 1.2.2, 2.2.2 and 3.2.2. Principle 1, Principle 2, Standard 4, Standard 6			ecological environment is very fragile. See ProDoc Sub-section: Gender Equality and Women's Empowerment Yunnan, Guizhou, Sichuan and Hubei have widespread minority cultures (that might trigger SES Standard 6), but factories are mainly located in industrialized regions where cultural heritage do not concentrate. PCI industries do not geographically overlap with indigenous settlements, though impacts (positive or negative) on such communities cannot be ruled out at this early stage.	that will be prepared during project implementation. The design and implementation of the project are consistent with the strategic requirements of the country's Law on ethnic unity and progress, which will ensure the enjoyment of the human rights of the potentially affected population in the demo places, especially ethnic minorities and/or vulnerable/disadvantaged groups.
[2.] Potential adverse impacts to habitats and/or ecosystems and ecosystem services (e.g. non-hazardous waste from phosphogypsum reuse), and some possible changes to the use of lands and resources in the phosphogypsum processing and reuse demo. Standard 1, Standard 8	I = 4 L = 3	Substantial	The phosphogypsum processing and reuse demo is mainly implemented in the existing facility. However, the demo may generate construction waste during the construction of supporting facilities or new facilities and generate non-hazardous waste from phosphogypsum reuse. These may have an impact on the surrounding environment that includes ecologically fragile areas. The proposed interventions may not be effective at the start and will continue or may even further negatively impact the surrounding ambient, bodies of what or land areas in the project areas.	The project has been designed such that demos will involve the conduct of a detailed feasibility study and ESIA (Activities 1.2.1.2, 2.2.1.2 and 3.2.1.4) on the local area before the demo site selection, and a site-specific ESMP will be put in place before implementation of each demo. The demos will be designed and constructed in accordance with national and international standards to ensure compliance with the SES and other relevant environmental ordinances/regulations, as outlined in the ESMF (and to be in the forthcoming ESIA/ESMPs). Additionally, the Project will observe the established regulatory framework for monitoring and assessing such risks that are aligned with the SES. China has regulations on the engineer project site for the restoration and protection of biodiversity, and manufactories are not allowed to be established in or around ecosystem function zone. In this case, the PhosChemEE Project will ensure that the demo sites will be located far away from ecological reserves and ecologically vulnerable areas.
[3.] The associated construction, operation, or decommissioning of the demo installations may have	I = 4 L = 3	Substantial	For example, inadequate disposal of waste during the operation of the	The ESMF and subsequent ESIA and ESMP will address all health and safety risks. The required management measures will be included in site-specific

potential health and safety risks to local communities and workers due to the transport, storage, and use and/or disposal of any hazardous or dangerous materials (e.g. explosives, fuel and other chemicals) that may be used during construction and operation. Standard 1, Standard 3, Standard 6, Standard 7, Standard 8			phosphogypsum processing reuse demo poses a threat to the environment. Specifically, if fluorine and arsenic contained in phosphogypsum are released into the atmosphere and soil environment, they will have adverse effects on surface/underground water plants and animals and produce health risks to local communities.	ESMPs and be conducted before project construction. Per those forthcoming plans, appropriate training will be provided to the demonstration enterprises to ensure that risks are properly addressed and managed. Training will also be provided to local governments in order to enhance their regulation capacity to control potential risks in the demonstration. A Grievance Mechanism will be designed (see ESMF) and will be implemented by the project, allowing communities and workers to request for interventions when facing issues with health and safety risks. The associated construction, operation, or decommissioning of the demo installations will strictly comply with the national standards[17] and international general standards on construction and waste management, as noted in the ESMF. Site-specific environmental and social assessments will be conducted for each demo to identify the critical health and safety risks. Where public access is available, appropriate engineering and administrative controls (e.g., detours, traffic calming, signs) will be considered and implemented in advance during the construction and operation of demos. Per those forthcoming plans, appropriate training will be provided to workers to ensure that they install and operate the installed system correctly and safely, and properly control and manage the release or disposal of waste. Training will also be provided to local governments in order to enhance their regulation capacity to control potential risks in the demonstration. This will minimize or avoid any community health risks and safety issues for the communities regarding construction work involved in the installation of the demos, and the minimization and management of waste generated from these demos (e.g., explosives, fuel and other chemicals).
[4.] The potential outcomes of the PhosChemEE Project will be sensitive or vulnerable to potential impacts of climate change, particularly in the identified project	I = 3 L = 3	Moderate	The demonstration activities of the project will be carried out in regions where there are risks of extreme weather that are exacerbated by climate change.	The ESIAs will assess activities for impact and sustainability within China national context, including its climate, and relevant ESMP has been outlined in the ESMF.

[17] Including Standard for pollution control on hazardous waste storage, Standard for pollution control on hazardous waste storage, etc.

sites in the partner provinces (Yunnan, Sichuan, Guizhou and Hubei) of the project where there are risks of extreme weather that can be exacerbated by climate change. Standard 2, Standard 3			The demos (either located outdoors or indoors) can be impacted by climate change-aggravated weather events that can cause extreme heat during summers or maybe flooding during rainy seasons.	The design of the demo will take climate-related events and risks into consideration by the preliminary environment risk assessment. Appropriate measures will be taken based on the specific climate-related events. For instance, the adverse impact of flood on demos can be eliminated by constructing diversion ditches, while sunshade roofs and cooling spray can reduce the impact of extreme high temperatures. The demos will also cover climate risk adaptation capacity building through training workshops. This training will be provided to the demo companies and local governments to enhance their capacities to deal with extreme climate events in general. Depending on the extent of the impacts of the adverse climate-related events, appropriate modifications in the installations (and budget) will be done. Potential reduction in the number of installations, or replacement with alternative demos will be done while considering the need to ensure the resulting interventions are still contributing to the realization of the project outcomes.
[5.] The new installed energy efficient systems in the phosrock mining and refining; and phosphate chemical manufacturers where the demonstrations will be implemented may not be operated and maintained properly and safely by the existing and newly hired operators. . Standard 3	I = 5 L = 2	Substantial	New energy efficient systems for improving the operations and processes of selected phosrock mining and refining; and phosphate chemical companies will be installed in these demo host companies.	Aside from conducting the required site-specific ESIAs/ESMPs, the installed systems will be designed, constructed, operated and decommissioned following national legal requirements, good international practice, and any relevant international obligations and standards. The operational and technical personnel of the companies hosting the demos will be adequately trained and certified in the proper and safe operation, maintenance of these systems. The demo installations will be designed by the qualified engineers and professionals in the PhosChemEE Project, and then be certified and approved by independent professionals not involved in the design process. During the construction, the demos will cover appropriate plans for construction supervision and quality assurance. Construction personnel will also have appropriate qualifications and training. Periodic safety inspections, monitoring and emergency preparedness will be included in demos in order to ensure the appropriate and safe operation and maintenance.

				The ESMF will outline detailed steps required to assess and manage this Substantial risk.
[6.] The Project includes activities, like phosphogypsum processing, that require significant consumption of raw materials, energy, and/or water; and involves significant extraction, diversion or containment of surface and/or ground water. That resource use, if not designed or implemented well, could lead to adverse impacts on ambient conditions in the project area and area of influence. Standard 1, Standard 8	I = 4 L = 3	Substantial	Phosphate chemicals production, as well as phosphogypsum processing consume significant amounts of water. Existing processes are energy inefficient, have considerable material losses, and wastewater generation. While the project will promote cleaner and more energy efficient phosphate chemicals production and processing of phosphogypsum that will either reduce or optimize the use of water, more processing of phosphogypsum may increase the water demand, and depending on where water is sourced, can result in significant extraction, diversion or containment of surface and/or groundwater.	Per the ESMF, the ESIA and ESMP will define the management measures for this risk. In the design of demos, in addition to the economic feasibility analysis of each demo, life-cycle analysis (LCA) will also be conducted to assess the raw materials, energy, and water used in new technologies and evaluate the potential of resource conservation. Based on LCA assessments, the project will be designed to seek cleaner and more energy-efficient phosphate chemicals production and processing of phosphogypsum. International practice for water and energy conservation will be applied in demos, especially the wastewater management and reuse technologies. Emergency accident pools will be designed in each demo in accordance with the requirement of contamination avoidance. Additionally, the demos will establish the resource efficiency monitoring mechanism, including using benchmarking data to the relative level of efficiency and reporting water and energy consumption periodically. This will ensure that the new and advanced technologies of phosphate chemicals production and phosphogypsum processing will reduce or optimize resource consumption, so that the project's water consumption does not have significant adverse impacts on communities, other users or the environment and ecosystems. Moreover, the Standards 1 & 8 resource efficiency requirements will be adequately adopted to avoid or reduce water usage so that the project's water consumption does not have significant adverse impacts on communities, other users or on the environment and ecosystems. Cumulative impacts of water use will be assessed, and appropriate mitigation measures implemented will be applied, such as water demand management, efficiency measures, benchmarking usage, alternative supplies, resource contamination avoidance, mitigation of impacts on downstream users, and water use offsets. Good international practice for water conservation and efficiency will be applied, including wastewater usage.

[7.] The PhosChemEE Project result in significant greenhouse gas emissions or may exacerbate climate change due to massive energy consumption in phosrock mining and refining, and in phosphate chemicals production. Standard 2	I = 4 L = 3	Substantial	Improving the efficiency of energy usage in phosrock mining and refining, and in phosphate chemicals production will reduce energy consumption and reduce GHG emissions. Increased processing of phosphogypsum can be made energy efficient but will make use of extra energy that will bring about extra GHG emissions (unless the energy source is renewable).	This risk will be included in the scope of the ESIAs/ESMPs, per the ESMF. LCA method will be conducted before project construction during the design of the demonstrations to identify potential project-related increases in GHG emissions that may exacerbate climate change. And then the potential sources of emissions related to project activities will be estimated to form a baseline for developing measures to reduce such emissions. The PhosChemEE Project aims to enable the extensive application of low carbon and energy-efficient technologies in the phosphate chemicals industry in China. To mitigate the potential increment of GHG emissions, the demos will apply these three solutions. The demos will promote the new, advanced and energy-efficient technologies, which will directly reduce emissions in phosphate ore processing and phosphorus chemical production. The demos will follow the principle of circular economy, including avoiding waste discharge, promoting the recycling and reusing of waste. Such measures will reduce the indirect emissions from intermediate material production and waste management. Additionally, GHG emissions of demos will be tracked and reported in accordance with provisions of the UNFCCC and GHG minimization measures are implemented.

	QUESTION 4: What is the overall project risk categorization?		
	Low Risk	☐	
	Moderate Risk	☐	
	Substantial Risk	☐	
	High Risk	√	The overall risk is High. An ESMF has been prepared during the PPG stage to ensure that the necessary steps are taken to manage the site-specific risks during implementation (during which site-specific assessments and management plans will be required).

-	**QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are triggered? (check all that apply)**				
	Question only required for Moderate, Substantial and High Risk projects				
	Is assessment required? (check if "yes")	√			***Status? (completed, planned)***

	if yes, indicate overall type and status		√	Targeted assessment(s)	Completed: stakeholder analysis, gender analysis Planned: TBC per the forthcoming ESMF
			√	ESIA (Environmental and Social Impact Assessment)	Planned
			√	SESA (Strategic Environmental and Social Assessment)	Planned
	Are management plans required? (check if "yes)	√			
	If yes, indicate overall type		√	Targeted management plans (e.g. Gender Action Plan, Emergency Response Plan, Waste Management Plan, others)	Completed: Gender Action Plan, Stakeholder Engagement Plan has been completed Planned: TBC per the forthcoming ESMF
			√	ESMP (Environmental and Social Management Plan which may include range of targeted plans)	Planned
			√	ESMF (Environmental and Social Management Framework)	Completed
	Based on identified risks, which Principles/Project-level Standards triggered?		**Comments (not required)**		
	Overarching Principle: Leave No One Behind				
	Human Rights	√			
	Gender Equality and Women's Empowerment	√	The site-specific environmental and social assessments of each demo site shall include the evaluation of impacts on gender equality and women's empowerment. The assessments shall identify mitigation measures to be carried out during demo implementation. A gender action plan is also prepared.		
	Accountability	☐			
	1. Biodiversity Conservation and Sustainable Natural Resource Management	√	All demos are subject to site-specific environmental and social assessments during their engineering design. The assessments shall include the assessment of biodiversity conservation and natural resource		

			~~management issues. The assessments will identify the required mitigation measures which will be carried out during demo implementation.~~
	~~***2. Climate Change and Disaster Risks***~~	~~√~~	
	~~***3. Community Health, Safety and Security***~~	~~√~~	~~The site-specific environmental and social assessments that will be conducted for each demo site shall include impact assessments on community health, safety, and working condition issues. The assessments will identify mitigation measures, which will be carried out during demo implementation.~~
	~~***4. Cultural Heritage***~~	~~√~~	
	~~***5. Displacement and Resettlement***~~	~~⊟~~	
	~~***6. Indigenous Peoples***~~	~~√~~	
	~~***7. Labour and Working Conditions***~~	~~√~~	
	~~***8. Pollution Prevention and Resource Efficiency***~~	~~√~~	~~The site-specific environmental and social assessments of each demo site shall include the evaluation of the pollution prevention aspects. The assessments shall identify mitigation measures to be carried out during demo implementation.~~

Final Sign Off

Signature	***Date***	***Description***
QA Assessor		UNDP staff member responsible for the project, typically a UNDP Programme Officer. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver		UNDP senior manager, typically the UNDP Deputy Country Director (DCD), Country Director (CD), Deputy Resident Representative (DRR), or Resident Representative (RR). The QA Approver cannot also be the QA Assessor. Final signature confirms they have "cleared" the SESP prior to submittal to the PAC.
PAC Chair		UNDP chair of the PAC. In some cases PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the project appraisal and considered in recommendations of the PAC.

SESP Attachment 1. Social and Environmental Risk Screening Checklist

Checklist Potential Social and Environmental Risks	
INSTRUCTIONS: The risk screening checklist will assist in answering Questions 2-6 of the Screening Template. Answers to the checklist questions help to (1) identify potential risks, (2) determine the overall risk categorization of the project, and (3) determine required level of assessment and management measures. Refer to the SES toolkit for further guidance on addressing screening questions.	
Overarching Principle: Leave No One Behind **Human Rights**	**Answer (Yes/No)**
P.1 Have local communities or individuals raised human rights concerns regarding the project (e.g. during the stakeholder engagement process, grievance processes, public statements)?	No
P.2 Is there a risk that duty-bearers (e.g. government agencies) do not have the capacity to meet their obligations in the project?	No
P.3 Is there a risk that rights-holders (e.g. project-affected persons) do not have the capacity to claim their rights?	Yes
Would the project potentially involve or lead to:	
P.4 adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	Yes
P.5 inequitable or discriminatory impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups, including persons with disabilities?[18]	Yes
P.6 restrictions in availability, quality of and/or access to resources or basic services, in particular to marginalized individuals or groups, including persons with disabilities?	No
P.7 exacerbation of conflicts among and/or the risk of violence to project-affected communities and individuals?	No
Gender Equality and Women's Empowerment	
P.8 Have women's groups/leaders raised gender equality concerns regarding the project, (e.g. during the stakeholder engagement process, grievance processes, public statements)?	No
Would the project potentially involve or lead to:	
P.9 adverse impacts on gender equality and/or the situation of women and girls?	No
P.10 reproducing discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	Yes
P.11 limitations on women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well being	No
P.12 exacerbation of risks of gender-based violence? For example, through the influx of workers to a community, changes in community and household power dynamics, increased exposure to unsafe public places and/or transport, etc.	No
Sustainability and Resilience: Screening questions regarding risks associated with sustainability and	

[18] Prohibited grounds of discrimination include race, ethnicity, sex, age, language, disability, sexual orientation, gender identity, religion, political or other opinion, national or social or geographical origin, property, birth or other status including as an indigenous person or as a member of a minority. References to "women and men" or similar is understood to include women and men, boys and girls, and other groups discriminated against based on their gender identities, such as transgender and transsexual people.

resilience are encompassed by the Standard-specific questions below	
Accountability	
Would the project potentially involve or lead to:	
P.13 exclusion of any potentially affected stakeholders, in particular marginalized groups and excluded individuals (including persons with disabilities), from fully participating in decisions that may affect them?	Yes
P.14 grievances or objections from potentially affected stakeholders?	Yes
P.15 risks of retaliation or reprisals against stakeholders who express concerns or grievances, or who seek to participate in or to obtain information on the project?	No
Project-Level Standards	
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management	
Would the project potentially involve or lead to:	
1.1 adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes	Yes
1.2 activities within or adjacent to critical habitats and/or environmentally sensitive areas, including (but not limited to) legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	Yes
1.3 changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	Yes
1.4 risks to endangered species (e.g. reduction, encroachment on habitat)?	No
1.5 exacerbation of illegal wildlife trade?	No
1.6 introduction of invasive alien species?	No
1.7 adverse impacts on soils?	Yes
1.8 harvesting of natural forests, plantation development, or reforestation?	No
1.9 significant agricultural production?	No
1.10 animal husbandry or harvesting of fish populations or other aquatic species?	No
1.11 significant extraction, diversion or containment of surface or ground water? For example, construction of dams, reservoirs, river basin developments, groundwater extraction	Yes
1.12 handling or utilization of genetically modified organisms/living modified organisms?[19]	No
1.13 utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)[20]	No
1.14 adverse transboundary or global environmental concerns?	Yes
Standard 2: Climate Change and Disaster Risks	

[19] See the Convention on Biological Diversity and its Cartagena Protocol on Biosafety.

[20] See the Convention on Biological Diversity and its Nagoya Protocol on access and benefit sharing from use of genetic resources.

Would the project potentially involve or lead to:	
2.1 areas subject to hazards such as earthquakes, floods, landslides, severe winds, storm surges, tsunami or volcanic eruptions?	No
2.2 outputs and outcomes sensitive or vulnerable to potential impacts of climate change or disasters? For example, through increased precipitation, drought, temperature, salinity, extreme events, earthquakes	Yes
2.3 increases in vulnerability to climate change impacts or disaster risks now or in the future (also known as maladaptive or negative coping practices)? For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding	Yes
2.4 increases of greenhouse gas emissions, black carbon emissions or other drivers of climate change?	Yes
Standard 3: Community Health, Safety and Security	
Would the project potentially involve or lead to:	
3.1 construction and/or infrastructure development (e.g. roads, buildings, dams)? (Note: the GEF does not finance projects that would involve the construction or rehabilitation of large or complex dams)	Yes
3.2 air pollution, noise, vibration, traffic, injuries, physical hazards, poor surface water quality due to runoff, erosion, sanitation?	Yes
3.3 harm or losses due to failure of structural elements of the project (e.g. collapse of buildings or infrastructure)?	No
3.4 risks of water-borne or other vector-borne diseases (e.g. temporary breeding habitats), communicable and noncommunicable diseases, nutritional disorders, mental health?	No
3.5 transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)?	Yes
3.6 adverse impacts on ecosystems and ecosystem services relevant to communities' health (e.g. food, surface water purification, natural buffers from flooding)?	Yes
3.7 influx of project workers to project areas?	Yes
3.8 engagement of security personnel to protect facilities and property or to support project activities?	No
Standard 4: Cultural Heritage	
Would the project potentially involve or lead to:	
4.1 activities adjacent to or within a Cultural Heritage site?	No
4.2 significant excavations, demolitions, movement of earth, flooding or other environmental changes?	Yes
4.3 adverse impacts to sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: projects intended to protect, and conserve Cultural Heritage may also have inadvertent adverse impacts)	Yes
4.4 alterations to landscapes and natural features with cultural significance?	No
4.5 utilization of tangible and/or intangible forms (e.g. practices, traditional knowledge) of Cultural Heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement	
Would the project potentially involve or lead to:	

5.1 temporary or permanent and full or partial physical displacement (including people without legally recognizable claims to land)?	No
5.2 economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	No
5.3 risk of forced evictions?[21]	
5.4 impacts on or changes to land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	Yes
Standard 6: Indigenous Peoples	
Would the project potentially involve or lead to:	
6.1 areas where indigenous peoples are present (including project area of influence)?	Yes
6.2 activities located on lands and territories claimed by indigenous peoples?	No
6.3 impacts (positive or negative) to the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in question)? If the answer to screening question 6.3 is "yes", then the potential risk impacts are considered significant and the project would be categorized as either Substantial Risk or High Risk	Yes
6.4 the absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	Yes
6.5 the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	No
6.6 forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources? Consider, and where appropriate ensure, consistency with the answers under Standard 5 above	No
6.7 adverse impacts on the development priorities of indigenous peoples as defined by them?	No
6.8 risks to the physical and cultural survival of indigenous peoples?	No
6.9 impacts on the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices? Consider, and where appropriate ensure, consistency with the answers under Standard 4 above.	No
Standard 7: Labour and Working Conditions	
Would the project potentially involve or lead to: (note: applies to project and contractor workers)	
7.1 working conditions that do not meet national labour laws and international commitments?	Yes
7.2 working conditions that may deny freedom of association and collective bargaining?	No
7.3 use of child labour?	No
7.4 use of forced labour?	No
7.5 discriminatory working conditions and/or lack of equal opportunity?	Yes

[21] Forced eviction is defined here as the permanent or temporary removal against their will of individuals, families or communities from the homes and/or land which they occupy, without the provision of, and access to, appropriate forms of legal or other protection. Forced evictions constitute gross violations of a range of internationally recognized human rights.

7.6 occupational health and safety risks due to physical, chemical, biological and psychosocial hazards (including violence and harassment) throughout the project life-cycle?	Yes
Standard 8: Pollution Prevention and Resource Efficiency	
Would the project potentially involve or lead to:	
8.1 the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or transboundary impacts?	Yes
8.2 the generation of waste (both hazardous and non-hazardous)?	Yes
8.3 the manufacture, trade, release, and/or use of hazardous materials and/or chemicals?	Yes
8.4 the use of chemicals or materials subject to international bans or phase-outs? For example, DDT, PCBs and other chemicals listed in international conventions such as the Montreal Protocol, Minamata Convention, Basel Convention, Rotterdam Convention, Stockholm Convention	No
8.5 the application of pesticides that may have a negative effect on the environment or human health?	No
8.6 significant consumption of raw materials, energy, and/or water?	Yes

~~**Checklist Potential Social and Environmental Risks**~~	
~~INSTRUCTIONS: The risk screening checklist will assist in answering Questions 2-6 of the Screening Template. Answers to the checklist questions help to (1) identify potential risks, (2) determine the overall risk categorization of the project, and (3) determine required level of assessment and management measures. Refer to the SES toolkit for further guidance on addressing screening questions.~~	
~~**Overarching Principle: Leave No One Behind**~~	~~**Answer**~~

Human Rights	(Yes/ No)
P.1 Have local communities or individuals raised human rights concerns regarding the project (e.g. during the stakeholder engagement process, grievance processes, public statements)?	No
P.2 Is there a risk that duty-bearers (e.g. government agencies) do not have the capacity to meet their obligations in the project?	No
P.3 Is there a risk that rights-holders (e.g. project-affected persons) do not have the capacity to claim their rights?	Yes
Would the project potentially involve or lead to:	
P.4 adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	Yes
P.5 inequitable or discriminatory impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups, including persons with disabilities?[22]	Yes
P.6 restrictions in availability, quality of and/or access to resources or basic services, in particular to marginalized individuals or groups, including persons with disabilities?	No
P.7 exacerbation of conflicts among and/or the risk of violence to project-affected communities and individuals?	No
Gender Equality and Women's Empowerment	
P.8 Have women's groups/leaders raised gender equality concerns regarding the project, (e.g. during the stakeholder engagement process, grievance processes, public statements)?	No
Would the project potentially involve or lead to:	
P.9 adverse impacts on gender equality and/or the situation of women and girls?	No
P.10 reproducing discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	Yes
P.11 limitations on women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well being	No
P.12 exacerbation of risks of gender-based violence? For example, through the influx of workers to a community, changes in community and household power dynamics, increased exposure to unsafe public places and/or transport, etc.	No
Sustainability and Resilience: Screening questions regarding risks associated with sustainability and resilience are encompassed by the Standard-specific questions below	
Accountability	
Would the project potentially involve or lead to:	
P.13 exclusion of any potentially affected stakeholders, in particular marginalized groups and excluded individuals (including persons with disabilities), from fully participating in decisions that may affect them?	No
P.14 grievances or objections from potentially affected stakeholders?	No
P.15 risks of retaliation or reprisals against stakeholders who express concerns or grievances, or who seek to participate in or to obtain information on the project?	No
Project-Level Standards	
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management	
Would the project potentially involve or lead to:	

[22] Prohibited grounds of discrimination include race, ethnicity, sex, age, language, disability, sexual orientation, gender identity, religion, political or other opinion, national or social or geographical origin, property, birth or other status including as an indigenous person or as a member of a minority. References to "women and men" or similar is understood to include women and men, boys and girls, and other groups discriminated against based on their gender identities, such as transgender and transsexual people.

1.1 adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes	Yes
1.2 activities within or adjacent to critical habitats and/or environmentally sensitive areas, including (but not limited to) legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	Yes
1.3 changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	Yes
1.4 risks to endangered species (e.g. reduction, encroachment on habitat)?	No
1.5 exacerbation of illegal wildlife trade?	No
1.6 introduction of invasive alien species?	No
1.7 adverse impacts on soils?	No
1.8 harvesting of natural forests, plantation development, or reforestation?	No
1.9 significant agricultural production?	No
1.10 animal husbandry or harvesting of fish populations or other aquatic species?	No
1.11 significant extraction, diversion or containment of surface or ground water? For example, construction of dams, reservoirs, river basin developments, groundwater extraction	Yes
1.12 handling or utilization of genetically modified organisms/living modified organisms?[23]	No
1.13 utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)[24]	No
1.14 adverse transboundary or global environmental concerns?	No
Standard 2: Climate Change and Disaster Risks	
Would the project potentially involve or lead to:	
2.1 areas subject to hazards such as earthquakes, floods, landslides, severe winds, storm surges, tsunami or volcanic eruptions?	No
2.2 outputs and outcomes sensitive or vulnerable to potential impacts of climate change or disasters? For example, through increased precipitation, drought, temperature, salinity, extreme events, earthquakes	Yes
2.3 increases in vulnerability to climate change impacts or disaster risks now or in the future (also known as maladaptive or negative coping practices)? For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding	No
2.4 increases of greenhouse gas emissions, black carbon emissions or other drivers of climate change?	Yes
Standard 3: Community Health, Safety and Security	
Would the project potentially involve or lead to:	
3.1 construction and/or infrastructure development (e.g. roads, buildings, dams)? (Note: the GEF does not finance projects that would involve the construction or rehabilitation of large or complex dams)	Yes

[23] See the Convention on Biological Diversity and its Cartagena Protocol on Biosafety.
[24] See the Convention on Biological Diversity and its Nagoya Protocol on access and benefit sharing from use of genetic resources.

3.2 air pollution, noise, vibration, traffic, injuries, physical hazards, poor surface water quality due to runoff, erosion, sanitation?	Yes
3.3 harm or losses due to failure of structural elements of the project (e.g. collapse of buildings or infrastructure)?	No
3.4 risks of water-borne or other vector-borne diseases (e.g. temporary breeding habitats), communicable and noncommunicable diseases, nutritional disorders, mental health?	No
3.5 transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)?	Yes
3.6 adverse impacts on ecosystems and ecosystem services relevant to communities' health (e.g. food, surface water purification, natural buffers from flooding)?	Yes
3.7 influx of project workers to project areas?	No
3.8 engagement of security personnel to protect facilities and property or to support project activities?	No
Standard 4: Cultural Heritage	
Would the project potentially involve or lead to:	
4.1 activities adjacent to or within a Cultural Heritage site?	No
4.2 significant excavations, demolitions, movement of earth, flooding or other environmental changes?	No
4.3 adverse impacts to sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: projects intended to protect, and conserve Cultural Heritage may also have inadvertent adverse impacts)	Yes
4.4 alterations to landscapes and natural features with cultural significance?	No
4.5 utilization of tangible and/or intangible forms (e.g. practices, traditional knowledge) of Cultural Heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement	
Would the project potentially involve or lead to:	
5.1 temporary or permanent and full or partial physical displacement (including people without legally recognizable claims to land)?	No
5.2 economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	No
5.3 risk of forced evictions?[25]	
5.4 impacts on or changes to land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	No
Standard 6: Indigenous Peoples	
Would the project potentially involve or lead to:	
6.1 areas where indigenous peoples are present (including project area of influence)?	Yes
6.2 activities located on lands and territories claimed by indigenous peoples?	No
6.3 impacts (positive or negative) to the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized	Yes

[25] Forced eviction is defined here as the permanent or temporary removal against their will of individuals, families or communities from the homes and/or land which they occupy, without the provision of, and access to, appropriate forms of legal or other protection. Forced evictions constitute gross violations of a range of internationally recognized human rights.

as indigenous peoples by the country in question)? If the answer to screening question 6.3 is "yes", then the potential risk impacts are considered significant and the project would be categorized as either Substantial Risk or High Risk	
6.4 the absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	Yes
6.5 the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	No
6.6 forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources? Consider, and where appropriate ensure, consistency with the answers under Standard 5 above	No
6.7 adverse impacts on the development priorities of indigenous peoples as defined by them?	No
6.8 risks to the physical and cultural survival of indigenous peoples?	No
6.9 impacts on the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices? Consider, and where appropriate ensure, consistency with the answers under Standard 4 above.	No
Standard 7: Labour and Working Conditions	
Would the project potentially involve or lead to: (note: applies to project and contractor workers)	
7.1 working conditions that do not meet national labour laws and international commitments?	No
7.2 working conditions that may deny freedom of association and collective bargaining?	No
7.3 use of child labour?	No
7.4 use of forced labour?	No
7.5 discriminatory working conditions and/or lack of equal opportunity?	Yes
7.6 occupational health and safety risks due to physical, chemical, biological and psychosocial hazards (including violence and harassment) throughout the project life-cycle?	Yes
Standard 8: Pollution Prevention and Resource Efficiency	
Would the project potentially involve or lead to:	
8.1 the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or transboundary impacts?	No
8.2 the generation of waste (both hazardous and non-hazardous)?	Yes
8.3 the manufacture, trade, release, and/or use of hazardous materials and/or chemicals?	Yes
8.4 the use of chemicals or materials subject to international bans or phase-outs? For example, DDT, PCBs and other chemicals listed in international conventions such as the Montreal Protocol, Minamata Convention, Basel Convention, Rotterdam Convention, Stockholm Convention	No
8.5 the application of pesticides that may have a negative effect on the environment or human health?	No
8.6 significant consumption of raw materials, energy, and/or water?	Yes

Annex 5: UNDP Risk Register

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
Implementation of the PhosChemEE Project (and most specifically Component 1	Social Political Financial	I = 5 L = 4 High	In the design stage, the following safeguards documents have been prepared, covering this risk and all other risks, to meet the SES requirements of a **High Risk** project: · a stakeholder analysis and comprehensive	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
which focuses on the upstream Phosphate mining and refining sub-sector) could lead to adverse impacts on enjoyment of the human rights (civil, political, economic, social, or cultural) of the affected population, potentially including ethnic minorities and/~~or~~ vulnerable/disadvantaged groups. This is most likely to occur under Outputs 1.2.2, 2.2.2 and 3.2.2, ~~in the rural areas where the on-the-ground project activities will be carried out.~~ Mining companies engaged in the project might not have the capacity and knowledge to uphold their duties as per UNDP SES requirements. This includes upholding duties and requirements on FPIC as per UNDP SES 6.			Stakeholder Engagement Plan, · a gender analysis and Gender Action Plan, · an Environmental and Social Management Framework (ESMF) with initial FPIC procedures (if confirmed as required), · a Grievance Redress Mechanism (draft/outline). The ESMF will outline the procedures for the (site-specific) Environmental and Social Impact Assessments (ESIAs) and Environmental and Social Management Plans (ESMPs) that will be prepared during project implementation. To adequately assess impact from upstream/policy and strategy-level interventions (including potential cumulative impacts) the project shall undertake a Strategic Environmental and Social Assessment (SESA). The design and implementation of the project are consistent with the strategic requirements of the country's Law on ethnic unity and progress, which will ensure the enjoyment of the human rights of the potentially affected population in the demo places, especially ethnic minorities and/or vulnerable/disadvantaged groups. During the implementation of the project, the MIIT and MNR, which both have adequate experiences in GEF project implementation and have established long-term cooperative relationships with various ministries, local departments, industry-related supporting units and relevant organizations, can establish a scientific and efficient collaborative working mechanism to ensure the policy implementation of the project. Extensive negotiation will also be conducted with local people with the cooperation of the MIIT, MNR and provincial government.		
During the construction of demo activities (Output 2.2.2).	Environmental	I = 4 L = 3 Substantial	The project has been designed such that demos involve the conduct of a detailed feasibility study and ESIA on the local area before the demo site selection, and a site-specific ESMP will be put in	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
and in support to the mining and refining sectors (under Component 1), Ppotential adverse impacts to habitats and/or ecosystems and ecosystem services (e.g. non-hazardous waste from phosphogypsum reuse) may occur, and some possible changes to the use of lands and resources in the phosphogypsum processing and reuse demo.			place before implementation of each demo. The detailed design of the demos and their engineering and construction shall be in accordance with national and international standards to ensure compliance with the SES and other relevant environmental ordinances and regulations, as to be outlined in the ESMF (and ESIA/ESMPs). Additionally, the Project will observe the established regulatory framework for monitoring and assessing such risks. China has regulations on the engineering of the project site for the restoration and protection of biodiversity, and manufactories are not allowed to be established in or around ecosystem function zone. In this case, the PhosChemEE Project will ensure that the demo sites will be located far away from ecological reserves and ecologically vulnerable areas. Second, demos will involve the conduct of a detailed feasibility study and ESIA on the local area before the demo site selection. For each demo, a site-specific ESMP will be put in place before implementation. Moreover, the demos will be constructed and implemented in accordance with national standards and international general standards to prevent pollution and ensure compliance with the SES and other relevant environmental ordinances/regulations.		
Inadequate disposal of waste during the operation of the phosphogypsum processing reuse demo (i.e. Outputs 3.2.1 and 3.2.2) poses a threat to the environment as well as the health and safety of workers and the community. The associated construction, operation, or decommissioning of the demo installations may	Operational Environmental	I = 4 L = 3 Substantial	The ESMF and subsequent ESIA and ESMP will address safety risks to local communities. The required management measures will be included in site-specific ESMPs and be conducted before project construction. Per those forthcoming plans, appropriate training will be provided to the demonstration enterprises to ensure that risks are properly addressed and managed. Training will also be provided to local governments in order to enhance their regulation capacity to control potential risks in the demonstration. A Grievance Mechanism will be designed (see ESMF) and will be implemented by the project, allowing communities and workers to request for interventions when facing issues with health and safety risks. The associated construction, operation, or decommissioning of the demo installations will strictly comply with the national standards[26] and international general standards on construction and waste management. Site-specific	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
have potential health and safety risks to local communities and workers due to the transport, storage, and use and/or disposal of any hazardous or dangerous materials (e.g. explosives, fuel and other chemicals) that may be used during construction and operation.			environmental and social assessments will be conducted for each demo to identify the critical health and safety risks. Where public access is available, appropriate engineering and administrative controls (e.g., detours, traffic calming, signs) will be considered and implemented in advance during the construction and operation of demos. Per those forthcoming plans, appropriate training will be provided to workers to ensure that they install and operate the installed system correctly and safely, and properly control and manage the release or disposal of waste. Training will also be provided to local governments in order to enhance their regulation capacity to control potential risks in the demonstration. This will minimize or avoid any community health risks and safety issues for the communities regarding construction work involved in the installation of the demos, and the minimization and management of waste generated from these demos (e.g., explosives, fuel and other chemicals).		
The potential outcomes of the PhosChemEE Project will be sensitive or vulnerable to potential impacts of climate change. Yunnan, Sichuan, Guizhou, and Hubei, where the demos are located, have risks of extreme weather that are exacerbated by climate change. Thus, some of the demonstration activities of the project (located outdoors) can be impacted by weather events.	Environmental	I = 3 L = 3 Moderate	The ESIA (as required and outlined in the ESMF) assesses activities for impact and sustainability within China national context, including its climatic vulnerability and associated impacts~~climate, and relevant ESMP has been outlined in the ESMF~~. The detailed design of the demos shall take climate-related events and risks into consideration by utilizing the findings of the risk assessment that shall be conducted as part of the ESIAs. Appropriate measures will be taken based on the specific climate-related events. For instance, the adverse impact of flood on demos can be eliminated by constructing diversion ditches, while sunshade roofs and cooling spray can reduce the impact of extreme high temperatures. The demos will also cover climate risk adaptation capacity building through training workshops (Activities 1.2.2.3, 2.2.2.2, 3.2.2.3). This training will be provided to the demo companies and local governments to enhance their capacities to deal with extreme climate events in general. Depending on the extent of the impacts of the adverse climate-related events, appropriate modifications in the installations (and budget) will	PMU MIIT MNR	No Change

[26] Including Standard for pollution control on hazardous waste storage, Standard for pollution control on hazardous waste storage, etc.

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
			be done. Potential reduction in the number of installations, or replacement with alternative demos will be done while considering the need to ensure the resulting interventions are still contributing to the realization of the project outcomes.		
~~The PhosChemEE Project will involve large-scale infrastructure development that might not be designed, built, operated or maintained safely.~~	~~Operational~~	~~I = 5~~ ~~L = 2~~ ~~Substantial~~	~~Aside from conducting the required site-specific ESIAs/ESMPs, the infrastructures (e.g., process buildings and access roads) shall be designed, constructed, operated and decommissioned following national legal requirements, good international practice, and any relevant international obligations and standards.~~ ~~The demo infrastructures shall be designed by qualified engineers and professionals who will be engaged by the PhosChemEE Project, and then be certified and approved by independent professionals not involved in the design process.~~ ~~During the construction phase of each demo, appropriate plans for construction supervision and quality assurance will be prepared and implemented. Construction personnel will also have appropriate qualifications and training.~~ ~~Periodic safety inspections, monitoring and emergency preparedness will be included in demos in order to ensure the appropriate and safe operation and maintenance.~~ ~~The ESMF will outline detailed steps required to assess and manage this Substantial risk.~~	~~PMU~~ ~~MIIT~~ ~~MNR~~	~~No Change~~
~~The PhosChemEE Project might pose potential risks and vulnerabilities related to occupational health and safety due to physical, chemical, biological, and radiological hazards during new infrastructure or system construction, operation.~~	~~Operational~~ ~~Environmental~~	~~I = 4~~ ~~L = 2~~ ~~Moderate~~	~~This risk will be assessed and included in the scope of the ESMF and subsequent ESIAs/ESMPs.~~ ~~The occupational health impact assessment will be conducted prior to the construction phase of the demos, which will make detailed identification and assessment of potential hazards and risks, particularly those that could result in serious injury, ill health or death and those identified through worker health surveillance.~~ ~~The construction/implementation process of demos will adopt preventive and protective measures (e.g., dust and noise reduction measures) to eliminate or minimize occupational health hazards and risks. Safety reminder signs will be implemented at worksites where occupational health hazards may arise, and employees will be provided personal protective equipment at no cost.~~ ~~The demo will include the conduct of the safety and health training at no cost to workers, including on the appropriate operation of new and~~	~~PMU~~ ~~MIIT~~ ~~MNR~~	~~No Change~~

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
			~~advanced facilities and the proper use of personal protective equipment. The emergency plan will be prepared for each demo, including emergency prevention, response arrangements and employment injury benefits and/or remedies for adverse occupational impacts.~~		
The Project includes activities both downstream/physical activities, (like phosphogypsum processing), as well as upstream/policy-level interventions that require/could lead to significant consumption of raw materials, energy, and/or water; and involves significant extraction, diversion or containment of surface and/or ground water. That resource use, if not designed or implemented well, could lead to adverse impacts on ambient conditions in the project area and area of influence.	Environmental Operational	I = 4 L = 3 Substantial	Per the ESMF, the ESIA and ESMP will define the management measures for this risk. In the design of demos, life-cycle analysis (LCA) will be conducted before each industrial process to assess the raw materials, energy, and water used in new technologies and evaluate the potential of resource conservation. Based on LCA assessments, the project will be designed to seek cleaner and more energy-efficient phosphate chemicals production and processing of phosphogypsum. International practice for water and energy conservation will be applied in demos, especially the wastewater management and reuse technologies. Emergency accident pools will be designed in each demo in accordance with the requirement of contamination avoidance. Additionally, the demos will establish the resource efficiency monitoring mechanism, including using benchmarking data to the relative level of efficiency and reporting water and energy consumption periodically. This will ensure that the new and advanced technologies of phosphate chemicals production and phosphogypsum processing will reduce or optimize resource consumption, so that the project's water consumption does not have significant adverse impacts on communities, other users or the environment and ecosystems. The potential for cumulative impacts relating to water and energy consumption will also be assessed in the project's SESA.	PMU MIIT MNR	No Change
The PhosChemEE Project result in significant greenhouse gas emissions or may exacerbate climate change due to massive energy	Environmental Operational	I = 4 L = 3 Substantial	As per the project's ESMF, t~~T~~his risk area will be included within the scope of the ESMF and ESIAs/ESMPs (for demo site related impacts) and the project's SESA for upstream and cumulative stanadard2 related impacts. LCA method will be conducted before project construction during the design of the demonstrations to identify potential project-related increases in GHG emissions that may	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
consumption in phosrock mining and refining, and in phosphate chemicals production.			exacerbate climate change. And then the potential sources of emissions related to project activities will be estimated to form a baseline for developing measures to reduce such emissions. The PhosChemEE Project aims to enable the extensive application of low carbon and energy-efficient technologies in the phosphate chemicals industry in China. To mitigate the potential increment of GHG emissions, the demos will apply these three solutions. First, the demos will promote the new, advanced and energy-efficient technologies, which will directly reduce emissions in phosphate ore processing and phosphorus chemical production. Second, Yunnan, Guizhou, Sichuan and Hubei, where the demos sites are located are rich in renewable energy resources. This will offset the potential extra GHG emissions from the overconsumption of energy induced by the new technologies. Last, the demos will follow the principle of circular economy, including avoiding waste discharge, promoting the recycling and reusing of waste. Such measures will reduce the indirect emissions from intermediate material production and waste management. Additionally, GHG emissions of demos will be tracked and reported in accordance with provisions of the UNFCCC and GHG minimization measures are implemented.		
Upstream project activities (primarily Outputs 1.1.1, 1.1.2, 2.1.1, 2.1.2, 2.1.6, 2.2.4, 2.2.5, 3.1.1, and 3.1.2) could result in adverse impacts to the local communities and the receptor environment if project-supported policy initiatives are not adequately screened and assessed. China accounts for more than	Environmental Operational	I-5 L-3 Substantial	The project will undertake the conduct of a SESA that shall cover all upstream activities, and those with potential cumulative impact related implications. Activities that will fall under the scope of the SESA include (but are not limited to): Outputs 1.1.1, 1.1.2, 2.1.1, 2.1.2, 2.1.6, 2.2.4, 2.2.5, 3.1.1, and 3.1.2)	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
80% of the world's total production capacity of Phosphate. As such, the project could inadvertently lead to issues of global and/or cumulative environmental concern if not managed adequately					
The phosphogypsum processing and reuse demos have been planned to be implemented in the existing facility. Legacy pollution issues may exists that have on-going environmental impacts which will have to be managed in line with UNDP SES requirements.	Environmental	I-4 L-2 Moderate	The site-specific ESIAs will include an assessment of historical pollution/legacy issues at demo sites which are being planned for existing facilities. Any historical/legacy pollution issues that are deemed to have on-going environmental concerns will be remediated and managed within the framework of the site-specific ESMPs that are to be developed as well.	PMU MIIT MNR	No Change
There is the possibility, that through some possible changes to the use of lands and resources in the phosphogypsum processing and reuse demo outputs of the project, access restrictions/economic displacement may occur as a	Operational	I-4 L-2 Moderate	Any access restrictions/economic displacement related issues are likely to occur only as an indirect result of project supported activities. However, SES5 related impacts cannot be ruled out at this stage of the project preparation. SES5 related risks will be assessed in detail at both a demonstration level (via the conduct of site specific ESIAs) and at a policy/upstream level (through the conduct of the project's SESA). Any impacts identified shall be managed through the relevant avenues. IF adverse SES5 related impacts are identified, then necessary SES tools may be required to be undertaken (this could include a Livelihood Action Plan, Land Acquisition Action Plan etc.).	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
result of indirect impacts emanating from the project's interventions.					
The project management office (PMO) cannot effectively organize and coordinate the participation of stakeholders (including relevant departments of local government, specific project implementation units, experts in various fields, etc.).	Organizational	I = 4 L = 3 Substantial	The MIIT and MNR have adequate experiences in GEF project implementation, and have established long-term cooperative relationships with various ministries, local departments, industry related supporting units and relevant organizations, and can establish a scientific and efficient collaborative working mechanism to ensure the policy implementation of the project. If coordination problems arise, the project team will have direct contact with the MNR and MIIT to ensure smooth coordination. Besides, investment from MIIT and MNR could facilitate building new program management team and increase management costs to ensure effective problem solving.	PMU MIIT MNR	No Change
Coordination of the relationship between project objectives and the implementation of comprehensive utilization technology.	Organizational Operational	I = 4 L = 1 Low	The MNR, MIIT and implementation units maintain close contact with relevant departments and can apply necessary adjustment promptly according to the adjustment of national policies, so as to ensure the implementation of this project and achieve the optimal effect. Workshops shall be carried out on a regular basis to adjust project technology employment so that the overall goal will be achieved.	PMU MIIT MNR	No Change
Unemployment caused by the reduction of chemical phosphate fertilizer production (possibly in cement manufacturing too)	Social Financial	I = 5 L = 4 High	In the process of realizing green and low-carbon transformation of PCI, the United Nations '2030 agenda for sustainable development' was benchmarked to explore the formation of a multi-objective collaborative green development model to achieve ecological compensation, reduce poverty, improve ecological diversity, and ensure agricultural and food security. Focus on green employment action in the PCI, improve the employment quality of the PCI, create new employment growth points in the comprehensive utilization of phosphogypsum, and reduce the unemployment risk caused by industrial transformation. Training programs will be carried out for displaced workers to relocate them to emerging industrial	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
			enterprises that will be facilitated by the project, i.e. renewable energy, phosphogypsum processing, waste water treatment, natural phosphate fertilizer production, etc.		
Allocated funds cannot support project implementation in time and in full.	Operational Financial	I = 4 L = 3 Substantial	Energy saving, green and low-carbon development and green mine construction are the important contents of China's ecological civilization goals, which make an important basis for guaranteeing supporting funds. The MNR and MIIT attach great importance to the implementation of this project, and provide relevant technology, resources, funds, and system support. Relevant project provinces will also make commitment to supporting funds for the project in the early stage of implementation. In the process of project design and implementation, it will continue to promote and attract the establishment of diversified financing modes including private enterprises to ensure the supporting funds are fully in place. Constant follow-up with the pertinent co-financers will be conducted either to secure the committed co-financing or negotiate the amount of co-financing. Where required the reallocation of budget will be done to support the implementation of affected activities. This may entail the delivery of alternative outputs that are still contributing to the achievement of the relevant project outcome.	PMU MIIT MNR	No Change
Low level of commitment of stakeholders (including government agencies and PCI companies) in the implementation of project activities.	Political Strategic	I = 4 L = 2 Moderate	Policy regulations will be published to ensure stakeholders participate in the project. Market mechanism for phosphogypsum replacing cement will be established to encourage participation. Incremental costs will be applied to increase subsidy for cement substitution and low-carbon mining technologies.	PMU MIIT MNR	No Change
Difficulties in the replication of successful results of the project.	Organizational Operational Financial	I = 5 L = 2 Substantial	PMO will organize regional and national workshops to ensure timely spread of experience and knowledge. In these workshops other regions could share pilot projects results and experiences. Research and seminars will be carried out to identify the critical problems in replication and proper solutions. PMO, will carry out direct consultations with target private sector entities to determine their concerns and explore ways to	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
			fulfill their initial commitments and ensure end-users better understanding about the use and benefits of green, EE and LC technology applications		
Availability of co-financing is delayed and negatively affects the implementation of the project activities	Financial Political	I = 4 L = 1 Low	The MIIT and MNR will ensure that committed co-financing from the national and local governments is available and during the required time. Constant follow-up with the project partners in the private sector in the planning and implementation of the activities that they are co-financing will ensure that their financial support is available in a timely manner If financing problems arise, the project team will have direct contact with the MNR and MIIT to potentially secure additional funding from other sources of funds. If necessary, the project activities may have to be modified or replaced to adapt to the available funds.	PMU MIIT MNR	No Change
Varying vested interests and objectives of PCI companies as well as other stakeholders in the local and central governments may prevent the effective organization and coordination of their participation and support of the project	Operational Political	I = 3 L = 2 Moderate	Policy as a regulative measure will be carried out to ensure PCI companies stick to the project, and financial support will be provided to companies to ensure their participation. In regular workshops, companies could express their problems and together with the MNR and MIIT find proper solutions when faced with difficulties.	PMU MIIT MNR	No Change
The project would potentially result in the generation of waste (non-hazardous).	Environmental	I = 3 L = 5 Substantial	During the design and implementation of the project, strict environmental and social assessments will be conducted, and relevant recommendations will be made to reduce related environmental and safety risks. The project will promote the application of new and advanced technologies for cleaner production, which are expected to reduce emissions, process wastes, effluents compared to the current processes. The demonstration component of the proposed project will be designed taking into consideration the need to prevent a potential for the release, in the environment, of hazardous materials resulting	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
			from their handling, storage and use for the demonstrations and replications that will be carried out under the project. As the phosphate chemical industry is concentrated in Yunnan, Guizhou, Sichuan province, and other underdeveloped and ecologically fragile areas, the potential waste generation will be controlled and evaluated in the design and implementation stages. During operation, advanced waste disposal methods will be used to treat waste generated in the project to minimize the impact of waste on the environment. For example, solidification treatment of fluorine-containing waste and highly-charged heavy metal waste generated in the process of phosphogypsum reuse will minimize the impact of phosphate chemical waste on the environment.		
Low local and international prices of phosrock and phosphate chemicals will discourage PCI companies to implement green, EE, and LC technology application projects	Financial Strategic	I = 4 L = 3 Substantial	MIIT and MNR will provide subsidies for green technologies and will carry out policies to favor companies in PCI with green technologies. Subsidizing policies should be flexible in order to provide proper incentives when market prices fluctuate.	PMU MIIT MNR	No Change
COVID-19 will negatively impact the social-economic development of the project areas.	Social Financial	I = 5 L = 4 High	The project development team will make use of China's effective institutional system to address the pandemic situation in a timely manner taking into consideration potential socio-economic impacts. Care will be taken in the design and implementation of the project especially when the pandemic persists to ensure that hardware installation works will be done safely in the demo sites with the health, safety, and welfare of the people in the demo sites and local communities in mind. If vaccination is not be ready, public health measures will be taken as is currently being done in China to ensure safety of the project participants and beneficiaries. If the pandemic persists during the project implementation phase, the project activities that will not involve on-the-ground work will be prioritized and implemented with the help of virtual means. The scheduling of, and action	PMU MIIT MNR	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
			planning for, the hardware-related activities will be reviewed regularly by keeping abreast with the status of the pandemic and the public health situation in the country, in general, and particularly in the project demo sites.		
Extreme climate events in the project areas will delay and negatively affect the installation, operation and monitoring of the demonstration activities.	Environmental Operational	I = 3 L = 1 Low	Yunnan, Sichuan, Guizhou, and Hubei have higher risks of extreme weather compared to the other provinces. This make them through the years experienced in dealing with extreme climate risks. Local governments shall strengthen their capacities to deal with extreme climate events in general. The project will also cover climate risk adaptation capacity building sector in technology training workshops. Depending on the extent of the impacts of the adverse climate –related events, appropriate modifications in the installations (and budget) will be done. Potential reduction in the number of installations, or replacement with alternative demos will be done while considering the need to ensure the resulting interventions are still contributing to the realization of the project outcomes.	PMU MIIT MNR	No Change

Annex 6: Overview of Project Staff and Technical Consultancies

Consultant	Time Input	Tasks, Inputs and Outputs
For Project Management		
Local / National contracting		
Project Manager/Coordinator *Rate: $1,200/week*	*43 weeks / over 5 years*	*The Project Manager (PM), together with the Lead Technical Advisor will be responsible for the overall management of the project, including the mobilization of all project inputs, supervision over project staff, consultants and sub-contractors.* *<u>Duties and Responsibilities</u>* • *Manage the overall conduct of the project.* • *Plan the activities of the project and monitor progress against the approved workplan.* • *Execute activities by managing personnel, goods and services, training and low-value grants, including drafting terms of reference and work specifications, and overseeing all contractors' work.* • *Monitor events as determined in the project monitoring plan, and update the plan as required.* • *Provide support for completion of assessments required by UNDP, spot checks and audits.* • *Manage requests for the provision of UNDP financial resources through funding advances, direct payments or reimbursement using the FACE form.* • *Monitor financial resources and accounting to ensure the accuracy and reliability of financial reports.* • *Monitor progress, watch for plan deviations and make course corrections when needed within project board-agreed tolerances to achieve results.* • *Ensure that changes are controlled, and problems addressed.* • *Perform regular progress reporting to the project board as agreed with the board, including measures to address challenges and opportunities.* • *Prepare and submit financial reports to UNDP on a quarterly basis.* • *Manage and monitor the project risks – including social and environmental risks - initially identified and submit new risks to the Project Board for consideration and decision on possible actions if required; update the status of these risks by maintaining the project risks log;* • *Capture lessons learned during project implementation.* • *Prepare revisions to the multi-year workplan, as needed, as well as annual and quarterly plans if required.* • *Prepare the inception report no later than one month after the inception workshop.* • *Ensure that the indicators included in the project results framework are monitored annually in advance of the GEF PIR submission deadline so that progress can be reported in the GEF PIR.* • *Prepare the GEF PIR;* • *Assess major and minor amendments to the project within the parameters set by UNDP-GEF;* • *Monitor implementation plans including the gender action plan, stakeholder engagement plan, and any environmental and social management plans;* • *Monitor and track progress against the GEF Core indicators.* • *Support the Mid-term review and Terminal Evaluation process.*

Consultant	Time Input	Tasks, Inputs and Outputs
		• *Add technical tasks as necessary*
Project Assistant *Rate: $1,200/week*	*60 weeks / over 5 years*	*Duties and Responsibilities* *Under the guidance and supervision of the Project Manager, the Project Assistant will carry out the following tasks:* • *Assist the Project Manager in day-to-day management and oversight of project activities;* • *Assist the M&E officer in matters related to M&E and knowledge resources management;* • *Assist in the preparation of progress reports;* • *Ensure all project documentation (progress reports, consulting and other technical reports, minutes of meetings, etc.) are properly maintained in hard and electronic copies in an efficient and readily accessible filing system, for when required by PB, TAC, UNDP, project consultants and other PMO staff;* • *Provide PMO-related administrative and logistical assistance.*
Project Accountant/Finance Assistant/Finance officer *Rate: $1,200/week*	*30 weeks / over 5 years*	*Duties and Responsibilities* • *Keep records of project funds and expenditures, and ensure all project-related financial documentation are well maintained and readily available when required by the Project Manager;* • *Review project expenditures and ensure that project funds are used in compliance with the Project Document and GoC financial rules and procedures;* • *Validate and certify FACE forms before submission to UNDP;* • *Provide necessary financial information as and when required for project management decisions;* • *Provide necessary financial information during project audit(s);* • *Review annual budgets and project expenditure reports, and notify the Project Manager if there are any discrepancies or issues;* • *Consolidate financial progress reports submitted by the responsible parties for implementation of project activities;* • *Liaise and follow up with the responsible parties for implementation of project activities in matters related to project funds and financial progress reports.*
International / Regional and global contracting		
International Cooperation officer *Rate: $700/day*	*20 days/ over 5 years*	*Duties and Responsibilities* • *Carry out international cooperation and exchange.*
Knowledge Management Officer *Rate: $3500/week*	*10 weeks / over 5 years*	*Duties and Responsibilities* • *Develop a communication strategy.* • *Develop a knowledge management plan for the project.* • *Lead the implementation of the approved communications strategy action plan.* • *Raise awareness of the project with target audiences for all project components.* • *Build on the project's digital presence through social media.* • *Produce quarterly newsletters or info-graphic for the project and disseminates to all project partners.*

Consultant	Time Input	Tasks, Inputs and Outputs
		• *Draft media advisories and press releases* • *Liaise with media for press releases and events, establish media connections and aim to get maximum media attention for press releases, special events, press interview either through broadcast or print media.* • *Compile contents for all planned communication efforts* • *Generate ideas and lead development and design of various publicity materials, including web content, brochures, and short stories.* • *Assist in collecting 'most significant change' stories to support annual reporting and editing project specific and related news updates.* • *Regularly update on the impact of the communication activities, including media outreach and online efforts* • *Share communication products and updates with UNDP CO, for regional and global visibility of the project.* • *Any other tasks related to communication activities in the project*
For Technical Assistance		
Component 1		
Local / National contracting		
Policy researcher *Rate: $1,250/week*	*30weeks/ over 5 years*	*Duties and Responsibilities* • Analyze the existing green energy-saving policies of phosphorus chemical industry. • Provide support for • research on existing policy and institutional frameworks in other countries. • Provide new green energy-saving policy suggestions. • Prepare and submit relevant analysis and evaluation reports. • *Conduct of research to update existing standards, policies and implementing rules and regulations to promote green technologies in the development and operations of phosphate mines.* • *Formulation and recommendation of new standards/technical guidelines and implementing rules and regulations for promoting and adopting green technologies in phosphate mines.*
Project Safeguards Officer *Rate: $1,250/week*	*12 weeks / over 5 years*	*Duties and Responsibilities* • *Monitor progress in development/implementation of the project ESMP/ESMF ensuring that UNDP's SES policy is fully met, and the reporting requirements are fulfilled;* • *Oversee/develop/coordinate implementation of all safeguard related plans;* • *Ensure social and environmental grievances are managed effectively and transparently;* • *Review the SESP annually, and update and revise corresponding risk log; mitigation/management plans as necessary;* • *Ensure full disclosure with concerned stakeholders;* • *Ensure environmental and social risks are identified, avoided, mitigated, and managed throughout project implementation;* • *Work with the M&E officer to ensure reporting, monitoring and evaluation fully address the safeguard issues of the project.*
International / Regional and global contracting		
Policy researcher *Rate: $700/day*	*50 days/ over 5 years*	*Duties and Responsibilities* • *Conduct of research studies on existing policies and institutional frameworks in other countries that can be used as*

Consultant	Time Input	Tasks, Inputs and Outputs
		reference for supporting and promoting low carbon development in phosrock mining and refining in China. • *Assessment of the ability and awareness level of relevant stakeholders and evaluation of needs and gaps to fill about green, energy efficient low carbon development of phosphate mining and refining in Guizhou, Yunnan, and Sichuan Provinces.*
Component 2		
Local / National contracting		
Policy and Standard Researcher *Rate: $1,250/wee*	*40 weeks / over 5 years*	*Duties and Responsibilities* • *Provide support for the Conduct of dual-carbon target-oriented analysis of the energy-savings, and green and low-carbon development path in phosphate chemicals production.* • *Provide support for provide suggestions on new policies and institutional frameworks for green, energy-saving and low-carbon development in waste management in the phosphate chemical production sub-sector.*
Project Safeguards Officer *Rate: $1,250/week*	*12 weeks / over 5 years*	*Duties and Responsibilities* • *Monitor progress in development/implementation of the project ESMP/ESMF ensuring that UNDP's SES policy is fully met, and the reporting requirements are fulfilled;* • *Oversee/develop/coordinate implementation of all safeguard related plans;* • *Ensure social and environmental grievances are managed effectively and transparently;* • *Review the SESP annually, and update and revise corresponding risk log; mitigation/management plans as necessary;* • *Ensure full disclosure with concerned stakeholders;* • *Ensure environmental and social risks are identified, avoided, mitigated, and managed throughout project implementation;* • *Work with the M&E officer to ensure reporting, monitoring and evaluation fully address the safeguard issues of the project.*
International / Regional and global contracting		
Technical Adviser *Rate: $700/day*	*100 days/ over 5 years*	*Duties and Responsibilities* • *Complete the goal-oriented analysis of the project.* • *Carry out international cooperation and exchange.* • *Prepare and submit relevant analysis and evaluation reports.* • *Provide support for the feasibility of the design of green, energy-saving and low-carbon technology demonstration projects in the production of phosphate chemicals.*
Project Safeguards Officer *Rate: $700/day*	*20 days/ over 5 years*	• *Implementation of an international comparative study on energy conservation and green and low-carbon development policies in phosphate chemicals production*
Component 3		
Local / National contracting		
Policy and Standard Researcher	*200 days/ over 5 years*	*Duties and Responsibilities* • *Analyze the existing green energy-saving policies of phosphorus chemical industry.*

Consultant	Time Input	Tasks, Inputs and Outputs
Rate: $250/day		• *Provide new green energy-saving policy suggestions.* • *Prepare and submit relevant analysis and evaluation reports.* • *Conduct of research to update existing policies and strengthen existing institutional mechanisms in China to enable the greening of waste management in the phosphate mining and refining sub-sector).* • *Conduct of research studies on existing policies and institutional frameworks in other countries that can be used as reference for supporting and promoting low carbon development in waste management in the phosphate mining and refining sub-sector.* • *Formulation and recommendation of new policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate mining and refining sub-sector.* • *Formulation and recommendation of new relevant schemes, standards，technical guidance，implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate mining and refining sub-sector.*
Project Safeguards Officer *Rate: $1,250/week*	*12 weeks / over 5 years*	*Duties and Responsibilities* • *Monitor progress in development/implementation of the project ESMP/ESMF ensuring that UNDP's SES policy is fully met, and the reporting requirements are fulfilled;* • *Oversee/develop/coordinate implementation of all safeguard related plans;* • *Ensure social and environmental grievances are managed effectively and transparently;* • *Review the SESP annually, and update and revise corresponding risk log; mitigation/management plans as necessary;* • *Ensure full disclosure with concerned stakeholders;* • *Ensure environmental and social risks are identified, avoided, mitigated, and managed throughout project implementation;* • *Work with the M&E officer to ensure reporting, monitoring and evaluation fully address the safeguard issues of the project.*
International / Regional and global contracting		
Technical Adviser *Rate: $700/day*	*50 days/ over 5 years*	*Duties and Responsibilities* • *Design and development of information management and sharing system.* • *Provide support for the feasibility of applying green and low-carbon waste management* • *Develop and approve engineering designs of green demos for waste management in the phosphate chemicals production sub-sector to increase energy efficiency processing of phosphogypsum and marketing of the gypsum product.* • *Formulate and recommend new relevant schemes, standards，technical guidance，implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate chemicals production sub-sector.*
Knowledge Management & Monitoring & Evaluation		
Project Gender Officer *Rate: $1,250/week*	*15 weeks / over 5 years*	*Duties and Responsibilities* • Monitor progress in implementation of the project Gender Action Plan ensuring that targets are fully met, and the reporting requirements are fulfilled; • Oversee/develop/coordinate implementation of all gender-related work;

Consultant	Time Input	Tasks, Inputs and Outputs
		• Review the Gender Action Plan annually, and update and revise corresponding management plans as necessary; • Work with the M&E officer and Safeguards Officer to ensure reporting, monitoring and evaluation fully address the gender issues of the project;
Project Monitoring and Evaluation Officer *Rate: $1,250/week*	*15 weeks / over 5 years*	*Duties and Responsibilities* • *Monitor project progress and participate in the production of progress reports ensuring that they meet the necessary reporting requirements and standards;* • *Ensure project's M&E meets the requirements of the Government, the UNDP Country Office, and UNDP-GEF; develop project-specific M&E tools as necessary;* • *Oversee and ensure the implementation of the project's M&E plan, including periodic appraisal of the Project's Theory of Change and Results Framework with reference to actual and potential project progress and results;* • *Oversee/develop/coordinate the implementation of the stakeholder engagement plan;* • *Oversee and guide the design of surveys/ assessments commissioned for monitoring and evaluating project results;* • *Facilitate mid-term and terminal evaluations of the project; including management responses;* • *Facilitate annual reviews of the project and produce analytical reports from these annual reviews, including learning and other knowledge management products;* • *Support project site M&E and learning missions;* • *Visit project sites as and when required to appraise project progress on the ground and validate written progress reports.*
Project Social and Environmental Safeguards Officer *Rate: $1,250/week*	*15 weeks / over 5 years*	*Duties and Responsibilities* • *Monitor progress in development/implementation of the project ESMP/ESMF ensuring that UNDPs SES policy is fully met, and the reporting requirements are fulfilled;* • *Oversee/develop/coordinate implementation of all safeguard related plans;* • *Ensure social and environmental grievances are managed effectively and transparently;* • *Review the SESP annually, and update and revise corresponding risk log; mitigation/management plans as necessary;* • *Ensure full disclosure with concerned stakeholders;* • *Ensure environmental and social risks are identified, avoided, mitigated and managed throughout project implementation;* • *Work with the M&E officer to ensure reporting, monitoring and evaluation fully address the safeguard issues of the project;*
International / Regional and global contracting		
Lead Technical Advisor *Rate: $2,000/week*	*20 weeks / over 5 years*	*The Lead Technical Adviser will be responsible for providing overall technical backstopping and management support to the Project.*

Annex 7: Stakeholder Engagement Plan

This proposed plan describes how the project's key stakeholders will be engaged during the project implementations. It should be note that this stakeholder engagement plan may be further elaborated or revised during the project's inception period to accommodate possible changes in stakeholders and will be finally approved at the project inception workshop. During the project development process (PPG phase), a stakeholder analysis was undertaken to identify key stakeholders, assess their interests in the project and define their roles and responsibilities in its implementation.

The roles of key stakeholders in the project' oversight and management, including the project steering committee is provided in the **Governance and Management Arrangements** section of the Project Document (i.e. Project Board members, Project Management Office), and are only briefly mentioned here.

Table 1 below summarizes the main engagements of the key stakeholders in the implementation of the project. It also shows the capacity building needs of each stakeholder as identified during the project development phase.

Table 1: Stakeholder Engagement and Training and Capacity Building Needs

Stakeholder	Engagement in the Project	Training and Capacity Building Needs
Ministry of Finance (MoF)	As the GEF Operational Focal Point MOF will receive the GEF grant for this project on behalf of the Chinese Government. MOF will be a core member of the Project Steering Committee. MOF will have a core role in providing guidance, coordination and supervision on the disbursement and expense of the GEF grant, performance evaluation, documentation, and promotion of project results. The project team will provide project documentation and reporting to MOF through UNDP.	There is a need to sensitize relevant staff and departments within MOF on the importance and necessity of funding efforts to achieve greener, energy efficient and low carbon operations in the PCI, and the benefits of doing this towards contributing to the achievement of country's NDC targets and improvements in the socio-economic development at the national and local levels.
United Nations Development Programme (UNDP)	As the GEF Agency for the project it is overall responsible for the project oversight. UNDP will ensure project quality control to ensure that the Implementing Partner achieve the set project objective and outcomes. As part of its oversight work, UNDP will ensure that the project implementation and management follows UNDP/GEF rules and regulations. UNDP is a core member of the Project Steering Committee. The project Implementing Partner will provide project documentation and reporting to UNDP as outlined in the project document or otherwise agreed.	
Ministry of Industry and Information Technology	MIIT is the project implementing partner and serves as a member of the project steering committee. MIIT will lead the implementation of the project's Component 2 activities, which are in phosphate chemicals	Within MIIT, it is necessary to carry out training on GEF project design, implementation and management, as well as on

(MIIT)	manufacturing; and part of the project's Component 3 activities on waste management in phosphate chemical production through its project office within the Energy Conservation and Comprehensive Utilization Division. It will ensure coordination between the relevant government agencies involved in the project at both national and local levels. MIIT will coordinate with the MNR, and supervise, the use of GEF funds allocated each year by the UNDP for all project components and carry out regular reporting of achievements in accordance with UNDP rules and procedures.	application of green and low carbon technologies that can bring about energy saving and emission reduction in PCI, particularly in phosphate chemicals production, as well as in the sustainable management of phosphate chemicals manufacturing waste.
Ministry of Natural Resources (MNR)	MNR is a responsible partner of the project and will assist the MIIT (project implementation partner). MNR will lead the implementation of the project's Component 1 activities, which are in phosphate mining and beneficiation; and part of the project's Component 3 activities in waste management in phosphate ore mining and refining through its project offices under the Division of International Cooperation and the Division of Mineral Resources Protection Supervision. It will assist the MIIT in ensuring close coordination between the relevant government agencies involved in the project at both national and local levels. MNR will coordinate with the MIIT the use of GEF funds allocated each year by the UNDP for Component 1 and part of Component 3 activities and carry out regular reporting of achievements through the MIIT in accordance with UNDP rules and procedures.	Within MNR, it is necessary to carry out training on GEF project design, implementation and management, as well as on application of green and low carbon technologies that can bring about energy saving and emission reduction in the PCI, particularly in phosphate ore mining and refining, as well as in the sustainable management of mining/refining waste.
National Development and Reform Commission (NDRC)	As it has done during the preparation phase of this project, the NDRC will participate in and provide guidance in the planning of the proposed project activities to ensure that the project outputs can guide and encourage the participating PCI entities to promote the comprehensive and conserving utilization of the country's phosphate ore resources, and also put forward suggestions for the planning and policy measures of resource conservation and comprehensive utilization in China.	No specific project related trainings are envisaged.
Ministry of Ecology and Environment (MEE)	As it has done during the PPG phase, MEE will participate in and guide the planning of this project activities during the implementation phase to ensure that the project outputs can provide intellectual support for China's participation in international climate change negotiations and the implementation of the United Nations Framework Convention on Climate Change (UNFCCC), and contributes to the achievement of the country's climate change mitigation and NDC targets.	No specific project related trainings are envisaged.
Local governments (Sichuan, Yunnan,	The Provincial departments of natural resources (DNRs) and industry and information technology (DIITs) are to provide financial support and technical and policy advice in the detailed design and implementation of the demos	The local government is the target of training for some of the project's capacity development activities. The training content

Guizhou, Hubei)	that will be carried out under the project. Municipal and county-level governments will directly participate in the project activities providing support (e.g., logistical) in the implementation and monitoring and evaluation of the demos and demo replications that will be carried out in their respective localities. During project implementation, the provincial DNRs and DIITs are responsible for supervising municipal and county-level work, including providing technical support, monitoring support, coordination with partners, and capacity-building.	includes, but not limited to, GEF project management, PCI knowledge, PCI energy conservation and emission reduction policies and technologies. At the same time, local government staff will also receive more training in the field.
Relevant Scientific Research Institutions	These are expected to provide technical support in the implementation of research, study, evaluation activities of the project, making use of their expertise and scientific knowledge in the areas of energy saving and green low-carbon technologies as applied to the comprehensive and sustainable utilization of phosphate ore resources in the PCI. These entities are expected to assist in clearly defining and verifying the baseline situation and market prospects regarding the PCI and its value chain. It is also expected to assist in coordinating the research activities of the project with ongoing and planned similar research work both at the national and provincial levels.	No specific project related trainings are envisaged.
Enterprises and Private Sector Organizations	These are expected to provide technical (research, engineering, construction, operation and maintenance) support to the project in the areas of low carbon and efficient comprehensive utilization of phosphate ore resources, cleaner and energy efficient production in the PCI, as well as circular economy type measures such as low-carbon reuse of phosphogypsum and phosphate mining/refining tailings. The participation of enterprises and the private sector in the project will ensure participation of the target beneficiaries, i.e., PCI entities, in guiding the efforts of the project to remove the financial and market barriers to greener, cleaner and more energy efficient PCI.	Enterprises and private sector organizations are the main beneficiaries of the project, and they will be the main targets of field training. The training they need to receive includes, but not limited to, GEF project management, PCI industry knowledge, PCI industry energy saving and emission reduction policies and technologies, and management, installation, operation and implementation of related facilities.
Non-governmental organizations	Non-governmental organizations at national, provincial and local levels shall participate in the stakeholder consultation processes during project implementation to ensure success of the promotional and capacity development activities of the project. They are also expected to facilitate effective coordination with the activities and projects of non-governmental organizations that contributes to the achievement of the project outcomes/outputs. These entities are also useful in the implementation of the capacity development and promotional activities of the project, where potential synergies can be used	No specific project related trainings are envisaged.

Based on the stakeholder analysis and the identified roles of each stakeholder, the following is the summary of the stakeholder plan, showing the means of engagement and engagement schedule.

Table 2: Stakeholder Plan

Project Stakeholder	Stakeholder Interest	Roles and Responsibilities in Implementation	Means of Engagement in the Project	Engagement Schedule during Implementation
GEF	GHG emission reduction	Project financing	Annual reporting	Yearly
UNDP	Sustainable Development	None. As GEF Agency, role is limited to Project oversight.	Regular consultation and follow-up meetings with implementing partner	Weekly
MIIT	Energy saving and emission reduction in phosphate chemicals production	MIIT is the project implementation partner and is the executive member of the project steering committee. It is responsible for the organization, management and implementation of the project activities, particularly those in phosphate chemicals production and waste management in that PCI sub-sector.	Project activity monitoring and reporting; daily project management, coordination with project partners, QPR and PIR reporting.	Daily
MNR	Energy saving and emission reduction in phosphate mining and beneficiation	MNR is a responsible partner of the project that will assist the MIIT in the project implementation and management. MNR is responsible for the organization, management and implementation of the project activities in phosphate mining, beneficiation and waste management in that PCI sub-sector.	Project activity monitoring and reporting; daily project management, coordination with project partners, QPR and PIR reporting.	Daily
National Development and Reform Commission (NDRC)	Promote national energy conservation and emission reduction	Provide suggestions for policies on energy saving and emission reduction in the PCI; and in the promotion of the implementation of the policies.	PSC Meeting	As scheduled

Project Stakeholder	Stakeholder Interest	Roles and Responsibilities in Implementation	Means of Engagement in the Project	Engagement Schedule during Implementation
Ministry of Ecology and Environment (MEE)	GHG emission reduction	Provide technical and policy support on climate change mitigation and GHG monitoring-related issues in the PCI.	PSC Meeting	As scheduled
Local governments, including Sichuan, Yunnan, Guizhou, Hubei etc.	Low carbon and energy saving development of local PCI industry	Participate in detailed project design and implementation, provide guidance and support in the implementation, monitoring and reporting of implemented project activities.	Project activity monitoring and reporting	Quarterly
Relevant scientific research institutions	The practice of PCI low-carbon energy-saving policy research and the best practice of PCI low-carbon energy-saving technology	Participation in the implementation of policy research and technical studies of the project.	Project activity monitoring and reporting	Quarterly
Enterprises and private sector organizations	Improve production efficiency, reduce energy consumption and reduce carbon emissions.	Detailed design, engineering, installation, operation and maintenance, as well as performance evaluation and reporting of demonstration activities	Implementation of Components 1, 2 and 3 activities.	As scheduled

Annex 8: Gender Analysis and Gender Action Plan

Gender equity has been mainstreamed on a number of components of the PhosChemEE Project and a range of activities have been designed and planned to ensure this. Table 1 below shows the success indicators and targets for these activities and a gender action plan with specific activities is listed in Table 2

Based on the initial surveys and assessment of potential gender issues of the project (as mentioned in the SESP), the PhosChemEE Project as a whole does not have too much of a negative impact on gender equity, and in some ways has a positive effect. Firstly, implementation of this project will improve the technical level and reduce the intensity of labor in the PCI, which will in turn improve the work environment, job adaptability and enhanced chances of job advancement for women who are in some form or another involved in the PCI. Education and professional training opportunities that the proposed project will provide to PCI employees will contribute to improving women's professional competitiveness in the workplace and expand women's scope of employment in the underdeveloped and agriculture-based regions of the country, some of which include the planned demo sites of the proposed project. Secondly, the implementation of this project will directly improve the quality of local ecological environment and indirectly improve women's and children's health. Thirdly, the implementation of this project will help ensure agriculture and food security through the reduction of rural pollution, and thus reduce the risk in women's major source of income in these regions.

However, since many of the on-the-ground activities the PhosChemEE Project are in the underdeveloped western regions of China, the potential negative impacts of the project may include potential occurrence of discriminations against women based on gender, especially regarding participation in the design and implementation of the project activities or access to opportunities and benefits. As a key measure to mitigate these impacts, the implementation of the PhosChemEE Project must ensure the conduct of a specific capacity development program on gender equality for the stakeholders in these regions of the country.

The professional job capacity development interventions of the project will be designed to ensure that there will be equal opportunities for all genders and ages, providing gender responsive capacity development trainings which account for gendered differences in capacity, determined through an initial assessment, as well as through the conduct of training activities at times, and in a manner that does not increase women's burden of work and that accounts for women's preferences in regards to how the training courses are delivered. By doing this, men and women in the project areas will benefit from these trainings, which will be designed with the view of rectifying any existing imbalances in technical qualifications of the individuals, and ultimately contributing to the improvement and promotion of gender equality. The training, and capacity building will improve the number of qualified women in China in technical areas such as the green low carbon transformation of phosphorus chemical industry, while enhancing the technique level and participation of women. The monitoring and evaluation of the project activities will include tracking a number of human development indicators, among them would be gender equity, as the number of trained women in the green low carbon transformation of phosphorus chemical industry.

Table 1 Gender-related Indicators and Targets

Indicators	Means of Verification
Number of local project developers, equipment suppliers and vendors trained on the design, installation, operation, and business models for the applications of low carbon and energy efficient technologies in PCI industries. The percentage of women in the mid-term project is **at least 25%**, and the female proportion of women is **at least 40%** at the end of the project.	Documentation on the training program including the profiles of the trainees; Documentation on the training course implementation including post training evaluation.
Number of enterprise level promoters, technicians, administrators trained on relevant subjects for successful management and operation of the low carbon and energy efficient technologies in PCI industries **(The female participation rate for training and publicity at the project implementation site is not less than 40%)**	Survey of enterprise technicians and administrators. Documentation on the training program including the profiles of the trainees; Documentation on the training course implementation including post training evaluation.
Number of trained people that are gainfully employed utilizing the knowledge and skills they learned from the training in newly created jobs. (The percentage of women in the mid-term project is **at least 20%**, and the female proportion of women is **at least 35%** at the end of the project)	Survey of trainees (survey conducted a year after the training program)

The targets for the above stated indicators will be set during the inception phase of the PhosChemEE Project. The implementing partner (Ministry of Industry and Information Technology (MIIT) and Ministry of Natural Resources (MNR)) will carry out surveys, which will cover project developers, equipment suppliers, technicians, and administrators, as well as local PCI enterprises. The end-user awareness data shall be collected through surveys and focus group discussions at the beginning and final year of the project. The results of survey will be on tracked and made public to monitor the achievement of gender equity mainstreaming in China, as well as enable timely detection of any gender-specific issues cropping up so that they are addressed promptly.

Table 2 PhosChemEE Gender Action Plan

Principles of Gender Equity	Gender Specific Activities	Indicators Included in M&E	Timeline	Possible Output/Impact
Capacity building, knowledge sharing and exchange	Ensure women representation in all capacity building workshops and trainings, as well as at any working group or committee for project implementation and monitoring. Identify design features that impact positively on women and implement them and remove those that negatively impact the project.	Number of staff (and % women) from national and local governments with enhanced capacity in climate management and zero carbon development, such as participating in the low/zero carbon energy technology project planning.	Year 1,2	A well represented governing structure that involves all groups of society is in place and participates meaningfully in the design of projects. All bottlenecks that prevent the full and meaningful participation of women and girls are identified and removed.
	Support women and girls (or children) to voice issues and concerns about decisions and process of the project that affect their lives.	The female participation rate for training, publicity, and consultation.	Year 1,2,3	Concerns of vulnerable groups, including women and girls, are addressed and their voices are integrated through participatory approaches during the implementation of the project activities. Additionally, this applies to any and all future low- or zero-carbon development initiatives and programs in China.
	Creation of a safe and inclusive environment for the vulnerable groups and women to enhance their participation.	Whether the training materials and project documents are developed with inclusive language.	Year 1,2	Significant women participation and representation are encouraged during capacity building trainings or workshops. Widely available training materials and project documents including written text as well as audio and visual developed with inclusive language (e.g. ethnic minority language) and appropriate illustrations without gender bias and stereotype.
	Improve information and knowledge	Number of trained people that	Year 1,2,3,4,5	Wider dissemination of information and

	access such as operation of low carbon and energy efficient technologies in PCI industries, financial services, market information, and techniques for the community (both males and females) by mobile, internet, and TV.	are applying the knowledge and skills they learned from the training in their current jobs. Number of trained people that are gainfully employed utilizing the knowledge and skills they learned from the training in newly created jobs.		knowledge related to low carbon and energy efficient technologies and zero carbon development in China's PCI industry.
Gender-balanced participation and female leadership	Facilitation of full representation of vulnerable groups including women, and their meaningful participation in projects and programs. Ensure adequate female representation is invited for surveys, focus groups and any training and workshops for the project. Both the project expert group and the project execution group contain a certain number of female members.	Number of staff (and % women) from national and local governments with enhanced capacity in climate management and zero carbon development, such as participating in the low/zero carbon energy technology project planning. The female participation rate for training, publicity, and consultation.	Year 1,2,3	This ensures that a well-represented expert and managing group is in place, as well as broad participation of vulnerable groups throughout project implementation.
	Improving women's access to new jobs created by low carbon and energy efficient technology application in PCI industries. Ensure the share of female labor force in new jobs created as a result of changing PCI industry's patterns.	Number of new employment opportunities created by the applications of low carbon and energy efficient technologies in PCI industries to bring about zero carbon development in PCI industries (with % of women)	Year 2,3,4,5	Women's time commitment to household chores may be reduced. Women and children's income and well-being improve due to new employment opportunities. The PCI industries have sufficient women working in them.
	Increasing the number of women receiving technical training and their participation in projects.	Number of enterprise level promoters, technicians, administrators, end-users trained	Year 1,2	Both men and women are familiar with and skilled in the operation and application of the low carbon and energy efficient

	Strengthening the technical capacity of women for low carbon and energy efficient technologies by providing training on maintenance and repair. Creation, or facilitation of the creation, of opportunities to empower and capacitate women including training needs are identified and tailor-made to suit projects being implemented. Ensure equitable allocation of resources for the implementation of appropriate implementation measures.	on relevant subjects for successful management and operation of low carbon and energy efficient technologies (The female participation rate for training and publicity at the project implementation site)		technologies designed for the project. Broad participation of vulnerable groups throughout project implementation.
	Strengthening sensitization of project stakeholders with regards to gender equality. Invest in developing competency and knowledge on gender mainstreaming for Government's counterparts, community leaders and partner agency as well as UNDP's project staff. Integration of gender session in each capacity-building training or workshop.	Whether gender equality mainstreamed in project activities, in particular in capacity-building training and workshops.	Year 1,2,3	Gender responsive policies are consistent and gender considerations are treated equally in all modules of the project and for all stakeholders. Targeted and trained stakeholders are well oriented on gender issues and more confident to mainstream gender throughout the whole project circle.
Gender Responsive Policy Coherence	Strengthening the regulation and supervision role of women's organizations in the project design and implementation processes.	Number of times women's organizations have been involved in project-related survey, consulting, training, workshop,	Year 1,2,3	Significant women participation and representation are encouraged during capacity building trainings or workshops. In particular, vulnerable groups are

	Invite representatives of women's organizations to participate in surveys and workshops related to project design and integrate their feedback into the project document. Strengthen the catalytic role of women's organizations in mobilizing women and girls in training such as low carbon and energy efficient technology training.	etc.		identified and involved in project activities.
	Designation and appointment of project staff to implement and monitor the gender mainstreaming activities accordingly and their performance appraisal is evaluated by accomplishment on the assigned gender tasks.		Year 1,2,3,4,5	
Monitoring, tracking, and reporting of policy implementation	Facilitating appropriate adjustments to activities and implementation approaches and accountability to gender equality.		Year 1,2,3,4,5	Gender balance practice is applied in project staff recruitment process and hire gender-oriented staff (both male and female) for project implementation. Project staff is assigned and designated to implement and monitor the gender mainstreaming activities accordingly and their performance appraisal is evaluated by accomplishment on the assigned gender tasks.
	Enable better planning and actions through disaggregated data.		Year 1,2,3,4,5	Data are collected and tabulated separately for women and men with specific indicators measuring changes to gender equality and empowerment for adjustments to activities

				and implementation approaches to better outcomes of gender equality. The case study or success story is collected as for gender-related findings at the intervention area to prove the results of gender mainstreaming. Evidence that adapts the better program accordingly, measure changes to gender equality, policy integration of gender considerations.

Annex 9: Procurement Plan

(for first year of implementation)

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
MIIT												
2	Services	Contract. Services – Companies	Design and evaluate the existing policies and institutional frameworks on energy conservation, green and low-carbon development in phosphate chemicals production from the perspective of the industrial chain to update and strengthen them.	Lumpsum	1	20000	20000	20000	July/22	August/22	September /22	July/23
2	Services	Contract. Services – Companies	Implementation of dual-carbon target-oriented analysis of the energy-savings, and green and low-carbon development path in phosphate chemicals production.	Lumpsum	1	5000	5000	5000	October/22	December/ 22	January/23	December/23
2	Services	Local consultant	A local consultant provide support for	day	20	125	2500	2500	October/22	December/ 22	January/23	December/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
		s	the Conduct of dual-carbon target-oriented analysis of the energy-savings, and green and low-carbon development path in phosphate chemicals production.									
2	Services	Training, Workshops and Confer	Organize conferences on the green and low-carbon development of the phosphorus chemical industry	lumpsum	1	7500	7500	7500	October/22	December/22	January/23	December/23
2	Services	Training, Workshops and Confer	Implementation of research on inter-regional green, and low-carbon development policies, cooperation and information sharing arrangements in phosphate chemicals production in key regions of China	Lumpsum	2	15000	30000	30000	July/22	August/22	September/22	July/23
2	Services	Training, Workshops and Confer	Organize conferences on the cross-industry energy saving, green and low-carbon policy seminars on phosphorus chemical industry to update and promote policy improvement	lumpsum	4	5000	20000	20000	July/22	August/22	September/22	July/23
2	Services	Contractual Services-Companie	Implementation of research on the inter-departmental and inter-industry policy	lumpsum	4	2500	10000	10000	July/22	August/22	September/22	July/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
		s	on energy conservation, and green and low-carbon development in the phosphorus chemical industry.									
2	Services	Contract. Services – Companies	Implementation of an international comparative study on energy conservation and green and low-carbon development policies in phosphate chemicals production.	lumpsum	1	17600	17600	17600	July/22	August/22	September /22	July/23
2	Services	Training, Workshops and Confer	Organize conferences on analyze the feasibility of new policies and institutional frameworks for the application of green, energy-saving and low-carbon technologies in the production of phosphate chemicals	lumpsum	1	5000	5000	5000	October/22	December/ 22	January/23	December/23
2	Services	Contractual Services-Companies	Formulate and recommend new policies and institutional frameworks on green, energy efficient low carbon technology application in phosphate chemicals production.	lumpsum	1	3800	3800	3800	October/22	December/ 22	January/23	December/23
2	Services	Training,	Organize conferences	lumpsum	1	10000	10000	10000	July/22	August/22	September	July/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
		Workshops and Confer	on phosphate chemical green technology training meeting								/22	
2	Services	Contractual Services-Companies	Implementation of research to update existing standards, policies and implementing rules and regulations to promote green technologies in phosphate chemicals production.	lumpsum	1	10000	10000	10000	July/22	August/22	September /22	July/23
2	Services	Contractual Services-Companies	Formulate, recommend and enforce new standards and technical guidelines for promoting green and low-carbon development in phosphate chemicals production.	lumpsum	1	1000	1000	1000	October/22	December/ 22	January/23	July/27
2	Services	Training, Workshops and Confer	Organize conferences on relevant standards and current rules and regulations of the phosphorus chemical industry to study and formulate new standards	lumpsum	1	3000	3000	3000	October/22	December/ 22	January/23	July/27
2	Services	Training, Workshops and Confer	Assess awareness levels of relevant stakeholders and assessment of needs to address in the area	lumpsum	4	1250	5000	5000	October/22	December/ 22	January/23	July/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			of green, energy efficient low carbon technologies in phosphate chemicals manufacturing in Yunnan, Guizhou, Sichuan, and Hubei Provinces.									
2	Services	Training, Workshops and Confer	Implementation of market analysis on the international trade trends for phosrock and phosphate chemicals, and the indirect carbon emission and ecological impacts in the international trading of these products.	lumpsum	3	2500	7500	7500	October/22	December/22	January/23	December/23
2	Services	Local consultants	A local consultant provide support for the feasibility of the design of green, energy-saving and low-carbon technology demonstration projects in the production of phosphate chemicals.	day	10	300	3000	3000	October/22	December/22	January/23	December/23
2	Services	Training, Workshops and Confer	Provide training on relevant capabilities for green, energy-saving and low-carbon technology demonstration	lumpsum	1	3000	3000	3000	October/22	December/22	January/23	December/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			projects in the production of phosphate chemicals									
2	Services	Contractual Services-Companies	Prepare an ESMF as well as an ESMP to manage and mitigate all identified risks associated with the design and implementation of the demo activities.	lumpsum	1	15000	15000	15000	October/22	December/22	January/23	December/23
2	Services	Training, Workshops and Confer	Organize conferences on existing financing mechanisms and evaluates needs and gaps in order to formulate financial plans.	lumpsum	1	2500	2500	2500	October/22	December/22	January/23	December/23
2	Services	Contractual Services-Companies	Evaluation of existing financing mechanisms and assessment of needs and gaps to develop financial schemes for financially support SMEs that manufacture phosphate chemicals.	lumpsum	1	5000	5000	5000	October/22	December/22	January/23	December/23
2	Services	Publication and printing	Publication and printing(training materials, brochures, etc.)	lumpsum	1	2500	2500	2500	July/22	August/22	September/22	July/23
2	Goods	Information Tech. Equipme	Communication and conferencing equipment (digital camera, overhead	Unit	8	650	5200	5200	July/22	August/22	September/22	July/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
		nt	projector and screen, etc.									
2	Goods	Supplies and stationery	Office Supplies	lumpsum	1	150	150	150	July/22	August/22	September /22	July/23
3	Services	Training, Workshops and Confer	Organize conferences on existing policies and institutional mechanisms for waste management in the phosphate chemical production sub-sector	lumpsum	1	2500	2500	2500	July/22	August/22	September /22	July/23
3	Services	Contractual Services-Companies	Implementation of research to update existing policies and strengthen existing institutional mechanisms in China to enable the greening of waste management in the phosphate chemicals production sub-sector.	lumpsum	1	5000	5000	5000	July/22	August/22	September /22	July/23
3	Services	Local consultants	A local consultant provide support for research on existing policy and institutional frameworks in other countries.	day	10	250	2500	2500	July/22	August/22	September /22	July/23
3	Services	Training, Workshops and Confer	Organize conferences on the research can be used as a reference to support and	lumpsum	1	2500	2500	2500	July/22	August/22	September /22	July/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			promote the low-carbon development of waste management in the phosphate chemical production sub-sector.									
3	Services	Local consultants	A local consultant provide support for provide suggestions on new policies and institutional frameworks for green, energy-saving and low-carbon development in waste management in the phosphate chemical production sub-sector.	day	10	250	2500	2500	October/22	December/22	January/23	December/23
3	Services	Training, Workshops and Confer	Organize conferences of new policies and institutional frameworks for green, energy-saving and low-carbon development in waste management in the phosphate chemical production sub-sector	lumpsum	1	2500	2500	2500	October/22	December/22	January/23	December/23
3	Services	Contractual Services-Companies	Formulate and recommend new policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate	lumpsum	1	5000	5000	5000	October/22	December/22	January/23	December/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			chemicals production sub-sector.									
3	Services	Local consultants	A local consultant provide support for the feasibility of applying green and low-carbon waste management technologies in the phosphate chemical production sub-industry.	day	20	250	5000	5000	July/22	August/22	September /22	July/23
3	Services	Training, Workshops and Confer	Organize conferences of discuss the impact of existing plans, standards and regulations on the promotion of green and low-carbon development in the phosphate chemical production sub-sector	lumpsum	1	5000	5000	5000	July/22	August/22	September /22	July/23
3	Services	Contractual Services-Companies	Implementation of research to update existing schemes, standards and regulations to promote the application of green and low carbon waste management technologies in the phosphate chemicals production sub-sector.	lumpsum	1	5000	5000	5000	July/22	August/22	September /22	July/23
3	Services	Local consultants	A local consultant provide support for the feasibility of a	day	10	250	2500	2500	July/22	August/22	September /22	July/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			cross-departmental, cross-sectoral and cross-industry policy linkage mechanism for waste management and comprehensive utilization of resources in the phosphate chemical production sub-sector									
3	Services	Contractual Services-Companies	Design and evaluate relevant departments and companies for waste management and comprehensive utilization of resources in the phosphate chemical production sub-sector to discuss and formulate a policy linkage mechanism	lumpsum	2	2500	5000	5000	July/22	August/22	September /22	July/23
3	Services	Contractual Services-Companies	Formulate and recommend new relevant schemes, standards technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate chemicals production sub-sector.	lumpsum	1	5000	5000	5000	October/22	December/ 22	January/23	December/23
3	Services	Contractu	Assess awareness	lumpsum	1	10000	10000	10000	October/22	December/	January/23	July/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
		al Services-Companies	levels of relevant stakeholders and assessment of needs and gaps to address in the area of waste management in the phosphate chemicals production sub-sector.							22		
3	Services	Contractual Services-Companies	Drafting and finalization of research reports on annual volume of waste production in phosphate chemical production sub-sector.	lumpsum	1	5000	5000	5000	October/22	December/22	January/23	December/23
3	Services	Contractual Services-Companies	Develop and approve engineering designs of green demos for waste management in the phosphate chemicals production sub-sector to increase energy efficiency processing of phosphogypsum and marketing of the gypsum product.	lumpsum	1	7500	7500	7500	July/22	August/22	September/22	December/23
3	Services	Contractual Services-Companies	Prepare and disseminate ESMF as well as an ESMP to manage and mitigate all identified risks associated with the design and implementation of the demo activities for waste management in	lumpsum	1	10000	10000	10000	October/22	December/22	January/23	December/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			the phosphate chemicals production sub-sector.									
3	Services	Training, Workshops and Confer	Other supplementary expenses for supplementary activities 'workshop	lumpsum	1	2000	2000	2000	July/22	August/22	September /22	July/23
4	Services	Local Consultants	A local consultant provide support for monitoring all risks and social environmental safeguard screening	lumpsum	1	6000	6000	6000	July/22	August/22	September /22	July/23
4	Services	Contractual services – Implementing Partner	Contractual services for inception workshop (M&E)and related management services.	lumpsum	1	37000	37000	37000	July/22	August/22	September /22	July/23
4	Goods	Information Tech. Equipment	Communication and conferencing equipment (digital camera, overhead projector and screen, etc.	Unit	5	1000	5000	5000	July/22	August/22	September /22	July/23
MNR												
1	Services	Local consultants	Conduct of research to update existing policies and strengthen existing institutional mechanisms in China to enable the greening of the phosphate mining and refining	day	10	500	5000	5000	July/22	August/22	September /22	June/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			sub-sector.									
1	Services	Local consultants	Formulation and recommendation of new policies on green, energy efficient low carbon development in phosrock mining and refining.	day	4	500	2000	2000	October/22	December/22	January/23	December/23
1	Services	Local consultants	Formulation and recommendation of new institutional frameworks on green, energy efficient low carbon development in phosrock mining and refining.	day	1	500	500	500	October/22	December/22	January/23	December/23
1	Services	Local consultants	Conduct of research to update existing standards, policies and implementing rules and regulations to promote green technologies in the development and operations of phosphate mines.	day	10	500	5000	5000	July/22	August/22	September/22	June/23
1	Services	Local consultants	Formulation and recommendation of new standards/technical guidelines and implementing rules and regulations for promoting and adopting green technologies in phosphate mines.	day	5	500	2500	2500	October/22	December/22	January/23	December/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
1	Services	Local consultants	Assessment of the ability and awareness level of relevant stakeholders and evaluation of needs and gaps to fill about green, energy efficient low carbon development of phosphate mining and refining in Guizhou, Yunnan, and Sichuan Provinces.	day	4	500	2000	2000	October/22	December/22	January/23	June/23
1	Services	Local consultants	Project chief and coordinator	day	80	500	40000	40000	July/22	August/22	September/22	June/23
1	Services	International consultants	Conduct of research studies on existing policies and institutional frameworks in other countries that can be used as reference for supporting and promoting low carbon development in phosrock mining and refining in China.	day	50	700	35000	35000	July/22	August/22	September/22	June/23
1	Services	Contractual Services-Companies	Conduct of research to update existing policies and strengthen existing institutional mechanisms in China	lumpsum	1	10000	10000	10000	July/22	August/22	September/22	June/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			to enable the greening of the phosphate mining and refining sub-sector.									
1	Services	Contractual Services-Companies	Conduct of research studies on existing policies and institutional frameworks in other countries that can be used as reference for supporting and promoting low carbon development in phosrock mining and refining in China.	lumpsum	1	10000	10000	10000	July/22	August/22	September /22	June/23
1	Services	Contractual Services-Companies	Formulation and recommendation of new policies on green, energy efficient low carbon development in phosrock mining and refining.	lumpsum	1	2500	2500	2500	October/22	December/ 22	January/23	December/23
1	Services	Contractual Services-Companies	Formulation and recommendation of new institutional frameworks on green, energy efficient low carbon development in phosrock mining and refining.	lumpsum	1	2500	2500	2500	October/22	December/ 22	January/23	December/23
1	Services	Contractual Services-Companies	Conduct of research to update existing standards, policies and implementing rules and regulations to promote green	lumpsum	1	5000	5000	5000	July/22	August/22	September /22	June/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			technologies in the development and operations of phosphate mines.									
1	Services	Contractual Services-Companies	Formulation and recommendation of new standards/technical guidelines and implementing rules and regulations for promoting and adopting green technologies in phosphate mines.	lumpsum	1	2500	2500	2500	October/22	December/22	January/23	December/23
1	Services	Contractual Services-Companies	Assessment of the ability and awareness level of relevant stakeholders and evaluation of needs and gaps to fill about green, energy efficient low carbon development of phosphate mining and refining in Guizhou, Yunnan, and Sichuan Provinces.	lumpsum	1	5000	5000	5000	October/22	December/22	January/23	June/23
1	Services	Contractual Services-Companies	Design and development of an online monitoring, reporting and verification (MRV) system, including the development of an information sharing mechanism, for	lumpsum	1	15000	15000	15000	July/22	August/22	September/22	June/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			energy-saving and GHG emission reduction for the entire PCI.									
1	Services	Contractual Services-Companies	Design and development of an up-to-date information management and exchange system on all aspects of low carbon phosphate mining and refining technologies and best practices.	lumpsum	1	7500	7500	7500	October/22	December/22	January/23	December/23
1	Services	Contractual Services-Companies	Development and approval of engineering designs of green demos for the mining and refining operations in the demo phosrock mines in Weng'an , Mabian, Guizhou and Sichuan	lumpsum	1	3000	3000	3000	October/22	December/22	January/23	December/23
1	Services	Contractual Services-Companies	Preparation of an ESMF as well as an ESMP to manage and mitigate all identified risks associated with the design and implementation of the demo activities.	lumpsum	1	1000	1000	1000	October/22	December/22	January/23	December/23
1	Services	Contractual Services-Companies	Evaluation of existing financing mechanisms and assessment of needs and gaps to fill for the development of schemes for	lumpsum	1	1000	1000	1000	October/22	December/22	January/23	December/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			financially supporting PCI mining and refining companies' green initiatives.									
1	Goods	Information Tech. Equipment	Communication and conferencing equipment (digital camera, overhead projector and screen, etc.	lumpsum	1	10,500	10,500	10,500	July/22	August/22	September /22	July/23
1	Goods	Supplies and stationery	Office Supplies	lumpsum	1	200	200	200	July/22	August/22	September /22	July/23
3	Services	Local consultants	Conduct of research to update existing policies and strengthen existing institutional mechanisms in China to enable the greening of waste management in the phosphate mining and refining sub-sector).	day	6	500	3000	3000	July/22	August/22	September /22	June/23
3	Services	Local consultants	Conduct of research studies on existing policies and institutional frameworks in other countries that can be used as reference for supporting and promoting low carbon development in waste	day	5	500	2500	2500	July/22	August/22	September /22	June/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			management in the phosphate mining and refining sub-sector.									
3	Services	Local consultants	Formulation and recommendation of new policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate mining and refining sub-sector.	day	2	500	1000	1000	October/22	December/22	January/23	December/23
3	Services	Local consultants	Conduct of research to update existing schemes, standards and regulations to promote the application of green and low carbon waste management technologies in the phosphate mining and refining sub-sector.	day	6	500	3000	3000	July/22	August/22	September/22	June/23
3	Services	Local consultants	Formulation and recommendation of new relevant schemes, standards technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization	day	3	500	1500	1500	October/22	December/22	January/23	December/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			technologies in the phosphate mining and refining sub-sector.									
3	Services	Local consultants	Assessment of the ability and awareness level of relevant stakeholders and assessment of needs and gaps to address in the area of waste management in the phosphate mining and refining sub-sector.	day	4	500	2000	2000	October/22	December/22	January/23	June/23
3	Services	Contractual Services-Companies	Conduct of research to update existing policies and strengthen existing institutional mechanisms in China to enable the greening of waste management in the phosphate mining and refining sub-sector).	lumpsum	1	5000	5000	5000	July/22	August/22	September/22	June/23
3	Services	Contractual Services-Companies	Conduct of research studies on existing policies and institutional frameworks in other countries that can be used as reference for supporting and promoting low carbon development in waste management in the phosphate mining and refining sub-sector.	lumpsum	1	5000	5000	5000	July/22	August/22	September/22	June/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
3	Services	Contractual Services-Companies	Formulation and recommendation of new policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate mining and refining sub-sector.	lumpsum	1	2500	2500	2500	October/22	December/22	January/23	December/23
3	Services	Contractual Services-Companies	Conduct of research to update existing schemes, standards and regulations to promote the application of green and low carbon waste management technologies in the phosphate mining and refining sub-sector.	lumpsum	1	5000	5000	5000	July/22	August/22	September/22	June/23
3	Services	Contractual Services-Companies	Conduct of research on the cross-ministry inter-departmental and cross-inter-industry policy linkage mechanisms for waste management and comprehensive utilization of resources in the phosphate mining and refining sub-sector.	lumpsum	1	5000	5000	5000	July/22	August/22	September/22	June/23
3	Services	Contractual Services-	Formulation and recommendation of new relevant	lumpsum	1	5000	5000	5000	October/22	December/22	January/23	December/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
		Companies	schemes, standards technical guidance, implementing rules and regulations for promoting and adopting green waste management and comprehensive resources utilization technologies in the phosphate mining and refining sub-sector.									
3	Services	Contractual Services-Companies	Assessment of the ability and awareness level of relevant stakeholders and assessment of needs and gaps to address in the area of waste management in the phosphate mining and refining sub-sector.	lumpsum	1	5000	5000	5000	October/22	December/22	January/23	June/23
3	Services	Contractual Services-Companies	Drafting and finalization of research reports on annual volume of waste production in the phosphate mining and refining sub-sector.	lumpsum	1	2500	2500	2500	October/22	December/22	January/23	December/23
3	Services	Contractual Services-Companies	Conduct of feasibility analysis to confirm the economic viability of the application and operation of new technologies for the improvement of green, energy efficient	lumpsum	1	5000	5000	5000	July/22	August/22	September/22	June/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			and low carbon in waste management in the phosphate mining and refining sub-sector.									
3	Services	Training, Workshops and Confer	Approval of the newly formulated and recommended policies and institutional framework on green, energy efficient low carbon development in waste management in the phosphate mining and refining sub-sector.	lumpsum	1	3000	3000	3000	October/22	December/22	January/24	March/24
3	Services	Training, Workshops and Confer	Development and approval of engineering designs of green demos for waste management in the phosphate mining and refining sub-sector to improve phosrock mining tailings recycling and reuse.	lumpsum	1	2500	2500	2500	October/22	December/22	January/23	December/23
3	Services	Training, Workshops and Confer	Preparation of an ESMF as well as an ESMP to manage and mitigate all identified risks	lumpsum	1	2500	2500	2500	July/22	August/22	September/22	December/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
			associated with the design and implementation of the demo activities for waste management in the phosphate mining and refining sub-sector.									
3	Services	Publication and printing	Publication and printing(training materials, brochures, etc.)	lumpsum	1	3000	3000	3000	July/22	August/22	September /22	July/23
3	Goods	Supplies and stationery	Office Supplies	lumpsum	1	200	200	200	July/22	August/22	September /22	July/23
4	Services	Local consultants	A local consultant provide support for monitoring all risks and social environmental safeguard screening	day	8	500	4000	4000	July/22	August/22	September /22	July/23
4	Services	Professional services	For monitoring all risks and social environmental safeguard screening	lumpsum	1	2500	2500	2500	July/22	August/22	September /22	July/23
4	Services	Contractual Services -Implementing Partner	Contractual services for inception workshop and related management services.	lumpsum	1	10000	10000	10000	July/22	August/22	September /22	July/23
4	Services	Training, Workshops and	For inception workshop.	lumpsum	1	1,800	1,800	1,800	July/22	August/22	September /22	July/23

Comp	Type of Supply	High Level Category	Description of goods, services or works	Unit of Measure	Qty	Est. Unit Price in USD	Est. Total Price in USD	Available budget in USD	Estimated start of Procurement	Est. completion of Procurement	Est. Start of Contract	Estimated Completion of Contract
		Confer										

Annex 10: GEF focal area specific annexes

(See Supplementary Annexes that include the Description of Demonstrations (Annex 10A), and GHG Emission Reduction Estimates (Annex 10B)

Annex 11: Additional Agreements (Co-Financing Letters)

Co-Financing Letters can be accessed Here

Annex 12: GEF Core indicators

Core Indicator 6	**Greenhouse gas emission mitigated**				***(Metric tons of CO_2e)***
		Expected metric tons of CO_2e (6.1+6.2)			
		PIF stage	Endorsement	MTR	TE
	Expected CO_{2e} (direct)	35,961,000	35,987,000*	429,000	35,987,000
	Expected CO_{2e} (indirect)				
	Total Expected CO_{2e}	35,961,000	35,987,000*	429,000	35,987,000
Indicator 6.1	Carbon sequestered or emissions avoided in the AFOLU sector				
		Expected metric tons of CO_2e			
		PIF stage	Endorsement	MTR	TE
	Expected CO_{2e} (direct)	N.A.	N.A.	N.A.	N.A.
	Expected CO_{2e} (indirect)	N.A.	N.A.	N.A.	N.A.
	Anticipated start year of accounting	N.A.	N.A.	N.A.	N.A.
	Duration of accounting	N.A.	N.A.	N.A.	N.A.
Indicator 6.2	Emissions avoided Outside AFOLU				
		Expected metric tons of CO_2e			
		Expected		Achieved	
		PIF stage	Endorsement	MTR	TE
	Expected CO_{2e} (direct)	35,961,000	35,987,000*		
	Expected CO_{2e} (indirect)				
	Anticipated start year of accounting		*2024*		
	Duration of accounting		*2027*		
Indicator 6.3	Energy saved				
		MJ			
		Expected		Achieved	
		PIF stage	Endorsement[27]	MTR	TE
			433.1x10[12]		
Indicator 6.4	Increase in installed renewable energy capacity per technology[28]				
	Technology	Capacity (MW)			
		Expected		Achieved	
		PIF stage	Endorsement	MTR	TE
Core Indicator 11	**Number of direct beneficiaries disaggregated by gender as co-benefit of GEF investment**				***(Number)***
		Number			
		Expected		Achieved	
		PIF stage	Endorsement	MTR	TE
	Female	600,000	600,000	150,000	600,000
	Male	60K -1,000K	600,000-1,000,000	150K -250K	600K-1,000K
	Total	1,200K-1,600K	1,200,000-1,600,000	300K-400K	1,200K-1,600K

*Cumulative by end of the five-year project implementation period.

[27] Based on the EOP target of 14.779 Mtce. 1 tce = 29,307 GJ https://www.planete-energies.com/en/content/ton-coal-equivalent-tce

[28] Energy savings in the PhosChemEE project are not obtained by installing RE technologies, but by improving the EE of the processes.

Annex 13: GEF 7 Taxonomy

Level 1	Level 2	Level 3	Level 4
☒ **Influencing models**			
	☒**Transform policy and regulatory environments**		
	☒**Strengthen institutional capacity and decision-making**		
	☒**Convene multi-stakeholder alliances**		
	☒ **Demonstrate innovative approaches**		
	☒**Deploy innovative financial instruments**		
☒ **Stakeholders**			
	☒**Indigenous Peoples**		
	☒**Private Sector**		
		☐Capital providers	
		☐Financial intermediaries and market facilitators	
		☐Large corporations	
		☒SMEs	
		☒Individuals/ Entrepreneurs	
		☐Non-Grant Pilot	
		☐Project Reflow	
	☐**Beneficiaries**		
	☒**Local Communities**		
	☒**Civil Society**		
		☒Community Based Organization	
		☒Non-Governmental Organization	
		☒Academia	
		☐Trade Unions and Workers Unions	
	☒**Type of Engagement**		
		☒Information Dissemination	
		☒Partnership	
		☒Consultation	
		☐Participation	
	☒**Communications**		
		☒Awareness Raising	
		☒Education	
		☒Public Campaigns	
		☐Behavior Change	

☒**Capacity, Knowledge and Research**			
	☐**Enabling Activities**		
	☒**Capacity Development**		
	☒**Knowledge Generation and Exchange**		
	☒**Targeted Research**		
	☒**Stakeholder Engagement Plan**		
☒ **Gender Equality**			
	☒ **Gender Mainstreaming**		
		☒Beneficiaries	
		☐Women groups	
		☐Sex-disaggregated indicators	
		☐Gender-sensitive indicators	
☒**Focal Areas/Theme**			
	☒ **Climate Change**		
		☒ **Climate Change Mitigation**	
			☐Agriculture, Forestry, and other Land Use
			☒ Energy Efficiency
			☐Sustainable Urban Systems and Transport
			☐Technology Transfer
			☐Renewable Energy
			☒Financing
			☐Enabling Activities
		☒**Climate Finance (Rio Markers)**	
			☒Climate Change Mitigation 1
			☐Climate Change Mitigation 2
			☐Climate Change Adaptation 1
			☐Climate Change Adaptation 2